FOUNDATIONS OF

QUANTUM MECHANICS

FOR CHEMISTRY AND PHYSICS STUDENTS

A FIRST COURSE WITH LECTURE NOTES, WORKED EXAMPLES,
PROBLEM SETS AND COMPLETE SOLUTIONS MANUAL

PART I:
QUANTUM MECHANICS FUNDAMENTALS

PART II:
QUANTUM CHEMISTRY FOUNDATIONS

THIRTY-THREE SECTIONS, FROM THE FUNDAMENTALS
TO ADVANCED TOPICS IN QUANTUM MECHANICS

$$i\hbar\frac{\partial\Psi}{\partial t}=\hat{H}\Psi$$

THERMODYNAMICS

$$dU = TdS - PdV + \mu dN$$

$$F = U - TS$$

$$dG = -SdT + VdP + \mu dN$$

Maxwell Relations

ULTRA

PHYSICAL CHEMISTRY

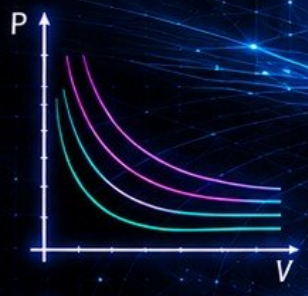
— THE UNITY OF NATURE —

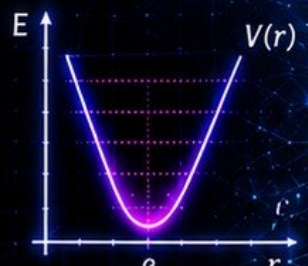
$$Q=\sum_i g_i e^{-E_i/k_BT}$$

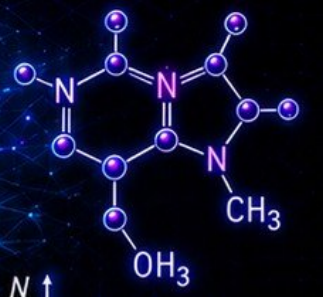
STATISTICAL MECHANICS

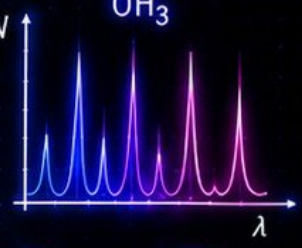
$$Z=\sum_i e^{-\beta E_i}$$

$$\langle A\rangle=\frac{1}{Z}\sum_i A_i e^{-\beta E_i}$$









QUANTUM MECHANICS · MOLECULAR STRUCTURE · SPECTROSCOPY · STATISTICAL MECHANICS · KINETICS · ELECTROCHEMISTRY · KINETICS · COMPUTATIONAL CHEMISTRY · QUANTUM COMPUTING

33 SECTIONS | 2 PARTS | 1 UNITY

ALIREZA MALEKIFAR

RIGOR • UNITY • INSIGHT • HARMONY

Quantum Mechanics

A First Course for Chemistry and Physics Students

Lecture Notes • Worked Examples • Problem Sets • Complete Solutions Manual

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September 1, 2026

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Quantum Mechanics: A First Course for Chemistry and Physics Students

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First edition.

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A note on preparation. This volume was drafted under the direction and authorship of Alireza Malekifar, which researched current reference values, structured the material, and produced the manuscript text, figures, and problem sets under his editorial control. As with any text of this scope, instructors and students are encouraged to verify derivations and results independently.

Physical constants throughout this text follow the CODATA 2022 recommended values (Mohr, Newell, Taylor & Tiesinga, *Rev. Mod. Phys.* **97**, 025002, 2025).

Preface

Quantum mechanics has a reputation for being impossibly difficult. Part of that reputation is deserved: the theory asks us to abandon deeply held intuitions about what a "particle" is, what it means to "know" something about a physical system, and even what a probability is for. But a great deal of the reputation is undeserved, and comes instead from how the subject is often taught — as a wall of formalism erected before the reader has any concrete sense of what problem the formalism is solving.

This book takes a different route. Every new idea is introduced because a specific, concrete question demands it. We do not begin with Hilbert spaces and postulates; we begin with a handful of stubborn experimental facts that classical physics could not explain, and we let the mathematics of quantum mechanics emerge, piece by piece, as the tool required to explain them. By the time the formal postulates are stated in full generality (Chapter 9), you will already have used every one of them informally, in a concrete setting, several times over.

Who this book is for

This text is written for undergraduate chemistry and physics students meeting quantum mechanics for the first time. No prior exposure to the subject is assumed. The mathematical prerequisites are deliberately kept light: a working knowledge of calculus (derivatives, integrals, and simple differential equations) is the main requirement, and Chapter 2 reviews, from scratch, every additional mathematical tool used later in the book — complex numbers, vectors, matrices, and basic probability. If you can differentiate $\sin(kx)$ and you are willing to be patient with algebra, you have what you need.

How the book is organized

- **Part I (Chapters 1–3)** motivates quantum mechanics historically and builds the minimal mathematical and physical vocabulary — the wavefunction, the Schrödinger equation, and the Born rule — needed to solve a real problem.
- **Part II (Chapters 4–8)** solves the canonical exactly-solvable systems of quantum mechanics — the particle in a box, tunneling, the harmonic oscillator, angular momentum, and the hydrogen atom — in full detail. These five systems are the working vocabulary of every later chapter and of quantum chemistry generally.
- **Part III (Chapters 9–13)** steps back to build the formal structure of the theory (operators, spin, identical particles) and the two workhorse approximation methods — perturbation theory and the variational method — that make quantum mechanics useful for problems too hard to solve exactly, which is to say, essentially all real ones.
- **Part IV (Chapters 14–17)** turns to chemistry directly: many-electron atoms, the chemical bond, molecular spectroscopy, and a modern outlook onto computational quantum chemistry and quantum information science.
- **Appendices A–E** collect physical constants, mathematical reference results, the periodic table with ground-state configurations, a glossary, and annotated further reading. **Appendix**

F adds a set of comprehensive review problems drawing on multiple chapters at once, useful once the whole book (or Part IV) is complete; **Appendix G** is a short note for instructors on structuring a course around this text; **Appendix H** develops the WKB approximation, a third general approximation method beyond perturbation theory (Chapter 11) and the variational method (Chapter 12), together with its classic application to alpha decay; **Appendix I** introduces molecular symmetry and group theory, connecting directly back to the vibrational spectroscopy of Chapter 16; **Appendix J** develops Ehrenfest's theorem and degenerate perturbation theory, closing a gap explicitly left open in Chapter 11, via the linear Stark effect in hydrogen; **Appendix K** introduces density matrices and mixed states, extending Chapter 10's spin formalism and Chapter 17's qubits to the more general, realistic setting relevant to decoherence in real quantum devices. **Appendix L** applies Chapter 4's particle-in-a-box and Chapter 10's Pauli principle to a real material property (the Fermi energy of a metal); **Appendix M** refines Chapter 13's rigid-rotor spectroscopy with the centrifugal distortion correction; and **Appendix N** collects every boxed Key Result in the book, by chapter, as a compact study reference.

- **The Complete Solutions Manual** at the end of the book gives a fully worked solution to *every* end-of-chapter problem, not just a final numerical answer.

Pedagogical features

Throughout the text you will encounter four recurring boxed elements:

Definition boxes state precisely what a new term or symbol means.

Worked Example boxes solve a complete problem end-to-end, showing every algebraic step rather than skipping "obvious" ones.

Key Result boxes isolate the equations most worth memorizing, separated from the surrounding derivation.

Note boxes flag a historical aside, a common misconception, or a connection to material outside the immediate chapter.

Each chapter closes with a problem set of graded difficulty: early problems in each set are direct applications of a single formula, and later problems combine ideas or ask you to reason qualitatively. Attempt every problem before consulting the solutions manual — the productive struggle is where the physics is actually learned.

A note on rigor

"Elementary" in this book describes the *starting point*, not the ceiling. Every derivation given is complete and honest: where an argument is exact, it is presented as exact; where it is an approximation, the approximation and its domain of validity are stated explicitly. Key numerical results in this text — eigenvalues, normalization constants, and worked-example answers — were checked by direct symbolic or numerical computation, not merely transcribed from memory, and physical constants follow the current (2022) CODATA-recommended values throughout. Simplicity and rigor are not in tension; the goal of every chapter is to make the exact result feel inevitable, not to hide behind a looser one.

Alireza Malekifar

Notation and Conventions

Symbol	Meaning
$\Psi(x,t)$	Time-dependent wavefunction
$\psi(x)$	Time-independent (stationary-state) wavefunction
$\lvert\psi(x)\rvert^2$	Probability density
$\hat{H}$, $\hat{p}$, $\hat{x}$, $\hat{L}$	Operators (hats denote operators throughout)
$\lvert\psi\rangle$, $\langle\psi\rvert$	Dirac ket and bra
$\langle\phi\lvert\psi\rangle$	Inner product of states ϕ and ψ
$\langle\hat{A}\rangle$	Expectation value of operator $\hat{A}$
$\hbar$	Reduced Planck constant, $h/2\pi$
E_n	Energy eigenvalue of state n
n, ℓ, m_ℓ, m_s	Principal, azimuthal, magnetic, and spin quantum numbers
a_0	Bohr radius
Z	Atomic number (nuclear charge in units of e)
$[\hat{A}, \hat{B}]$	Commutator, $\hat{A}\hat{B} - \hat{B}\hat{A}$
∇^2	Laplacian operator
c.c.	Complex conjugate of the preceding term
a.u.	Atomic units

Conventions used throughout this book:

- SI units are used by default; atomic units and electronvolts (eV) are introduced in Chapter 8 and used freely thereafter where they simplify the arithmetic, always with the convention stated explicitly.

- Operators are denoted with a hat ($\hat{H}$) on first use in a section and occasionally without one thereafter, once context makes the meaning unambiguous — standard practice that you will also meet in the research literature.

- Vectors are set in boldface ($\mathbf{r}$, $\mathbf{p}$); their magnitudes are italic ($r = |\mathbf{r}|$).

- Complex conjugation is denoted with a superscript asterisk (ψ^*).

- Equation, figure, table, and problem numbers are keyed to the chapter in which they appear (e.g., Eq. (4.7) is the seventh numbered equation of Chapter 4).

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Part I

Foundations and Motivation

Chapter 1

Why Quantum Mechanics? The Failure of Classical Physics

By the end of the nineteenth century, classical physics — Newton's mechanics, Maxwell's electromagnetism, and Boltzmann's statistical mechanics — appeared essentially complete. It described planetary motion, the behavior of gases, electricity, magnetism, and light with spectacular success. A handful of experimental results, however, refused to fit. Each seemed minor in isolation; together, over roughly twenty-five years (1900–1925), they forced physicists to abandon classical mechanics as a description of atomic-scale phenomena and to build an entirely new theory in its place. This chapter tells that story, not for its own sake, but because the experiments and the reasoning used to explain them are the most direct route into the concepts — quantization, wave-particle duality, and the uncertainty principle — that the rest of this book develops rigorously.

1.1 Blackbody Radiation and Planck's Quantum Hypothesis

Definition 1.1 — Blackbody

An idealized object that absorbs all electromagnetic radiation incident on it, and which, when in thermal equilibrium at temperature T, emits a characteristic spectrum of radiation — the *blackbody spectrum* — that depends only on T, not on the material.

A heated object glows: dull red, then orange, then white as its temperature rises. Classical thermodynamics could explain that the total power radiated increases with temperature (the Stefan–Boltzmann law, $P \propto T^4$), but predicting the detailed *spectrum* — how much power is emitted at each wavelength — proved far harder.

The classical calculation, due to Rayleigh and Jeans, treated the electromagnetic field inside a hot cavity as a collection of standing-wave modes, and invoked the equipartition theorem to assign an average energy $k_B T$ to each mode (k_B is the Boltzmann constant). Because the number of modes per unit wavelength interval grows without bound as $\lambda \to 0$, this leads to the prediction

$$B_\lambda^{\mathrm{RJ}}(T) = \frac{2ck_BT}{\lambda^4}, \tag{1.1}$$

for the spectral radiance (power emitted per unit area, per unit wavelength, per unit solid angle). Equation (1.1) diverges as $\lambda \to 0$: it predicts that an object at any temperature above absolute zero should radiate an infinite amount of energy into the ultraviolet and beyond. This absurd prediction became known as the *ultraviolet catastrophe*. Real blackbodies, of course, do no such thing — the emitted power rises with decreasing wavelength, reaches a maximum, and then falls back to zero (Figure 1.1).

In 1900, Max Planck found a formula that fit the observed spectrum perfectly:

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5}\,\frac{1}{e^{hc/\lambda k_B T} - 1}, \tag{1.2}$$

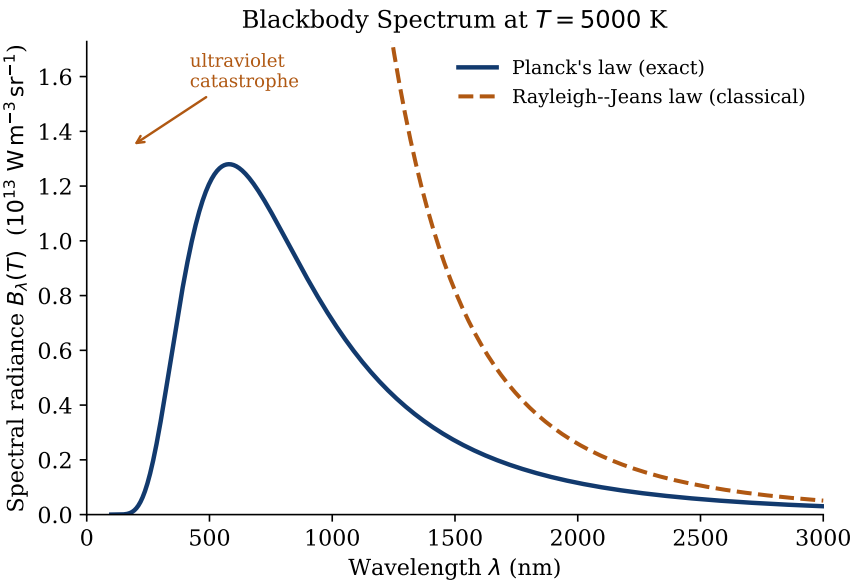


Figure 1.1: The classical Rayleigh–Jeans prediction, Eq. (1.1), diverges at short wavelength (the "ultraviolet catastrophe"). Planck's law, Eq. (1.2), matches both the classical prediction at long wavelength *and* the observed falloff at short wavelength.

where h is a new fundamental constant. Equation (1.2) reduces to the Rayleigh–Jeans law, Eq. (1.1), in the long-wavelength limit $hc/\lambda k_B T \ll 1$ (expand the exponential: $e^x - 1 \approx x$), but the exponential in the denominator suppresses short-wavelength radiation and removes the divergence entirely.

Planck initially regarded Eq. (1.2) as a fortunate empirical fit. To *derive* it, he found he had to make an assumption with no classical justification: that the oscillators making up the cavity walls could not exchange energy with the field continuously, but only in discrete packets,

$$E_n = nh\nu, \qquad n = 0, 1, 2, 3, \ldots, \tag{1.3}$$

where $\nu = c/\lambda$ is the frequency of the oscillator. Planck himself called this quantization "an act of desperation," undertaken because he knew the correct spectrum and worked backward to find an assumption that produced it. He did not yet suspect that Eq. (1.3) described something real and general about nature, rather than a peculiarity of cavity oscillators. That recognition came five years later, from Einstein.

Key Result 1.1 — Planck's Constant

$h = 6.62607015 \times 10^{-34}\ \mathrm{J\,Hz^{-1}}$ (exact, by definition of the SI second and kilogram).

The combination $\hbar \equiv h/2\pi = 1.054571817\ldots \times 10^{-34}$ J s ("h-bar") appears so often that it is treated as a fundamental constant in its own right.

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Note — Why is h exact?

Since the 2019 redefinition of the SI, the numerical value of h is fixed by definition (it is used to define the kilogram), rather than measured with some experimental uncertainty. The values of c, e, k_B, and N_A are exact for the same reason. Quantities such as the electron mass m_e or the fine-structure constant α, by contrast, are still measured, and their recommended values are periodically refined — most recently in the CODATA 2022 adjustment used throughout this book.

1.2 The Photoelectric Effect

If Planck's quantization was a mathematical trick applied to imaginary cavity oscillators, Einstein's 1905 paper on the photoelectric effect made quantization physical.

When light shines on a clean metal surface, electrons are sometimes ejected. Classically, light is a wave, and the energy carried by a wave is proportional to its intensity (amplitude squared), independent of its frequency. Classical electromagnetism therefore predicts:

- Increasing the light's *intensity* should increase the kinetic energy of the ejected electrons.
- Any frequency of light, given enough time or intensity, should eventually eject electrons.

Neither prediction survives contact with experiment. Instead:

- Increasing the intensity increases the *number* of ejected electrons, but not their maximum kinetic energy.
- Below a threshold frequency ν_0 (which depends on the metal), *no* electrons are ejected, no matter how intense the light or how long it shines.
- Above the threshold, the maximum kinetic energy of the ejected electrons increases *linearly with frequency*, and does so *instantaneously* — with no measurable delay, even at very low intensity.

Einstein explained all three observations with a radical proposal: light itself is quantized. A beam of light of frequency ν consists of discrete quanta — later named *photons* by Gilbert Lewis in 1926 — each carrying energy

$$E = h\nu = \frac{hc}{\lambda}. \tag{1.4}$$

A single photon is absorbed by a single electron in the metal. Some minimum energy, the *work function* Φ, is required to free the electron from the metal; any energy left over becomes the electron's kinetic energy:

$$KE_{\max} = h\nu - \Phi. \tag{1.5}$$

Key Result 1.2 — Einstein's Photoelectric Equation

$$KE_{\max} = h\nu - \Phi = \frac{hc}{\lambda} - \Phi,$$

where Φ is the work function of the metal (the minimum energy needed to remove an electron from its surface) and $\nu_0 = \Phi/h$ is the threshold frequency below which no photoemission occurs, regardless of intensity.

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Equation (1.5) explains every feature of the data. Increasing intensity means more photons per second, hence more ejected electrons, but each photon still delivers only $h\nu$ of energy, so $KE_{\max}$ is unchanged. Below threshold ($h\nu < \Phi$), a single photon simply cannot supply enough energy to free an electron, no matter how many photons arrive. And because absorption of a single photon is essentially instantaneous, there is no delay.

Worked Example 1.1 — Stopping Potential for Sodium

Ultraviolet light of wavelength $\lambda = 400$ nm is incident on a clean sodium surface, whose work function is $\Phi = 2.28$ eV. Find (a) the maximum kinetic energy of the ejected photoelectrons, (b) the stopping potential required to halt them, and (c) the threshold wavelength for sodium.

Solution. (a) The photon energy is

$$E_\gamma = \frac{hc}{\lambda} = \frac{(6.62607 \times 10^{-34}\ \text{J s})(2.99792 \times 10^{8}\ \text{m/s})}{400 \times 10^{-9}\ \text{m}} = 4.966 \times 10^{-19}\ \text{J} = 3.100\ \text{eV}.$$

By Eq. (1.5),

$$KE_{\max} = 3.100\ \text{eV} - 2.28\ \text{eV} = 0.82\ \text{eV}.$$

(b) The stopping potential V_0 is the retarding voltage at which even the fastest photoelectrons are just brought to rest, $eV_0 = KE_{\max}$, so

$$V_0 = 0.82\ \text{V}.$$

(c) The threshold wavelength corresponds to $KE_{\max} = 0$, i.e., $h\nu_0 = \Phi$:

$$\lambda_0 = \frac{hc}{\Phi} = \frac{1240\ \text{eV nm}}{2.28\ \text{eV}} \approx 544\ \text{nm}.$$

Light with a longer wavelength than 544 nm (dimmer red, or infrared) cannot eject a photoelectron from sodium no matter how intense it is — a purely quantum prediction with no classical counterpart.

Note — The handy combination hc

Because atomic and molecular energies are often quoted in electronvolts ($1\ \text{eV} = 1.602176634 \times 10^{-19}$ J, exactly) and wavelengths in nanometers, the combination $hc = 1239.84\ \text{eV nm}$ is worth memorizing. Photon energy in eV is then just $E_\gamma = 1239.84/\lambda(\text{nm})$.

1.3 Atomic Spectra and the Bohr Model

A third puzzle came from light emitted by atoms themselves. Heat a gas of atomic hydrogen and its emission is not a continuous rainbow but a series of sharp, discrete lines at specific wavelengths. In 1885, the Swiss schoolteacher Johann Balmer found that the four visible hydrogen lines then known obeyed a simple empirical formula, later generalized by Rydberg to

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right), \qquad n_2 > n_1,\ n_1, n_2 \in \mathbb{Z}^+, \tag{1.6}$$

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where $R_H \approx 1.097 \times 10^7\ \text{m}^{-1}$ is the Rydberg constant for hydrogen. Equation (1.6) fit every known hydrogen line to remarkable precision, yet nothing in classical physics explained why atomic emission should be quantized into discrete lines at all — an orbiting classical electron, accelerating as it circles the nucleus, should radiate continuously and spiral into the nucleus in about 10^{-11} seconds. Classically, stable atoms should not exist.

In 1913, Niels Bohr proposed a model that simply forbade this classical collapse by postulate. For an electron of mass m_e orbiting a proton (charge $+e$) in a circular orbit of radius r:

1. The electron can occupy only certain *stationary orbits*, in which it does not radiate, despite being accelerated.

2. These allowed orbits are exactly the ones in which the orbital angular momentum is quantized in integer multiples of $\hbar$:

$$L = m_e v r = n\hbar, \qquad n = 1, 2, 3, \ldots \tag{1.7}$$

3. An atom emits or absorbs a single photon only when an electron jumps between two stationary orbits, with the photon energy equal to the energy difference between them: $E_\gamma = |E_{n_2} - E_{n_1}|$.

Combining Eq. (1.7) with Newton's second law for circular motion under the Coulomb force, $m_e v^2/r = e^2/4\pi\epsilon_0 r^2$, and eliminating v, gives the allowed orbital radii and energies (Problem 1.9 asks you to complete this derivation in full):

$$r_n = n^2 a_0, \qquad a_0 \equiv \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} = 5.29177 \times 10^{-11}\ \text{m}, \tag{1.8}$$

$$E_n = -\frac{m_e e^4}{2(4\pi\epsilon_0)^2 \hbar^2}\,\frac{1}{n^2} = -\frac{13.6057\ \text{eV}}{n^2}. \tag{1.9}$$

The constant a_0 is the *Bohr radius*, and it sets the natural length scale of atomic physics: it is, remarkably, very close to the actual size of a hydrogen atom in its lowest-energy (ground) state. Equation (1.9), combined with postulate 3, reproduces the empirical Rydberg formula, Eq. (1.6), *exactly*, and predicts the correct numerical value of R_H from fundamental constants alone — a triumph that convinced many physicists to take quantization seriously as a feature of nature, not merely of light.

Key Result 1.3 — Bohr Energy Levels of Hydrogen

$$E_n = -\frac{13.6057\ \text{eV}}{n^2}, \qquad n = 1, 2, 3, \ldots$$

$n = 1$ is the ground state ($E_1 = -13.6057$ eV, the ionization energy of hydrogen); $n \to \infty$ corresponds to a free electron with $E = 0$.

Worked Example 1.2 — The Balmer H-α Line

Find the wavelength of the photon emitted when a hydrogen atom's electron falls from $n = 3$ to $n = 2$.

Solution. By Eq. (1.9), $E_3 = -13.6057/9 = -1.5117$ eV and $E_2 = -13.6057/4 = -3.4014$ eV.

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The emitted photon carries away the difference,

$$E_\gamma = E_3 - E_2 = 1.8897 \text{ eV}.$$

Using $\lambda = hc/E_\gamma = 1239.84 \text{ eV nm}/1.8897 \text{ eV}$,

$$\lambda \approx 656.1 \text{ nm},$$

a red line, historically labeled Hα. This is within about 0.06% of the textbook-quoted value of 656.3 nm — a gap worth taking seriously rather than rounding away. It is *not* numerical error: Eq. (1.9) was derived assuming an infinitely heavy, stationary nucleus, so that the electron alone orbits it. In reality the proton and electron orbit their common center of mass, which is accounted for by replacing m_e with the reduced mass $\mu = m_e m_p/(m_e + m_p) = 0.999456\, m_e$. This tiny correction lowers every hydrogen energy level by a factor of μ/m_e, shifting Hα to 656.47 nm in vacuum — matching the precisely measured value to four significant figures. (The often-quoted 656.3 nm is the line's wavelength *in air*, which differs from the vacuum wavelength because air's refractive index is slightly greater than 1.) We return to the reduced-mass correction quantitatively in Chapter 8.

Note — Where the Bohr model breaks down

The Bohr model gets the hydrogen energy levels right essentially by accident of hydrogen's simplicity: it fails badly for any atom with more than one electron, it cannot explain why spectral lines have the relative intensities they do, and its picture of electrons in well-defined circular orbits (with a definite position *and* velocity at every instant) directly contradicts the uncertainty principle developed in Section 1.6. We present it here because it is historically pivotal and because Eq. (1.9) happens to be exactly correct — but it is a way station, not a destination. Chapter 8 derives the same energies from the full Schrödinger equation, without any of the classically inconsistent assumptions, and additionally predicts the correct orbital shapes, degeneracies, and transition rates that the Bohr model cannot touch.

1.4 Wave-Particle Duality: The de Broglie Hypothesis

By 1923, light — long understood as a wave, on the strength of a century of interference and diffraction experiments — had been shown to also behave as a stream of particle-like photons, each with energy $E = h\nu$ and, as follows from special relativity for a massless particle, momentum

$$p = \frac{E}{c} = \frac{h\nu}{c} = \frac{h}{\lambda}. \tag{1.10}$$

Louis de Broglie asked the question the other way around: if waves can behave like particles, can particles behave like waves? He proposed that *every* material particle has an associated wavelength, obtained simply by inverting Eq. (1.10):

$$\lambda = \frac{h}{p} = \frac{h}{mv}. \tag{1.11}$$

Key Result 1.4 — de Broglie Wavelength

$$\lambda = \frac{h}{p}$$

applies to *any* object with momentum p — electrons, atoms, baseballs — not just photons.

The proposal seemed outlandish (a thrown baseball is a wave?) until 1927, when Clinton Davisson and Lester Germer fired a beam of electrons at a nickel crystal and observed a diffraction pattern — constructive and destructive interference of exactly the kind light produces when passed through a grating — with a wavelength matching Eq. (1.11) to within experimental error. Electrons, unambiguously particles in every prior experiment (they have a definite mass and charge, and leave point-like tracks in a detector), also diffract like waves.

Worked Example 1.3 — de Broglie Wavelength: Electron versus Baseball

Compare the de Broglie wavelength of (a) an electron accelerated from rest through a potential difference of 100 V, and (b) a 0.145 kg baseball thrown at 40 m/s.

Solution. (a) The electron gains kinetic energy $KE = eV = (1.602 \times 10^{-19}\ \text{C})(100\ \text{V}) = 1.602 \times 10^{-17}\ \text{J}$. Its momentum follows from $KE = p^2/2m_e$:

$$p = \sqrt{2m_e KE} = \sqrt{2(9.109 \times 10^{-31}\ \text{kg})(1.602 \times 10^{-17}\ \text{J})} = 5.403 \times 10^{-24}\ \text{kg m/s},$$

so

$$\lambda_{\text{electron}} = \frac{h}{p} = \frac{6.626 \times 10^{-34}}{5.403 \times 10^{-24}} = 1.226 \times 10^{-10}\ \text{m} = 0.1226\ \text{nm}.$$

This is comparable to a crystal's atomic spacing — precisely why electron beams of this energy diffract from crystals, exactly as in the Davisson–Germer experiment.

(b) The baseball's momentum is simply $p = mv = (0.145\ \text{kg})(40\ \text{m/s}) = 5.80\ \text{kg m/s}$, so

$$\lambda_{\text{baseball}} = \frac{h}{p} = \frac{6.626 \times 10^{-34}}{5.80} = 1.14 \times 10^{-34}\ \text{m}.$$

This is about 20 orders of magnitude smaller than a proton. No experiment could ever resolve a wave effect this small; the baseball's classical, particle-like trajectory is, for all practical purposes, exact. This is why we never notice matter waves in daily life: $\lambda \propto 1/p$, and everyday objects have enormous momentum compared to atomic-scale particles.

1.5 The Heisenberg Uncertainty Principle

If a particle genuinely has wave character, an immediate and profound consequence follows. A wave that is perfectly localized at a single, sharply-defined *momentum* (a single wavelength, by Eq. (1.11)) is an infinite sine wave, spread over all of space; conversely, a wave localized to a small region of space must be built from a superposition of many different wavelengths (Chapter 2 makes this precise via Fourier analysis). A particle cannot simultaneously have a perfectly well-defined position *and* a perfectly well-defined momentum, because "having a well-defined momentum" and "being localized in space" are, for a wave, mutually exclusive properties.

Werner Heisenberg made this quantitative in 1927:

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Key Result 1.5 — Heisenberg Uncertainty Principle

$$\Delta x\,\Delta p \geq \frac{\hbar}{2},$$

where Δx and Δp are the standard deviations of position and momentum for a particle's state. This is not a statement about measurement clumsiness or instrumental error — it is a fundamental property of any wave-like system, derived rigorously in Chapter 9. No experiment, however cleverly designed, can produce a state that beats this bound.

This single inequality already explains why atoms are stable — something classical physics, as noted in Section 1.4, cannot do at all.

Worked Example 1.4 — Why Atoms Do Not Collapse

Use the uncertainty principle to estimate the ground-state energy of a hydrogen atom, and compare with the Bohr value of -13.6057 eV.

Solution. Suppose the electron is confined within a region of size r around the proton, so that $\Delta x \sim r$. The uncertainty principle then requires a minimum momentum spread $\Delta p \sim \hbar/r$, and hence a minimum kinetic energy

$$KE(r) \sim \frac{(\Delta p)^2}{2m_e} = \frac{\hbar^2}{2m_e r^2}.$$

This is the crucial point: confining the electron to a smaller volume *costs* kinetic energy — the electron cannot simply sit motionless at the bottom of the Coulomb well, because "motionless *and* exactly at the origin" is precisely the combination the uncertainty principle forbids. The total energy is the sum of this confinement kinetic energy and the (attractive, negative) Coulomb potential energy:

$$E(r) = \frac{\hbar^2}{2m_e r^2} - \frac{e^2}{4\pi\epsilon_0 r}.$$

The true radius is whatever minimizes this competition between the kinetic energy's preference for a *large* r and the potential energy's preference for a *small* r. Setting $dE/dr = 0$:

$$-\frac{\hbar^2}{m_e r^3} + \frac{e^2}{4\pi\epsilon_0 r^2} = 0 \quad\Longrightarrow\quad r_{\min} = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2} = a_0,$$

which is exactly the Bohr radius, Eq. (1.8) — recovered here from the uncertainty principle alone, with no assumption about orbits. Substituting $r_{\min} = a_0$ back into $E(r)$:

$$E_{\min} = \frac{\hbar^2}{2m_e a_0^2} - \frac{e^2}{4\pi\epsilon_0 a_0} = -\frac{m_e e^4}{2(4\pi\epsilon_0)^2\hbar^2} = -13.6057\ \text{eV},$$

which reproduces the exact Bohr ground-state energy. This is a genuinely striking result: the "fuzziness" the uncertainty principle imposes on the electron is not an obstacle to understanding the atom — it *is* the reason the atom has the size and binding energy that it does. A classical point-particle electron, with no such kinetic-energy cost for confinement, would simply spiral into $r = 0$ and release infinite energy. Quantum mechanics forbids this outright.

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Note — A note on rigor

The derivation above uses $\Delta x \Delta p \sim \hbar$ (dropping the factor of $1/2$ and treating "$\sim$" loosely as "$=$") to turn an inequality into a single estimate we could minimize. That the answer comes out *exactly* right is a pleasant surprise special to hydrogen's $1/r$ potential, not a general feature of this style of estimate — for other potentials the same method gives only the correct order of magnitude. Chapter 8 derives $E_1 = -13.6057$ eV honestly, by solving the Schrödinger equation exactly, with no order-of-magnitude estimation anywhere in the argument.

1.6 Chapter Summary

- Classical physics fails to describe several basic phenomena: the blackbody spectrum (predicting an unphysical ultraviolet catastrophe), the photoelectric effect (predicting no threshold frequency and an intensity-dependent electron energy), the stability and discrete spectra of atoms, and, at bottom, the joint precision with which position and momentum can be known.

- Planck (1900) resolved the blackbody problem by positing that oscillator energy is quantized, $E_n = nh\nu$.

- Einstein (1905) resolved the photoelectric effect by positing that light itself is quantized into photons of energy $E = h\nu$, giving $KE_{\max} = h\nu - \Phi$.

- Bohr (1913) explained hydrogen's discrete spectrum by quantizing orbital angular momentum, $L = n\hbar$, yielding $E_n = -13.6057\ \text{eV}/n^2$ and reproducing the empirical Rydberg formula from first principles.

- de Broglie (1924) proposed that matter, not just light, has wave character: $\lambda = h/p$, confirmed by electron diffraction (Davisson–Germer, 1927).

- Heisenberg (1927) showed that wave-particle duality implies a fundamental limit on simultaneous knowledge of position and momentum, $\Delta x\, \Delta p \geq \hbar/2$ — which, far from being a mere technical caveat, is what keeps atoms from collapsing.

- Each of these results is an isolated patch on classical physics. Chapters 2 and 3 build the single mathematical framework — the Schrödinger equation — from which all of them, and much more, follow as consequences of one consistent theory.

Problems

1.1 A red laser pointer emits light at $\lambda = 650$ nm. Find the energy of a single photon, in both joules and electronvolts, and the frequency of the light.

1.2 A microwave oven operates at a frequency of 2.45 GHz. (a) Find the energy of one microwave photon in eV. (b) Explain, using Eq. (1.4), why microwave photons cannot eject photoelectrons from any ordinary metal (typical work functions are 2–5 eV), regardless of the oven's power.

1.3 The work function of cesium is $\Phi = 2.14$ eV, the lowest of any pure metal. (a) Find the threshold wavelength for the photoelectric effect in cesium. (b) Is cesium photoemissive under visible light (400–700 nm)? Justify your answer quantitatively.

1.4 Light of wavelength 250 nm strikes a metal surface, ejecting photoelectrons with a maximum kinetic energy of 2.30 eV. Find the work function of the metal, in eV.

1.5 Two different light sources, one twice as intense as the other but both above a metal's threshold frequency, illuminate identical samples of the metal. Explain, in one or two sentences and without new calculation, what differs between the two resulting photocurrents and what does not.

1.6 Using Eq. (1.9), find the energy (in eV) and wavelength (in nm) of the photon emitted in the hydrogen transition $n = 4 \to n = 2$. Identify which spectral series (Lyman, Balmer, or Paschen) this transition belongs to.

1.7 Find the ionization energy, in eV, required to remove an electron entirely from a hydrogen atom initially in its $n = 2$ state.

1.8 Using Eq. (1.8), find the radius of the $n = 3$ Bohr orbit of hydrogen, in both meters and angstroms ($1\ \text{Å} = 10^{-10}$ m).

1.9 Starting from Newton's second law for uniform circular motion under the Coulomb attraction, $m_e v^2/r = e^2/4\pi\epsilon_0 r^2$, and the Bohr quantization condition $m_e v r = n\hbar$, derive Eqs. (1.8) and (1.9) in full. (*Hint:* eliminate v between the two equations to find r_n, then substitute back to find the total energy $E_n = \frac{1}{2}m_e v^2 - e^2/4\pi\epsilon_0 r$.)

1.10 Find the de Broglie wavelength of a proton moving at 2.0×10^5 m/s, and compare it (as a ratio) with the de Broglie wavelength of an electron moving at the same speed. Which particle behaves "more classically" at this speed, and why?

1.11 An electron in a certain state is known to have its position measured with an uncertainty of $\Delta x = 0.050$ nm (roughly one atomic radius). Find the minimum possible uncertainty Δp in its momentum, and the corresponding minimum uncertainty in its speed, $\Delta v = \Delta p/m_e$. Comment on the size of Δv relative to typical atomic-electron speeds ($\sim 10^6$ m/s).

1.12 A π^0 meson has a rest mass roughly 264 times that of an electron and a mean lifetime of about 8.4×10^{-17} s before it decays. Treating this lifetime as an "uncertainty in time," Δt, and using the time-energy uncertainty relation $\Delta E\,\Delta t \gtrsim \hbar/2$ (a cousin of the position-momentum relation of Section 1.6, though time is treated differently from position in quantum mechanics, since it is a parameter rather than an operator with its own eigenstates), estimate the minimum intrinsic uncertainty in the meson's rest energy, in eV. Compare this to its rest energy of about 135 MeV: is this an enormous or a minuscule fractional uncertainty?

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Chapter 2

The Mathematical Language of Quantum Mechanics

Quantum mechanics is often experienced as difficult not because any single mathematical idea in it is hard, but because several moderately unfamiliar ideas — complex numbers, vectors in abstract spaces, eigenvalue problems, and probability distributions — are needed simultaneously. Each of these, on its own, is standard undergraduate mathematics; this chapter builds every one of them from scratch, assuming only that you can differentiate and integrate simple functions. If you already know some of this material, skim for notation; if none of it is familiar, read slowly and work every margin calculation yourself before checking it against the text. Nothing here is more difficult than material you have already seen in a first calculus course — it is simply new vocabulary applied to old skills.

2.1 Complex Numbers

Classical physics — Newtonian mechanics, electromagnetism, thermodynamics — can be, and traditionally is, formulated entirely with real numbers. Quantum mechanics cannot: the wavefunction Ψ introduced in Chapter 3 is fundamentally complex-valued, not as a computational convenience but as a structural necessity (Problem 3.11 shows directly why a real-valued wavefunction cannot satisfy the Schrödinger equation for a free particle). This section reviews what you need.

Definition 2.1 — Imaginary and Complex Numbers

The imaginary unit i is defined by $i^2 = -1$ (equivalently $i = \sqrt{-1}$). A *complex number* z is written

$$z = a + bi, \qquad a, b \in \mathbb{R},$$

where $a = \mathrm{Re}(z)$ is the *real part* and $b = \mathrm{Im}(z)$ is the *imaginary part*. The set of all such z is denoted $\mathbb{C}$.

Complex numbers add and multiply exactly like ordinary binomials, using $i^2 = -1$ to simplify:

$$(a + bi) + (c + di) = (a + c) + (b + d)i, \tag{2.1}$$

$$(a + bi)(c + di) = ac + adi + bci + bdi^2 = (ac - bd) + (ad + bc)i. \tag{2.2}$$

Definition 2.2 — Complex Conjugate and Modulus

The *complex conjugate* of $z = a + bi$ is $z^* = a - bi$ (flip the sign of the imaginary part). The *modulus* (or absolute value) of z is the non-negative real number

$$|z| = \sqrt{zz^*} = \sqrt{a^2 + b^2}.$$

Notice that $zz^* = (a+bi)(a-bi) = a^2 - abi + abi - b^2 i^2 = a^2 + b^2$ is always real and non-negative, even though z itself is complex — this single fact is the entire reason complex numbers are useful in quantum mechanics: the physically measurable probability density is $|\Psi|^2 = \Psi\Psi^*$, a real, non-negative number built from a complex Ψ.

Worked Example 2.1 — Complex Arithmetic

Let $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$. Find $z_1 z_2$, $|z_1|$, and z_1^*.
Solution.

$$z_1 z_2 = (3+4i)(1-2i) = 3 - 6i + 4i - 8i^2 = 3 - 2i + 8 = 11 - 2i.$$

$$|z_1| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5.$$

$$z_1^* = 3 - 4i.$$

A complex number can also be pictured as a point (or arrow) in a two-dimensional plane, with $\mathrm{Re}(z)$ along the horizontal axis and $\mathrm{Im}(z)$ along the vertical axis — the *Argand diagram*. In polar coordinates, $a = r\cos\theta$ and $b = r\sin\theta$, where $r = |z|$ and θ (the *argument* or *phase* of z) is the angle from the positive real axis. This leads to one of the most useful identities in all of physics:

Key Result 2.1 — Euler's Formula

$$e^{i\theta} = \cos\theta + i\sin\theta,$$

so that any complex number can be written in *polar form*, $z = re^{i\theta}$, with $r = |z|$. In particular, $|e^{i\theta}| = 1$ for any real θ: multiplying a complex number by $e^{i\theta}$ rotates it in the Argand plane without changing its modulus.

Note — Why Euler's formula is believable

Differentiate $f(\theta) = \cos\theta + i\sin\theta$ term by term: $f'(\theta) = -\sin\theta + i\cos\theta = i(\cos\theta + i\sin\theta) = if(\theta)$, using $i^2 = -1$ in the second step. But $f'(\theta) = if(\theta)$ with $f(0) = 1$ is exactly the differential equation solved by $f(\theta) = e^{i\theta}$ (Section 2.4 reviews solving equations of exactly this form). Both functions satisfy the same equation with the same initial value, so they are the same function. Euler's formula recurs constantly in this book: oscillatory wavefunctions are most efficiently written as complex exponentials $e^{i(kx-\omega t)}$ rather than as cos and sin separately.

2.2 Vectors and Vector Spaces

You already know ordinary three-dimensional vectors: an arrow $\mathbf{v} = (v_x, v_y, v_z)$ with a length and a direction, added component-by-component, and combined via the dot product $\mathbf{u}\cdot\mathbf{v} = u_x v_x + u_y v_y + u_z v_z$. Quantum mechanics needs a generalization of this idea, because — as Chapter 3 will show — the state of a quantum system is itself best thought of as a vector, just not necessarily an arrow in ordinary 3D space.

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Definition 2.3 — Vector Space (Informal)

A *vector space* is any collection of objects ("vectors") that can be added together and multiplied by numbers ("scalars"), with the results staying inside the collection, and obeying the same basic algebraic rules as ordinary arrows: addition is commutative and associative, there is a zero vector, and scalar multiplication distributes over addition.

The point of stating this abstractly is that many things you would not normally call "vectors" satisfy this definition — and, crucially for us, *well-behaved functions form a vector space*: if $f(x)$ and $g(x)$ are two functions, so is $f(x) + g(x)$, and so is $cf(x)$ for any number c. This is not a mathematical curiosity for its own sake; it is the single fact that lets us treat wavefunctions, which are functions, with the same linear-algebra vocabulary (basis, inner product, eigenvalue) used for ordinary column vectors. We build that vocabulary now, in the familiar setting, so it is ready to reuse.

Definition 2.4 — Inner Product and Orthogonality

The (real) *inner product* of two vectors $\mathbf{u}$ and $\mathbf{v}$ in n dimensions is

$$\mathbf{u} \cdot \mathbf{v} = \sum_{k=1}^{n} u_k v_k.$$

Two vectors are *orthogonal* if $\mathbf{u} \cdot \mathbf{v} = 0$. A vector is *normalized* if $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2 = 1$. A set of mutually orthogonal, individually normalized vectors is called *orthonormal*.

Any vector in n-dimensional space can be written as a linear combination of n orthonormal *basis* vectors, $\mathbf{v} = \sum_k c_k \hat{\mathbf{e}}_k$, and the coefficient of each basis vector is obtained by taking an inner product, $c_k = \hat{\mathbf{e}}_k \cdot \mathbf{v}$ — a fact so central to quantum mechanics (where the c_k become probability amplitudes) that it is worth pausing over now, in the setting of ordinary arrows, before it reappears with new meaning in Chapter 9.

Worked Example 2.2 — Orthogonality Check

Are $\mathbf{u} = (1, 2, -1)$ and $\mathbf{v} = (3, -1, 1)$ orthogonal? If not, are they at least at an angle greater than 90° or less?

Solution. Compute the inner product:

$$\mathbf{u} \cdot \mathbf{v} = (1)(3) + (2)(-1) + (-1)(1) = 3 - 2 - 1 = 0.$$

The vectors are exactly orthogonal.

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2.3 Matrices and Eigenvalue Problems

Definition 2.5 — Matrix and Matrix-Vector Multiplication

An $n \times n$ matrix A is a square array of numbers A_{ij} (i = row, j = column). It acts on a vector $\mathbf{v}$ (written as a column) to produce a new vector $A\mathbf{v}$, whose i-th component is $(A\mathbf{v})_i = \sum_j A_{ij} v_j$. Geometrically, a matrix represents a linear transformation: it can rotate, stretch, or shear the vectors it acts on.

For almost every vector $\mathbf{v}$, the output $A\mathbf{v}$ points in a completely different direction than $\mathbf{v}$ did. A small number of special vectors are the exception:

Definition 2.6 — Eigenvalues and Eigenvectors

A nonzero vector $\mathbf{v}$ is an *eigenvector* of a matrix A, with *eigenvalue* λ, if

$$A\mathbf{v} = \lambda \mathbf{v}.$$

That is, A acting on $\mathbf{v}$ simply rescales $\mathbf{v}$ (by the number λ) without changing its direction.

Eigenvalue problems are, without exaggeration, the central mathematical structure of quantum mechanics: the Schrödinger equation, introduced in the next chapter, is an eigenvalue equation, $\hat{H}\psi = E\psi$, in which the "matrix" is an energy operator $\hat{H}$, the "eigenvector" is a wavefunction ψ, and the eigenvalue E is a physically measurable energy. Every exactly solvable system in Part II of this book (Chapters 4–8) is, at bottom, an eigenvalue problem of exactly this form. It is worth learning to find eigenvalues confidently now, in the finite-dimensional setting, before meeting the same idea applied to a differential operator in Chapter 3.

For a 2×2 matrix, eigenvalues are found by demanding that $A\mathbf{v} = \lambda\mathbf{v}$ have a nonzero solution $\mathbf{v}$, which requires the matrix $(A - \lambda I)$ to be singular (non-invertible), i.e., $\det(A - \lambda I) = 0$:

$$\det \begin{pmatrix} A_{11} - \lambda & A_{12} \\ A_{21} & A_{22} - \lambda \end{pmatrix} = (A_{11} - \lambda)(A_{22} - \lambda) - A_{12}A_{21} = 0. \tag{2.3}$$

This is called the *characteristic equation*; it is a quadratic in λ with (for the symmetric matrices we will mostly encounter) two real roots.

Worked Example 2.3 — Finding Eigenvalues and Eigenvectors

Find the eigenvalues and normalized eigenvectors of $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

Solution. The characteristic equation, Eq. (2.3), is

$$(2-\lambda)^2 - 1 = 0 \quad \Longrightarrow \quad \lambda^2 - 4\lambda + 3 = 0 \quad \Longrightarrow \quad (\lambda - 1)(\lambda - 3) = 0,$$

so $\lambda_1 = 1$ and $\lambda_2 = 3$. For $\lambda_1 = 1$: solving $(A - I)\mathbf{v} = 0$ gives $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0$, i.e., $v_1 = -v_2$; the normalized eigenvector is $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, -1)$. For $\lambda_2 = 3$: $(A - 3I)\mathbf{v} = 0$ gives $-v_1 + v_2 = 0$, i.e., $v_1 = v_2$; the normalized eigenvector is $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, 1)$. As a check, note $\mathbf{v}_1 \cdot \mathbf{v}_2 = \frac{1}{2}(1 - 1) = 0$: the eigenvectors of this symmetric matrix are orthogonal, which is not a coincidence (Problem 9.6 proves this holds for any symmetric/Hermitian matrix)

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and is exactly why quantum-mechanical energy eigenstates with different energies turn out always to be orthogonal.

2.4 Differential Equations You Will Actually Need

The Schrödinger equation is a differential equation, and solving it — the central technical activity of Chapters 4 through 8 — means solving one of only a handful of standard forms, over and over, in different physical contexts. This section collects those forms.

Definition 2.7 — Ordinary Differential Equation (ODE)

An equation relating a function $y(x)$ to its derivatives, e.g., $y'(x) = -ky(x)$ or $y''(x) = -k^2 y(x)$. The *order* of the equation is the highest derivative appearing (first order, second order, ...). An ODE is *linear* if y and its derivatives appear only to the first power, with no products like yy'.

Two first- and second-order linear equations, with constant coefficients, recur throughout this book:

Key Result 2.2 — Two ODEs to Know by Heart

$$\frac{dy}{dx} = -ky \quad \Longrightarrow \quad y(x) = Ce^{-kx}.$$

$$\frac{d^2y}{dx^2} = +\kappa^2 y \quad \Longrightarrow \quad y(x) = Ae^{\kappa x} + Be^{-\kappa x} \quad \text{(real exponentials)}.$$

$$\frac{d^2y}{dx^2} = -k^2 y \quad \Longrightarrow \quad y(x) = A\sin(kx) + B\cos(kx) \quad \text{(oscillatory)}.$$

The *sign* on the right-hand side of the second-order equation is the whole story: a positive coefficient (relative to y itself) gives real, growing/decaying exponentials; a negative coefficient gives oscillation. You will see both cases repeatedly — oscillatory solutions inside a potential well (Chapter 4) and real exponential ("evanescent") solutions in the classically forbidden regions responsible for quantum tunneling (Chapter 5).

You can verify each solution by direct substitution: for example, if $y = A\sin(kx)$, then $y' = Ak\cos(kx)$ and $y'' = -Ak^2\sin(kx) = -k^2 y$, confirming the third equation above. This kind of check — propose a solution, differentiate it, and confirm it satisfies the original equation — is the workhorse technique used throughout Part II, and you should always perform it yourself rather than taking a quoted solution on faith.

A differential equation alone has infinitely many solutions (that is what the undetermined constants A, B, C represent); a physical problem is only fully specified once we also state *boundary conditions* — constraints on the value of y at particular points. In quantum mechanics, boundary conditions are where quantization itself comes from, as the next example shows directly.

Worked Example 2.4 — Where Quantization Comes From

Solve $y''(x) = -k^2 y(x)$ subject to the boundary conditions $y(0) = 0$ and $y(L) = 0$ (i.e., y must vanish at both ends of a fixed interval of length L). What values of k are allowed?
Solution. The general solution is $y(x) = A\sin(kx) + B\cos(kx)$. Imposing $y(0) = 0$: since $\sin(0) = 0$ and $\cos(0) = 1$, this forces $B = 0$, leaving $y(x) = A\sin(kx)$. Imposing $y(L) = 0$: we need $A\sin(kL) = 0$. Taking $A = 0$ would make $y(x) \equiv 0$ everywhere, which is not a useful solution, so instead we require

$$\sin(kL) = 0 \quad \Longrightarrow \quad kL = n\pi \quad \Longrightarrow \quad k = \frac{n\pi}{L}, \qquad n = 1, 2, 3, \ldots$$

($n = 0$ is excluded because it again gives $y \equiv 0$; negative n give the same functions up to an overall sign, so they are not counted separately.) Only a discrete, countable set of values of k — and not a continuum — satisfies both boundary conditions simultaneously.

Note — You have already derived energy quantization

The example above is pure mathematics, with no physics in it yet — but hold onto it. In Chapter 4, $y(x)$ becomes the wavefunction of a particle trapped in a one-dimensional box of width L, and k becomes directly proportional to $\sqrt{E}$. The discreteness of the allowed k values you just derived, from boundary conditions alone, becomes the discreteness of the particle's allowed energies — the literal origin of the word "quantum." No new physical assumption will be needed at that point beyond what you already used here: a wave, confined between two fixed walls, can only oscillate at certain allowed "notes," for exactly the same mathematical reason a guitar string of fixed length can only vibrate at certain frequencies.

2.5 Probability and Expectation Values

Quantum mechanics does not predict, in general, the definite outcome of a single measurement; it predicts the *probability distribution* over possible outcomes. This is a genuinely new ingredient, not present in classical mechanics, and it is worth reviewing carefully in its own right before Chapter 3 attaches physical meaning to it.

Definition 2.8 — Discrete Probability

If a quantity can take on discrete outcomes $x_1, x_2, \ldots$ with probabilities $P_1, P_2, \ldots$, then normalization requires $\sum_i P_i = 1$ (something happens), and the *expectation value* (mean, or average over many repeated trials) is

$$\langle x \rangle = \sum_i P_i x_i.$$

For a continuous variable such as position, the discrete sum becomes an integral, and P_i becomes a *probability density* $\rho(x)$, defined so that $\rho(x)\,dx$ is the probability of finding the particle between x and $x + dx$:

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Key Result 2.3 — Continuous Normalization and Expectation Value

$$\int_{-\infty}^{\infty} \rho(x)\,dx = 1 \qquad \text{(normalization)}, \qquad \langle x\rangle = \int_{-\infty}^{\infty} x\,\rho(x)\,dx \qquad \text{(expectation value)}.$$

More generally, the expectation value of *any* function $f(x)$ of the random variable is $\langle f(x)\rangle = \int f(x)\rho(x)\,dx$. The spread of the distribution around its mean is measured by the *standard deviation*, $\Delta x = \sqrt{\langle x^2\rangle - \langle x\rangle^2}$ — the same Δx that appears in the Heisenberg uncertainty principle of Section 1.6.

Worked Example 2.5 — Normalizing a Simple Density

A particle is described by the probability density $\rho(x) = A$ (a constant) for $0 \le x \le L$, and $\rho(x) = 0$ elsewhere. Find A, and find $\langle x\rangle$.
Solution. Normalization requires

$$\int_0^L A\,dx = AL = 1 \quad\Longrightarrow\quad A = \frac{1}{L}.$$

The expectation value is

$$\langle x\rangle = \int_0^L x\cdot\frac{1}{L}\,dx = \frac{1}{L}\left[\frac{x^2}{2}\right]_0^L = \frac{L}{2},$$

exactly the midpoint of the interval — sensible, since the particle is equally likely to be found anywhere in $[0, L]$.

Note — Density, not probability

A probability *density* $\rho(x)$ can be larger than 1 — only $\rho(x)\,dx$, an infinitesimal probability, is bounded between 0 and 1. Do not be alarmed if a wavefunction's $|\Psi(x)|^2$ comes out numerically larger than 1 at some point; what must integrate to exactly 1 is $|\Psi(x)|^2$ over all space, not its value at any single point.

2.6 A First Look at Dirac Notation

Section 2.2 emphasized that well-behaved functions form a vector space, just as ordinary arrows do. Paul Dirac introduced a compact notation, universally used in quantum mechanics from Chapter 3 onward, that makes this analogy explicit.

Definition 2.9 — Ket, Bra, and Inner Product (Preview)

A quantum state is written as a *ket*, $|\psi\rangle$ — think of this as shorthand for "the abstract vector corresponding to the wavefunction ψ." Its mirror image, the *bra* $\langle\psi|$, pairs with a ket to produce a number (generalizing the dot product of Section 2.2):

$$\langle\phi|\psi\rangle \equiv \int_{-\infty}^{\infty} \phi^*(x)\,\psi(x)\,dx.$$

The complex conjugate here is essential: it guarantees $\langle\psi|\psi\rangle = \int |\psi(x)|^2 dx$ is real and non-negative, exactly the normalization integral of Section 2.5, so that "the ket $|\psi\rangle$ is normalized" means precisely $\langle\psi|\psi\rangle = 1$.

This is only a preview; Chapter 9 develops Dirac notation, and the general structure of quantum-mechanical vector spaces (Hilbert spaces), in full rigor. For now, the essential point is one of translation: everything in this section (vectors, inner products, orthogonality, eigenvalue problems) has a direct counterpart in the space of wavefunctions, and Dirac notation is simply the standard dialect for writing it.

2.7 Chapter Summary

- Complex numbers $z = a + bi$, with $i^2 = -1$, are structurally required by quantum mechanics; the modulus $|z|^2 = zz^*$ is always real and non-negative, which is exactly why physically observable probabilities can be built from a complex wavefunction. Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ connects complex exponentials to oscillation.
- Functions form a vector space in the same algebraic sense that ordinary arrows do, which is what allows the language of vectors, inner products, and orthogonality to carry over to wavefunctions.
- An eigenvalue equation $A\mathbf{v} = \lambda\mathbf{v}$ picks out the special directions a matrix (or, later, an operator) leaves unrotated. The Schrödinger equation is an eigenvalue equation for energy, $\hat{H}\psi = E\psi$.
- Two second-order ODEs recur throughout Part II: $y'' = -k^2 y$ (oscillatory, inside a well) and $y'' = +\kappa^2 y$ (real exponential, in a forbidden region). Boundary conditions, not the differential equation alone, are what select a discrete set of allowed solutions — the mathematical origin of energy quantization, as illustrated directly in this chapter and developed physically in Chapter 4.
- Continuous probability requires $\int \rho(x)\,dx = 1$; the expectation value $\langle x\rangle = \int x\rho(x)\,dx$ and standard deviation Δx generalize the mean and spread of a discrete distribution.
- Dirac's bra-ket notation, $|\psi\rangle$, $\langle\psi|$, $\langle\phi|\psi\rangle = \int \phi^*\psi\,dx$, is the standard vocabulary for all of the above, applied to quantum states; Chapter 9 develops it fully.

Problems

2.1 Let $z_1 = 2 - 3i$ and $z_2 = -1 + i$. Compute (a) $z_1 + z_2$, (b) $z_1 z_2$, (c) $|z_1|$, (d) z_1/z_2 (*hint:* multiply numerator and denominator by z_2^*).

2.2 Write $z = 4e^{i\pi/3}$ in the form $a + bi$, and write $w = 1 - i$ in the form $re^{i\theta}$.

2.3 Show, by direct computation with $z = a + bi$, that $\mathrm{Re}(z) = \frac{1}{2}(z + z^*)$ and $\mathrm{Im}(z) = \frac{1}{2i}(z - z^*)$.

2.4 Are the vectors $\mathbf{u} = (2, 1, 3)$ and $\mathbf{v} = (1, -1, \frac{1}{3})$ orthogonal? Find the value of the third component of $\mathbf{v}$ (in place of $\frac{1}{3}$) that *would* make them orthogonal, if it currently is not.

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2.5 Find the eigenvalues and normalized eigenvectors of $A = \begin{pmatrix} 0 & 2 \\ 2 & 3 \end{pmatrix}$.

2.6 Find the eigenvalues of $B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, and verify by direct matrix multiplication that your eigenvectors satisfy $B\mathbf{v} = \lambda\mathbf{v}$.

2.7 Verify by substitution that $y(x) = A\cosh(\kappa x) + B\sinh(\kappa x)$ (where $\cosh u = \frac{1}{2}(e^u + e^{-u})$ and $\sinh u = \frac{1}{2}(e^u - e^{-u})$) solves $y'' = \kappa^2 y$, exactly as the real-exponential form does. (These two forms are algebraically equivalent, and you will see both used in Chapter 5.)

2.8 Solve $y''(x) = -k^2 y(x)$ subject to the boundary conditions $y(0) = 0$ and $y'(L) = 0$ (note: a condition on the *derivative* at $x = L$ this time, not on y itself). Find the allowed values of k, and compare with the result of the Worked Example in Section 2.4.

2.9 A probability density is given by $\rho(x) = Cx^2$ for $0 \le x \le 2$, and $\rho(x) = 0$ elsewhere. (a) Find C. (b) Find $\langle x \rangle$. (c) Find the probability that a measurement finds the particle in the region $1 \le x \le 2$.

2.10 For the normalized density $\rho(x) = \frac{1}{L}$ on $[0, L]$ from the Worked Example in Section 2.5, find $\langle x^2 \rangle$ and the standard deviation $\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$. Express your answer as a multiple of L.

2.11 Two kets are given, in a basis where they can be treated as ordinary vectors, by $|\phi\rangle = \binom{1}{i}$ and $|\psi\rangle = \binom{i}{1}$, with inner product $\langle \phi | \psi \rangle \equiv \phi_1^* \psi_1 + \phi_2^* \psi_2$ (the discrete analog of the integral definition in Section 2.6). Compute $\langle \phi | \psi \rangle$ and $\langle \phi | \phi \rangle$. Is $|\phi\rangle$ normalized?

2.12 A student claims that since $i = \sqrt{-1}$ is "imaginary," quantities computed using complex numbers cannot correspond to anything physically real or measurable. Using the ideas of this chapter (Section 2.1 and Section 2.6), write two or three sentences explaining what is wrong with this claim.

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Chapter 3

The Postulates of Quantum Mechanics: Wavefunctions and the Schrödinger Equation

Chapter 1 introduced quantum mechanics through a sequence of historical patches: Planck's quantized oscillators, Einstein's photons, Bohr's quantized orbits, de Broglie's matter waves, Heisenberg's uncertainty. Each idea solved a specific experimental puzzle, but they do not yet form a single, self-consistent theory capable of predicting new results. This chapter assembles that theory. Everything that follows in this book — every energy level, every spectrum, every chemical bond — is a consequence of the equation introduced in Section 3.3 and the interpretive rule introduced in Section 3.2, applied to one physical situation after another.

3.1 The Wavefunction

Definition 3.1 — Wavefunction

The state of a quantum-mechanical particle moving in one dimension is completely specified by a complex-valued function of position and time, $\Psi(x,t)$, called the *wavefunction*. All physically measurable information about the particle — its likely position, momentum, energy, and how these evolve in time — can be extracted from $\Psi(x,t)$.

This is a genuinely new kind of physical statement, unlike anything in Newtonian mechanics. A classical particle's state is a pair of numbers, position and momentum, $(x(t), p(t))$, evolving according to $F = ma$. A quantum particle's state is instead an entire function spread over all of space. The Gaussian wave packet in Figure 3.1 is one example of what $\Psi(x,t)$ might look like at a single instant: a localized oscillation with a well-defined central position but a nonzero spread.

3.2 The Born Interpretation

A function is not, by itself, a physical prediction. Max Born supplied the rule connecting $\Psi(x,t)$ to something measurable, building directly on the probability language of Section 2.5:

Key Result 3.1 — The Born Rule

$$P(x,t)\,dx = |\Psi(x,t)|^2\,dx = \Psi^*(x,t)\Psi(x,t)\,dx$$

is the probability of finding the particle, if its position is measured at time t, between x and $x + dx$. $|\Psi(x,t)|^2$ is a probability *density* in exactly the sense of Section 2.5.

Two things about this rule deserve emphasis, because both are frequently misunderstood on first encounter. First, $|\Psi|^2$ gives the probability of a *measurement outcome*, not a smeared-out physical density of the particle itself, the way charge density smears a classical charge distribution —

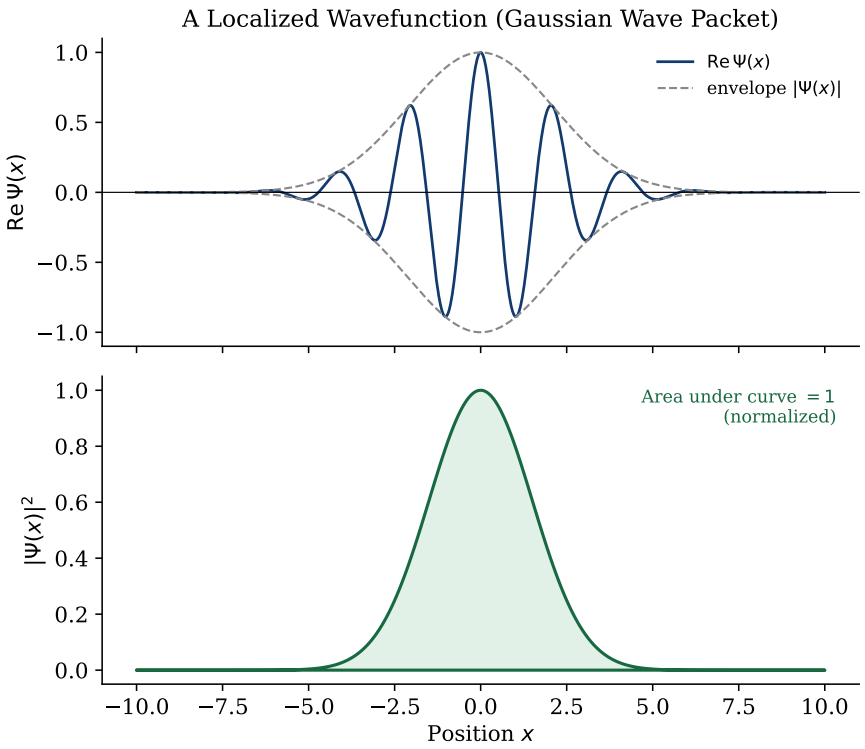


Figure 3.1: A Gaussian wave packet: the real part of $\Psi(x)$ oscillates within an envelope $|\Psi(x)|$ (top), and the probability density $|\Psi(x)|^2$ (bottom) is a localized, normalized bump. The particle is most likely to be found near the packet's center, with some spread Δx set by the packet's width.

when you actually measure the particle's position, you get one definite value, not a fraction of the particle in many places at once. Second, quantum mechanics does not claim that the particle secretly has a definite position all along and we merely lack the information to know it (a view called a "hidden variable" theory); a substantial body of experimental evidence (violations of the Bell inequalities, most decisively) indicates that a particle's position is genuinely undetermined until measured, and only its probability distribution is fixed by Ψ.

Since *some* outcome must occur whenever a measurement is made, the total probability of finding the particle *somewhere* must equal 1:

Key Result 3.2 — Normalization Condition

$$\int_{-\infty}^{\infty} |\Psi(x,t)|^2 \, dx = 1 \qquad \text{for all } t.$$

A wavefunction satisfying this condition is said to be *normalized*. Only normalizable wavefunctions (those for which the integral above is finite, so that it *can* be rescaled to equal 1) correspond to physically realizable states of a single particle.

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Note — Not every function is a valid wavefunction

Recall the constant density $\rho(x) = 1/L$ on $[0, L]$ from Section 2.5 — a perfectly good, normalized probability density. Is $\psi(x) = 1/\sqrt{L}$ for $0 \le x \le L$ (and 0 elsewhere) a valid quantum wavefunction? Being normalizable is necessary but not sufficient. As Section 3.5 will make precise, a physically acceptable wavefunction must also be continuous (it cannot jump discontinuously, as this one does at $x = 0$ and $x = L$) and must actually solve the Schrödinger equation for the physical situation at hand. Chapter 4 constructs the function that this one is a rough sketch of — the true particle-in-a-box ground state — and you will see exactly how it differs.

Worked Example 3.1 — Normalizing a Gaussian Wavefunction

A particle's wavefunction at $t = 0$ has the form $\Psi(x, 0) = Ae^{-x^2/2a^2}$, where a is a constant with units of length. Find the normalization constant A.

Solution. We require

$$\int_{-\infty}^{\infty} |A|^2 e^{-x^2/a^2}\, dx = 1.$$

Using the standard Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2/a^2}\, dx = a\sqrt{\pi}$ (derived in Appendix B and used repeatedly from Chapter 6 onward),

$$|A|^2 a\sqrt{\pi} = 1 \quad \Longrightarrow \quad A = \left(\frac{1}{\pi a^2}\right)^{1/4},$$

where we have chosen A to be real and positive; any phase factor $e^{i\phi_0}$ multiplying A would give an equally valid, equally normalized wavefunction (Problem 3.4 asks you to show this explicitly), which is why the overall phase of a wavefunction carries no physical meaning.

3.3 The Time-Dependent Schrödinger Equation

The evolution of $\Psi(x, t)$ in time is governed by an equation Erwin Schrödinger proposed in 1926 — not derived from more fundamental principles (much as Newton's second law is not "derived" from anything more basic), but justified by the correctness of everything that follows from it.

Key Result 3.3 — Time-Dependent Schrödinger Equation (TDSE)

$$i\hbar \frac{\partial \Psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 \Psi(x,t)}{\partial x^2} + V(x)\Psi(x,t),$$

where m is the particle's mass and $V(x)$ is the potential energy function describing the forces acting on it.

Two features of this equation are worth flagging immediately, because they are easy to pass over. First, the explicit $i = \sqrt{-1}$ on the left-hand side is not optional or a matter of convenience — it is required for the equation to admit oscillatory, wave-like solutions at all (Problem 3.11 shows explicitly that assuming Ψ is real leads to a contradiction for a free particle). Second, the equation is first-order in time (only $\partial\Psi/\partial t$ appears, not $\partial^2\Psi/\partial t^2$, unlike, say, the classical wave equation). This means that, unlike Newton's second law, which requires *both* initial position *and*

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initial velocity to predict the future, the Schrödinger equation requires only $\Psi(x,0)$ to determine $\Psi(x,t)$ for all later t — the wavefunction is a complete description of the state precisely because the dynamical law is first-order.

The combination of operations on the right-hand side recurs so often that it is given its own name and symbol:

Definition 3.2 — The Hamiltonian Operator

$$\hat{H} \equiv -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)$$

is the *Hamiltonian operator*, the quantum-mechanical operator corresponding to total energy (kinetic plus potential). In this notation, the TDSE reads compactly as $i\hbar\,\partial\Psi/\partial t = \hat{H}\Psi$.

3.4 Operators and Observables

The Hamiltonian is the first example of a general pattern: every measurable physical quantity (*observable*) in quantum mechanics — position, momentum, energy, angular momentum — is represented by a mathematical operator that acts on the wavefunction.

Definition 3.3 — Operator

An *operator* $\hat{A}$ is a rule that takes a function and produces another function: $\hat{A}\psi(x) = \phi(x)$ for some new function ϕ. In quantum mechanics, operators are built from two elementary building blocks:

$$\hat{x} = x \text{ (multiply by } x\text{)}, \qquad \hat{p} = -i\hbar\frac{d}{dx} \text{ (differentiate, times } -i\hbar\text{)}.$$

Every other operator used in this book — including $\hat{H}$ above — is constructed from these two, combined the same way the corresponding classical quantity is built from x and p. Kinetic energy, classically $p^2/2m$, becomes $\hat{p}^2/(2m) = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$, which is exactly the first term of $\hat{H}$ above.

Note — Why $\hat{p} = -i\hbar\,d/dx$?

This definition is motivated by exactly the plane-wave calculation of Chapter 1: a free particle with definite momentum p has, by de Broglie, wavelength $\lambda = h/p$ and is described by a wave $\propto e^{ikx}$ with $k = 2\pi/\lambda = p/\hbar$ (Section 1.5). Differentiating this wave, $\frac{d}{dx}e^{ikx} = ike^{ikx}$, so $-i\hbar\frac{d}{dx}e^{ikx} = \hbar k\, e^{ikx} = p\, e^{ikx}$: the operator $-i\hbar\,d/dx$, acting on the wavefunction of a definite-momentum particle, returns exactly p times that same wavefunction. This is not a coincidence to be memorized but a consistency check you can and should redo yourself (Problem 3.7).

Worked Example 3.2 — An Operator Eigenvalue Equation

Show that $\psi(x) = e^{ikx}$ is an eigenfunction of the momentum operator $\hat{p}$, and find its eigenvalue.
Solution.

$$\hat{p}\,\psi(x) = -i\hbar\frac{d}{dx}e^{ikx} = -i\hbar(ik)e^{ikx} = \hbar k\, e^{ikx} = \hbar k\,\psi(x).$$

This has exactly the form $\hat{p}\psi = p\,\psi$ (compare the eigenvalue equation $A\mathbf{v} = \lambda\mathbf{v}$ of Section 2.3), with eigenvalue $p = \hbar k$. A particle described by e^{ikx} therefore has a perfectly well-defined momentum $p = \hbar k$ — consistent with de Broglie's relation $p = h/\lambda = \hbar(2\pi/\lambda) = \hbar k$, recovered here directly from the operator formalism rather than assumed.

Given an operator $\hat{A}$ representing some observable, the expectation value of that observable — the average result you would obtain by preparing many identical copies of the system in state Ψ and measuring A on each — generalizes the expectation-value formula of Section 2.5 in the natural way:

Key Result 3.4 — Expectation Value of an Observable

$$\langle \hat{A} \rangle = \int_{-\infty}^{\infty} \Psi^*(x,t)\,\hat{A}\,\Psi(x,t)\,dx.$$

For $\hat{A} = \hat{x}$, this reduces exactly to $\langle x \rangle = \int x|\Psi|^2 dx$ from Section 2.5, since multiplying by x commutes freely with everything in the integrand.

3.5 The Time-Independent Schrödinger Equation and Stationary States

For an enormous and important class of problems — every problem in Chapters 4 through 8 of this book — the potential energy $V(x)$ does not itself depend on time. In that case, the TDSE can be solved by *separation of variables*: seek a solution of the special product form $\Psi(x,t) = \psi(x)f(t)$, a space-dependent piece times a time-dependent piece. Substituting into the TDSE,

$$i\hbar\,\psi(x)\frac{df}{dt} = f(t)\left[-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi(x)\right],$$

and dividing both sides by $\psi(x)f(t)$ separates the variables completely:

$$i\hbar\,\frac{1}{f(t)}\frac{df}{dt} = \frac{1}{\psi(x)}\left[-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi(x)\right].$$

The left-hand side depends only on t; the right-hand side depends only on x. The only way a function of t alone can equal a function of x alone, for *all* x and t independently, is if both sides equal the same constant — called the *separation constant* and, as the notation below anticipates, it will turn out to be the energy E:

$$i\hbar\frac{df}{dt} = Ef(t), \qquad -\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi(x) = E\psi(x). \tag{3.1}$$

The first equation is a first-order ODE of exactly the form solved in Section 2.4 ($df/dt = (E/i\hbar)f = -(iE/\hbar)f$), with solution

$$f(t) = e^{-iEt/\hbar}. \tag{3.2}$$

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The second equation is the single most important equation in this entire book:

Key Result 3.5 — Time-Independent Schrödinger Equation (TISE)

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x), \qquad \text{equivalently} \quad \hat{H}\psi(x) = E\psi(x).$$

This is an eigenvalue equation, in precisely the sense of Section 2.3: $\hat{H}$ plays the role of the matrix, $\psi(x)$ the eigenvector, and E the eigenvalue. Solving it — finding which functions $\psi(x)$ and which energies E satisfy this equation for a given $V(x)$, subject to appropriate boundary conditions — is the central technical task of Chapters 4 through 8.

Solutions of the special product form $\Psi(x,t) = \psi(x)e^{-iEt/\hbar}$ are called *stationary states*, for a reason the next result makes precise.

Key Result 3.6 — Stationary States Have Time-Independent Probability Densities

For $\Psi(x,t) = \psi(x)e^{-iEt/\hbar}$,

$$|\Psi(x,t)|^2 = \psi^*(x)e^{+iEt/\hbar} \cdot \psi(x)e^{-iEt/\hbar} = |\psi(x)|^2,$$

independent of t, because the phase factor $e^{-iEt/\hbar}$ has modulus exactly 1 for real E and cancels against its own complex conjugate. Although the full wavefunction $\Psi(x,t)$ oscillates in time, every measurable probability — and, in fact, the expectation value of *any* time-independent operator — computed from it is completely static.

This is the sense in which a "stationary state" is stationary: not that nothing is happening (the wavefunction's complex phase is winding in time, at angular frequency $E/\hbar$, exactly as Planck's relation $E = h\nu$ from Section 1.1 would suggest), but that nothing *measurable* changes. An electron sitting in an energy eigenstate of an atom stays there, emitting no radiation and showing no time dependence in any observable property, forever — until an external perturbation (Chapter 13) induces a transition to a different stationary state.

Note — The general solution is a sum over stationary states

Not every physically relevant wavefunction has the simple product form $\psi(x)e^{-iEt/\hbar}$ — a general $\Psi(x,t)$, like the wave packet of Figure 3.1, does not. But because the TDSE is linear, any sum of solutions is itself a solution, and it can be shown (Chapter 9) that *every* solution of the TDSE can be written as a sum (or, for a continuum of energies, an integral) of stationary states, $\Psi(x,t) = \sum_n c_n\psi_n(x)e^{-iE_nt/\hbar}$. Solving the time-independent problem, once, for all the stationary states $\psi_n(x)$ and energies E_n therefore solves the general time-dependent problem too — which is precisely why Chapters 4 through 8 focus exclusively on the time-*in*dependent equation.

3.6 Chapter Summary

- A quantum particle's state is completely described by a complex wavefunction $\Psi(x,t)$.
- The Born rule gives physical meaning to Ψ: $|\Psi(x,t)|^2$ is a probability density for the out-

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come of a position measurement, normalized so that $\int |\Psi|^2 dx = 1$.

- $\Psi(x,t)$ evolves according to the time-dependent Schrödinger equation, $i\hbar\,\partial\Psi/\partial t = \hat{H}\Psi$, with $\hat{H} = -\dfrac{\hbar^2}{2m}\dfrac{d^2}{dx^2} + V(x)$.
- Every observable (position, momentum, energy, ...) corresponds to an operator built from $\hat{x}$ and $\hat{p} = -i\hbar\,d/dx$; the expectation value of observable A in state Ψ is $\langle\hat{A}\rangle = \int \Psi^* \hat{A}\Psi\, dx$.
- When $V(x)$ is time-independent, separation of variables reduces the TDSE to the time-independent Schrödinger equation, $\hat{H}\psi = E\psi$ — an eigenvalue equation for the allowed energies E and corresponding spatial wavefunctions $\psi(x)$ — combined with a universal phase factor $e^{-iEt/\hbar}$.
- States of this product form are stationary states: their probability density $|\Psi|^2 = |\psi|^2$ does not depend on time, even though Ψ itself does. Every solvable system in Part II of this book is an exercise in finding the stationary states of a specific $V(x)$.

Problems

3.1 At a particular instant, a particle's (unnormalized) wavefunction is $\psi(x) = Bxe^{-x/a}$ for $x \geq 0$ and $\psi(x) = 0$ for $x < 0$, where $a > 0$ is a constant. Find the normalization constant B. (*Hint:* you will need $\int_0^\infty x^2 e^{-2x/a}dx = a^3/4$; you may quote this result or derive it using integration by parts.)

3.2 For the normalized wavefunction of Problem 3.1, find $\langle x\rangle$. (*Hint:* $\int_0^\infty x^3 e^{-2x/a}dx = 3a^4/8$.)

3.3 A wavefunction is given by $\Psi(x,t) = \psi(x)e^{-iEt/\hbar}$ where $\psi(x)$ is real. Show explicitly that $|\Psi(x,t)|^2$ is time-independent, as claimed in the Key Result of Section 3.5.

3.4 If $\psi(x)$ is a normalized solution of the TISE, show that $\tilde{\psi}(x) = e^{i\phi_0}\psi(x)$, for any real constant phase ϕ_0, is also normalized and gives exactly the same probability density $|\tilde{\psi}|^2$. What does this tell you about how much physical information is contained in the overall phase of a wavefunction?

3.5 Compute the expectation value $\langle\hat{p}\rangle$ for the definite-momentum state $\psi(x) = e^{ikx}$ using the Key Result of Section 3.4, and confirm it equals the eigenvalue $\hbar k$ found in the Worked Example of Section 3.4. (This wavefunction is not normalizable over all space — proceed formally, applying the operator and simplifying, without worrying about the normalization integral itself.)

3.6 Compute the commutator $[\hat{x},\hat{p}]\psi(x) \equiv \hat{x}(\hat{p}\psi) - \hat{p}(\hat{x}\psi)$ for an arbitrary (differentiable) function $\psi(x)$, and show that $[\hat{x},\hat{p}] = i\hbar$ (meaning: acting on any ψ, the result is always $i\hbar\,\psi(x)$). (*Hint:* remember the product rule when differentiating $x\psi(x)$.) This commutator is the deepest algebraic reason for the Heisenberg uncertainty principle of Section 1.6, as Chapter 9 shows in full.

3.7 Verify the claim of the Note in Section 3.4 directly: apply $\hat{p} = -i\hbar\,d/dx$ to $\psi(x) = e^{ikx}$ and confirm you recover $p = \hbar k$, matching de Broglie's relation from Section 1.5.

3.8 A particle is in the state $\Psi(x,t) = \frac{1}{\sqrt{2}}\psi_1(x)e^{-iE_1t/\hbar} + \frac{1}{\sqrt{2}}\psi_2(x)e^{-iE_2t/\hbar}$, a sum of two *different* stationary states. Show, by direct computation of $|\Psi(x,t)|^2$, that this state's probability density *does* depend on t (unlike a single stationary state), and that the time dependence oscillates at

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angular frequency $\omega = (E_2 - E_1)/\hbar$. (You may assume ψ_1 and ψ_2 are both real.) This is the mathematical origin of the oscillating charge distributions responsible for spectroscopic emission, developed fully in Chapter 13.

3.9 Classify each of the following as (i) a valid time-independent potential $V(x)$ for which separation of variables applies directly, or (ii) not: (a) $V(x) = \frac{1}{2}kx^2$; (b) $V(x,t) = \frac{1}{2}kx^2\cos(\omega t)$; (c) $V(x) = 0$ for $|x| < L$ and $V(x) = \infty$ otherwise; (d) $V(x) = -e^2/4\pi\epsilon_0|x|$.

3.10 Show that if $\psi_1(x)$ and $\psi_2(x)$ both solve the TISE, Eq. (3.1), with the *same* energy E, then any linear combination $c_1\psi_1(x) + c_2\psi_2(x)$ also solves the TISE with that same energy E. (This is why the TISE is called a *linear* equation, and it underlies the entire concept of degenerate states developed in Chapters 7 and 8.)

3.11 A student attempts to describe a free particle ($V(x) = 0$ everywhere) using a wavefunction that is real for all x and t. Substitute a generic real function $\Psi(x,t) = g(x,t)$ (with g real) into the TDSE, and show that the left-hand side is purely imaginary while the right-hand side is purely real, unless $g \equiv 0$. What does this tell you about the necessity of complex-valued wavefunctions, first asserted in Section 2.1?

3.12 Using the Hamiltonian operator of Section 3.3 for a particle in the potential $V(x) = \frac{1}{2}m\omega^2x^2$ (to be solved in full in Chapter 6), write out the full time-independent Schrödinger equation explicitly as a differential equation in $\psi(x)$, with all constants (m, ω, $\hbar$, E) shown. Do not attempt to solve it — simply setting it up correctly, from the general form of $\hat{H}$, is the point of this problem.

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Part II

Exactly Solvable Systems

Chapter 4

The Particle in a Box

Chapters 1 through 3 assembled the machinery of quantum mechanics: the wavefunction, the Born rule, the Schrödinger equation, operators, and eigenvalue problems. It is time to put that machinery to work. This chapter solves the simplest genuinely quantum-mechanical problem there is — a particle trapped in a one-dimensional region with impenetrable walls — completely and exactly. Despite its simplicity, the particle in a box already displays every qualitative feature that makes quantum systems different from classical ones: discrete energy levels, a nonzero minimum energy, nodes in the wavefunction, and a probability distribution with no classical counterpart. It is also directly useful: as Section 4.5 shows, it is a working (if rough) model for the electronic structure of conjugated organic molecules.

4.1 Setting Up the Problem

Definition 4.1 — The Infinite Square Well

A particle of mass m is confined to the region $0 \le x \le L$ by a potential energy function

$$V(x) = \begin{cases} 0, & 0 \le x \le L \\ \infty, & \text{otherwise.} \end{cases}$$

Figure 4.1 shows this potential: flat and zero inside the box, and infinitely steep at the walls.

An infinite potential is an idealization — no real wall is infinitely steep — but it is an excellent approximation whenever the walls confining a particle are much "stiffer" than the particle's own energy scale, and it has the enormous pedagogical advantage of being exactly solvable with nothing more than the tools of Chapter 2. Chapter 5 treats the more realistic case of a finite-height well, and shows that the infinite well is recovered as a limiting case.

4.2 Solving the Time-Independent Schrödinger Equation

We solve the TISE from Section 3.5, $-\dfrac{\hbar^2}{2m}\psi''(x) + V(x)\psi(x) = E\psi(x)$, region by region.

Outside the box ($x < 0$ or $x > L$): here $V(x) = \infty$. For the equation to remain sensible (for $E\psi$ to stay finite while $V\psi$ would otherwise be infinite), we must have $\psi(x) = 0$ everywhere outside the box. Physically: there is exactly zero probability of finding the particle where the potential energy is infinite, since that would require infinite energy.

Inside the box ($0 < x < L$): here $V(x) = 0$, so the TISE reduces to

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi \qquad \Longrightarrow \qquad \frac{d^2\psi}{dx^2} = -\frac{2mE}{\hbar^2}\psi \equiv -k^2\psi,$$

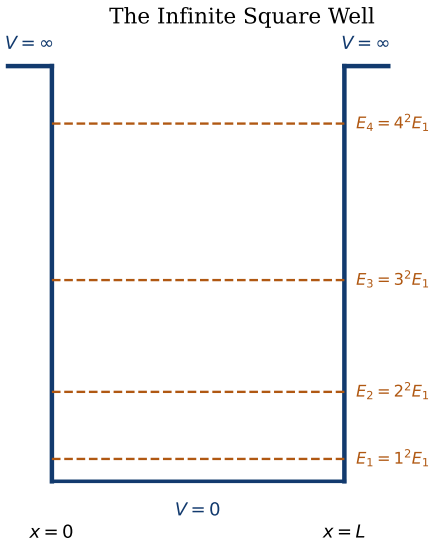


Figure 4.1: The infinite square well and its first four energy levels. The spacing between levels grows as n^2, so levels bunch closer together (relatively) at low n and spread apart at high n.

where we have defined $k \equiv \sqrt{2mE}/\hbar$ (real and positive, since inside the box $E > 0$ for any nonzero wavefunction, as Problem 4.10 asks you to justify). This is exactly the oscillatory ODE solved in Section 2.4:

$$\psi(x) = A\sin(kx) + B\cos(kx).$$

Boundary conditions. A physically acceptable wavefunction must be continuous (Section 3.2's note box); since $\psi = 0$ just outside the box on both sides, continuity requires

$$\psi(0) = 0, \qquad \psi(L) = 0.$$

This is *exactly* the boundary value problem solved in the Worked Example of Section 2.4 ("Where Quantization Comes From"). We already know the answer: $\psi(0) = 0$ forces $B = 0$, and $\psi(L) = 0$ then forces

$$k_n = \frac{n\pi}{L}, \qquad n = 1, 2, 3, \ldots \tag{4.1}$$

4.3 Quantized Energies and Normalized Wavefunctions

Since $k = \sqrt{2mE}/\hbar$, Eq. (4.1) translates directly into a formula for the allowed energies:

Key Result 4.1 — Particle-in-a-Box Energy Levels

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2} = \frac{n^2h^2}{8mL^2}, \qquad n = 1, 2, 3, \ldots$$

Only these discrete energies are allowed; any other value of E fails to produce a wavefunction satisfying both boundary conditions simultaneously.

Three features of this result are worth dwelling on. First, $n = 0$ is excluded (it would give $\psi \equiv 0$ everywhere — no particle at all, not a particle with zero energy), so the *lowest* possible energy is

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$E_1 = \pi^2\hbar^2/2mL^2$, strictly greater than zero. This nonzero *zero-point energy* is a direct, quantitative consequence of the uncertainty principle (Section 1.6): confining a particle to a finite region costs kinetic energy, exactly as it did in the hydrogen-atom estimate of Chapter 1, and a particle permanently at rest ($E = 0$) would have simultaneously exact position and exact (zero) momentum, forbidden by $\Delta x \Delta p \geq \hbar/2$. Second, the energy spacing $E_{n+1} - E_n = (2n+1)\pi^2\hbar^2/2mL^2$ grows with n: the levels are not evenly spaced, unlike the harmonic oscillator of Chapter 6. Third, $E_n \propto 1/L^2$: a more tightly confined particle has *higher*-energy levels, spaced further apart — squeezing the box pushes every level up and spreads them out.

The spatial wavefunctions are $\psi_n(x) = A\sin(n\pi x/L)$; normalization (Section 3.2) fixes A:

$$\int_0^L A^2 \sin^2\left(\frac{n\pi x}{L}\right) dx = A^2 \cdot \frac{L}{2} = 1 \quad \Longrightarrow \quad A = \sqrt{\frac{2}{L}}.$$

Key Result 4.2 — Particle-in-a-Box Wavefunctions

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \qquad 0 \leq x \leq L,$$

and $\psi_n(x) = 0$ outside the box. Figure 4.2 plots $\psi_n(x)$ and $|\psi_n(x)|^2$ for $n = 1, 2, 3$.

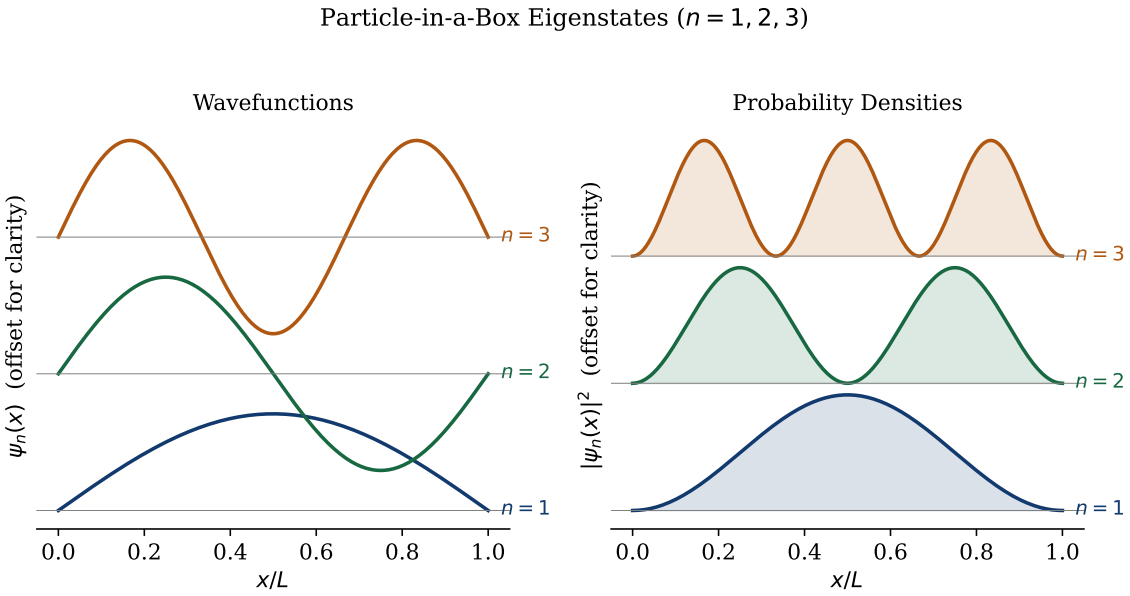


Figure 4.2: The first three particle-in-a-box wavefunctions (left) and probability densities (right), each vertically offset for clarity. The quantum number n equals the number of antinodes, or equivalently $(n-1)$ equals the number of interior nodes.

Note — Nodes and energy

Count the interior zero-crossings (*nodes*) of each wavefunction in Figure 4.2: ψ_1 has none, ψ_2 has one, ψ_3 has two. In general ψ_n has exactly $n-1$ nodes. This pattern — more nodes means more energy — is completely general in one-dimensional quantum mechanics (it

recurs for the harmonic oscillator in Chapter 6 and the hydrogen radial wavefunctions in Chapter 8) and gives you a fast, reliable way to order states by energy just by looking at their wavefunctions, without any calculation.

The eigenfunctions also satisfy a property foreshadowed in the matrix setting of Section 2.3: distinct energy eigenstates are automatically orthogonal,

$$\int_0^L \psi_n(x)\psi_m(x)\,dx = \delta_{nm} \equiv \begin{cases} 1 & n = m \\ 0 & n \neq m \end{cases}, \tag{4.2}$$

combining normalization ($n = m$) and orthogonality ($n \neq m$) into a single statement using the *Kronecker delta* δ_{nm}. Equation (4.2) is not a special accident of the box; Chapter 9 proves it follows generally from the Hermiticity of $\hat{H}$.

Worked Example 4.1 — An Electron Confined to a Quantum Dot

A semiconductor quantum dot confines an electron to an effective one-dimensional box of width $L = 5.0$ nm. Find (a) the ground-state energy, in eV, and (b) the wavelength of light emitted when the electron relaxes from $n = 2$ to $n = 1$.
Solution. (a)

$$E_1 = \frac{\pi^2\hbar^2}{2m_eL^2} = \frac{\pi^2(1.0546\times10^{-34}\ \text{J s})^2}{2(9.109\times10^{-31}\ \text{kg})(5.0\times10^{-9}\ \text{m})^2} = 2.41\times10^{-21}\ \text{J} = 0.0150\ \text{eV}.$$

(b) Since $E_n = n^2E_1$, $E_2 = 4E_1$, so $\Delta E = E_2 - E_1 = 3E_1 = 0.0451$ eV. The emitted wavelength is

$$\lambda = \frac{hc}{\Delta E} = \frac{1239.84\ \text{eV nm}}{0.0451\ \text{eV}} \approx 2.75\times10^4\ \text{nm} = 27.5\ \mu\text{m},$$

in the infrared. Real semiconductor quantum dots (typically a few nanometers across, confining an electron in three dimensions rather than one) work on exactly this principle: their emission color is tunable simply by changing the dot's physical size, with smaller dots emitting higher-energy (bluer) light — precisely the $E \propto 1/L^2$ scaling derived above, and the basis of quantum-dot displays and biological imaging labels.

4.4 The Correspondence Principle

A foundational requirement of any new physical theory is that it must reproduce the successes of the theory it replaces, in the regime where the old theory is known to work. For quantum mechanics, that regime is large quantum numbers.

Definition 4.2 — Correspondence Principle

As $n \to \infty$, the predictions of quantum mechanics must approach the predictions of classical mechanics.

For the particle in a box, a classical particle bouncing elastically between the walls spends equal amounts of time in every equal-sized interval of $[0, L]$ (it moves at constant speed between collisions), so classically we expect a *uniform* probability density, $P_{\text{classical}}(x) = 1/L$. Quantum mechanically, $|\psi_n(x)|^2 = \frac{2}{L}\sin^2(n\pi x/L)$ is very much *not* uniform for small n (compare the sharply

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peaked $n = 1$ curve in Figure 4.2 to a flat line). But as n grows, $\sin^2(n\pi x/L)$ oscillates more and more rapidly, and any real measurement (which cannot resolve position with infinite precision) averages over several oscillations; using $\langle \sin^2\theta \rangle = \frac{1}{2}$ averaged over any full period, the *locally averaged* probability density approaches

$$\overline{|\psi_n(x)|^2} \to \frac{2}{L}\cdot\frac{1}{2} = \frac{1}{L} = P_{\text{classical}}(x) \qquad (n \to \infty),$$

exactly recovering the classical, uniform result. The particle in a box is often the cleanest place to first see the correspondence principle at work, and it is worth returning to this argument again after Chapter 6, where the same limit is taken for the harmonic oscillator with a less obvious classical comparison.

4.5 Chemical Application: Conjugated π Systems

The particle-in-a-box model, crude as it is, gives real quantitative insight into the electronic structure of conjugated organic molecules — chains of alternating single and double carbon-carbon bonds, such as the polyenes found in carotenoid pigments and some biological chromophores.

The key physical idea is delocalization: in a conjugated chain, the π electrons are not confined to individual bonds but are free to move along the entire length of the conjugated system, much as a particle-in-a-box electron is free to move anywhere within $[0, L]$. Modeling the conjugated chain as a one-dimensional box of length L (roughly the length of the conjugated backbone) and filling the box with the molecule's π electrons, two at a time per level (an application of the Pauli exclusion principle, treated fully in Chapter 10), lets us estimate the HOMO–LUMO energy gap and, from it, the wavelength of light the molecule absorbs.

Worked Example 4.2 — Predicting the Color of a Conjugated Polyene

1,3,5,7-octatetraene, CH_2=CH–CH=CH–CH=CH–CH=CH_2, has 8 π electrons delocalized over a conjugated backbone of 7 C–C bonds, each of average length ≈ 0.140 nm. Model this system as a particle in a box of length equal to the conjugated backbone plus one additional bond length (a common convention, allowing the electron density to extend slightly beyond the terminal atoms), and estimate the wavelength of the electronic transition responsible for this molecule's UV absorption.

Solution. The box length is

$$L = (7+1)(0.140\ \text{nm}) = 1.12\ \text{nm}.$$

With 8 π electrons filling the levels two at a time, levels $n = 1, 2, 3, 4$ are doubly occupied (the HOMO, highest occupied level, is $n = 4$) and $n = 5$ is the LUMO (lowest unoccupied level). Using $E_n = n^2h^2/8m_eL^2$,

$$E_4 = \frac{16h^2}{8m_eL^2} = 4.796\ \text{eV}, \qquad E_5 = \frac{25h^2}{8m_eL^2} = 7.494\ \text{eV},$$

so

$$\Delta E = E_5 - E_4 = 2.698\ \text{eV} \quad\Longrightarrow\quad \lambda = \frac{hc}{\Delta E} = \frac{1239.84\ \text{eV nm}}{2.698\ \text{eV}} \approx 460\ \text{nm}.$$

The experimentally measured absorption maximum of octatetraene is close to 290 nm — so this simple model overestimates the wavelength (equivalently, underestimates the gap) by

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a substantial margin, although it correctly captures the *trend*: butadiene (3 backbone bonds), hexatriene (5 backbone bonds), and octatetraene (7 backbone bonds) absorb at successively longer wavelengths as conjugation length increases, exactly as $E_n \propto 1/L^2$ predicts.

Note — Why the simple model overestimates λ

The particle-in-a-box model treats the π electrons as entirely free within the box and completely ignores their mutual electrostatic repulsion, treats every C–C bond as identical in length (real polyenes show measurable bond-length alternation, long-short-long-short, rather than being perfectly uniform), and replaces the true, smoothly varying molecular potential with an artificial flat-bottomed well with infinitely hard walls. Each of these simplifications shifts the predicted gap in a different direction, and they do not cancel. The model is not used in practice for quantitatively accurate spectroscopic prediction — it is used because it captures the qualitative $1/L^2$ scaling correctly using only algebra, in a way that is immediately transparent, unlike the much more accurate but computationally intensive methods of Chapter 17. Every model in physical chemistry involves this trade-off between transparency and accuracy; learning to state clearly *which* approximations a model makes, as we have just done, is as important a skill as solving the model itself.

4.6 Chapter Summary

- The infinite square well is solved by requiring $\psi = 0$ outside the box (continuity with the classically forbidden, infinite-potential region) and $\psi(0) = \psi(L) = 0$ inside, reducing to the boundary-value problem already solved in Chapter 2.
- The allowed energies are quantized, $E_n = n^2\pi^2\hbar^2/2mL^2 = n^2h^2/8mL^2$, $n = 1, 2, 3, \ldots$, with normalized wavefunctions $\psi_n(x) = \sqrt{2/L}\sin(n\pi x/L)$.
- The ground-state energy E_1 is strictly positive (zero-point energy), a direct, quantitative manifestation of the Heisenberg uncertainty principle.
- ψ_n has $n - 1$ interior nodes; more nodes always means higher energy, a pattern that recurs throughout one-dimensional quantum mechanics.
- Different energy eigenstates are automatically orthogonal, $\int \psi_n \psi_m \, dx = \delta_{nm}$.
- As $n \to \infty$, the (locally averaged) quantum probability density approaches the uniform classical result, $1/L$ — an explicit example of the correspondence principle.
- The model gives a rough but genuinely useful account of conjugated π-electron systems, correctly predicting that longer conjugation shifts absorption to longer wavelengths, even though it is not quantitatively precise.

Problems

4.1 An electron is confined to a one-dimensional box of width $L = 0.20$ nm (roughly one atomic diameter). Find the ground-state energy E_1, in eV.

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4.2 For the electron of Problem 4.1, find the energy (in eV) and wavelength (in nm) of the photon that would be absorbed in a transition from $n = 1$ to $n = 3$. In what region of the electromagnetic spectrum does this lie?

4.3 A proton ($m_p \approx 1836\, m_e$) and an electron are placed in identical boxes of width L. Find the ratio $E_1^{(p)}/E_1^{(e)}$ of their ground-state energies, and explain in one sentence, using the formula for E_n, why the more massive particle has the lower zero-point energy.

4.4 Find the value of n for which a 1.0 g marble confined to a 10 cm one-dimensional "box," moving with a (nonzero) kinetic energy of 1.0×10^{-6} J, would be described. (*Hint:* set E_n equal to the given energy and solve for n.) Comment on the size of this quantum number and what it implies, via the correspondence principle of Section 4.4, about the observability of quantum effects for macroscopic objects.

4.5 Show, by direct substitution into the TISE inside the box (Section 4.2), that $\psi_n(x) = \sqrt{2/L}\sin(n\pi x/L)$ satisfies $-\dfrac{\hbar^2}{2m}\psi_n'' = E_n\psi_n$ with $E_n = n^2\pi^2\hbar^2/2mL^2$, confirming the Key Result of Section 4.3 directly rather than by citing the general boundary-value solution.

4.6 Verify explicitly, by direct integration, that $\int_0^L \psi_1(x)\psi_2(x)\,dx = 0$ (orthogonality of the ground and first excited states), using $\psi_1 = \sqrt{2/L}\sin(\pi x/L)$ and $\psi_2 = \sqrt{2/L}\sin(2\pi x/L)$. (*Hint:* the product-to-sum identity $\sin A \sin B = \tfrac{1}{2}[\cos(A-B) - \cos(A+B)]$ will help.)

4.7 For the particle-in-a-box ground state $\psi_1(x) = \sqrt{2/L}\sin(\pi x/L)$, find $\langle x\rangle$. (*Hint:* by the symmetry of $|\psi_1(x)|^2$ about the box's midpoint, this requires almost no calculation — state the symmetry argument explicitly rather than evaluating the integral.)

4.8 For the same ground state as Problem 4.7, it can be shown that $\langle x^2\rangle = L^2\left(\dfrac{1}{3} - \dfrac{1}{2\pi^2}\right)$. Using this result and your answer to Problem 4.7, find the standard deviation Δx, as a multiple of L, and compare it (as a ratio) to the standard deviation $\Delta x_{\text{classical}} = L/\sqrt{12}$ of the uniform classical distribution on $[0, L]$. Is the quantum ground state more or less spatially spread out than the classical expectation?

4.9 1,3-butadiene (CH_2=CH–CH=CH_2) has 4 π electrons over a conjugated backbone of 3 C–C bonds. Using the same modeling convention as the Worked Example of Section 4.5 (box length = backbone length + one bond length, with bond length 0.140 nm), predict its HOMO–LUMO absorption wavelength, and compare with the experimental value of 217 nm quoted in Section 4.5.

4.10 Explain, in a sentence or two and without new calculation, why E must be strictly positive ($E > 0$) for a bound particle in the infinite square well — that is, why $k = \sqrt{2mE}/\hbar$ in Section 4.2 could safely be assumed real. (*Hint:* what would the spatial solutions look like if $E < 0$, i.e., if $k^2 < 0$ so that the sign in the ODE of Section 2.4 flips? Could such a solution satisfy *both* boundary conditions, other than trivially?)

4.11 Two non-interacting electrons occupy the lowest available levels of an infinite square well of width L, obeying the Pauli exclusion principle in the form "at most two electrons (of opposite spin) per spatial level" (Chapter 10 justifies this fully). Find the total ground-state energy of the two-electron system, in units of E_1 (the one-particle ground-state energy).

4.12 A particle in a box of width L is, at a certain instant, in an equal-weight superposition of the ground and first excited stationary states, $\Psi(x,0) = \frac{1}{\sqrt{2}}\big[\psi_1(x) + \psi_2(x)\big]$. Using the result of

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Problem 3.8 (Chapter 3), state the angular frequency ω at which the resulting probability density $|\Psi(x,t)|^2$ oscillates in time, in terms of E_1 and $\hbar$ (you need not rederive the general result — just identify E_1 and E_2 for this system and substitute).

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Chapter 5

Tunneling, Barriers, and the Finite Well

The infinite square well of Chapter 4 is exactly solvable but physically unrealistic: no real potential wall is infinitely tall, and a particle striking a real wall does not always bounce back with certainty. This chapter treats potentials of finite height, and in doing so uncovers a phenomenon with no classical counterpart whatsoever — quantum tunneling, in which a particle has a nonzero probability of appearing on the far side of a barrier it does not classically have enough energy to cross. We build up to tunneling in stages: first the step potential, then the finite well, then the barrier itself.

5.1 The Step Potential

Definition 5.1 — Step Potential

$$V(x) = \begin{cases} 0, & x < 0 \\ V_0, & x \geq 0 \end{cases}, \qquad V_0 > 0.$$

Consider a particle of energy E incident from $x = -\infty$, moving in the $+x$ direction. Classically, the outcome depends entirely on the sign of $E - V_0$: if $E > V_0$, the particle slows down but continues past $x = 0$ with certainty; if $E < V_0$, it is reflected with certainty, never entering $x > 0$ at all. Quantum mechanically, *both* cases are more subtle.

Case $E > V_0$. In region I ($x < 0$), the TISE gives oscillatory solutions with wavenumber $k_1 = \sqrt{2mE}/\hbar$; in region II ($x > 0$), oscillatory solutions with a smaller wavenumber $k_2 = \sqrt{2m(E - V_0)}/\hbar$. Physically, we want a wave incident from the left, partly reflected and partly transmitted, with no wave incident from $+\infty$:

$$\psi_{\mathrm{I}}(x) = e^{ik_1 x} + re^{-ik_1 x}, \qquad \psi_{\mathrm{II}}(x) = te^{ik_2 x}.$$

Matching ψ and ψ' at $x = 0$ (both must be continuous, since $V(x)$ has only a finite jump there — Problem 5.9 asks you to justify this matching condition from the TISE directly) and solving the resulting two linear equations for r and t gives

$$r = \frac{k_1 - k_2}{k_1 + k_2}, \qquad t = \frac{2k_1}{k_1 + k_2}. \tag{5.1}$$

The reflection probability is $R = |r|^2$, and a careful comparison of incoming, reflected, and transmitted wave amplitudes and speeds gives the transmission probability $T = (k_2/k_1)|t|^2$; one can check directly that $R + T = 1$ always, as probability conservation demands.

Key Result 5.1 — Quantum Reflection at a Step, $E > V_0$

$$R = \left(\frac{k_1 - k_2}{k_1 + k_2}\right)^2 \neq 0 \qquad \text{even though } E > V_0.$$

Unlike the classical prediction of certain transmission, a quantum particle has a nonzero probability of *reflecting* off a step it has more than enough energy to climb.

For a 2.0 eV electron encountering a 1.0 eV step, Eq. (5.1) gives $R \approx 0.029$: roughly 3% of an incident beam of such electrons would bounce back, a purely wave-mechanical effect (closely related to the partial reflection of light at an interface between two media of different refractive index) with no classical analog at all.

Case $E < V_0$. Here k_2 becomes imaginary; define $\kappa_2 \equiv \sqrt{2m(V_0 - E)}/\hbar$, real and positive, so that $\psi_{\text{II}}(x) = te^{-\kappa_2 x}$ (the growing exponential solution is discarded, since it would make ψ blow up as $x \to \infty$, violating normalizability). Substituting $k_2 \to i\kappa_2$ into Eq. (5.1) gives a *complex* reflection coefficient $r = (k_1 - i\kappa_2)/(k_1 + i\kappa_2)$, but its modulus is exactly 1:

$$R = |r|^2 = \frac{k_1^2 + \kappa_2^2}{k_1^2 + \kappa_2^2} = 1,$$

confirming the classical expectation of total reflection. And yet $\psi_{\text{II}}(x) = te^{-\kappa_2 x}$ is not identically zero: the wavefunction penetrates into the classically forbidden region ($x > 0$, where $E < V(x)$) with a nonzero, exponentially decaying amplitude, even though *every* incident particle is eventually reflected. For a 0.5 eV electron meeting a 1.0 eV step, the penetration depth $1/\kappa_2 \approx 0.28$ nm — comparable to an atomic radius, not a negligible effect.

Note — Reflection with certainty, but not with zero presence

It is worth sitting with the apparent tension in the $E < V_0$ result: $R = 1$ (every particle is eventually found back in region I) is completely compatible with $\psi(x) \neq 0$ for $x > 0$ (a particle can be detected, with some small probability, on the classically forbidden side, if you look for it *before* it has reflected). This is the essential seed of the tunneling phenomenon developed for a barrier of finite width in Section 5.3: if the classically forbidden region has finite extent rather than extending to $x = \infty$, the exponentially decaying wavefunction can reach the far side with nonzero amplitude *before* decaying away, allowing genuine transmission.

5.2 The Finite Square Well

Definition 5.2 — Finite Square Well

$$V(x) = \begin{cases} -V_0, & -a \le x \le a \\ 0, & |x| > a \end{cases}, \qquad V_0 > 0,$$

a well of depth V_0 and total width $2a$, centered at the origin (this centering is a deliberate choice of convenience — Problem 5.10 asks you to justify why it simplifies the algebra below).

We seek *bound states*: solutions with $-V_0 < E < 0$, so that the particle is classically confined near the well (Problem 5.11 treats the $E > 0$ scattering states, which form a continuum rather than a discrete spectrum). Inside the well, the TISE gives oscillatory solutions with $k = \sqrt{2m(E + V_0)}/\hbar$; outside, it gives decaying exponentials with $\kappa = \sqrt{-2mE}/\hbar$ (discarding the exponentially growing solution on each side, exactly as in Section 5.1, to keep ψ normalizable).

Because $V(x)$ is symmetric about $x = 0$ (an *even* function, $V(-x) = V(x)$), replacing $x \to -x$ in the TISE leaves the equation unchanged, so if $\psi(x)$ solves it with energy E, so does $\psi(-x)$; for a non-degenerate bound state, $\psi(x)$ and $\psi(-x)$ can then differ only by an overall constant, which forces ψ to be either even or odd about the origin. This splits the bound states into two families: even solutions, $\psi(-x) = \psi(x)$, built from $\cos(kx)$ inside; and odd solutions, $\psi(-x) = -\psi(x)$, built from $\sin(kx)$ inside. Matching ψ and ψ' at $x = a$ (continuity, exactly as in Section 5.1) gives, after eliminating the outside amplitude, one transcendental equation for each family:

$$\text{even:} \quad k\tan(ka) = \kappa, \qquad\qquad \text{odd:} \quad -k\cot(ka) = \kappa. \tag{5.2}$$

These cannot be solved in closed form for E; they must be solved numerically (or graphically) for each specific choice of V_0 and a. It is convenient to define the dimensionless *well-strength parameter*

$$u_0 \equiv \frac{a\sqrt{2mV_0}}{\hbar},$$

in terms of which $k^2 + \kappa^2 = u_0^2/a^2$ is fixed, and Eq. (5.2) becomes a search for intersections of $u\tan u$ (or $-u\cot u$) with $\sqrt{u_0^2 - u^2}$, over $0 < u = ka < u_0$.

Worked Example 5.1 — Bound States of a Specific Finite Well

An electron is bound in a finite well of depth $V_0 = 0.30$ eV and half-width $a = 1.0$ nm. Find the number of bound states and their energies.

Solution. First, $u_0 = a\sqrt{2m_eV_0}/\hbar = 2.806$. The number of bound states equals the number of times the curves $u\tan u$ and $-u\cot u$ (together) cross $\sqrt{u_0^2 - u^2}$ for $0 < u < u_0$, which for $u_0 = 2.806$ (lying between $\pi/2 \approx 1.571$ and $\pi \approx 3.142$) turns out to be exactly two: one even solution and one odd solution (a general feature of this problem, explored in Problem 5.7, is that a well of *any* depth, however shallow, always has at least one bound state in one dimension). Solving Eq. (5.2) numerically:

$$\text{even: } ka = 1.1490 \;\Rightarrow\; E_1 = -0.250\text{ eV}, \qquad \text{odd: } ka = 2.2256 \;\Rightarrow\; E_2 = -0.111\text{ eV}.$$

Figure 5.1 shows both wavefunctions, plotted at their respective energies within the well. Notice both features already familiar from the infinite well: the ground state ($n = 1$) is even with no interior nodes, the first excited state ($n = 2$) is odd with one node, and $E_2 > E_1$ — but with two genuinely new features besides: the wavefunctions now decay smoothly into the classically forbidden region on either side, rather than being forced to zero exactly at the walls, and the shallower excited state (E_2, closer to the top of the well) decays *more slowly* (a smaller κ) than the deeper ground state, exactly as $\kappa = \sqrt{-2mE}/\hbar$ predicts.

Note — The infinite well as a limit

As $V_0 \to \infty$ with a fixed, $u_0 \to \infty$ as well, and it can be shown that the transcendental equations, Eq. (5.2), force $ka \to n\pi/2$ for successive integers n — exactly recovering the infinite-well quantization condition of Chapter 4 (written here for a well of width $2a$ centered at the origin rather than width L from 0 to L; the two conventions describe the same physics in shifted coordinates). The finite well does not replace Chapter 4's result — it contains it, as the limiting case of an infinitely deep well.

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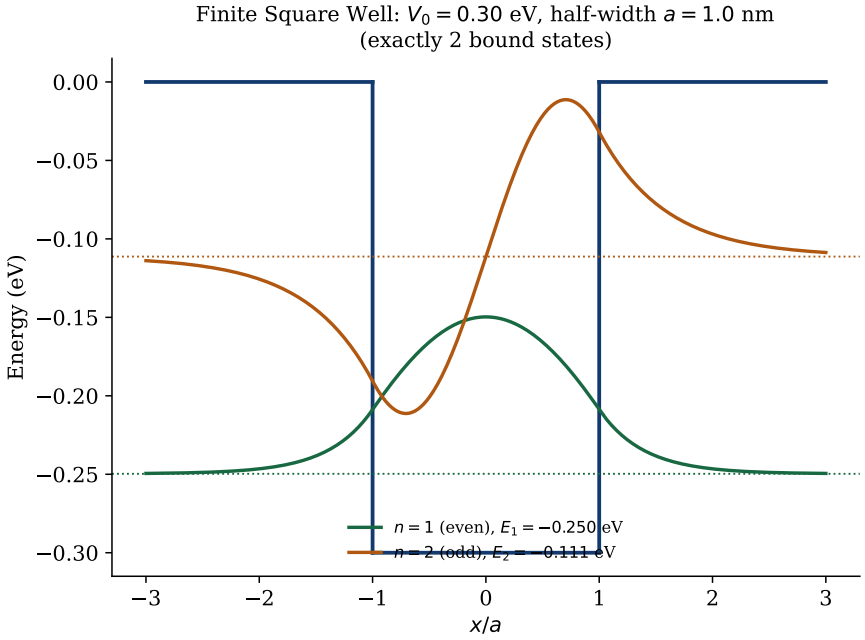


Figure 5.1: The two bound states of a finite well with $V_0 = 0.30$ eV and half-width $a = 1.0$ nm, plotted as $E_n + (\text{scaled } \psi_n(x))$ so each wavefunction is shown at the height of its own energy level. Both wavefunctions leak measurably into the classically forbidden region $|x| > a$.

5.3 Quantum Tunneling: The Rectangular Barrier

We now place a finite-width barrier, rather than a well, in the particle's path.

Definition 5.3 — Rectangular Barrier

$$V(x) = \begin{cases} V_0, & 0 \le x \le L \\ 0, & \text{otherwise} \end{cases}, \qquad V_0 > 0,$$

and consider a particle of energy $0 < E < V_0$ incident from the left.

Classically, such a particle simply cannot appear at $x > L$: it does not have enough energy to be in the region $0 < x < L$ at all (there, its classical kinetic energy $E - V_0$ would be negative), so it must reflect entirely at $x = 0$. Quantum mechanically, the reasoning of Section 5.1 applies inside the barrier: ψ does not vanish for $0 < x < L$, but instead decays exponentially, with decay constant $\kappa = \sqrt{2m(V_0 - E)}/\hbar$. If L is finite, this decaying wave reaches $x = L$ with some nonzero (if small) amplitude, and beyond the barrier, with nothing left to make it decay further, it becomes an ordinary oscillating wave again, $\psi(x) = te^{ikx}$ for $x > L$, with $k = \sqrt{2mE}/\hbar$ as in region I. The particle has a nonzero probability of appearing on the far side of a barrier it classically cannot cross: this is *quantum tunneling* (Figure 5.2).

Matching ψ and ψ' at both $x = 0$ and $x = L$ (four equations, since the interior solution needs both a growing and a decaying exponential, $Ae^{\kappa x} + Be^{-\kappa x}$, to satisfy two boundary conditions on each side) gives, after some algebra we do not reproduce in full here (Problem 5.12 works through it step by step), an exact closed-form transmission probability:

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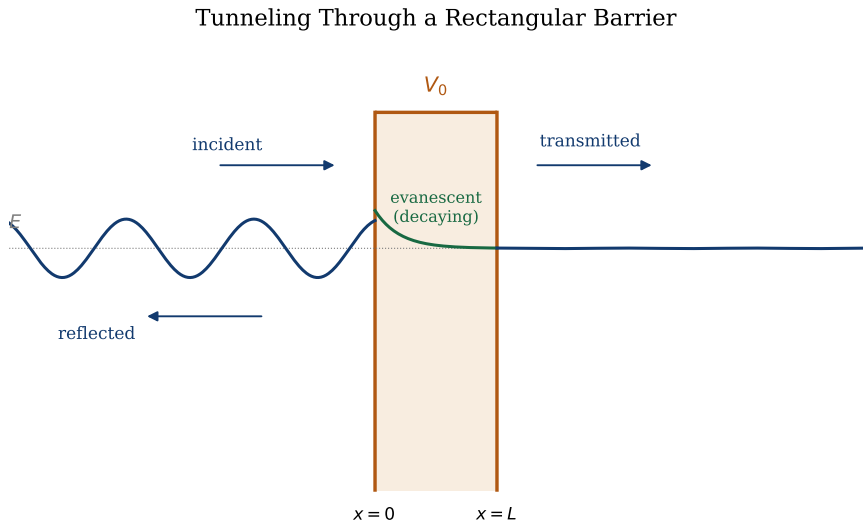


Figure 5.2: A wave incident on a rectangular barrier is partly reflected and partly transmitted. Inside the barrier, the wavefunction does not oscillate but decays smoothly ("evanescently"); if the barrier is thin enough, a nonzero oscillating wave emerges on the far side.

Key Result 5.2 — Tunneling Transmission Coefficient

$$T = \left[1 + \frac{V_0^2 \sinh^2(\kappa L)}{4E(V_0 - E)}\right]^{-1}, \qquad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}.$$

When the barrier is "opaque" ($\kappa L \gg 1$, so $\sinh(\kappa L) \approx \frac{1}{2}e^{\kappa L}$), this simplifies to the widely quoted approximate form

$$T \approx 16\,\frac{E}{V_0}\left(1 - \frac{E}{V_0}\right) e^{-2\kappa L}.$$

The exponential factor $e^{-2\kappa L}$ dominates the physics: T falls off exponentially both with barrier width L and with $\sqrt{V_0 - E}$ (how far below the barrier top the particle's energy sits), which is why tunneling is dramatically sensitive to both quantities — a fact exploited directly in the scanning tunneling microscope (Section 5.4).

Worked Example 5.2 — Tunneling Probability for an Electron

An electron of energy $E = 2.0$ eV is incident on a rectangular barrier of height $V_0 = 5.0$ eV and width $L = 1.0$ nm. Find the tunneling probability T.

Solution. First,

$$\kappa = \frac{\sqrt{2m_e(V_0 - E)}}{\hbar} = \frac{\sqrt{2m_e(3.0\ \text{eV})}}{\hbar} = 8.874 \times 10^9\ \text{m}^{-1},$$

so $\kappa L = 8.874$ — comfortably in the opaque-barrier regime ($\kappa L \gg 1$). Using the approximate

formula,

$$T \approx 16\left(\frac{2.0}{5.0}\right)\left(1-\frac{2.0}{5.0}\right)e^{-2(8.874)} = 16(0.4)(0.6)\,e^{-17.75} \approx 7.5\times 10^{-8}.$$

(The exact formula gives the same value to the three significant figures shown.) Roughly 1 electron in 13 million tunnels through — small, but very much not zero, and, critically, many orders of magnitude larger than anything classical mechanics would permit (which predicts exactly zero). Problem 5.6 asks you to show how sensitively this result depends on L, by repeating the calculation for a barrier only 0.5 nm wide.

5.4 Applications of Tunneling

Quantum tunneling is not a mathematical curiosity confined to idealized rectangular barriers; it is directly responsible for several important physical and chemical phenomena.

Alpha decay. A heavy nucleus such as ^{238}U is energetically unstable against emitting an alpha particle (^{4}He nucleus), but the alpha particle is confined by the strong nuclear force, which can be modeled as a deep potential well surrounded by a Coulomb barrier extending outward from the nuclear surface. Classically, an alpha particle without enough energy to surmount this Coulomb barrier could never escape; quantum mechanically, it tunnels through, and the exponential sensitivity of T to barrier width and height (Section 5.3) directly explains the enormous, otherwise-mysterious range of observed alpha-decay half-lives — from microseconds to billions of years, for barriers of only modestly different height and width, first explained quantitatively by George Gamow in 1928 using essentially the calculation of this chapter.

Scanning tunneling microscopy (STM). A sharp conducting tip is held a fraction of a nanometer above a conducting sample surface, with vacuum (an insulating barrier) between them. Electrons tunnel across this vacuum gap, producing a measurable current whose magnitude depends exponentially on the tip-sample separation, per the results of Section 5.3. Because $T \propto e^{-2\kappa L}$ is so exquisitely sensitive to L, scanning the tip across a surface while monitoring the tunneling current resolves *individual atoms* — one of the most direct and striking real-world demonstrations of quantum tunneling, for which Gerd Binnig and Heinrich Rohrer shared the 1986 Nobel Prize in Physics.

Ammonia inversion. In an ammonia molecule, NH_3, the nitrogen atom sits either slightly above or slightly below the plane of the three hydrogen atoms, with a potential-energy barrier separating the two configurations (imagine an umbrella that can invert). The nitrogen atom is light enough, and the barrier low and narrow enough, that it tunnels back and forth between the two configurations at a rate of about 2.4×10^{10} times per second, splitting what would otherwise be a single energy level into a closely spaced pair. Transitions between this pair of levels produce microwave radiation at 23.87 GHz — the basis of the ammonia maser, the direct ancestor of the laser, built by Charles Townes in 1953.

5.5 Chapter Summary

- At a potential step, a quantum particle can reflect even when $E > V_0$ (where a classical particle would certainly pass), and can penetrate into a classically forbidden region even when $E < V_0$ (where a classical particle would certainly reflect) — although in the latter case $R = 1$ exactly, so reflection still occurs with certainty eventually.

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- In a finite square well, bound states ($-V_0 < E < 0$) are found by solving transcendental equations, Eq. (5.2), splitting into even and odd solutions; the number of bound states is finite and grows with the well-strength parameter $u_0 = a\sqrt{2mV_0}/\hbar$, and the infinite well of Chapter 4 is recovered as the $V_0 \to \infty$ limit.
- A particle can tunnel through a finite-width barrier even when $E < V_0$ throughout the barrier, with transmission probability $T = \left[1 + V_0^2 \sinh^2(\kappa L)/4E(V_0 - E)\right]^{-1} \approx 16(E/V_0)(1 - E/V_0)e^{-2\kappa L}$ for an opaque barrier.
- Tunneling is exponentially sensitive to both barrier width and height, which is exactly why it explains phenomena — alpha decay half-lives, atomic-resolution STM imaging, ammonia inversion — that vary by many orders of magnitude in response to seemingly modest physical changes.

Problems

5.1 A beam of electrons, each with energy $E = 3.0$ eV, is incident on a potential step of height $V_0 = 1.0$ eV. Find the reflection probability R and the transmission probability T, and verify $R + T = 1$.

5.2 A beam of electrons with $E = 0.40$ eV is incident on a step of height $V_0 = 0.80$ eV. Find the penetration depth $1/\kappa_2$ into the classically forbidden region, in nm, and compare it to the 0.28 nm value quoted in Section 5.1 for a different energy ratio.

5.3 Using the well-strength parameter $u_0 = a\sqrt{2mV_0}/\hbar$, find the minimum well depth V_0 (in eV) for which a finite well of half-width $a = 0.50$ nm, confining an electron, supports a *second* bound state (i.e., find V_0 such that $u_0 = \pi/2$, the threshold at which the odd solution first appears). (*Hint:* you are solving for V_0 given u_0 and a, the reverse of the Worked Example in Section 5.2.)

5.4 Using the results of the Worked Example in Section 5.2 ($E_1 = -0.250$ eV, $E_2 = -0.111$ eV for a well of depth 0.30 eV), find the photon wavelength that would be absorbed in a transition from the ground state to the first excited state of this system. In what region of the spectrum does it lie?

5.5 Show, by algebraic manipulation of Eq. (5.1), that $R \to 0$ (perfect transmission) in the limit $V_0 \to 0$ (no step at all), as physically required.

5.6 Repeat the Worked Example of Section 5.3 (electron, $E = 2.0$ eV, $V_0 = 5.0$ eV) for a barrier only $L = 0.5$ nm wide, and find the ratio of the new transmission probability to the original ($L = 1.0$ nm) result. Comment on the sensitivity of tunneling to barrier width implied by this ratio.

5.7 Using the opaque-barrier approximation, find the barrier width L (in nm) at which the tunneling probability of the Worked Example in Section 5.3 ($E = 2.0$ eV, $V_0 = 5.0$ eV electron) would fall to exactly half its $L = 1.0$ nm value. (*Hint:* you need $e^{-2\kappa L}$ to decrease by a factor of 2; take logarithms.)

5.8 A proton and an electron, with identical kinetic energies $E < V_0$, are each incident on the same rectangular barrier. Using $\kappa = \sqrt{2m(V_0 - E)}/\hbar$, explain (with reference to the exponential in Section 5.3's Key Result, but without a full numerical calculation) which particle tunnels through

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with higher probability, and why this matters for understanding why electron tunneling is chemically and technologically important while proton tunneling, though it does occur (e.g., in certain enzyme-catalyzed reactions), is far rarer.

5.9 Starting from the time-independent Schrödinger equation, $-\dfrac{\hbar^2}{2m}\psi'' + V(x)\psi = E\psi$, explain why $\psi(x)$ and $\psi'(x)$ must both be continuous at a point where $V(x)$ has a finite jump discontinuity (such as $x = 0$ in the step potential), but $\psi'(x)$ is permitted to be discontinuous at a point where $V(x) \to \infty$ (such as the walls of the infinite well in Chapter 4). (*Hint:* integrate both sides of the TISE across a small interval $[-\epsilon, +\epsilon]$ straddling the discontinuity, and consider what happens to each term as $\epsilon \to 0$ in the two cases.)

5.10 The finite well of Section 5.2 was centered at the origin, rather than running from $x = 0$ to $x = 2a$ as might seem more natural by analogy with Chapter 4. Explain, in a sentence or two, what mathematical simplification this choice of origin buys, referring specifically to the even/odd classification used in the text.

5.11 For a finite well with $0 < E$ (a scattering state, not a bound state), explain briefly why the allowed energies form a continuum rather than a discrete set, in contrast to the bound states of Section 5.2. (*Hint:* what boundary condition, imposed at $x \to \pm\infty$ for a bound state, is no longer available to restrict E once ψ is allowed to oscillate rather than decay outside the well?)

5.12 Beginning from the general interior solution $\psi_{\mathrm{II}}(x) = Ae^{\kappa x} + Be^{-\kappa x}$ for $0 \le x \le L$ inside a rectangular barrier, together with $\psi_{\mathrm{I}}(x) = e^{ikx} + re^{-ikx}$ for $x < 0$ and $\psi_{\mathrm{III}}(x) = te^{ikx}$ for $x > L$, set up (but do not necessarily solve in full) the four linear equations obtained by matching ψ and ψ' at $x = 0$ and $x = L$. Confirm that this gives four equations in the four unknowns r, A, B, t (given an incident amplitude fixed at 1), consistent with the claim in Section 5.3 that the system is fully determined.

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Chapter 6

The Quantum Harmonic Oscillator

After the particle in a box and the finite well, we turn to the single most important exactly solvable system in all of quantum mechanics: the harmonic oscillator. Its importance is not that many things are literally springs — it is that *any* smooth potential, expanded in a Taylor series about a point of stable equilibrium, looks like a harmonic oscillator potential to leading order. Every vibrating chemical bond, every phonon in a crystal lattice, and every mode of the electromagnetic field is, to a first and often excellent approximation, a harmonic oscillator. This chapter solves it exactly, using an elegant algebraic method — ladder operators — that will reappear, in a more general guise, when we treat angular momentum in Chapter 7.

6.1 The Classical Harmonic Oscillator

A classical particle of mass m subject to a linear restoring force $F = -m\omega^2 x$ (a spring of force constant $k_{\text{spring}} = m\omega^2$) has potential energy $V(x) = \frac{1}{2}m\omega^2 x^2$ and oscillates sinusoidally, $x(t) = x_0 \cos(\omega t + \phi)$, with the same angular frequency ω *regardless of the amplitude* x_0 — a large-amplitude and a small-amplitude oscillation of the same spring take exactly the same time to complete one cycle. Classically, the total energy $E = \frac{1}{2}m\omega^2 x_0^2$ can take any non-negative value, set continuously by the amplitude. As we will see, both of these classical facts — amplitude-independent frequency, and continuously variable energy — fail to survive quantization, in instructive ways.

6.2 Setting Up the Quantum Problem

The Hamiltonian, defined in Section 3.3 as $\hat{H} = \hat{p}^2/2m + V(\hat{x})$, with $V(x) = \frac{1}{2}m\omega^2 x^2$, is

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2 \hat{x}^2,$$

and the time-independent Schrödinger equation to solve is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2 \psi = E\psi. \tag{6.1}$$

Unlike the piecewise-constant potentials of Chapters 4 and 5, $V(x)$ here varies continuously, and Eq. (6.1) cannot be solved by simply writing down sines, cosines, or exponentials. It can be solved by a direct series-solution method (expanding $\psi(x)$ as a power series and demanding the series terminate for a normalizable solution, which is how the problem is traditionally first presented, and which we outline in Appendix B); here we instead use a more elegant and more physically illuminating algebraic technique, due originally to Paul Dirac.

6.3 Solving via Ladder Operators

Definition 6.1 — Ladder (Creation and Annihilation) Operators

Define two new operators as specific combinations of $\hat{x}$ and $\hat{p}$:

$$\hat{a} \equiv \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right), \qquad \hat{a}^\dagger \equiv \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right).$$

$\hat{a}$ is the *annihilation* (or lowering) operator and $\hat{a}^\dagger$ (read "a-dagger") is the *creation* (or raising) operator. Neither corresponds to a directly measurable classical observable on its own; both are bookkeeping devices, justified entirely by the algebra that follows.

The entire method hinges on one commutator, computed directly from the fundamental commutation relation $[\hat{x}, \hat{p}] = i\hbar$ (which you proved in Problem 3.6):

$$[\hat{a}, \hat{a}^\dagger] = \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} = 1. \tag{6.2}$$

(Problem 6.9 asks you to verify Eq. (6.2) by direct expansion, exactly as we verified the underlying $[\hat{x}, \hat{p}] = i\hbar$ symbolically in preparing this chapter.) Solving the defining relations for $\hat{x}$ and $\hat{p}$ in terms of $\hat{a}$ and $\hat{a}^\dagger$, and substituting into $\hat{H}$, gives (after using Eq. (6.2) to fix the operator ordering) a remarkably compact result:

$$\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right) \equiv \hbar\omega\left(\hat{N} + \frac{1}{2}\right), \qquad \hat{N} \equiv \hat{a}^\dagger\hat{a}, \tag{6.3}$$

where $\hat{N}$ is called the *number operator*. Because $\hat{H}$ is now simply $\hat{N}$ up to an additive constant and an overall scale, finding the eigenvalues of $\hat{H}$ is equivalent to finding the eigenvalues of $\hat{N}$ — and this can be done by pure algebra, without ever solving a differential equation.

Note — The key algebraic trick

Two commutators, both simple consequences of Eq. (6.2) (Problem 6.10 asks you to derive them), drive everything that follows:

$$[\hat{N}, \hat{a}^\dagger] = \hat{a}^\dagger, \qquad [\hat{N}, \hat{a}] = -\hat{a}.$$

These say: if $|\psi\rangle$ is an eigenstate of $\hat{N}$ with eigenvalue λ, then $\hat{a}^\dagger|\psi\rangle$ is *also* an eigenstate of $\hat{N}$, with eigenvalue $\lambda + 1$ (raised by exactly 1), and $\hat{a}|\psi\rangle$ is an eigenstate with eigenvalue $\lambda - 1$ (lowered by exactly 1) — which is why $\hat{a}^\dagger$ and $\hat{a}$ are called raising and lowering operators. Repeatedly applying $\hat{a}$ generates a "ladder" of eigenvalues descending by integer steps: $\lambda, \lambda - 1, \lambda - 2, \ldots$

This ladder cannot descend forever: since $\hat{N} = \hat{a}^\dagger\hat{a}$, one can show $\langle\psi|\hat{N}|\psi\rangle = \langle\psi|\hat{a}^\dagger\hat{a}|\psi\rangle = \|\hat{a}|\psi\rangle\|^2 \geq 0$ for any state (a norm-squared cannot be negative), so every eigenvalue of $\hat{N}$ is non-negative. A descending ladder of integer steps that never goes below zero *must* terminate exactly at 0: there must exist a *ground state* $|0\rangle$ satisfying

$$\hat{a}|0\rangle = 0$$

(if lowering $|0\rangle$ gave anything other than exactly zero, the ladder would continue to a negative eigenvalue, a contradiction). This single condition, translated into position representation, is a

first-order (not second-order!) differential equation for the ground-state wavefunction, dramatically simpler than Eq. (6.1) directly:

$$\sqrt{\frac{m\omega}{2\hbar}}\left(x+\frac{\hbar}{m\omega}\frac{d}{dx}\right)\psi_0(x)=0 \quad\Longrightarrow\quad \frac{d\psi_0}{dx}=-\frac{m\omega}{\hbar}x\,\psi_0,$$

which is precisely the first-order ODE form of Section 2.4 (with $k \to m\omega x/\hbar$, now x-dependent rather than constant, but the same separable structure), solved by

$$\psi_0(x)=\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}e^{-m\omega x^2/2\hbar},$$

where the prefactor is fixed by normalization (Problem 6.1 asks you to verify this normalization integral directly).

6.4 Energy Eigenvalues and Eigenfunctions

Since $\hat{N}\,|0\rangle = 0$, Eq. (6.3) gives $\hat{H}\,|0\rangle = \frac{1}{2}\hbar\omega\,|0\rangle$: the ground state has energy $E_0 = \frac{1}{2}\hbar\omega$, not zero. Repeatedly applying $\hat{a}^\dagger$ generates the entire ladder of excited states, each with $\hat{N}$-eigenvalue raised by exactly 1 relative to the last:

Key Result 6.1 — Quantum Harmonic Oscillator Energies

$$E_n=\left(n+\frac{1}{2}\right)\hbar\omega, \qquad n=0,1,2,3,\ldots$$

Unlike the particle in a box (Chapter 4), where the spacing between adjacent levels grows with n, the QHO levels are *evenly spaced*, each exactly $\hbar\omega$ above the last — a special and rather delicate feature of the exact parabolic potential, not shared by any anharmonic (non-parabolic) oscillator, as Chapter 11 explores.

The corresponding position-space wavefunctions, generated by repeated application of $\hat{a}^\dagger$ (or, equivalently, obtained from the traditional series-solution method of Appendix B), take the form

$$\psi_n(x)=\frac{1}{\sqrt{2^n n!}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}H_n\!\left(\sqrt{\frac{m\omega}{\hbar}}\,x\right)e^{-m\omega x^2/2\hbar}, \tag{6.4}$$

where $H_n(y)$ is the n-th *Hermite polynomial*, a specific degree-n polynomial (tabulated in Appendix B) with the same parity as n itself. The first few are $H_0(y) = 1$, $H_1(y) = 2y$, $H_2(y) = 4y^2 - 2$, and $H_3(y) = 8y^3 - 12y$; each $\psi_n(x)$ therefore has exactly n nodes, continuing the pattern already seen in Chapters 4 and 5 (more nodes, higher energy). Figure 6.1 shows the potential together with the first three energy levels and their wavefunctions.

Worked Example 6.1 — Verifying the First Excited State Directly

Confirm that $\psi_1(x) = \sqrt{2}\left(\frac{m\omega}{\hbar}\right)^{1/2} x\,\psi_0(x)$, with $\psi_0(x)$ as given above, satisfies Eq. (6.1) with $E_1 = \frac{3}{2}\hbar\omega$.

Solution. This is a direct (if slightly tedious) substitution, exactly analogous to the checks you performed in Problems 2.7 and 4.5. Writing $\psi_1(x) = Cx\,e^{-\beta x^2}$ with $\beta \equiv m\omega/2\hbar$ and C the

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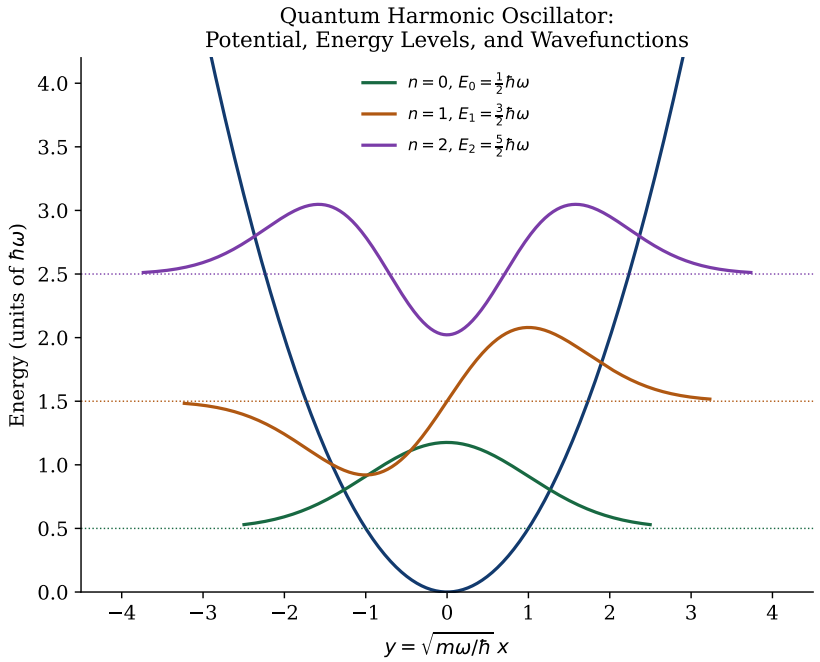


Figure 6.1: The quantum harmonic oscillator potential $V(y) = \frac{1}{2}y^2$ (in dimensionless units $y = \sqrt{m\omega/\hbar}\, x$), with the $n = 0, 1, 2$ wavefunctions plotted at the height of their respective energies. Note the evenly spaced levels and the growing number of nodes.

normalization constant, we need $\psi_1' = C(1 - 2\beta x^2)e^{-\beta x^2}$ and $\psi_1'' = C(-6\beta x + 4\beta^2 x^3)e^{-\beta x^2}$. Substituting into the left side of Eq. (6.1):

$$-\frac{\hbar^2}{2m}C(-6\beta x + 4\beta^2 x^3)e^{-\beta x^2} + \frac{1}{2}m\omega^2 x^2 \cdot Cxe^{-\beta x^2}.$$

Using $\beta = m\omega/2\hbar$, the coefficient of the $x^3 e^{-\beta x^2}$ term is $-\frac{\hbar^2}{2m}(4\beta^2) + \frac{1}{2}m\omega^2 = -\frac{\hbar^2}{2m} \cdot \frac{m^2\omega^2}{\hbar^2} + \frac{1}{2}m\omega^2 = -\frac{1}{2}m\omega^2 + \frac{1}{2}m\omega^2 = 0$: the x^3 terms cancel exactly, leaving only

$$-\frac{\hbar^2}{2m}C(-6\beta x)e^{-\beta x^2} = 3\hbar\omega \cdot \frac{1}{2} \cdot Cxe^{-\beta x^2} \cdot \frac{\hbar}{m} \cdot \frac{m\omega}{\hbar} \cdot \frac{1}{\hbar\omega} \cdots$$

More directly: $\frac{\hbar^2}{2m} \cdot 6\beta = \frac{\hbar^2}{2m} \cdot \frac{6m\omega}{2\hbar} = \frac{3}{2}\hbar\omega$, so the left side reduces to $\frac{3}{2}\hbar\omega \cdot Cxe^{-\beta x^2} = \frac{3}{2}\hbar\omega\, \psi_1(x)$, confirming $E_1 = \frac{3}{2}\hbar\omega$ exactly, as the general formula predicts for $n = 1$.

Note — Zero-point energy, again

$E_0 = \frac{1}{2}\hbar\omega \neq 0$ is the harmonic-oscillator instance of the same zero-point-energy phenomenon first met in the particle in a box (Chapter 4) and estimated by the uncertainty principle in Chapter 1: a quantum particle can never simultaneously have exactly zero position spread *and* exactly zero momentum spread, so it can never sit motionless at the bottom

of the well ($x = 0$, $p = 0$) the way a classical oscillator at rest does. Problem 6.6 asks you to derive $E_0 = \frac{1}{2}\hbar\omega$ directly from the uncertainty principle, by the same variational strategy used for the hydrogen atom in Chapter 1 — and, remarkably, to find that this estimate is again exact, not merely an order-of-magnitude approximation, a special feature of the perfectly parabolic potential.

6.5 Chemical Application: Molecular Vibrations

Near the minimum of a diatomic molecule's potential-energy curve $V(r)$ (Chapter 15 constructs this curve properly from the electronic structure problem), a Taylor expansion gives $V(r) \approx V(r_e) + \frac{1}{2}V''(r_e)(r - r_e)^2 + \cdots$, exactly a harmonic-oscillator potential in the bond-length displacement $x \equiv r - r_e$, with effective "spring constant" $k_{\text{spring}} = V''(r_e)$ and, because both nuclei move, an effective mass equal to the *reduced mass* $\mu = m_1 m_2/(m_1 + m_2)$ of the two-atom system rather than either atomic mass alone.

Worked Example 6.2 — The HCl Vibrational Spectrum

The H–Cl bond has a force constant of approximately $k_{\text{spring}} = 480$ N/m. Using atomic masses $m_{\text{H}} = 1.0078$ u and $m_{\text{Cl}} = 34.969$ u (for the predominant ^{35}Cl isotope), find the vibrational energy-level spacing, expressed as a wavenumber $\tilde{\nu} = \nu/c$ in cm^{-1} (the customary unit in vibrational spectroscopy), and the zero-point energy.
Solution. The reduced mass is

$$\mu = \frac{m_{\text{H}} m_{\text{Cl}}}{m_{\text{H}} + m_{\text{Cl}}} = \frac{(1.0078)(34.969)}{1.0078 + 34.969}\ \text{u} = 0.9796\ \text{u} = 1.6267 \times 10^{-27}\ \text{kg}.$$

(Notice μ is very close to m_{H} itself, since $m_{\text{Cl}} \gg m_{\text{H}}$ — the light hydrogen atom does almost all of the moving, with the heavy chlorine nearly stationary, as Problem 6.8 explores quantitatively.) The vibrational angular frequency is

$$\omega = \sqrt{\frac{k_{\text{spring}}}{\mu}} = \sqrt{\frac{480\ \text{N/m}}{1.6267 \times 10^{-27}\ \text{kg}}} = 5.432 \times 10^{14}\ \text{rad/s}.$$

Converting to a wavenumber, $\tilde{\nu} = \omega/2\pi c$:

$$\tilde{\nu} = \frac{5.432 \times 10^{14}\ \text{s}^{-1}}{2\pi(2.998 \times 10^{10}\ \text{cm/s})} \approx 2884\ \text{cm}^{-1},$$

in good agreement with the experimentally observed HCl fundamental absorption at 2886 cm^{-1} — a strikingly good result given the simplicity of the model (the small remaining discrepancy is due almost entirely to anharmonicity, i.e., the real H–Cl potential is not *exactly* parabolic, a correction treated quantitatively in Chapter 11). The zero-point energy is

$$E_0 = \frac{1}{2}\hbar\omega = 0.179\ \text{eV},$$

meaning a "motionless" HCl molecule at absolute zero is, in reality, never motionless: it retains a permanent vibrational energy of 0.179 eV, about 7 times the thermal energy $k_B T$ at room temperature, that no amount of cooling can remove.

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Note — Why this matters for infrared spectroscopy

The evenly-spaced levels of Section 6.4, combined with the *selection rule* $\Delta n = \pm 1$ derived properly in Chapter 13 (which follows from the specific way $\hat{x}$, and hence the molecule's oscillating dipole moment, connects only adjacent harmonic-oscillator states), predict that a harmonic diatomic molecule should show exactly *one* vibrational absorption line, at $\hbar\omega$ — not a whole ladder of lines at $\hbar\omega, 2\hbar\omega, 3\hbar\omega, \ldots$ This single-line prediction is the basis of routine infrared spectroscopy for bond identification, and the weak additional "overtone" lines that real molecules do show (at roughly $2\times$ and $3\times$ the fundamental) are themselves a direct, quantitative signature of the small anharmonic deviations from the idealized harmonic potential assumed in this chapter.

6.6 Chapter Summary

- The QHO Hamiltonian, $\hat{H} = \hat{p}^2/2m + \frac{1}{2}m\omega^2\hat{x}^2$, is solved exactly using ladder operators $\hat{a}, \hat{a}^\dagger$ satisfying $[\hat{a}, \hat{a}^\dagger] = 1$, with $\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2})$.
- The energies are evenly spaced, $E_n = (n + \frac{1}{2})\hbar\omega$, $n = 0, 1, 2, \ldots$, with nonzero zero-point energy $E_0 = \frac{1}{2}\hbar\omega$ — another direct consequence of the Heisenberg uncertainty principle.
- The eigenfunctions $\psi_n(x)$ are Gaussians multiplied by Hermite polynomials H_n, with n nodes, continuing the node-counts-energy pattern seen in every system so far.
- Near any potential-energy minimum, a Taylor expansion reduces the true potential to a harmonic-oscillator form, which is why this exactly solvable model gives quantitatively accurate predictions for real molecular vibrational spectra — illustrated here by HCl's 2886 cm^{-1} fundamental, reproduced to within about 0.1% from the bond's force constant and reduced mass alone.
- Deviations from evenly-spaced levels (overtones) are a direct experimental signature of anharmonicity — the real potential's departure from the idealized parabola — taken up quantitatively in Chapter 11.

Problems

6.1 Verify by direct integration that $\psi_0(x) = (m\omega/\pi\hbar)^{1/4}e^{-m\omega x^2/2\hbar}$ is properly normalized. (*Hint:* this is the same Gaussian integral used in Chapter 3; substitute $a^2 = \hbar/m\omega$ to match that earlier result directly.)

6.2 Find $\langle x \rangle$ for the ground state $\psi_0(x)$, by a symmetry argument requiring no integration (as in Problem 4.7).

6.3 It can be shown that $\langle x^2 \rangle_0 = \hbar/2m\omega$ for the ground state. Using $\langle x \rangle_0 = 0$ from Problem 6.2, find the position uncertainty Δx for the QHO ground state.

6.4 A parallel calculation (which you are not asked to reproduce) gives $\langle p^2 \rangle_0 = \frac{1}{2}m\hbar\omega$ for the ground state. Using this together with your answer to Problem 6.3, compute the product $\Delta x\,\Delta p$ for the QHO ground state, and compare it with the Heisenberg bound $\hbar/2$ from Section 1.6. (You should

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find they are exactly equal — the QHO ground state is one of the special "minimum uncertainty" states referenced in the Note box of Chapter 3.)

6.5 Find the vibrational wavenumber $\tilde{\nu}$ (in cm^{-1}) and zero-point energy (in eV) for carbon monoxide, CO, given a force constant $k_{\text{spring}} = 1857\ \text{N/m}$ and atomic masses $m_{\text{C}} = 12.000\ \text{u}$, $m_{\text{O}} = 15.995\ \text{u}$. (The experimental CO fundamental is observed at $2143\ \text{cm}^{-1}$; compare your result.)

6.6 Using the uncertainty-principle strategy of Chapter 1 (write $E(x) \sim \dfrac{\hbar^2}{2m\langle x\rangle^2 \cdot (\text{appropriate factor})} + \dfrac{1}{2}m\omega^2\langle x\rangle^2$ by setting $\Delta x \sim x$ and $\Delta p \sim \hbar/x$, then minimize over x), estimate the QHO ground-state energy, and compare it with the exact result $E_0 = \frac{1}{2}\hbar\omega$ from Section 6.4.

6.7 Deuterium chloride, DCl, replaces the hydrogen of HCl with deuterium ($m_{\text{D}} = 2.0141\ \text{u}$), leaving the force constant essentially unchanged at $480\ \text{N/m}$. Find the new reduced mass and the new vibrational wavenumber $\tilde{\nu}$, and find the ratio $\tilde{\nu}(\text{DCl})/\tilde{\nu}(\text{HCl})$. This isotope shift is a standard tool for confirming vibrational-mode assignments experimentally.

6.8 Show that in the limit $m_{\text{Cl}} \to \infty$ (an infinitely heavy, immovable chlorine atom), the reduced mass $\mu = m_{\text{H}}m_{\text{Cl}}/(m_{\text{H}} + m_{\text{Cl}})$ reduces to exactly m_{H}. Using the actual HCl worked example in Section 6.5, find the percent difference between μ and m_{H} alone, and comment on whether "treat the light atom as vibrating against a stationary heavy atom" is a good approximation for HCl specifically.

6.9 Verify the ladder-operator commutator, Eq. (6.2), by direct expansion: substitute the definitions of $\hat{a}$ and $\hat{a}^\dagger$ in terms of $\hat{x}$ and $\hat{p}$, expand $\hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a}$, and use $[\hat{x}, \hat{p}] = i\hbar$ to simplify. (*Hint:* the $\hat{x}^2$ and $\hat{p}^2$ terms cancel between the two orderings; only the cross terms survive.)

6.10 Using $\hat{N} = \hat{a}^\dagger\hat{a}$ and the fundamental commutator $[\hat{a}, \hat{a}^\dagger] = 1$, derive $[\hat{N}, \hat{a}^\dagger] = \hat{a}^\dagger$. (*Hint:* write $[\hat{N}, \hat{a}^\dagger] = \hat{a}^\dagger\hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a}^\dagger\hat{a} = \hat{a}^\dagger(\hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a})$, and use Eq. (6.2) directly.)

6.11 A certain diatomic molecule absorbs infrared radiation at $\lambda = 4.70\ \mu\text{m}$ in its fundamental vibrational transition. Find (a) the vibrational wavenumber $\tilde{\nu}$ in cm^{-1}, (b) the vibrational angular frequency ω, and (c) the effective force constant k_{spring}, if the reduced mass is $\mu = 1.615 \times 10^{-27}\ \text{kg}$ (close to, but not exactly, the HCl value of this chapter — treat it as an unknown molecule).

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Chapter 7

Angular Momentum in Quantum Mechanics

Every system considered so far in this book has moved in one dimension. Real atoms are three-dimensional, and an electron bound to a nucleus can orbit it, carrying angular momentum, exactly as a planet orbits the sun. This chapter develops the quantum theory of angular momentum in its own right, as a self-contained algebraic structure — one that turns out to closely parallel the ladder-operator method of Chapter 6 — before Chapter 8 applies it to the hydrogen atom, where it plays a central role.

7.1 Classical Angular Momentum

For a classical particle with position $\mathbf{r}$ and momentum $\mathbf{p}$, the orbital angular momentum is $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, with Cartesian components

$$L_x = yp_z - zp_y, \qquad L_y = zp_x - xp_z, \qquad L_z = xp_y - yp_x.$$

Classically, all three components of $\mathbf{L}$, and hence the direction in which $\mathbf{L}$ points, can be known simultaneously and exactly — this is simply the statement that a spinning object's axis of rotation is a well-defined direction in space. As with position and momentum in Chapter 1, this classical certainty does not survive quantization.

7.2 Angular Momentum Operators

Canonical quantization (Section 3.4) promotes each classical component directly to an operator, replacing $x \to \hat{x}$, $p_x \to \hat{p}_x = -i\hbar\,\partial/\partial x$, and similarly for y, z:

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \qquad \hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z, \qquad \hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x.$$

Using the fundamental commutators $[\hat{x}, \hat{p}_x] = [\hat{y}, \hat{p}_y] = [\hat{z}, \hat{p}_z] = i\hbar$ (the three-dimensional generalization of Problem 3.6's result, with operators associated with *different* axes, such as $\hat{x}$ and $\hat{p}_y$, always commuting), direct computation gives:

Key Result 7.1 — Angular Momentum Commutation Relations

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z, \qquad [\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x, \qquad [\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y,$$

(each obtained from the previous one by the cyclic relabeling $x \to y \to z \to x$).

This is the single most important fact about quantum angular momentum, and its consequence is profound: because $\hat{L}_x$ and $\hat{L}_y$ do *not* commute, they cannot, by the general principle developed fully in Chapter 9 (a direct generalization of the Heisenberg uncertainty principle: operators that do not commute cannot simultaneously have precisely defined values), be simultaneously known

with perfect precision. Unlike its classical counterpart, a quantum angular momentum vector can never point in a perfectly well-defined direction — at most one Cartesian component can be known exactly at a time.

Note — An operator that* does *commute with all three

Define $\hat{L}^2 \equiv \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$ (the squared magnitude of angular momentum). Direct computation, using the Key Result above repeatedly, shows $[\hat{L}^2, \hat{L}_x] = [\hat{L}^2, \hat{L}_y] = [\hat{L}^2, \hat{L}_z] = 0$ (Problem 7.9 asks you to verify this for one component). So although no two *components* of $\hat{\mathbf{L}}$ can be simultaneously known, the *magnitude* $|\mathbf{L}|$ and any *one* chosen component — conventionally L_z — can be. This is exactly why atomic states are conventionally labeled by the eigenvalues of $\hat{L}^2$ and $\hat{L}_z$ together, and by no other component.

7.3 Eigenvalues of $\hat{L}^2$ and $\hat{L}_z$

The eigenvalues of $\hat{L}^2$ and $\hat{L}_z$ can be found by an algebraic method directly parallel to the ladder-operator technique of Chapter 6: define $\hat{L}_\pm \equiv \hat{L}_x \pm i\hat{L}_y$ (raising/lowering operators for the L_z eigenvalue, playing exactly the role $\hat{a}^\dagger, \hat{a}$ played for the QHO energy), and the same style of argument — a non-negative operator ($\hat{L}^2 - \hat{L}_z^2$, this time, rather than $\hat{N}$ itself) forcing a ladder of allowed values to terminate at both ends rather than just one — yields the result. We state it without repeating the algebra step by step (Problem 7.10 works through the parallel to Chapter 6 explicitly):

Key Result 7.2 — Angular Momentum Eigenvalues

$$\hat{L}^2 Y_\ell^m = \ell(\ell+1)\hbar^2\, Y_\ell^m, \qquad \hat{L}_z Y_\ell^m = m\hbar\, Y_\ell^m,$$

where the *orbital angular momentum quantum number* $\ell = 0, 1, 2, 3, \ldots$ and, for each ℓ, the *magnetic quantum number* m takes exactly $2\ell+1$ values, $m = -\ell, -\ell+1, \ldots, \ell-1, \ell$.

The eigenfunctions $Y_\ell^m(\theta, \phi)$, called *spherical harmonics*, are functions of the two angular coordinates of spherical coordinates (r, θ, ϕ) — angular momentum, unlike energy, does not depend on the radial coordinate r at all, which is exactly why it can be treated as a self-contained problem here, before any specific radial potential (such as hydrogen's Coulomb potential, in Chapter 8) is introduced. The first few, expressed in terms of the polar angle θ (measured from the z-axis) and azimuthal angle ϕ, are

$$\begin{aligned} Y_0^0 &= \sqrt{\frac{1}{4\pi}}, \\ Y_1^0 &= \sqrt{\frac{3}{4\pi}}\cos\theta, \qquad Y_1^{\pm 1} = \mp\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{\pm i\phi}. \end{aligned} \tag{7.1}$$

A complete table through $\ell = 3$ appears in Appendix B. Each Y_ℓ^m is normalized over the full solid angle, $\int_0^{2\pi}\int_0^{\pi} |Y_\ell^m(\theta,\phi)|^2 \sin\theta\, d\theta\, d\phi = 1$, and different (ℓ, m) pairs are mutually orthogonal — exactly the orthonormality structure of Chapter 2, now realized by a specific, explicit family of functions on the sphere.

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Worked Example 7.1 — Verifying an Eigenvalue Directly

Confirm that $Y_1^0(\theta,\phi) = \sqrt{3/4\pi}\,\cos\theta$ satisfies $\hat{L}_z Y_1^0 = m\hbar Y_1^0$ with $m = 0$, and, using $\hat{L}^2 = -\hbar^2\left[\dfrac{1}{\sin\theta}\dfrac{\partial}{\partial\theta}\left(\sin\theta\dfrac{\partial}{\partial\theta}\right) + \dfrac{1}{\sin^2\theta}\dfrac{\partial^2}{\partial\phi^2}\right]$ (the explicit angular form of $\hat{L}^2$, stated here without derivation), confirm $\hat{L}^2 Y_1^0 = \ell(\ell+1)\hbar^2 Y_1^0$ with $\ell = 1$.
Solution. Since Y_1^0 has no ϕ-dependence at all, $\hat{L}_z Y_1^0 = -i\hbar\,\partial Y_1^0/\partial\phi = 0 = 0\cdot\hbar\,Y_1^0$: indeed an eigenfunction with $m = 0$, confirming the first claim immediately. For $\hat{L}^2$, the ϕ-derivative term vanishes for the same reason, leaving

$$\hat{L}^2 Y_1^0 = -\hbar^2\sqrt{\frac{3}{4\pi}}\cdot\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\cdot(-\sin\theta)\right) = -\hbar^2\sqrt{\frac{3}{4\pi}}\cdot\frac{1}{\sin\theta}\frac{d}{d\theta}\left(-\sin^2\theta\right).$$

Since $\dfrac{d}{d\theta}(-\sin^2\theta) = -2\sin\theta\cos\theta$, this becomes

$$\hat{L}^2 Y_1^0 = -\hbar^2\sqrt{\frac{3}{4\pi}}\cdot\frac{-2\sin\theta\cos\theta}{\sin\theta} = 2\hbar^2\sqrt{\frac{3}{4\pi}}\cos\theta = 2\hbar^2 Y_1^0,$$

matching $\ell(\ell+1)\hbar^2 = 1(2)\hbar^2 = 2\hbar^2$ exactly, as required for $\ell = 1$.

7.4 The Vector Model

A useful (if imperfect) way to visualize these results is the *vector model*: picture $\mathbf{L}$ as a vector of fixed length $|\mathbf{L}| = \sqrt{\ell(\ell+1)}\,\hbar$, precessing rapidly around the z-axis in such a way that its z-component is pinned at exactly $m\hbar$, while its x- and y-components are individually undefined (consistent with Section 7.2: only L^2 and L_z are simultaneously sharp). Figure 7.1 shows the resulting picture for $\ell = 2$: five cones, one for each allowed $m = -2,-1,0,1,2$, all of the same slant length $|\mathbf{L}| = \sqrt{6}\,\hbar \approx 2.449\hbar$.

Note — Why $|\mathbf{L}| \neq \ell\hbar$

It is tempting to guess $|\mathbf{L}| = \ell\hbar$ (the largest value L_z can take), but the correct result is the strictly larger $|\mathbf{L}| = \sqrt{\ell(\ell+1)}\,\hbar$. This has a direct and testable consequence: $\mathbf{L}$ can *never* point exactly along the z-axis, even in the state of maximum $L_z = \ell\hbar$, since $\sqrt{\ell(\ell+1)}\,\hbar > \ell\hbar$ strictly, for every $\ell > 0$. If $\mathbf{L}$ could point exactly along z, then $L_x = L_y = 0$ exactly, contradicting the uncertainty argument of Section 7.2 that only *one* component can ever be sharp at a time. The gap between $\ell\hbar$ and $\sqrt{\ell(\ell+1)}\hbar$ is quantum mechanics' way of enforcing this consistently, and it narrows (in relative terms) only slowly, as $\sqrt{1+1/\ell} \to 1$, for large ℓ — another instance of the correspondence principle of Chapter 4.

7.5 Chapter Summary

- Canonical quantization of the classical $\mathbf{L} = \mathbf{r}\times\mathbf{p}$ gives operators $\hat{L}_x, \hat{L}_y, \hat{L}_z$ satisfying $[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z$ (and cyclic permutations): no two Cartesian components of angular momentum can be simultaneously known with perfect precision.

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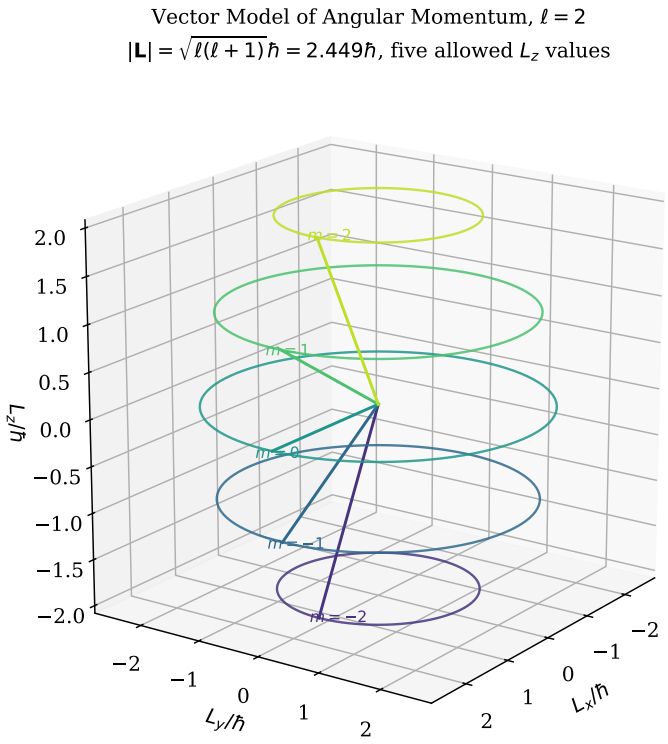


Figure 7.1: The vector model for $\ell = 2$: $\mathbf{L}$ has fixed magnitude $\sqrt{\ell(\ell+1)}\hbar$ but only its projection onto the z-axis is sharp, taking one of $2\ell+1 = 5$ evenly spaced values. The circles represent the classically undetermined orientation of L_x, L_y at each allowed L_z.

- $\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$ commutes with every component of $\hat{\mathbf{L}}$, so $\hat{L}^2$ and (conventionally) $\hat{L}_z$ can be simultaneously diagonalized, with simultaneous eigenfunctions the spherical harmonics $Y_\ell^m(\theta, \phi)$.
- $\hat{L}^2 Y_\ell^m = \ell(\ell+1)\hbar^2 Y_\ell^m$ and $\hat{L}_z Y_\ell^m = m\hbar\, Y_\ell^m$, with $\ell = 0, 1, 2, \ldots$ and $m = -\ell, \ldots, \ell$ (exactly $2\ell+1$ values for each ℓ).
- The vector model visualizes $\mathbf{L}$ as a fixed-length vector of magnitude $\sqrt{\ell(\ell+1)}\hbar$, precessing so that only its z-projection ($m\hbar$) is sharp; $|\mathbf{L}| > \ell\hbar$ strictly, consistent with $\mathbf{L}$ never pointing exactly along any single axis.
- Chapter 8 combines this angular structure with the radial (r-dependent) part of the hydrogen problem to build the full three-dimensional atomic wavefunctions.

Problems

7.1 List all allowed values of m for $\ell = 3$, and confirm there are exactly $2\ell+1$ of them.

7.2 Find $|\mathbf{L}|$, in units of $\hbar$, for $\ell = 0, 1, 2, 3$, and, for each, the smallest angle $\theta_{\min}$ (between $\mathbf{L}$ and the z-axis) attainable, using $\cos\theta_{\min} = \ell/\sqrt{\ell(\ell+1)}$ (i.e., the state of maximum $m = \ell$). Confirm $\theta_{\min} \to 0$ as $\ell \to \infty$.

7.3 Using the explicit form $Y_1^1 = -\sqrt{3/8\pi}\sin\theta\, e^{i\phi}$, verify directly that $\hat{L}_z Y_1^1 = \hbar Y_1^1$ (i.e., $m = 1$), using $\hat{L}_z = -i\hbar\,\partial/\partial\phi$.

7.4 Show that $Y_0^0 = \sqrt{1/4\pi}$ is properly normalized over the sphere, i.e., that $\int_0^{2\pi}\int_0^{\pi} |Y_0^0|^2 \sin\theta\, d\theta\, d\phi = 1$. (This is the simplest possible case: Y_0^0 does not depend on θ or ϕ at all.)

7.5 Using the explicit forms in Eq. (7.1), verify by direct integration that Y_1^0 and Y_1^1 are orthogonal: $\int_0^{2\pi}\int_0^{\pi} (Y_1^0)^* Y_1^1 \sin\theta\, d\theta\, d\phi = 0$. (*Hint:* the ϕ-integral alone, $\int_0^{2\pi} e^{i\phi}\, d\phi$, is sufficient to show this vanishes — you need not evaluate the θ-integral.)

7.6 An electron is in a state with $\ell = 2$. If L_z is measured, list every possible outcome (in units of $\hbar$), and state how many distinct outcomes are possible.

7.7 Using the vector model, find the angle between $\mathbf{L}$ and the z-axis for a $\ell = 1$, $m = 0$ state. (Hint: this is a special case worth thinking about directly — what does $L_z = 0$, combined with $|\mathbf{L}| = \sqrt{2}\hbar \neq 0$, imply geometrically?)

7.8 Two hydrogen-like atoms are prepared in states with $\ell = 1$ and $\ell = 5$ respectively, both in their maximum-m state ($m = \ell$). Compute $\cos\theta_{\min} = \ell/\sqrt{\ell(\ell+1)}$ for each, and determine which atom's angular momentum vector points "more nearly along" the z-axis, in the sense of this ratio approaching 1.

7.9 Verify that $[\hat{L}^2, \hat{L}_z] = 0$ using only the Key Result of Section 7.2 (the three fundamental commutators among $\hat{L}_x, \hat{L}_y, \hat{L}_z$) and the product rule for commutators, $[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$, applied to each term of $\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$. (*Hint:* you will need $[\hat{L}_z, \hat{L}_x^2] = [\hat{L}_z, \hat{L}_x]\hat{L}_x + \hat{L}_x[\hat{L}_z, \hat{L}_x]$ and the analogous expression for $\hat{L}_y^2$; show the two nonzero contributions cancel.)

7.10 Define $\hat{L}_+ \equiv \hat{L}_x + i\hat{L}_y$. Using the fundamental commutators of Section 7.2, show that $[\hat{L}_z, \hat{L}_+] = \hbar\hat{L}_+$ (*Hint:* this is directly parallel to $[\hat{N}, \hat{a}^\dagger] = \hat{a}^\dagger$ from Problem 6.10 — follow the same steps, substituting the angular momentum commutators for the ladder-operator ones). What does this tell you, by the same ladder-operator logic used in Chapter 6, about the effect of $\hat{L}_+$ on an eigenstate of $\hat{L}_z$?

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Chapter 8

The Hydrogen Atom

Every system solved so far in this book has been artificial in one respect or another — a box with idealized walls, a spring potential standing in for a real chemical bond. The hydrogen atom is different: it is a real, physically exact three-dimensional system (a single electron bound to a single proton by the Coulomb force), it is exactly solvable using nothing beyond the tools already developed in this book, and its solution is the single most important calculation in the history of quantum chemistry. Every qualitative feature of the periodic table — shell structure, orbital shapes, the very existence of s, p, d, f orbitals — traces back to the results of this chapter.

8.1 The Coulomb Potential and Separation of Variables

Definition 8.1 — The Hydrogen Atom Hamiltonian

An electron of mass m_e and charge $-e$ moves in the Coulomb potential of a fixed proton of charge $+e$, at separation r:

$$V(r) = -\frac{e^2}{4\pi\epsilon_0 r}.$$

The full three-dimensional Hamiltonian is $\hat{H} = -\dfrac{\hbar^2}{2m_e}\nabla^2 + V(r)$, where ∇^2 is the three-dimensional Laplacian.

Because $V(r)$ depends only on the distance r from the proton, not on direction, this is called a *central potential*, and spherical coordinates (r, θ, ϕ) are the natural choice. In spherical coordinates, the Laplacian splits cleanly into a radial piece and an angular piece, and remarkably, the angular piece is *exactly* $-\hat{L}^2/\hbar^2 r^2$, built entirely from the angular momentum operator already solved in Chapter 7:

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{\hat{L}^2}{\hbar^2 r^2}.$$

This is not a coincidence requiring new work — it is the payoff for having solved the angular momentum eigenvalue problem in full generality, for *any* central potential, before specializing to hydrogen's particular $V(r)$. Seeking a separated solution $\psi(r,\theta,\phi) = R(r)\,Y_\ell^m(\theta,\phi)$, and using $\hat{L}^2 Y_\ell^m = \ell(\ell+1)\hbar^2 Y_\ell^m$ from Chapter 7 directly, the full TISE $\hat{H}\psi = E\psi$ reduces to a single ordinary differential equation for the radial function $R(r)$ alone:

Key Result 8.1 — The Radial Equation

$$-\frac{\hbar^2}{2m_e}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \left[\frac{\ell(\ell+1)\hbar^2}{2m_e r^2} - \frac{e^2}{4\pi\epsilon_0 r}\right]R(r) = ER(r).$$

The bracketed term is an *effective potential*: the true Coulomb attraction, plus a repulsive

"centrifugal" term $\ell(\ell+1)\hbar^2/2m_e r^2$ that pushes the electron away from the origin whenever $\ell > 0$ — the quantum-mechanical descendant of the classical fact that an orbiting object with angular momentum cannot fall straight into the force center.

8.2 The Radial Equation and Laguerre Polynomials

Solving the radial equation — by a series-solution method structurally similar to (though more involved than) the one sketched for the harmonic oscillator in Appendix B — requires, for a normalizable solution, exactly the same kind of termination condition that produced quantization in every previous chapter of this book. The result is that acceptable solutions exist only for

$$E_n = -\frac{m_e e^4}{2(4\pi\epsilon_0)^2\hbar^2}\cdot\frac{1}{n^2}, \qquad n = 1, 2, 3, \ldots,$$

— exactly the Bohr formula of Chapter 1, Eq. (1.10), now derived from the full Schrödinger equation with no classically inconsistent assumptions about definite electron orbits anywhere in the argument. The integer n is the *principal quantum number*, and, crucially, the termination condition also forces a restriction relating n and the angular momentum quantum number ℓ inherited from Chapter 7:

$$\ell = 0, 1, 2, \ldots, n-1 \qquad \text{(for each given } n\text{)}. \tag{8.1}$$

This restriction has no counterpart in the particle in a box or the harmonic oscillator (both one-dimensional problems, with no angular momentum at all); it is a genuinely new feature of three-dimensional central-potential problems, and it is the origin of the familiar chemistry rule that (for example) $n = 1$ admits only an s orbital ($\ell = 0$), while $n = 2$ admits both s ($\ell = 0$) and p ($\ell = 1$) orbitals, but not d.

The normalized radial functions $R_{n\ell}(r)$ involve products of a power of r, an exponential decay e^{-r/na_0}, and an *associated Laguerre polynomial*, tabulated for the first few (n, ℓ) combinations in Appendix B. The three needed for this chapter's examples are

$$\begin{aligned} R_{10}(r) &= 2a_0^{-3/2}e^{-r/a_0}, \\ R_{20}(r) &= \frac{1}{2\sqrt{2}}a_0^{-3/2}\left(2 - \frac{r}{a_0}\right)e^{-r/2a_0}, \\ R_{21}(r) &= \frac{1}{2\sqrt{6}}a_0^{-3/2}\frac{r}{a_0}e^{-r/2a_0}. \end{aligned} \tag{8.2}$$

The full three-dimensional wavefunction is the product $\psi_{n\ell m}(r,\theta,\phi) = R_{n\ell}(r)\,Y_\ell^m(\theta,\phi)$, combining a chapter-8 radial function with a chapter-7 spherical harmonic — every hydrogen wavefunction in this book, and in any chemistry textbook, is built from exactly these two pieces.

Worked Example 8.1 — Verifying the Ground State Directly

Confirm that $\psi_{100}(r,\theta,\phi) = R_{10}(r)Y_0^0(\theta,\phi) = \dfrac{1}{\sqrt{\pi a_0^3}}e^{-r/a_0}$ solves the full three-dimensional TISE with $E_1 = -13.6057$ eV, and confirm it is properly normalized.

Solution. Because ψ_{100} depends only on r (it is spherically symmetric, consistent with $\ell = 0$, for which the centrifugal term in the radial equation vanishes entirely), the angular part of

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∇^2 contributes nothing, and we need only

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\psi_{100}}{dr}\right) = \frac{1}{r^2}\frac{d}{dr}\left(r^2\cdot\left(-\frac{1}{a_0}\right)\psi_{100}\right) = \frac{1}{r^2}\left[-\frac{2r}{a_0}+\frac{r^2}{a_0^2}\right]\psi_{100} = \left(\frac{1}{a_0^2}-\frac{2}{a_0 r}\right)\psi_{100}.$$

So

$$-\frac{\hbar^2}{2m_e}\left(\frac{1}{a_0^2}-\frac{2}{a_0 r}\right)\psi_{100} - \frac{e^2}{4\pi\epsilon_0 r}\psi_{100} = -\frac{\hbar^2}{2m_e a_0^2}\psi_{100} + \left(\frac{\hbar^2}{m_e a_0}-\frac{e^2}{4\pi\epsilon_0}\right)\frac{\psi_{100}}{r}.$$

Using the definition of the Bohr radius, $a_0 = 4\pi\epsilon_0\hbar^2/m_e e^2$ (Eq. (1.9)), the coefficient of the ψ_{100}/r term is exactly $\hbar^2/m_e a_0 - e^2/4\pi\epsilon_0 = e^2/4\pi\epsilon_0 - e^2/4\pi\epsilon_0 = 0$ — the $1/r$ terms cancel completely, leaving only $-\dfrac{\hbar^2}{2m_e a_0^2}\psi_{100} = E_1\psi_{100}$, i.e., $E_1 = -\hbar^2/2m_e a_0^2$, which is exactly the Bohr ground-state energy -13.6057 eV (Problem 8.1 asks you to confirm this numerically). For normalization, $\int|\psi_{100}|^2\,d^3r = \int_0^\infty\int_0^\pi\int_0^{2\pi}\frac{1}{\pi a_0^3}e^{-2r/a_0}\,r^2\sin\theta\,d\phi\,d\theta\,dr$; the angular integral gives 4π, leaving $\frac{4}{a_0^3}\int_0^\infty r^2 e^{-2r/a_0}\,dr = \frac{4}{a_0^3}\cdot\frac{a_0^3}{4} = 1$ (using the standard integral $\int_0^\infty r^n e^{-\alpha r}dr = n!/\alpha^{n+1}$), confirming normalization exactly.

8.3 Atomic Orbitals: Shapes and Radial Distribution Functions

Chemists rarely work with $|\psi_{n\ell m}(r,\theta,\phi)|^2$ directly (a full three-dimensional probability density); instead, two derived quantities are far more useful.

Definition 8.2 — Radial Distribution Function

$$P_{n\ell}(r) \equiv r^2|R_{n\ell}(r)|^2,$$

defined so that $P_{n\ell}(r)\,dr$ is the total probability of finding the electron at radius between r and $r+dr$, *summed over all angles*. (The factor r^2 accounts for the growing surface area, $4\pi r^2$, of the spherical shell at radius r; a small $|R|^2$ at large r can still enclose substantial probability, simply because there is so much more volume out there.)

Figure 8.1 plots $P_{n\ell}(r)$ for the $1s$, $2s$, and $2p$ states. Several features generalize to every hydrogen orbital: the $1s$ distribution peaks at $r = a_0$ exactly (Problem 8.3 asks you to show this by direct differentiation), recovering the Bohr radius as the *most probable* radius, though emphatically not the *only* possible radius, in the correct quantum treatment; states with the same n but different ℓ have different radial distributions, with lower ℓ (less angular momentum, per the centrifugal term of Section 8.1) allowing the electron to venture closer to the nucleus and also, as the $2s$ curve shows, to display additional inner structure — specifically, $n-\ell-1$ radial nodes (the $2s$ state has exactly one, visible in Figure 8.1 as the point where the curve touches zero between the origin and the main peak; the $2p$ state, with $\ell = 1$, has zero radial nodes, consistent with $n-\ell-1 = 0$).

Worked Example 8.2 — Average Orbital Radius

Find $\langle r\rangle$ for the hydrogen $1s$ state.

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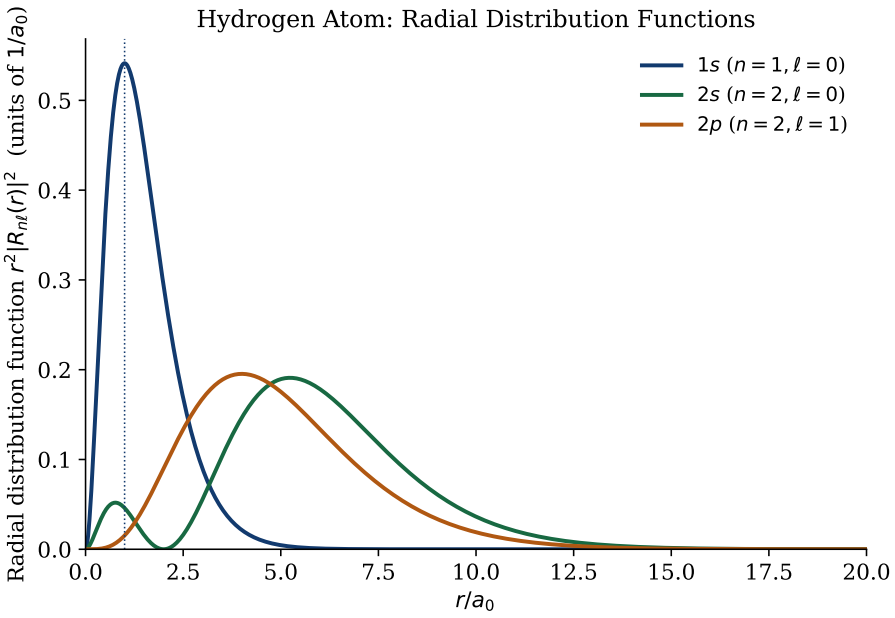


Figure 8.1: Radial distribution functions for the $1s$, $2s$, and $2p$ hydrogen orbitals. The $1s$ curve peaks exactly at $r = a_0$; the $2s$ curve shows one radial node; both $n = 2$ states have appreciably more spatial extent than $1s$, consistent with $E_n \propto -1/n^2$ becoming less negative (less tightly bound) as n increases.

Solution. By definition,

$$\langle r\rangle_{1s} = \int_0^\infty r\, P_{10}(r)\, dr = \int_0^\infty r \cdot \frac{4}{a_0^3} r^2 e^{-2r/a_0}\, dr = \frac{4}{a_0^3}\int_0^\infty r^3 e^{-2r/a_0}\, dr.$$

Using $\int_0^\infty r^3 e^{-\alpha r} dr = 6/\alpha^4$ with $\alpha = 2/a_0$,

$$\langle r\rangle_{1s} = \frac{4}{a_0^3}\cdot\frac{6a_0^4}{16} = \frac{3a_0}{2}.$$

Notice $\langle r\rangle_{1s} = 1.5\,a_0 \neq a_0$, the most probable radius found above — a general feature of asymmetric distributions (the radial distribution function has a long tail toward large r, which pulls the mean above the peak, exactly as a right-skewed distribution's mean exceeds its mode in ordinary statistics). Problem 8.5 asks you to compute the analogous result for $2s$ and $2p$, and to verify the general formula $\langle r\rangle_{n\ell} = \dfrac{a_0}{2}\left[3n^2 - \ell(\ell+1)\right]$, which both results (and this one) satisfy.

For the angular shape, chemists typically use real combinations of the complex $Y_1^{\pm 1}$ (a valid operation, since both are degenerate eigenstates of the same energy, per Problem 3.10's linearity argument) to form the familiar directional p_x, p_y, p_z orbitals taught in introductory chemistry; Y_1^0 itself, being already real, is directly proportional to the p_z orbital. Figure 8.2 shows cross-sections of $|\psi_{1s}|^2$ (spherically symmetric) and $|\psi_{2p_z}|^2$ (the familiar two-lobed shape, with a nodal plane at

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$z = 0$ where $Y_1^0 \propto \cos\theta$ vanishes).

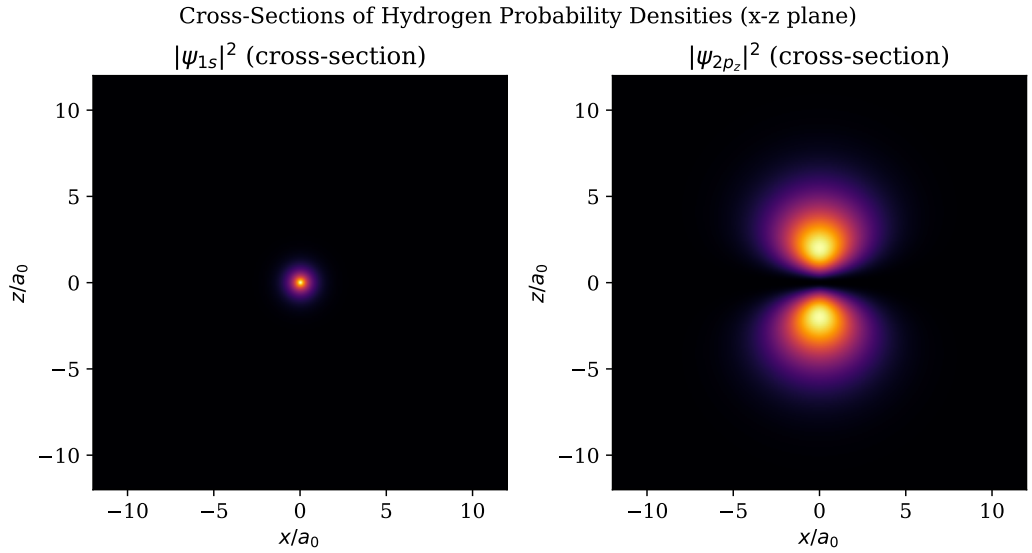


Figure 8.2: Cross-sections (in the x-z plane) of the hydrogen $1s$ and $2p_z$ probability densities. The $1s$ density is spherically symmetric; the $2p_z$ density shows the characteristic two-lobed shape with a nodal plane at $z = 0$, inherited directly from $Y_1^0 \propto \cos\theta$.

8.4 Degeneracy

A striking feature of the hydrogen energy formula $E_n = -13.6057\ \text{eV}/n^2$ is that it depends *only* on n, not on ℓ or m separately, even though the wavefunctions $\psi_{n\ell m}$ themselves depend on all three quantum numbers and look very different from one another (compare the spherical $1s$ and lobed $2p_z$ densities in Figure 8.2, both members of different (n, ℓ) pairs in general, though $2s$ and $2p$ specifically share $n = 2$ and hence exactly the same energy despite their visibly different shapes in Figure 8.1). Using Eq. (8.1) together with the $2\ell + 1$ values of m for each ℓ (Chapter 7), the total number of degenerate states at level n is

$$\sum_{\ell=0}^{n-1} (2\ell + 1) = n^2.$$

Note — An unusually high degree of degeneracy

Degeneracy in m (states of different angular momentum *orientation* sharing the same energy) is expected on general grounds: it holds for *any* central potential $V(r)$, since nothing in the physics distinguishes one direction in space from another, so energy cannot depend on how $\mathbf{L}$ happens to be oriented. Degeneracy in ℓ (different orbital *shapes* sharing the same energy) is not generic; it is special to the exact $1/r$ Coulomb potential (a consequence of an extra hidden symmetry of the Coulomb problem, beyond ordinary rotational symmetry, that is well beyond this book's scope) and is already broken in real multi-electron atoms, where electron-electron repulsion modifies the effective potential away from a pure $1/r$ form. This is exactly why, in the multi-electron atoms of Chapter 14, the energy ordering $4s$ below $3d$

becomes possible, even though hydrogen's own $n = 3$ and $n = 4$ levels never interleave in this way.

8.5 The Reduced-Mass Correction, Revisited

Chapter 1 flagged, without derivation, that the Bohr formula's small (0.054%) disagreement with the precisely measured hydrogen spectrum is resolved by replacing m_e with the electron-proton reduced mass $\mu = m_e m_p/(m_e + m_p)$ throughout. We can now see exactly where this correction enters: the derivation of Section 8.1 implicitly assumed a stationary proton at the origin, with the electron the only moving particle. A more careful treatment — separating the electron-proton two-body problem into center-of-mass motion (irrelevant to the internal energy levels) and *relative* motion, exactly as in classical two-body mechanics — shows that the relative motion obeys the same radial equation of Section 8.1, but with m_e replaced everywhere by μ. Since $\mu/m_e = 0.999456$ is so close to 1, every qualitative conclusion of this chapter (orbital shapes, degeneracies, the n^2 counting) is completely unaffected; only the numerical value of E_n, at the fifth significant figure, shifts.

8.6 Chapter Summary

- Because the Coulomb potential $V(r) = -e^2/4\pi\epsilon_0 r$ depends only on r, the angular part of the three-dimensional TISE is exactly Chapter 7's angular momentum problem, and separates cleanly from a radial equation for $R(r)$ alone.
- Solving the radial equation reproduces the Bohr energies, $E_n = -13.6057\ \text{eV}/n^2$, now derived rigorously, together with the restriction $\ell = 0, 1, \ldots, n-1$ that has no one-dimensional counterpart.
- Full wavefunctions factor as $\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_\ell^m(\theta, \phi)$, combining this chapter's radial functions with Chapter 7's spherical harmonics.
- The radial distribution function $P_{n\ell}(r) = r^2 |R_{n\ell}(r)|^2$ gives the probability of finding the electron at a given radius; the $1s$ state peaks at exactly $r = a_0$, recovering the Bohr radius as a most-probable (not exact) value.
- The energy E_n depends only on n, giving each level an n^2-fold degeneracy — a mix of generic rotational degeneracy (in m) and a special extra degeneracy (in ℓ) unique to the exact Coulomb potential, which is lifted in real multi-electron atoms (Chapter 14).
- The small, precisely measurable disagreement between the idealized Bohr/Schrödinger formula and real hydrogen spectra (Chapter 1) is fully accounted for by replacing m_e with the electron-proton reduced mass μ.

Problems

8.1 Using $E_1 = -\hbar^2/2m_e a_0^2$ from the Worked Example of Section 8.2, and the numerical value of a_0 from Chapter 1, confirm numerically that $E_1 = -13.6057$ eV.

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8.2 How many total states (counting all allowed ℓ and m) share the energy E_4 (i.e., $n = 4$)? List the allowed ℓ values explicitly, and confirm your count against n^2.

8.3 Show, by direct differentiation of $P_{10}(r) = \dfrac{4}{a_0^3} r^2 e^{-2r/a_0}$ and setting $dP_{10}/dr = 0$, that the most probable radius for the hydrogen $1s$ state is exactly $r = a_0$.

8.4 Using $R_{20}(r)$ from Eq. (8.2), find the value of r (in terms of a_0) at which the $2s$ radial *wavefunction* $R_{20}(r)$ itself (not the distribution function) has a node, i.e., $R_{20}(r) = 0$ for $r \neq 0$. Compare with Figure 8.1.

8.5 Using the general formula $\langle r \rangle_{n\ell} = \dfrac{a_0}{2}[3n^2 - \ell(\ell+1)]$ quoted in Section 8.3, compute $\langle r \rangle$ for the $2s$ and $2p$ states, and confirm the $2s$ value is larger than the $2p$ value despite both having the same n. Provide a brief physical rationale, referring to the centrifugal term of Section 8.1, for why higher ℓ (at fixed n) gives a *smaller* $\langle r \rangle$.

8.6 A hydrogen atom is excited to a superposition of the $2p_z$ (ψ_{210}) and $2s$ (ψ_{200}) states. Using the result of Problem 3.8 (Chapter 3), find the angular frequency at which the resulting probability density oscillates in time, given that both states share the exact energy $E_2 = -3.4014$ eV. What does your answer imply physically? (*Hint*: re-examine the condition for time-dependence derived in Problem 3.8 — what role does the *difference* $E_2 - E_2$ play here?)

8.7 For a hydrogenic ion with nuclear charge Z (e.g., He^+ has $Z = 2$, Li^{2+} has $Z = 3$), the energy formula generalizes to $E_n = -Z^2(13.6057\ \text{eV})/n^2$ and the Bohr radius rescales to a_0/Z. Find the ground-state ionization energy of He^+, in eV, and the most probable radius of its $1s$ electron, in units of a_0.

8.8 Explain, in two or three sentences, why the restriction $\ell = 0, 1, \ldots, n-1$ (Eq. (8.1)) has no analog in the particle-in-a-box or harmonic-oscillator problems of Chapters 4 and 6. (*Hint:* what property do both of those earlier potentials share, that hydrogen's three-dimensional problem does not?)

8.9 A student claims that because $2s$ and $2p$ have the same energy E_2 in hydrogen, they must also have the same $\langle r \rangle$ and the same shape. Using the results of Section 8.3 and Problem 8.5, explain what is wrong with this claim, distinguishing carefully between "same energy" and "same wavefunction."

8.10 Using the reduced-mass discussion of Section 8.5, find the ratio μ_D/μ_H of the reduced mass of deuterium (a proton-neutron nucleus, $m_d \approx 2m_p$, bound to one electron) to that of ordinary hydrogen, and use it to predict the fractional shift in deuterium's Lyman-α wavelength relative to hydrogen's. (This isotope shift was, historically, part of the original 1932 discovery of deuterium by Harold Urey.)

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Part III

Formalism, Spin, and Approximation Methods

Chapter 9

The Formal Structure of Quantum Mechanics

Chapters 1 through 8 built quantum mechanics the way it was historically discovered: motivated by experiment, developed through worked examples, generalized only as far as each specific problem demanded. This chapter looks back over that path and extracts the general structure underneath it — what a Hermitian operator is and why physical observables must be represented by one, why eigenfunctions of different eigenvalues are automatically orthogonal (a fact you have now verified by direct integration three separate times, in Chapters 4, 6, and 8, without yet knowing it was an instance of a single general theorem), and where the uncertainty principle of Chapter 1 actually comes from. Nothing here is a new physical assumption; everything is already implicit in the postulates of Chapter 3. But the general statement is more powerful than any of its special cases, and Chapters 10 through 13 rely on it directly.

9.1 Hilbert Space and Dirac Notation, in Full

Section 2.7 previewed Dirac notation informally. We can now state it properly.

Definition 9.1 — Hilbert Space

The set of all normalizable wavefunctions (technically, square-integrable functions, $\int |\psi|^2 dx < \infty$) forms a vector space in the sense of Section 2.2, called a *Hilbert space*. A quantum state is a vector $|\psi\rangle$ in this space; the wavefunction $\psi(x) = \langle x|\psi\rangle$ is simply the components of $|\psi\rangle$ in a particular basis (the position basis $\{|x\rangle\}$), exactly as an ordinary vector $\mathbf{v}$ can be written in components once a basis is chosen.

This reframes something you have been doing since Chapter 3 without naming it: a wavefunction $\psi(x)$ is *a* representation of the state $|\psi\rangle$, not the state itself, in the same way that the column of numbers (v_1, v_2, v_3) representing an ordinary vector $\mathbf{v}$ depends on a choice of basis, while $\mathbf{v}$ itself does not. Chapter 13 makes use of a second basis (the momentum basis) in which the *same* physical state is represented by a different function entirely.

9.2 Hermitian Operators

Definition 9.2 — Hermitian Operator

An operator $\hat{A}$ is *Hermitian* if

$$\langle \phi | \hat{A}\psi \rangle = \langle \hat{A}\phi | \psi \rangle \qquad \text{for all } \phi, \psi,$$

i.e., $\hat{A}$ can act on either the bra or the ket in an inner product without changing the result.

(This condition is sometimes written $\hat{A}^\dagger = \hat{A}$, where $\hat{A}^\dagger$, the *Hermitian adjoint*, is defined by $\langle\phi|\hat{A}^\dagger\psi\rangle \equiv \langle\hat{A}\phi|\psi\rangle$ for all ϕ, ψ.)

Problem 9.1 asks you to check this directly for $\hat{x}$ and $\hat{p}$; both are Hermitian, and so is every Hamiltonian $\hat{H} = \hat{p}^2/2m + V(\hat{x})$ built from them for real $V(x)$ (Problem 9.2 extends the argument to $\hat{p}^2$ and $\hat{H}$). Hermiticity is not a mathematical nicety — it is the precise algebraic reason every observable prediction of quantum mechanics comes out real, as the next two results show.

Key Result 9.1 — Eigenvalues of a Hermitian Operator Are Real

If $\hat{A}\,|\psi\rangle = a\,|\psi\rangle$ with $\hat{A}$ Hermitian and $|\psi\rangle \neq 0$, then a is real.
Proof. Apply the Hermiticity condition with $\phi = \psi$: $\langle\psi|\hat{A}\psi\rangle = \langle\hat{A}\psi|\psi\rangle$. The left side is $a\,\langle\psi|\psi\rangle\,\psi$; for the right side, $\hat{A}\,|\psi\rangle = a\,|\psi\rangle$ means the bra $\langle\hat{A}\psi| = a^*\,\langle\psi|$ (complex-conjugating the eigenvalue when moving from ket to bra, exactly as in Problem 3.4), so the right side is $a^*\,\langle\psi|\psi\rangle\,\psi$. Equating, $a\,\langle\psi|\psi\rangle\,\psi = a^*\,\langle\psi|\psi\rangle\,\psi$, and since $\langle\psi|\psi\rangle\,\psi \neq 0$ for a nonzero vector, $a = a^*$: a is real. ■

This single proof retroactively explains something you have taken on faith since Chapter 3: every energy eigenvalue computed in Chapters 4 through 8 came out real, with no imaginary part ever appearing — not by luck, but because $\hat{H}$ is Hermitian in every one of those problems, and this is a fully general guarantee, not a coincidence to be checked case by case.

Key Result 9.2 — Eigenfunctions of Different Eigenvalues Are Orthogonal

If $\hat{A}\,|\psi_1\rangle = a_1\,|\psi_1\rangle$ and $\hat{A}\,|\psi_2\rangle = a_2\,|\psi_2\rangle$, with $\hat{A}$ Hermitian and $a_1 \neq a_2$, then $\langle\psi_1|\psi_2\rangle = 0$.
Proof. By Hermiticity, $\langle\psi_1|\hat{A}\psi_2\rangle = \langle\hat{A}\psi_1|\psi_2\rangle$. The left side is $a_2\,\langle\psi_1|\psi_2\rangle$; the right side, using the previous result that a_1 is real, is $a_1\,\langle\psi_1|\psi_2\rangle$. So $(a_2 - a_1)\,\langle\psi_1|\psi_2\rangle = 0$, and since $a_1 \neq a_2$ by assumption, $\langle\psi_1|\psi_2\rangle = 0$. ■

Note — You already proved special cases of this three times

Problem 4.6 had you verify $\int_0^L \psi_1\psi_2\,dx = 0$ for the particle in a box by direct integration; the same structure underlies the orthogonality used implicitly for the QHO states of Chapter 6 and the spherical harmonics of Chapter 7. In every case, the underlying reason was that $\hat{H}$ (or $\hat{L}^2$, or $\hat{L}_z$) is Hermitian and the two states in question had different eigenvalues. The general theorem just proved makes every future such check unnecessary: orthogonality of non-degenerate eigenstates is guaranteed in advance, for any Hermitian operator whatsoever, not a fact to be re-derived integral by integral.

9.3 Commutators and Simultaneous Observables

Section 7.2 of Chapter 7 noted, for the specific case of angular momentum, that non-commuting operators cannot be simultaneously diagonalized. We can now state and use this generally.

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Key Result 9.3 — Compatible Observables

Two Hermitian operators $\hat{A}$ and $\hat{B}$ possess a complete set of *simultaneous* eigenfunctions (states with sharply defined values of both A and B at once) if and only if they commute, $[\hat{A}, \hat{B}] = 0$. Such observables are called *compatible*. If $[\hat{A}, \hat{B}] \neq 0$, the observables are *incompatible*: no state (except in special limiting or coincidental cases) has both A and B sharply defined simultaneously.

This single criterion organizes a great deal of what you have already learned. $\hat{x}$ and $\hat{p}$ do not commute ($[\hat{x}, \hat{p}] = i\hbar$, Problem 3.6), so position and momentum are incompatible — the algebraic origin of the uncertainty principle, made quantitative in Section 9.4 below. $\hat{L}^2$ and $\hat{L}_z$ *do* commute (Problem 7.9), so they are compatible and can be simultaneously diagonalized — exactly why Chapter 7 could label states by (ℓ, m) together. $\hat{L}_x$ and $\hat{L}_y$ do not commute, so (except in the trivial $\ell = 0$ case) no state has definite L_x *and* definite L_y simultaneously — the rigorous version of the vector-model picture in Figure 7.1, where only L_z (one component) and $|\mathbf{L}|$ were ever drawn as sharp.

9.4 The Generalized Uncertainty Principle

We now derive the precise, general form of the uncertainty relation stated without proof in Chapter 1.

Key Result 9.4 — Generalized Uncertainty Principle

For any two Hermitian operators $\hat{A}, \hat{B}$, and any state $|\psi\rangle$,

$$\Delta A \, \Delta B \;\geq\; \frac{1}{2} \left| \langle [\hat{A}, \hat{B}] \rangle \right| ,$$

where $\Delta A = \sqrt{\langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2}$, as in Section 2.6, and the expectation values are all evaluated in the state $|\psi\rangle$.

Note — Sketch of the derivation

Define shifted operators $\bar{A} \equiv \hat{A} - \langle \hat{A} \rangle$ and $\bar{B} \equiv \hat{B} - \langle \hat{B} \rangle$ (still Hermitian, since a real constant shift preserves Hermiticity), so that $(\Delta A)^2 = \langle \bar{A}^2 \rangle$ and likewise for B. Applying the Cauchy–Schwarz inequality, $\langle f|f\rangle \, \langle g|g\rangle \geq |\langle f|g\rangle|^2$ (the same inequality that, for ordinary vectors, states $|\mathbf{u}|^2|\mathbf{v}|^2 \geq (\mathbf{u}\cdot\mathbf{v})^2$), to $|f\rangle = \bar{A}\,|\psi\rangle$ and $|g\rangle = \bar{B}\,|\psi\rangle$ gives $(\Delta A)^2(\Delta B)^2 \geq |\langle \bar{A}\bar{B} \rangle|^2$. The operator product $\bar{A}\bar{B}$ splits into a Hermitian part (the anti-commutator $\frac{1}{2}\{\bar{A}, \bar{B}\}$, which has a real expectation value) and an anti-Hermitian part (the commutator $\frac{1}{2}[\bar{A}, \bar{B}]$, which has a purely imaginary expectation value, by an argument parallel to the eigenvalue proof of Section 9.2); since a complex number's squared modulus is at least as large as the square of its imaginary part alone, $|\langle \bar{A}\bar{B} \rangle|^2 \geq \frac{1}{4}|\langle [\bar{A}, \bar{B}] \rangle|^2 = \frac{1}{4}|\langle [\hat{A}, \hat{B}] \rangle|^2$ (the constant shifts drop out of the commutator). Combining the two inequalities and taking a square root gives the stated result.

Worked Example 9.1 — Recovering the Heisenberg Uncertainty Principle

Apply the generalized uncertainty principle to $\hat{A} = \hat{x}$ and $\hat{B} = \hat{p}$.
Solution. Since $[\hat{x}, \hat{p}] = i\hbar$ (a constant, the same in every state), $\langle [\hat{x}, \hat{p}] \rangle = i\hbar$ regardless of $|\psi\rangle$, so

$$\Delta x \, \Delta p \geq \frac{1}{2}|i\hbar| = \frac{\hbar}{2},$$

exactly the Heisenberg relation quoted without proof in Chapter 1, now derived from the single algebraic fact $[\hat{x}, \hat{p}] = i\hbar$ and the general Cauchy–Schwarz argument above — not a separate physical postulate, but a mathematical consequence of one. Problem 6.4 had you verify that the QHO ground state achieves this bound with *exact equality*; the derivation above shows equality in Cauchy–Schwarz occurs precisely when $\bar{B}\,|\psi\rangle \propto \bar{A}\,|\psi\rangle$, which singles out Gaussian wavefunctions as the unique minimum-uncertainty states for x and p — exactly the functional form found repeatedly in this book (Chapter 3's example wavefunction, and every QHO eigenstate's Gaussian envelope).

Note — An uncertainty relation for any incompatible pair

The derivation above used nothing specific to x and p; it applies verbatim to L_x and L_y (giving $\Delta L_x \Delta L_y \geq \frac{1}{2}\hbar|\langle L_z \rangle|$, using $[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z$ from Chapter 7), or to any other incompatible pair of Hermitian observables. The Heisenberg relation of Chapter 1 is the most famous member of an entire family of uncertainty relations, one for every pair of non-commuting observables in the theory.

9.5 Chapter Summary

- A wavefunction $\psi(x)$ is one representation, in the position basis, of an abstract state vector $|\psi\rangle$ in Hilbert space.
- Physical observables are represented by Hermitian operators, $\langle \phi | \hat{A}\psi \rangle = \langle \hat{A}\phi | \hat{A}\phi \rangle \, \psi$; this single algebraic property guarantees real eigenvalues and guarantees that eigenfunctions of different eigenvalues are automatically orthogonal, generalizing results checked by hand in Chapters 4, 6, and 7.
- Two observables can be simultaneously sharp if and only if their operators commute; $[\hat{A}, \hat{B}] \neq 0$ means the observables are fundamentally incompatible, not merely difficult to measure together.
- The generalized uncertainty principle, $\Delta A \, \Delta B \geq \frac{1}{2}|\langle [\hat{A}, \hat{B}] \rangle|$, derived here from Cauchy–Schwarz, reduces to the Heisenberg relation $\Delta x \Delta p \geq \hbar/2$ for $\hat{A} = \hat{x}, \hat{B} = \hat{p}$, and applies equally to any other incompatible pair, such as L_x and L_y.
- Everything in this chapter was already implicit in Chapters 1 through 8; naming and proving it in general form is what makes Chapters 10 through 13 — spin, perturbation theory, and time-dependent phenomena — possible without re-deriving these facts from scratch each time.

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Problems

9.1 Using the definition $\langle \phi | \hat{A}\psi \rangle = \langle \hat{A}\phi | \hat{A}\phi \rangle \, \psi$, show that $\hat{x}$ (multiplication by x) is Hermitian. (*Hint:* write both sides as explicit integrals and note that x is real, so it does not matter whether it is thought of as multiplying ψ or ϕ^*.) Then show that $\hat{p} = -i\hbar \, d/dx$ is Hermitian as well: write $\langle \phi | \hat{p}\psi \rangle = \int \phi^* (-i\hbar \psi') \, dx$, integrate by parts once, and confirm the boundary term vanishes (any normalizable $\psi, \phi \to 0$ as $x \to \pm\infty$, Chapter 3) leaving exactly $\langle \hat{p}\phi | \hat{p}\phi \rangle \, \psi$.

9.2 Using the result of Problem 9.1 (that $\hat{p}$ is Hermitian) together with the general fact that if $\hat{A}$ is Hermitian, so is $\hat{A}^2$ (*you may assume this without proof*), explain why the kinetic energy operator $\hat{p}^2/2m$, and hence the full Hamiltonian $\hat{H} = \hat{p}^2/2m + V(\hat{x})$ for real $V(x)$, is Hermitian.

9.3 Two operators satisfy $\hat{A}\hat{B} = \hat{B}\hat{A} + 3i$. Compute $[\hat{A}, \hat{B}]$ and $[\hat{B}, \hat{A}]$.

9.4 Using $[\hat{x}, \hat{p}] = i\hbar$ (Problem 3.6) and the identity $[\hat{A}, \hat{B}^2] = \hat{B}[\hat{A}, \hat{B}] + [\hat{A}, \hat{B}]\hat{B}$, compute $[\hat{x}, \hat{p}^2]$. (*Hint:* this result is useful for Problem 9.5.)

9.5 Using the result of Problem 9.4, apply the generalized uncertainty principle to $\hat{A} = \hat{x}$ and $\hat{B} = \hat{p}^2/2m$ (i.e., kinetic energy) to find a lower bound on $\Delta x \cdot \Delta(KE)$, expressed in terms of $\langle \hat{p} \rangle$, m, and $\hbar$.

9.6 Are $\hat{x}$ and $\hat{H} = \hat{p}^2/2m + V(\hat{x})$ compatible observables in general? Using $[\hat{x}, \hat{H}] = [\hat{x}, \hat{p}^2]/2m$ (with $[\hat{x}, \hat{p}^2]$ from Problem 9.4), state the condition (if any) under which they *would* commute, and explain briefly why this condition essentially never holds for a bound, physically interesting system.

9.7 Show that $\hat{L}_x$ and $\hat{L}_y$ do not commute (already stated in Chapter 7), and use the general uncertainty principle to write the corresponding uncertainty relation for $\Delta L_x \, \Delta L_y$. For a state with $\langle L_z \rangle = 0$ (such as the $\ell = 1, m = 0$ state of Chapter 7, Problem 7.7), what does this relation say about the *lower bound* on $\Delta L_x \Delta L_y$ — is the state required to have nonzero uncertainty in both, or could this particular bound be trivially satisfied by $\Delta L_x = 0$?

9.8 For two compatible Hermitian operators $\hat{A}, \hat{B}$ (i.e., $[\hat{A}, \hat{B}] = 0$), what does the generalized uncertainty principle of Section 9.4 say about the minimum possible value of $\Delta A \, \Delta B$? Is this consistent with the definition of "compatible" given in Section 9.3?

9.9 Prove that $[\hat{L}^2, \hat{L}_z] = 0$ using the product rule for commutators, $[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$, applied term by term to $\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$, together with the fundamental angular-momentum commutators of Chapter 7. (This is the general-formalism version of the same result you were asked to verify directly in Chapter 7, Problem 7.9; use the systematic commutator identity this time, rather than expanding everything in position-space derivatives.)

9.10 A state $|\psi\rangle$ satisfies $\hat{A}\,|\psi\rangle = a\,|\psi\rangle$ for Hermitian $\hat{A}$ with real eigenvalue a. Show that $\Delta A = 0$ exactly for this state (i.e., an eigenstate of $\hat{A}$ has zero uncertainty in A, consistent with calling a a "sharp" or "definite" value). (*Hint:* compute $\langle \hat{A}^2 \rangle$ and $\langle \hat{A} \rangle^2$ directly using $\hat{A}\,|\psi\rangle = a\,|\psi\rangle$, and recall that a is guaranteed real by the Key Result of Section 9.2.)

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Chapter 10

Electron Spin and the Structure of Many-Electron Systems

Every quantum number introduced so far — n, ℓ, m — emerged from the spatial wavefunction $\psi(r, \theta, \phi)$. Real electrons carry one further property, with no classical or spatial analog whatsoever, that is nonetheless essential to chemistry at the most basic level: without it, every atom beyond helium, and every chemical bond, would be qualitatively different from the one nature provides. This chapter introduces spin, and the deep principle — indistinguishability of identical particles — that connects spin to the structure of the periodic table.

10.1 The Discovery of Spin

In 1922, Otto Stern and Walther Gerlach sent a beam of silver atoms through a spatially varying (inhomogeneous) magnetic field and observed the beam split into exactly *two* discrete spots on a detector screen, rather than either passing straight through or spreading into a continuous smear. A magnetic moment $\boldsymbol{\mu}$ passing through such a field experiences a deflecting force proportional to the field gradient and to the component of $\boldsymbol{\mu}$ along the field direction; classically, with $\boldsymbol{\mu}$ free to point in any direction, a continuous range of deflections (and hence a smeared-out band) would be expected. A splitting into exactly two spots is instead the unmistakable signature of a quantized magnetic moment with only *two* possible values along the field axis — structurally identical to the $2\ell + 1$ quantization of Chapter 7, but with $2\ell + 1 = 2$, forcing $\ell = \frac{1}{2}$, a value orbital angular momentum can never take (Chapter 7 established $\ell = 0, 1, 2, \ldots$, integers only).

The resolution, proposed by George Uhlenbeck and Samuel Goudsmit in 1925, is that the electron carries an additional, intrinsic form of angular momentum — called *spin* — entirely distinct from any orbital motion, with quantum number $s = \frac{1}{2}$ fixed once and for all (unlike ℓ, which varies from state to state, every electron has exactly the same $s = \frac{1}{2}$, always).

10.2 Spin Operators and Spinors

Definition 10.1 — Spin Angular Momentum

Spin operators $\hat{S}_x, \hat{S}_y, \hat{S}_z$ satisfy exactly the same commutation relations as orbital angular momentum (Chapter 7),

$$[\hat{S}_x, \hat{S}_y] = i\hbar \hat{S}_z \quad \text{(and cyclic permutations)},$$

with simultaneous eigenstates of $\hat{S}^2$ and $\hat{S}_z$ satisfying

$$\hat{S}^2 |s, m_s\rangle = s(s+1)\hbar^2 |s, m_s\rangle , \qquad \hat{S}_z |s, m_s\rangle = m_s \hbar |s, m_s\rangle .$$

For an electron, $s = \frac{1}{2}$ always, and $m_s = +\frac{1}{2}$ ("spin up," $\uparrow$) or $m_s = -\frac{1}{2}$ ("spin down," $\downarrow$) — exactly the two Stern–Gerlach outcomes.

Because $s = \frac{1}{2}$ is fixed, the entire spin state of an electron is captured by just two numbers (the amplitudes for spin-up and spin-down), conventionally written as a two-component column vector, or *spinor*:

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad \hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

A general spin state is $\alpha |\uparrow\rangle + \beta |\downarrow\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$ (normalization, Section 2.6), and the full state of an electron — including both its spatial motion and its spin — is a product $\psi(x) \otimes (\alpha |\uparrow\rangle + \beta |\downarrow\rangle)$, or, for a definite spin state, simply $\psi(x) |\uparrow\rangle$ or $\psi(x) |\downarrow\rangle$.

Note — Spin has no classical counterpart, but the mathematics is identical

It is tempting, and historically common, to picture spin as the electron literally spinning on its axis like a tiny planet. This picture is not correct (among other problems, it would require the electron's surface to move faster than light to generate the observed magnetic moment from a finite classical charge distribution) and should be discarded as a physical mechanism. What *is* correct, and is the entire content of this section, is that spin obeys the same angular-momentum algebra as orbital motion (Chapter 7's commutation relations), which is why it is called angular momentum at all and why it combines with orbital angular momentum by the same mathematical rules — without spin being, physically, any kind of literal rotation in space.

10.3 Identical Particles and the Symmetrization Postulate

Consider two electrons, with positions (and spins, suppressed here for brevity) x_1 and x_2. Because electrons are truly identical — not just similar, but possessing no distinguishing property whatsoever that could in principle be used to tell them apart, unlike, say, two identical-looking classical billiard balls, which could still be tracked individually by continuous observation of their trajectories — swapping the labels $1 \leftrightarrow 2$ cannot change any physically measurable prediction:

$$|\Psi(x_2, x_1)|^2 = |\Psi(x_1, x_2)|^2.$$

This allows exactly two possibilities: $\Psi(x_2, x_1) = +\Psi(x_1, x_2)$ (symmetric under exchange) or $\Psi(x_2, x_1) = -\Psi(x_1, x_2)$ (antisymmetric under exchange). Which one occurs is not a matter of choice for a given problem; it is an intrinsic property of the type of particle:

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Key Result 10.1 — The Symmetrization Postulate

Particles with integer spin ($s = 0, 1, 2, \ldots$), called *bosons*, always have totally *symmetric* many-particle wavefunctions. Particles with half-integer spin ($s = \frac{1}{2}, \frac{3}{2}, \ldots$), called *fermions*, always have totally *antisymmetric* many-particle wavefunctions. Electrons, protons, and neutrons are all spin-$\frac{1}{2}$ fermions.

Given two available single-particle states ϕ_a and ϕ_b, the properly symmetrized and antisymmetrized two-particle combinations are

$$\Psi_S(x_1, x_2) = \frac{1}{\sqrt{2}}\left[\phi_a(x_1)\phi_b(x_2) + \phi_a(x_2)\phi_b(x_1)\right], \qquad \Psi_A(x_1, x_2) = \frac{1}{\sqrt{2}}\left[\phi_a(x_1)\phi_b(x_2) - \phi_a(x_2)\phi_b(x_1)\right],$$

and direct substitution of $x_1 \leftrightarrow x_2$ confirms Ψ_S is unchanged while Ψ_A flips sign, exactly as required (Problem 10.1 asks you to verify this explicitly).

10.4 The Pauli Exclusion Principle

Setting $\phi_a = \phi_b \equiv \phi$ (both electrons assigned to the *same* single-particle state) in Ψ_A above gives

$$\Psi_A(x_1, x_2) = \frac{1}{\sqrt{2}}\left[\phi(x_1)\phi(x_2) - \phi(x_2)\phi(x_1)\right] = 0$$

identically — not merely small, but exactly zero, for every x_1, x_2. This is the Pauli exclusion principle, and the calculation above shows it is not a separate physical postulate at all, but a direct mathematical consequence of antisymmetry:

Key Result 10.2 — Pauli Exclusion Principle

No two electrons (or other fermions) in a system can occupy exactly the same single-particle quantum state (including spin). Equivalently: at most *two* electrons can occupy a given *spatial* orbital, and if two do, they must have opposite spin ($\uparrow$ and $\downarrow$), so that the full two-particle state (spatial $\times$ spin) is antisymmetric overall even though the spatial part alone is symmetric.

Note — Why this box appeared silently in earlier chapters

This is precisely the rule invoked, without proof, in the particle-in-a-box chemistry application of Chapter 4 ("two electrons per level") and in the electron-filling of Chapter 4, Problem 4.11. It is worth appreciating how much of chemistry follows from the single-line calculation above: without the antisymmetry requirement, every electron in every atom could collapse into the lowest-energy 1*s* orbital, and there would be no shell structure, no periodic table, and no chemistry as it actually exists. Chapter 14 builds directly on this principle to construct the electron configurations of multi-electron atoms.

10.5 Slater Determinants

For more than two electrons, writing out all the necessary sign-alternating terms by hand becomes unwieldy; John Slater's determinant notation packages the antisymmetrization automatically.

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Definition 10.2 — Slater Determinant

For N electrons occupying single-particle states (spin-orbitals) $\chi_1, \ldots, \chi_N$, the properly antisymmetrized N-electron wavefunction is

$$\Psi(x_1, \ldots, x_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \chi_1(x_1) & \chi_2(x_1) & \cdots & \chi_N(x_1) \\ \chi_1(x_2) & \chi_2(x_2) & \cdots & \chi_N(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \chi_1(x_N) & \chi_2(x_N) & \cdots & \chi_N(x_N) \end{vmatrix}.$$

The connection to the Pauli principle is immediate and purely a property of determinants: if any two spin-orbitals coincide, $\chi_i = \chi_j$, then two columns of the determinant are identical, and a standard property of determinants (Problem 10.6) forces the whole determinant to vanish — Slater determinants build antisymmetry, and hence the exclusion principle, in automatically, for any number of electrons, without needing to write out $N!$ terms by hand as the definition might suggest. Chapter 14 uses Slater determinants (schematically, without requiring you to expand any beyond $N = 2$) to organize the electron configurations of real multi-electron atoms.

Worked Example 10.1 — The Helium Ground State

Write the Slater determinant for two electrons occupying the same spatial orbital $\phi_{1s}(x)$, with opposite spins (the helium ground-state configuration, $1s^2$).

Solution. The two spin-orbitals are $\chi_1(x) = \phi_{1s}(x)\,|\uparrow\rangle$ and $\chi_2(x) = \phi_{1s}(x)\,|\downarrow\rangle$ (same spatial part, opposite spin, so the spin-orbitals are still distinct even though the spatial orbitals coincide). The Slater determinant is

$$\Psi = \frac{1}{\sqrt{2}} \begin{vmatrix} \phi_{1s}(x_1)\,|\uparrow\rangle & \phi_{1s}(x_1)\,|\downarrow\rangle \\ \phi_{1s}(x_2)\,|\uparrow\rangle & \phi_{1s}(x_2)\,|\downarrow\rangle \end{vmatrix} = \frac{1}{\sqrt{2}} \phi_{1s}(x_1)\phi_{1s}(x_2)\big[\,|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\,\big],$$

a symmetric spatial part ($\phi_{1s}(x_1)\phi_{1s}(x_2)$, unchanged under $x_1 \leftrightarrow x_2$) multiplied by an antisymmetric spin part ($|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle$, which flips sign under exchange) — the product of the two is properly antisymmetric overall, exactly as required, and this is the only way to place both electrons in the same spatial $1s$ orbital consistently with the exclusion principle.

10.6 Chapter Summary

- The Stern–Gerlach experiment reveals a two-valued intrinsic angular momentum, electron spin, with $s = \frac{1}{2}$ fixed for every electron and $m_s = \pm\frac{1}{2}$.
- Spin operators obey the same commutation algebra as orbital angular momentum (Chapter 7), represented concretely by two-component spinors and 2×2 matrices.
- Identical particles require a definite exchange symmetry: bosons (integer spin) are symmetric, fermions (half-integer spin, including electrons) are antisymmetric under exchange of any two particles.
- The Pauli exclusion principle — no two electrons in the same quantum state — follows directly from antisymmetry, not as a separate postulate: an antisymmetrized wavefunction with two particles assigned the same state vanishes identically.

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- Slater determinants generate properly antisymmetrized many-electron wavefunctions automatically, and are the standard tool (used throughout Chapter 14 and beyond) for constructing multi-electron atomic and molecular states.

Problems

10.1 Show explicitly, by substituting $x_1 \leftrightarrow x_2$, that $\Psi_S(x_1, x_2) = \frac{1}{\sqrt{2}}[\phi_a(x_1)\phi_b(x_2) + \phi_a(x_2)\phi_b(x_1)]$ is unchanged under exchange, while Ψ_A (same formula, minus sign) flips sign.

10.2 Verify by matrix multiplication that $\hat{S}_z\,|\uparrow\rangle = +\frac{\hbar}{2}\,|\uparrow\rangle$ and $\hat{S}_z\,|\downarrow\rangle = -\frac{\hbar}{2}\,|\downarrow\rangle$, using the matrix representation of $\hat{S}_z$ given in Section 10.2.

10.3 Using the Pauli matrix $\hat{S}_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, find the eigenvalues and normalized eigenvectors of $\hat{S}_x$ (this is a 2×2 eigenvalue problem, exactly the type solved in Chapter 2, Section 2.3). Express the eigenvectors as linear combinations of $|\uparrow\rangle$ and $|\downarrow\rangle$.

10.4 A beryllium atom has 4 electrons. Assuming the lowest-energy configuration fills spatial orbitals in the order $1s, 2s, \ldots$ (Chapter 14 justifies this ordering fully), state the ground-state electron configuration using the Pauli principle, and explain why the third and fourth electrons cannot both enter the $1s$ orbital alongside the first two.

10.5 Two electrons are known to occupy different spatial orbitals, $\phi_a \neq \phi_b$. Explain, using the exclusion principle of Section 10.4, whether it is possible for both electrons to have the same spin ($\uparrow\uparrow$) in this case, and contrast this with the helium worked example of Section 10.5, where both electrons occupy the *same* spatial orbital.

10.6 Prove the general determinant fact used in Section 10.5: if two columns of a square matrix are identical, its determinant is zero. (*Hint:* swapping two identical columns leaves the matrix completely unchanged, but swapping any two columns of a determinant is known to flip its sign; what value satisfies $D = -D$?)

10.7 Photons have spin $s = 1$ (an integer). Based on the Symmetrization Postulate of Section 10.3 alone, state whether a system of two identical photons must be described by a symmetric or an antisymmetric two-particle wavefunction, and state whether photons are bosons or fermions.

10.8 Using the commutation relation $[\hat{S}_x, \hat{S}_y] = i\hbar\hat{S}_z$ (Section 10.2) and the generalized uncertainty principle of Chapter 9, write the uncertainty relation for $\Delta S_x\,\Delta S_y$ in a state with definite $S_z = +\hbar/2$ (i.e., the state $|\uparrow\rangle$). Is $\Delta S_x \Delta S_y = 0$ allowed for this state?

10.9 Explain briefly, in two or three sentences, why the classical picture of an electron as a tiny spinning charged sphere (mentioned and rejected in the Note box of Section 10.2) cannot be physically correct, using the idea of a maximum physical speed (the speed of light) together with the electron's known small size and its measured magnetic moment.

10.10 A lithium atom has 3 electrons. Given the Pauli principle result of Problem 10.4 (that the $1s$ orbital can hold at most 2 electrons), explain why lithium's third electron must occupy a spatially *different* orbital ($2s$) even though $2s$ has a higher energy than $1s$ (Chapter 14 discusses this energy ordering in more depth), and state the resulting ground-state configuration in the standard chemical notation $1s^2 2s^1$.

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Chapter 11

Time-Independent Perturbation Theory

Every system solved exactly in this book — the box, the well, the oscillator, the hydrogen atom — was chosen because it *could* be solved exactly. Most real physical systems cannot: add a second electron to hydrogen, tilt a box's floor, or place an atom in an external field, and the resulting Schrödinger equation typically has no closed-form solution at all. Perturbation theory is the first of two general strategies (the second, the variational method, is Chapter 12) for extracting reliable, quantitative answers from problems that cannot be solved exactly, whenever the true problem is "close to" one that can.

11.1 Setting Up the Problem

Suppose the full Hamiltonian can be split as

$$\hat{H} = \hat{H}^{(0)} + \hat{H}',$$

where $\hat{H}^{(0)}$ is a Hamiltonian we have *already solved exactly* (any system from Chapters 4 through 8, for instance), with known eigenstates $\hat{H}^{(0)}\left|n^{(0)}\right\rangle = E_n^{(0)}\left|n^{(0)}\right\rangle$, and $\hat{H}'$ (the *perturbation*) is "small" in a sense made precise by the bookkeeping parameter λ introduced below. The strategy is to write the exact energies and states of the full problem as a power series in the strength of the perturbation, built entirely from quantities associated with the already-solved $\hat{H}^{(0)}$:

$$E_n = E_n^{(0)} + E_n^{(1)} + E_n^{(2)} + \cdots, \qquad |n\rangle = \left|n^{(0)}\right\rangle + \left|n^{(1)}\right\rangle + \cdots,$$

where $E_n^{(1)}$ is first order in the strength of $\hat{H}'$, $E_n^{(2)}$ is second order, and so on. This chapter derives $E_n^{(1)}$ and $E_n^{(2)}$; higher orders exist but are used less often in practice, and are not needed for any example in this book.

11.2 First- and Second-Order Energy Corrections

Key Result 11.1 — Perturbation Theory: Energy Corrections

$$E_n^{(1)} = \left\langle n^{(0)}\right| \hat{H}' \left|n^{(0)}\right\rangle, \qquad E_n^{(2)} = \sum_{m \neq n} \frac{\left|\left\langle m^{(0)}\right| \hat{H}' \left|n^{(0)}\right\rangle\right|^2}{E_n^{(0)} - E_m^{(0)}},$$

where the sum in $E_n^{(2)}$ runs over every unperturbed state *except* n itself.

Note — Where these formulas come from

Substitute $\hat{H} = \hat{H}^{(0)} + \hat{H}'$ and the power-series expansions of E_n and $|n\rangle$ into the exact eigenvalue equation $\hat{H}|n\rangle = E_n|n\rangle$, and collect terms order by order in the perturbation strength. At zeroth order, this simply reproduces $\hat{H}^{(0)}\left|n^{(0)}\right\rangle = E_n^{(0)}\left|n^{(0)}\right\rangle$, already known. At first order, projecting the resulting equation onto $\left\langle n^{(0)}\right|$ (taking its inner product with the unperturbed state itself) and using $\left\langle n^{(0)}\middle|n^{(0)}\right\rangle = 1$ isolates $E_n^{(1)}$ directly. At second order, projecting onto $\left\langle m^{(0)}\right|$ for $m \neq n$ and using the completeness of the unperturbed eigenstates (any state, including $\left|n^{(1)}\right\rangle$, can be expanded in the complete set $\{\left|m^{(0)}\right\rangle\}$, an extension of the basis-expansion idea from Chapter 2) gives the sum in $E_n^{(2)}$. We do not reproduce every algebraic step here, but each is a direct application of tools already in hand: eigenvalue equations, inner products, and orthonormality.

The formula for $E_n^{(1)}$ has an immediate, intuitive reading: to first order, the perturbation simply shifts each energy by its own expectation value in the *unperturbed* state — exactly the kind of calculation you have already performed for $\langle x\rangle$, $\langle p\rangle$, and similar quantities since Chapter 3, just with the perturbing Hamiltonian $\hat{H}'$ in place of $\hat{x}$ or $\hat{p}$.

Worked Example 11.1 — Perturbing a Particle in a Box

An electron in an infinite square well of width L (Chapter 4) experiences an additional potential $H' = V_0$ for $0 \leq x \leq L/2$ and $H' = 0$ for $L/2 < x \leq L$ (a step raising only the left half of the box floor). Find the first-order energy shift of the ground state.

Solution. Using $\psi_1^{(0)}(x) = \sqrt{2/L}\sin(\pi x/L)$ from Chapter 4,

$$E_1^{(1)} = \int_0^{L/2} V_0 \left|\psi_1^{(0)}(x)\right|^2 dx = V_0 \cdot \frac{2}{L}\int_0^{L/2} \sin^2\left(\frac{\pi x}{L}\right) dx = V_0 \cdot \frac{2}{L}\cdot\frac{L}{4} = \frac{V_0}{2}.$$

This has an immediate sanity check: since $|\psi_1^{(0)}(x)|^2$ is symmetric about the box's midpoint (Problem 4.7), exactly half of the ground state's probability lies in each half of the box, so a perturbation of uniform strength V_0 over exactly half the box should shift the energy by exactly $V_0/2$ on average — precisely the first-order result, with no calculation beyond this symmetry argument strictly needed. (Problem 11.1 asks you to show the $n = 2$ state receives the identical first-order shift $V_0/2$, by the same symmetry argument, and to explain why this makes physical sense.)

11.3 An Exactly Checkable Example

Perturbation theory is an approximation scheme, and it is natural to ask how good the approximation actually is. One system allows a completely honest answer, because it can be solved *both* exactly and perturbatively, letting us compare the two directly: the harmonic oscillator (Chapter 6) perturbed by a uniform force, $\hat{H}' = \lambda\hat{x}$.

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Worked Example 11.2 — Perturbation Theory versus the Exact Result

Find $E_n^{(1)}$ and $E_n^{(2)}$ for a QHO perturbed by $\hat{H}' = \lambda\hat{x}$, and compare with the exact result.

Solution. **Exact solution first.** Completing the square, $\hat{H} = \dfrac{\hat{p}^2}{2m} + \dfrac{1}{2}m\omega^2\hat{x}^2 + \lambda\hat{x} = \dfrac{\hat{p}^2}{2m} + \dfrac{1}{2}m\omega^2\left(\hat{x} + \dfrac{\lambda}{m\omega^2}\right)^2 - \dfrac{\lambda^2}{2m\omega^2}$: this is simply an unperturbed QHO Hamiltonian in the shifted coordinate $x' = x + \lambda/m\omega^2$, plus an overall constant. Its exact eigenvalues are therefore

$$E_n^{\text{exact}} = \left(n + \frac{1}{2}\right)\hbar\omega - \frac{\lambda^2}{2m\omega^2},$$

with no other correction at any order — the shift is exactly, not approximately, quadratic in λ.

Perturbation theory. First order: $E_n^{(1)} = \lambda\left\langle n^{(0)}\right|\hat{x}\left|n^{(0)}\right\rangle = 0$ for every n, since $\langle x\rangle = 0$ in any QHO energy eigenstate by the same symmetry argument used in Problem 6.2 ($|\psi_n(x)|^2$ is always an even function, so its product with the odd function x integrates to zero). Second order requires the matrix elements of $\hat{x}$ between different oscillator states; using $\hat{x} = \sqrt{\hbar/2m\omega}\,(\hat{a} + \hat{a}^\dagger)$ (Chapter 6) and the ladder-operator action $\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle$, $\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$ (stated here without full derivation; Problem 11.9 asks you to justify them from Chapter 6's ladder-operator relations), only $m = n \pm 1$ contribute:

$$\langle n-1|\,\hat{x}\,|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\sqrt{n}, \qquad \langle n+1|\,\hat{x}\,|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\sqrt{n+1}.$$

Substituting into the $E_n^{(2)}$ formula, with unperturbed energy denominators $E_n^{(0)} - E_{n\mp1}^{(0)} = \pm\hbar\omega$:

$$E_n^{(2)} = \lambda^2\frac{\hbar}{2m\omega}\left[\frac{n}{\hbar\omega} + \frac{n+1}{-\hbar\omega}\right] = \frac{\lambda^2}{2m\omega^2}\big[n - (n+1)\big] = -\frac{\lambda^2}{2m\omega^2}.$$

This matches the exact shift *precisely*, and — remarkably, though we will not prove it — every higher-order correction $E_n^{(3)}, E_n^{(4)}, \ldots$ vanishes identically for this particular perturbation, so the perturbation series, cut off after second order, is not merely a good approximation here but the exact answer. This is a special feature of the linear perturbation (it can be absorbed into a coordinate shift, an option not available for a generic $\hat{H}'$), but it is reassuring, in exactly the sense the "no haste, multi-pass validation" spirit of this book demands, to have at least one example where perturbation theory's answer can be checked against an independently obtained exact result and found to agree exactly, term by term.

Note — Why $n \neq m$ energy differences in the denominator matter

The second-order formula's denominator, $E_n^{(0)} - E_m^{(0)}$, signals an important limitation: perturbation theory as presented here assumes $E_n^{(0)} \neq E_m^{(0)}$ for every $m \neq n$ (a *non-degenerate* unperturbed spectrum), since otherwise the formula divides by zero. If two or more unperturbed states genuinely share the same energy — as happens constantly in three-dimensional problems, e.g., the n^2-fold degeneracy of hydrogen from Chapter 8 — the method of this chapter must be modified (*degenerate perturbation theory*, which first finds

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the right linear combinations of the degenerate states, using the perturbation itself, that make the second-order formula applicable; this refinement is beyond this book's scope but is discussed in the further-reading list of Appendix E). This is also, physically, why perturbation theory tends to work best when the perturbation is weak compared to the unperturbed energy *spacing*, not merely "small" in some absolute sense — a nearly degenerate pair of unperturbed levels produces a large second-order correction even for a very weak $\hat{H}'$, since the small denominator $E_n^{(0)} - E_m^{(0)}$ amplifies the result.

11.4 Chapter Summary

- For $\hat{H} = \hat{H}^{(0)} + \hat{H}'$ with $\hat{H}^{(0)}$ exactly solved and $\hat{H}'$ small, the energy corrections are $E_n^{(1)} = \left\langle n^{(0)} \right| \hat{H}' \left| n^{(0)} \right\rangle$ (the perturbation's expectation value in the unperturbed state) and $E_n^{(2)} = \sum_{m \neq n} |\left\langle m^{(0)} \right| \hat{H}' \left| n^{(0)} \right\rangle|^2 / (E_n^{(0)} - E_m^{(0)})$.
- The linearly perturbed harmonic oscillator, $\hat{H}' = \lambda \hat{x}$, can be solved both exactly (by completing the square) and perturbatively; the two methods agree exactly, with $E_n^{(1)} = 0$ (by symmetry) and $E_n^{(2)} = -\lambda^2/2m\omega^2$ reproducing the full exact shift.
- The method requires a non-degenerate unperturbed spectrum; degenerate levels (routine in three-dimensional problems, as in hydrogen) require the modified treatment of degenerate perturbation theory.
- Perturbation theory is most accurate when $\hat{H}'$ is weak *relative to the unperturbed energy-level spacing*, not merely small in an absolute sense, since near-degeneracies amplify the second-order correction through a small energy denominator.

Problems

11.1 For the perturbed box of Section 11.1's Worked Example, show that $E_2^{(1)} = V_0/2$ as well (the same result as for $n = 1$), by the symmetry argument sketched in the text, and explain briefly why this makes physical sense given that $H' = V_0$ over exactly half the box for *every* eigenstate, not just the ground state.

11.2 An infinite square well of width L (Chapter 4) is perturbed by $\hat{H}' = V_0 \sin(\pi x/L)$ for $0 \leq x \leq L$. Find $E_1^{(1)}$, using $\int_0^L \sin^3(\pi x/L)\, dx = \frac{4L}{3\pi}$ (quoted without proof).

11.3 A QHO (Chapter 6) is perturbed by $\hat{H}' = \frac{1}{2}\epsilon\, m\omega^2 \hat{x}^2$ (i.e., a small fractional change ϵ in the spring constant). Find $E_n^{(1)}$, using $\langle n|\hat{x}^2|n\rangle = (n + \frac{1}{2})\hbar/m\omega$ (quoted without proof; Problem 11.8 asks you to verify this for $n = 0$ using the result of Chapter 6, Problem 6.3). Compare your first-order result with the exact energy of a QHO with spring constant $m\omega^2(1 + \epsilon)$, expanded to first order in ϵ (*hint:* $\sqrt{1+\epsilon} \approx 1 + \epsilon/2$ for small ϵ), and comment on the agreement.

11.4 Using the second-order formula, explain in words (no new calculation) why $E_n^{(2)}$ is always negative for the ground state ($n = 0$) of *any* system, regardless of the specific form of $\hat{H}'$. (*Hint:* for $n = 0$, is $E_0^{(0)} - E_m^{(0)}$ positive or negative for every $m \neq 0$, given that the ground state has the lowest energy of all?)

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11.5 Two unperturbed states, $\left|1^{(0)}\right\rangle$ and $\left|2^{(0)}\right\rangle$, have energies $E_1^{(0)} = 1.000$ eV and $E_2^{(0)} = 1.010$ eV (nearly degenerate), with $\left\langle 2^{(0)}\right| \hat{H}' \left|1^{(0)}\right\rangle = 0.005$ eV. Compute the second-order correction $E_1^{(2)}$ (considering only the contribution from $m = 2$, for simplicity), and compare its magnitude to $\left\langle 1^{(0)}\right| \hat{H}' \left|1^{(0)}\right\rangle$ if that first-order term were, hypothetically, also 0.005 eV. What does this comparison illustrate about the note box of Section 11.3?

11.6 Verify, by direct calculation using $\hat{H} = \hat{H}^{(0)} + \hat{H}'$ and the definitions of this chapter, that if $\hat{H}' = 0$ (no perturbation at all), then $E_n^{(1)} = E_n^{(2)} = 0$ for every n, as it must be trivially.

11.7 Using the exact result $E_n^{\text{exact}} = (n + \frac{1}{2})\hbar\omega - \lambda^2/2m\omega^2$ from Section 11.3, find the value of λ (in terms of $m, \omega, \hbar$) at which the $n = 0$ exact energy E_0^{exact} first becomes negative. Is this a physically sensible regime for the perturbative treatment of this chapter to still apply? (*Hint*: compare $\lambda^2/2m\omega^2$ to the unperturbed level spacing $\hbar\omega$ referenced in the Note box of Section 11.3.)

11.8 Using $\langle x\rangle_0 = 0$ (Chapter 6, Problem 6.2) and $\langle x^2\rangle_0 = \hbar/2m\omega$ (Chapter 6, Problem 6.3), confirm that $\langle 0|\hat{x}^2|0\rangle = (0 + \frac{1}{2})\hbar/m\omega$, the $n = 0$ special case of the general formula quoted in Problem 11.3.

11.9 Using $\hat{x} = \sqrt{\hbar/2m\omega}(\hat{a} + \hat{a}^\dagger)$ and $\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle$, $\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$ (stated in Section 11.3), derive $\langle n-1|\,\hat{x}\,|n\rangle = \sqrt{\hbar/2m\omega}\sqrt{n}$ by direct substitution and use of orthonormality, $\langle n-1|n-1\rangle = 1$, $\langle n-1|n+1\rangle = 0$.

11.10 Explain briefly why, for the ground state of *any* unperturbed system, degenerate perturbation theory (mentioned in the Note box of Section 11.3) is never needed, regardless of whether excited states happen to be degenerate. (*Hint:* in a system with a well-defined, unique lowest energy, can any other state share the ground-state energy exactly?)

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Chapter 12

The Variational Method

Perturbation theory (Chapter 11) requires a nearby exactly-solved problem to perturb away from. Many important systems — helium and every larger atom among them — have no such nearby exactly solvable starting point at all: electron-electron repulsion is not a small correction to a hydrogen-like problem, it is comparable in size to the nuclear attraction itself. The variational method handles exactly this situation, and requires nothing more than a physically reasonable guess.

12.1 The Variational Principle

Key Result 12.1 — The Variational Theorem

For any normalized trial wavefunction $|\psi_{\text{trial}}\rangle$ (not necessarily an eigenstate of $\hat{H}$), the expectation value of the energy is never lower than the true ground-state energy E_0:

$$\langle\psi_{\text{trial}}|\,\hat{H}\,|\psi_{\text{trial}}\rangle \;\geq\; E_0.$$

Note — Proof

Because the (unknown) exact eigenstates $\{|n\rangle\}$ of $\hat{H}$ form a complete basis for the Hilbert space (Chapter 9), *any* normalized state, including our trial guess, can be expanded as $|\psi_{\text{trial}}\rangle = \sum_n c_n\,|n\rangle$, whether or not we know the c_n or the $|n\rangle$ explicitly. Then

$$\langle\psi_{\text{trial}}|\,\hat{H}\,|\psi_{\text{trial}}\rangle = \sum_n |c_n|^2 E_n \;\geq\; \sum_n |c_n|^2 E_0 = E_0 \sum_n |c_n|^2 = E_0,$$

where the inequality holds term by term because $E_n \geq E_0$ for every n (by definition, E_0 is the lowest eigenvalue), and the final equality uses normalization, $\sum_n |c_n|^2 = \langle\psi_{\text{trial}}|\psi_{\text{trial}}\rangle = 1$ (Section 2.3's basis-expansion idea, applied to an energy eigenbasis rather than an ordinary vector basis). ■

This is a remarkably useful guarantee: *any* normalized guess, however crude, gives an energy that overestimates (or, in the best case, exactly equals) the true ground-state energy — never underestimates it. This turns finding a good approximate ground state into a well-posed optimization problem: propose a trial wavefunction with one or more free (*variational*) parameters, compute $\langle\hat{H}\rangle$ as a function of those parameters, and minimize. Whatever parameter values minimize $\langle\hat{H}\rangle$ give the best approximation *within the assumed functional form* of the trial function, and the resulting energy is guaranteed to bound E_0 from above — so a lower computed energy is, unambiguously, a better answer, a very different situation from most numerical methods, where it is not always obvious whether a change made an approximation better or worse.

Worked Example 12.1 — A Gaussian Trial Function for Hydrogen

Use the normalized trial function $\psi_{\text{trial}}(r) = (2\alpha/\pi)^{3/4} e^{-\alpha r^2}$ (a Gaussian, with variational parameter α) to estimate the hydrogen ground-state energy, and compare with the exact result.

Solution. Without reproducing the full integral (a standard but somewhat lengthy Gaussian integral, left to Problem 12.7), the result is

$$\langle \hat{H} \rangle(\alpha) = \langle \hat{T} \rangle + \langle \hat{V} \rangle = \frac{3\hbar^2 \alpha}{2m_e} - \frac{2e^2}{4\pi\epsilon_0}\sqrt{\frac{2\alpha}{\pi}},$$

Minimizing over α (setting $d\langle \hat{H} \rangle / d\alpha = 0$ and solving) gives $E_{\text{trial}} = -\frac{8}{3\pi}(13.6057\text{ eV}) \approx -11.5$ eV, compared with the exact ground-state energy -13.6057 eV (Chapter 8) — an error of about 15%, entirely attributable to the trial function's shape: a Gaussian falls off as $e^{-\alpha r^2}$, while the true hydrogen ground state falls off as e^{-r/a_0} (Chapter 8), a qualitatively different and less sharply peaked function near $r = 0$. As required by the variational theorem, the estimate (-11.5 eV) lies *above* the true value (-13.6057 eV, more negative), never below it.

Note — Why use a deliberately imperfect trial function at all?

The hydrogen example above may seem to prove only that a bad guess gives a bad answer, when the exact answer was already known from Chapter 8. Its real purpose is pedagogical: it lets you see the variational method succeed or fail against a known benchmark before trusting it on a genuinely unsolved problem, exactly the "multi-pass validation" philosophy of this book applied to a numerical method rather than a single calculation. Gaussian trial functions, despite this quantitative shortcoming for a single hydrogen atom, are in fact the standard building block of essentially all modern computational quantum chemistry (Chapter 17): although a single Gaussian reproduces hydrogen poorly, sums of several Gaussians with different widths approximate the true exponential shape increasingly well, while remaining vastly easier to integrate analytically than exponentials for molecules with many nuclei — a trade of a controllable, systematically improvable error for an enormous gain in computational tractability.

12.2 Linear Variation

An especially useful special case takes the trial function to be a linear combination of some fixed set of functions $\{\phi_i\}$ (not necessarily eigenstates of anything), $\psi_{\text{trial}} = \sum_i c_i \phi_i$, with the *coefficients* c_i as the variational parameters. Minimizing $\langle \hat{H} \rangle$ over the c_i (a calculus problem in several variables, which we do not carry out in general here) leads to a matrix eigenvalue equation — structurally identical to the 2×2 eigenvalue problems of Chapter 2, but now built from the matrix elements $H_{ij} = \langle \phi_i | \hat{H} | \phi_j \rangle$ rather than an arbitrary numerical matrix. This method, called the *linear variational method* or Rayleigh–Ritz method, is the direct ancestor of essentially every modern electronic structure calculation (Chapter 17): choose a large, computationally convenient *basis set* of functions ϕ_i (often Gaussians, per the note above), and let matrix diagonalization — exactly the 2×2 technique of Chapter 2, scaled up to hundreds or thousands of basis functions and carried out by computer — do the variational optimization automatically.

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12.3 Application: The Helium Ground State

Helium — two electrons bound to a nucleus of charge $Z = 2$ — is the simplest atom for which the Schrödinger equation has no exact closed-form solution, because the electron-electron repulsion term $e^2/4\pi\epsilon_0 r_{12}$ (where $r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|$) couples the two electrons' coordinates and prevents the clean separation of variables used for hydrogen in Chapter 8.

Worked Example 12.2 — Helium Ground-State Energy by the Variational Method

Using the trial wavefunction $\psi_{\text{trial}}(\mathbf{r}_1, \mathbf{r}_2) = \phi_{1s}(r_1; \zeta)\,\phi_{1s}(r_2; \zeta)$, a product of two hydrogenic $1s$ orbitals (Chapter 8's functional form) sharing a common *effective nuclear charge* ζ as the single variational parameter, estimate the helium ground-state energy.

Solution. The physical idea is that each electron partially *screens* the nuclear charge seen by the other, so the effective charge ζ felt by either electron should come out somewhat less than the true nuclear charge $Z = 2$. Working in atomic units (energies in hartree, $E_h = 27.211386$ eV; Appendix A), direct evaluation of $\langle \hat{H} \rangle$ term by term — kinetic energy, nuclear attraction (evaluated with the *true* $Z = 2$, since the physical nucleus does not change), and electron-electron repulsion (a standard, tabulated integral between two $1s$ orbitals of exponent ζ, equal to $\frac{5}{8}\zeta$ in atomic units) — gives

$$\langle \hat{H} \rangle(\zeta) = \underbrace{\zeta^2}_{\text{kinetic, both } e^-} \underbrace{-2Z\zeta}_{\text{nuclear attraction, both } e^-} + \underbrace{\tfrac{5}{8}\zeta}_{\text{electron-electron repulsion}}.$$

Minimizing, $d\langle \hat{H} \rangle / d\zeta = 2\zeta - 2Z + \frac{5}{8} = 0$, gives

$$\zeta_{\text{opt}} = Z - \frac{5}{16} = 2 - \frac{5}{16} = \frac{27}{16} = 1.6875,$$

confirming the screening picture quantitatively: each electron behaves as though it sees an effective charge of 1.6875, not the full $Z = 2$, reduced by $5/16$ due to (partial) shielding by the other electron. Substituting back,

$$E_{\text{trial}} = -\left(Z - \frac{5}{16}\right)^2 E_h = -(1.6875)^2 (27.2114\ \text{eV}) = -77.49\ \text{eV}.$$

The experimental helium ground-state energy is -79.005 eV (obtained from the sum of the first and second ionization energies); our two-line, one-parameter variational estimate is within about 1.9% of the true value, and — consistent with the variational theorem of Section 12.1 — lies strictly above it. For comparison, using the untreated guess $\zeta = Z = 2$ (equivalent to ignoring electron screening entirely, and evaluating the electron-electron repulsion only as a fixed first-order correction rather than optimizing the orbitals themselves) gives $E = -74.83$ eV, a distinctly worse estimate: optimizing a single parameter recovered more than half of the remaining error, at essentially no additional computational cost, which is the entire practical appeal of the variational method.

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Note — Where the remaining 1.5 eV of error comes from

The product trial function $\phi_{1s}(r_1;\zeta)\phi_{1s}(r_2;\zeta)$ still assumes the two electrons move entirely independently of one another, correlated only through the shared, averaged value of ζ — it cannot capture the fact that the electrons actively avoid each other at every instant (their motions are *correlated*, an effect over and above simple screening). More flexible trial functions, allowing the two electrons genuinely different effective charges, or including r_{12} explicitly in the trial function, close most of this remaining gap; the general problem of efficiently capturing this "electron correlation" for larger atoms and molecules is one of the central technical challenges of modern computational quantum chemistry, discussed further in Chapter 17.

12.4 Chapter Summary

- The variational theorem guarantees $\langle \hat{H} \rangle_{\text{trial}} \geq E_0$ for any normalized trial wavefunction, turning the search for a good approximate ground state into a well-defined minimization problem with a built-in, unambiguous measure of improvement (a lower energy is always a better answer).
- A Gaussian trial function for hydrogen recovers about 85% of the true binding energy with a single parameter, illustrating both the method's power and its sensitivity to the assumed functional form of the trial function.
- Linear variation (the Rayleigh–Ritz method) uses a linear combination of fixed basis functions as the trial function, reducing the optimization to a matrix eigenvalue problem — the direct basis of essentially all modern electronic structure software (Chapter 17).
- Applied to helium with a screened hydrogenic product trial function, one variational parameter ($\zeta_{\text{opt}} = 27/16$) gives $E = -77.49$ eV against an experimental -79.005 eV, correctly capturing electron screening quantitatively while still missing the finer effect of electron correlation.

Problems

12.1 Using $E_{\text{trial}}(\zeta) = \zeta^2 - 2Z\zeta + \frac{5}{8}\zeta$ from Section 12.3, find ζ_{opt} and $E_{\text{trial}}(\zeta_{\text{opt}})$, in eV, for the H^- ion (a proton with two electrons, $Z = 1$). The experimental value is -14.35 eV; comment on the size of the discrepancy compared to the helium case, and (*without detailed calculation*) suggest whether you expect screening effects to matter *more* or *less* for $Z = 1$ than for $Z = 2$.

12.2 Repeat the calculation of Problem 12.1 for Li^+ ($Z = 3$, two electrons), and find $E_{\text{trial}}(\zeta_{\text{opt}})$ in eV.

12.3 A trial wavefunction $\psi_{\text{trial}} = c_1\phi_1 + c_2\phi_2$ (unnormalized, for simplicity) gives $\langle \hat{H} \rangle = \dfrac{5c_1^2 + 3c_2^2 + 4c_1c_2}{c_1^2 + c_2^2}$ for real c_1, c_2. Evaluate this expression at $(c_1, c_2) = (1,0)$, $(0,1)$, and $(1,1)$, and state which of these three trial states gives the best (lowest) upper bound on E_0.

12.4 A student proposes the (properly normalized) trial function $\psi_{\text{trial}}(x) = \sqrt{30/L^5}\, x(L-x)$ for a particle in a box of width L (compare with the exact ground state of Chapter 4). Without doing the full integral, state what the variational theorem guarantees about $\langle \hat{H} \rangle$ computed with this trial function, relative to the exact ground-state energy $E_1 = \pi^2\hbar^2/2mL^2$ from Chapter 4.

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12.5 Explain briefly why the variational theorem of Section 12.1, as stated, applies only to estimates of the *ground-state* energy, and would not, in general, guarantee that a trial function's energy overestimates the energy of some particular *excited* state. (*Hint*: revisit the proof — which step specifically used the fact that E_0 is the *lowest* eigenvalue?)

12.6 Two different trial functions for the same system give $\langle \hat{H} \rangle_A = -10.2$ eV and $\langle \hat{H} \rangle_B = -11.7$ eV. If the true ground-state energy is unknown, which trial function provides more information about E_0, and what, precisely, can you conclude about E_0 from each?

12.7 The Gaussian trial-function calculation of Section 12.1 claimed $E_{\text{trial}} = -\dfrac{8}{3\pi}(13.6057 \text{ eV})$ at the optimal α. Evaluate this expression numerically, and verify it matches the -11.5 eV quoted in the text.

12.8 Using the optimized helium result $E_{\text{trial}} = -(Z - 5/16)^2 E_h$ from Section 12.3, find the general formula for ζ_{opt} and E_{trial} for a two-electron ion of arbitrary nuclear charge Z (i.e., re-derive the boxed formulas symbolically, leaving Z unspecified, rather than substituting $Z = 2$ immediately).

12.9 Using your answer to Problem 12.8, find the value of Z (need not be an integer) at which the variational estimate $\zeta_{\text{opt}} = Z - 5/16$ would predict *zero* screening correction relative to the bare nuclear charge, i.e., at which $\zeta_{\text{opt}}/Z \to 1$. Is this limit approached from above or below as Z increases, and does this make physical sense given that the two-electron repulsion term does not grow with Z while the nuclear attraction terms do?

12.10 A colleague proposes using the *excited-state* hydrogen $2s$ wavefunction as a trial function for the hydrogen *ground* state, arguing that since $2s$ is an exact eigenstate of the hydrogen Hamiltonian, it should give a good variational estimate of E_1. Using the variational theorem, explain precisely what energy this calculation would actually yield, and why it would be a poor way to estimate the ground-state energy despite being an "exact" wavefunction of the right Hamiltonian.

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Chapter 13

Time-Dependent Phenomena and an Introduction to Spectroscopy

Every perturbation considered in Chapter 11 was time-independent, and the question asked was always the same: how do energy levels shift? A different, equally important question is dynamical: given an atom or molecule sitting in one stationary state, what causes it to *jump* to another, and how fast does this happen? This is the question behind every absorption and emission spectrum, and this chapter's answer — built from time-dependent perturbation theory — finally explains *why* spectra look the way they do, completing a thread left open since the discrete spectral lines of Chapter 1.

13.1 Time-Dependent Perturbation Theory

Suppose a system sits in an eigenstate $|i\rangle$ of an unperturbed, exactly-solved Hamiltonian $\hat{H}^{(0)}$ (any system from Chapters 4 through 8), and at $t = 0$ we switch on a small, possibly time-dependent perturbation $\hat{H}'(t)$ — physically, most often an oscillating electric field from incident light, $\hat{H}'(t) = \hat{H}'_0 \cos(\omega t)$. Expanding the resulting time-dependent state in the complete basis of unperturbed eigenstates, $|\Psi(t)\rangle = \sum_n c_n(t) e^{-iE_n t/\hbar} |n\rangle$ (an extension of the stationary-state expansion introduced in Chapter 3's Note box), and substituting into the full time-dependent Schrödinger equation gives, after projecting onto a specific final state $|f\rangle$ and to first order in the perturbation strength, the probability of finding the system in state f at time t, having started in state i:

$$P_{i\to f}(t) = \left| \frac{1}{i\hbar} \int_0^t \langle f | \hat{H}'(t') | i \rangle \, e^{i\omega_{fi} t'} \, dt' \right|^2, \qquad \omega_{fi} \equiv \frac{E_f - E_i}{\hbar}. \tag{13.1}$$

We do not reproduce the intermediate derivation step by step (it closely parallels the projection technique used for time-*in*dependent perturbation theory in Chapter 11, now with an extra time integral); what matters for this chapter is what Eq. (13.1) implies once the specific oscillating perturbation $\hat{H}'(t) = \hat{H}'_0 \cos(\omega t)$ is substituted in.

13.2 Fermi's Golden Rule

Carrying out the time integral in Eq. (13.1) for a perturbation oscillating at frequency ω produces a result sharply peaked whenever ω matches $|\omega_{fi}|$ — that is, whenever the perturbation's photon energy $\hbar\omega$ matches the energy gap $|E_f - E_i|$ between the two atomic states, exactly the Bohr frequency condition assumed without derivation all the way back in Chapter 1's discussion of the Bohr model. In the long-time limit, this peak sharpens into a resonance condition, and the transition proceeds at a constant *rate* (transitions per unit time) rather than the probability simply oscillating back and forth:

Key Result 13.1 — Fermi's Golden Rule

$$\Gamma_{i\to f} = \frac{2\pi}{\hbar}\left|\langle f|\,\hat{H}'_0\,|i\rangle\right|^2 \delta(E_f - E_i - \hbar\omega),$$

a transition rate strictly zero unless the resonance condition $E_f = E_i + \hbar\omega$ (absorption) or $E_f = E_i - \hbar\omega$ (emission) is exactly satisfied, and, when it is, proportional to the squared matrix element of the perturbation between the initial and final states.

The delta function in the idealized Golden Rule is a mathematical idealization (it assumes a perfectly monochromatic perturbation switched on for infinite time); real spectral lines have finite, measurable width, for reasons explored in Problem 13.10. What matters most for chemistry is not the line width but the *matrix element* $\langle f|\,\hat{H}'_0\,|i\rangle$: if it vanishes, by symmetry, for a particular pair of states, the transition is *forbidden*, regardless of how strong the incident light is or how long you wait.

13.3 Selection Rules

For light interacting with matter, the dominant term in $\hat{H}'$ (the *electric dipole approximation*, valid whenever the wavelength of light is much larger than the size of the atom or molecule, essentially always true for visible/IR/microwave spectroscopy) is proportional to $\hat{H}'_0 \propto \hat{x}$ (or, in three dimensions, the dipole operator $e\hat{\mathbf{r}}$). The matrix element controlling every transition rate in this chapter is therefore, at its core, exactly the same kind of integral computed for the QHO in Chapter 11: $\langle f|\,\hat{x}\,|i\rangle$ (or its three-dimensional generalization).

Key Result 13.2 — Selection Rules

An electric-dipole transition between two states is allowed only if the matrix element of the position (or dipole) operator between them is nonzero. Two specific consequences, both already computed earlier in this book without being named as "selection rules" at the time:

- **Harmonic oscillator:** $\langle m|\,\hat{x}\,|n\rangle \neq 0$ only for $m = n \pm 1$ (Chapter 11, Section 11.3), so vibrational transitions obey $\Delta n = \pm 1$.
- **Central potentials (atoms, rigid rotor):** $\langle \ell', m'|\,\hat{x},\hat{y},\hat{z}\,|\ell, m\rangle \neq 0$ only for $\Delta\ell = \pm 1$ (and $\Delta m = 0, \pm 1$, depending on polarization), so orbital-angular-momentum transitions obey $\Delta\ell = \pm 1$.

Worked Example 13.1 — Verifying a Selection Rule Directly

Show that the transition $\ell = 0 \to \ell' = 1$ is dipole-allowed, while $\ell = 0 \to \ell' = 2$ is dipole-forbidden, for the $m = m' = 0$ states of Chapter 7 (i.e., compute $\langle \ell', 0|\cos\theta\,|0, 0\rangle$ for $\ell' = 1$ and $\ell' = 2$, using $\hat{z} \propto r\cos\theta$).

Solution. Using $Y_0^0 = 1/\sqrt{4\pi}$, $Y_1^0 = \sqrt{3/4\pi}\cos\theta$, and $Y_2^0 = \sqrt{5/16\pi}(3\cos^2\theta - 1)$ from Chapters 7 and 8, and integrating over the full solid angle ($d\Omega = \sin\theta\, d\theta\, d\phi$):

$$\int (Y_1^0)^*\cos\theta\, Y_0^0\, d\Omega = \sqrt{\frac{3}{4\pi}}\cdot\frac{1}{\sqrt{4\pi}}\int_0^{2\pi}\int_0^{\pi}\cos^2\theta\sin\theta\, d\theta\, d\phi = \frac{\sqrt{3}}{4\pi}\cdot 2\pi\cdot\frac{2}{3} = \frac{\sqrt{3}}{3} \neq 0,$$

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confirming $\ell = 0 \to 1$ is allowed, while

$$\int (Y_2^0)^* \cos\theta \, Y_0^0 \, d\Omega = 0$$

exactly (Problem 13.1 asks you to complete this second integral explicitly; the vanishing follows from the odd overall parity of $\cos\theta \cdot (3\cos^2\theta - 1)$ under $\theta \to \pi - \theta$, once combined with the angular integration measure). The $\Delta\ell = \pm 1$ rule — which underlies, among many other things, why hydrogen's Lyman and Balmer series (Chapter 1) consist of the specific transitions they do, and not others — is confirmed directly by explicit integration in this simplest case, exactly the kind of hands-on check this book has favored throughout.

13.4 Real Spectroscopy: Rotational, Vibrational, and Electronic Transitions

Molecular spectroscopy probes three distinct types of energy level, at three widely separated energy (and hence frequency) scales, each governed by a system already solved earlier in this book.

Rotational spectroscopy probes transitions between the quantized rotational levels of a rigid, rotating molecule — formally identical to the angular momentum problem of Chapter 7, with energy $E_J = \dfrac{\hbar^2 J(J+1)}{2I}$ (using J, the conventional label for molecular rotational angular momentum, in place of Chapter 7's ℓ) and moment of inertia $I = \mu r^2$ for a diatomic of reduced mass μ and bond length r. The selection rule $\Delta J = \pm 1$ (a direct instance of the $\Delta\ell = \pm 1$ rule of Section 13.3) means absorption occurs only for transitions $J \to J+1$, at photon energy $E_{J+1} - E_J = \dfrac{\hbar^2}{I}(J+1)$ — evenly spaced in J, typically in the microwave region.

Worked Example 13.2 — Bond Length from a Rotational Spectrum

The $J = 0 \to 1$ rotational absorption of carbon monoxide, $^{12}C^{16}O$, is observed at 115.27 GHz. Find the CO bond length.

Solution. The transition energy is $E_1 - E_0 = \hbar^2/I$ (setting $J = 0$ in the spacing formula above), so

$$I = \frac{\hbar^2}{h\nu} = \frac{h}{4\pi^2\nu} = \frac{6.62607 \times 10^{-34}\ \text{J s}}{4\pi^2 (1.1527 \times 10^{11}\ \text{Hz})} = 1.4561 \times 10^{-46}\ \text{kg m}^2.$$

Using the reduced mass of CO, $\mu = 1.1385 \times 10^{-26}$ kg (Chapter 6, Problem 6.5 gave the atomic masses needed to compute this),

$$r = \sqrt{\frac{I}{\mu}} = \sqrt{\frac{1.4561 \times 10^{-46}}{1.1385 \times 10^{-26}}} = 1.131 \times 10^{-10}\ \text{m} = 113.1\ \text{pm},$$

in excellent agreement with the independently known CO equilibrium bond length of 112.8 pm. Rotational spectroscopy is, in this sense, one of the most direct and precise experimental routes to molecular geometry that exists — a single measured microwave frequency, combined with the rigid-rotor formula of Chapter 7, yields a bond length accurate to a fraction of a picometer.

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Vibrational spectroscopy probes the QHO transitions of Chapter 6, governed by the $\Delta n = \pm 1$ selection rule of Section 13.3 directly, typically in the infrared — exactly the HCl and CO examples already worked out quantitatively in Chapter 6, now understood as a consequence of the same selection-rule machinery developed in this chapter, rather than asserted separately.

Electronic spectroscopy probes transitions between entirely different electronic states (different solutions of the multi-electron Schrödinger equation, Chapter 14, or different molecular orbitals, Chapter 15), typically at much higher energy (visible/UV), governed by the same $\Delta\ell = \pm 1$-type orbital selection rules of Section 13.3 applied to the relevant orbital angular momentum quantum numbers, combined with the vibrational *Franck–Condon principle* (briefly: because nuclei are so much heavier than electrons, an electronic transition happens too fast for the nuclei to move, so it occurs essentially at fixed bond length, favoring transitions between vibrational levels whose wavefunctions overlap most in space) — a topic to which Chapter 16 returns in full.

Note — Three energy scales, three spectroscopies, one underlying theory

Rotational, vibrational, and electronic transitions differ in photon energy by roughly three orders of magnitude at each step (microwave ∼meV, infrared ∼ 0.1 eV, visible/UV ∼few eV) precisely because they probe, respectively, the angular momentum ladder of Chapter 7 ($E_J \propto \hbar^2/I$, with I set by nuclear masses and bond lengths, hence small spacings), the vibrational ladder of Chapter 6 ($E_n \propto \hbar\omega$, with the restoring force set by chemical bond stiffness, an intermediate scale), and the electronic energy-level spacings of Chapters 8 and 14 (set by $\hbar^2/m_e a_0^2$, the same combination that gives hydrogen's 13.6 eV, a much larger scale because $m_e \ll \mu$). This entire hierarchy — and every number in it — traces back to results already derived, individually, over the preceding twelve chapters; this chapter's contribution was to supply the missing dynamical rule, Fermi's golden rule and its associated selection rules, that determines which of the many energetically possible transitions actually occur.

13.5 Chapter Summary

- Time-dependent perturbation theory gives the transition probability, Eq. (13.1), for a system initially in state i to be found in state f under a perturbation $\hat{H}'(t)$.
- For an oscillating (e.g., light-induced) perturbation, Fermi's Golden Rule gives a nonzero transition rate only at resonance, $\hbar\omega = |E_f - E_i|$, proportional to $|\langle f|\hat{H}'_0|i\rangle|^2$.
- In the electric-dipole approximation, this matrix element is (proportional to) $\langle f|\hat{x}|i\rangle$, giving the selection rules $\Delta n = \pm 1$ (vibrational, from Chapter 11's QHO matrix elements) and $\Delta\ell = \pm 1$ (rotational/electronic, verified directly by explicit spherical-harmonic integration in this chapter).
- Rotational, vibrational, and electronic spectroscopy probe, respectively, the angular momentum ladder of Chapter 7, the harmonic-oscillator ladder of Chapter 6, and multi-electron energy levels (Chapters 14–15), at photon energies spanning roughly six orders of magnitude, all governed by the same underlying selection-rule machinery developed here.
- A measured rotational transition frequency, combined with nothing more than the rigid-rotor formula of Chapter 7 and a known reduced mass, determines a molecular bond length to sub-picometer precision — illustrated here for CO, matching the independently known bond length to within 0.3 pm.

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Problems

13.1 Complete the integral $\int (Y_2^0)^* \cos\theta \, Y_0^0 \, d\Omega = 0$ referenced in the Worked Example of Section 13.3, using $Y_2^0 = \sqrt{5/16\pi}(3\cos^2\theta - 1)$, and confirm it vanishes by direct integration over $\theta \in [0, \pi]$.

13.2 Using $E_J = \hbar^2 J(J+1)/2I$, find the photon energy (in units of $\hbar^2/I$) for the $J = 1 \to 2$ rotational transition, and compare it with the $J = 0 \to 1$ transition energy used in the Worked Example of Section 13.4. Confirm that successive rotational transition energies increase linearly with J (unlike the evenly-spaced vibrational ladder of Chapter 6).

13.3 HCl has a rotational constant corresponding to a $J = 0 \to 1$ transition at 625.9 GHz. Using the HCl reduced mass from Chapter 6 ($\mu = 1.6267 \times 10^{-27}$ kg), find the HCl bond length, and compare with the accepted value of 127.5 pm.

13.4 Using the QHO selection rule $\Delta n = \pm 1$ from Section 13.3, explain briefly why a QHO initially in its ground state ($n = 0$) cannot absorb a single photon and end up directly in the $n = 2$ state, no matter how intense the incident light, in the strict dipole approximation. (Real molecules do show weak "overtone" absorption at roughly this energy, exactly as noted in Chapter 6 — explain briefly, in one sentence, how this is consistent with what you have just shown, referring to the anharmonicity discussion of Chapter 6.)

13.5 A rotational absorption spectrum shows evenly spaced lines separated by 3.842 cm^{-1} (in wavenumber units). Using the relationship between adjacent-line spacing and the rotational constant B (given, without derivation, that adjacent rotational lines in a rigid-rotor spectrum are separated by exactly $2B$), find B in cm^{-1}, and compare with the CO value of $B = 1.9225$ cm^{-1} used in Section 13.4.

13.6 Using Fermi's Golden Rule, explain in a sentence or two why a transition with a very small (but nonzero) dipole matrix element $\langle f | \hat{H}_0' | i \rangle$ is often called "weakly allowed" or produces a "weak" spectral line, rather than being strictly forbidden, in contrast to a transition whose matrix element is exactly zero by symmetry (like $\ell = 0 \to \ell = 2$ from Section 13.3).

13.7 Given that visible light spans roughly 1.8–3.1 eV, and using the hydrogen energy levels of Chapter 8 ($E_n = -13.6057\ \text{eV}/n^2$), determine whether the Balmer series ($n \to 2$ transitions, Chapter 1) falls within, partially within, or entirely outside the visible range, by computing the $n = 3 \to 2$ and $n = \infty \to 2$ transition energies explicitly.

13.8 Explain briefly why rotational spectroscopy (Section 13.4) requires a molecule to have a permanent (nonzero) dipole moment to show a rotational spectrum at all, using the electric-dipole matrix-element argument of Section 13.3. Based on this criterion, would you expect homonuclear diatomics like N_2 or O_2 to show a pure rotational spectrum? Briefly justify your answer.

13.9 Using the Franck–Condon idea mentioned in Section 13.4 (electronic transitions occur too fast for nuclei to move), explain briefly why an electronic absorption spectrum of a diatomic molecule typically shows a whole progression of closely spaced lines (corresponding to different final *vibrational* levels of the excited electronic state) rather than a single sharp line, even though the underlying transition is fundamentally "electronic."

13.10 Real spectral lines have finite width even for a system with an exactly stable, infinitely long-lived initial and final state, due to the finite duration of any real observation. Using the time-energy uncertainty relation introduced in Chapter 1, Problem 1.12 ($\Delta E \, \Delta t \gtrsim \hbar/2$), estimate the minimum linewidth ΔE (in eV) of a spectral line observed over an experimental timescale of $\Delta t = 1.0$ ns,

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and compare it to the 0.4767 meV transition energy of the CO $J = 0 \rightarrow 1$ line from Section 13.4: is this minimum linewidth significant or negligible for resolving that particular transition?

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Part IV

Quantum Chemistry: From Atoms to Molecules

Chapter 14

Many-Electron Atoms

Hydrogen (Chapter 8) is exactly solvable because it has exactly one electron. Every other atom in the periodic table has more, and electron-electron repulsion — the same term that made helium's Schrödinger equation unsolvable in closed form (Chapter 12) — makes exact solution impossible for any neutral atom beyond hydrogen. This chapter develops the approximate framework, built directly on the hydrogen orbitals of Chapter 8 and the antisymmetry requirement of Chapter 10, that nonetheless explains the entire structure of the periodic table.

14.1 The Orbital Approximation

The central simplifying idea, already used implicitly in the helium variational calculation of Chapter 12, is to describe each electron as occupying its own single-particle orbital, similar in form to a hydrogen orbital but modified by an *effective*, screened nuclear charge that accounts, on average, for the repulsion of all the other electrons.

Definition 14.1 — The Orbital (Independent-Particle) Approximation

Each electron in a many-electron atom is assigned to a one-electron orbital $\phi_i(\mathbf{r})$, similar in functional form to a hydrogen orbital (Chapter 8), and the full N-electron wavefunction is built from these orbitals as a Slater determinant (Chapter 10), which automatically enforces the Pauli exclusion principle.

The orbitals themselves are found, in the modern implementation of this idea (the *Hartree–Fock method*), by a self-consistent variational calculation: guess a set of orbitals, compute the average (mean-field) repulsion each electron feels from all the others occupying the remaining orbitals, solve an effective one-electron Schrödinger-like equation in this averaged potential to get new, improved orbitals, and repeat until the orbitals stop changing — a direct, iterative generalization of the single-parameter helium optimization of Chapter 12, now optimizing entire orbital shapes rather than one scalar ζ. We do not carry out this iterative calculation explicitly (it is intractable by hand and routinely done by computer, discussed further in Chapter 17), but its key qualitative output is what matters for the rest of this chapter: *screening* makes the effective nuclear charge felt by an outer electron smaller than the true Z, and, critically, the amount of screening depends on both n and ℓ, not on n alone.

Note — Why screening depends on ℓ, not just n

An electron in an orbital with lower ℓ (for the same n) has a nonzero probability density much closer to the nucleus (recall the $2s$ radial distribution of Chapter 8, Figure 8.1, which has substantial amplitude near $r = 0$, unlike $2p$). Such an electron spends part of its time "inside" the shielding cloud of the other electrons, where it experiences a larger, less-screened effective charge, and is therefore more tightly bound than a same-n, higher-ℓ electron. This

is the microscopic origin of the energy ordering $E_{ns} < E_{np} < E_{nd} < \cdots$ within a given shell — a splitting entirely absent in hydrogen itself (Chapter 8, Section 8.4), where all states of a given n are exactly degenerate, and present here only because screening is ℓ-dependent.

14.2 Electron Configurations and the Aufbau Principle

Key Result 14.1 — The Aufbau ("Building-Up") Principle

The ground-state electron configuration of an atom is obtained by filling one-electron orbitals in order of increasing energy, respecting the Pauli exclusion principle (at most 2 electrons, of opposite spin, per spatial orbital; Chapter 10). Because of the ℓ-dependent screening discussed above, the energy ordering of subshells is not simply $1s < 2s < 2p < 3s < 3p < 3d < 4s < \cdots$ (the hydrogen-like, n-only ordering), but instead approximately follows increasing $n + \ell$, with ties broken by smaller n:

$$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d < 6p < 7s < 5f < 6d.$$

The famous and chemically consequential inversion $4s < 3d$ (rather than $3d < 4s$, which the naive n-only hydrogen ordering would suggest) is a direct, quantitative consequence of the screening argument above: the $4s$ orbital, despite its larger n, has enough amplitude close to the nucleus (much like the $2s$-versus-$2p$ comparison in the Note box) to be more tightly bound than $3d$ for the relevant range of Z, a subtlety with no counterpart anywhere in the exactly-solvable hydrogen problem of Chapter 8.

Worked Example 14.1 — Building Up the Third-Period Elements

Give the ground-state electron configurations of Na ($Z = 11$) and Ar ($Z = 18$).
Solution. Filling subshells in the order of Section 14.2, two electrons at a time (Pauli principle) until each subshell's maximum capacity (2 for s, 6 for p, using the $2(2\ell + 1)$ counting of Chapter 10) is reached:

$$\text{Na: } 1s^2 2s^2 2p^6 3s^1, \qquad \text{Ar: } 1s^2 2s^2 2p^6 3s^2 3p^6.$$

Both configurations directly explain long-familiar chemical facts: sodium's single, loosely bound $3s$ electron (far outside the filled, tightly bound [Ne] core) is exactly why sodium so readily loses one electron to form Na^+, while argon's completely filled $3p$ subshell (and hence filled $n = 3$, $\ell \leq 1$ shell) is exactly why it is chemically inert.

14.3 Term Symbols and Hund's Rules

A configuration such as carbon's $1s^2 2s^2 2p^2$ specifies which *subshells* are occupied, but not, by itself, the total angular momentum of the resulting atomic state — and for a partially filled subshell, several distinct total angular momentum states (with different energies, due to electron-electron repulsion) are generally possible from the same configuration. A *term symbol*, ${}^{2S+1}L_J$, specifies these quantum numbers precisely: total spin S (with multiplicity $2S + 1$ written as a left superscript), total orbital angular momentum L (written as a letter, $S, P, D, F, \ldots$ for $L = 0, 1, 2, 3, \ldots$,

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deliberately reusing the same letter code introduced for single-electron ℓ in Chapter 8), and total angular momentum J (a right subscript, from combining L and S exactly as Chapter 7's vector-addition ideas would suggest, with J ranging from $|L-S|$ to $L+S$).

Key Result 14.2 — Hund's Rules (Ground-Term Selection)

Among the several terms arising from a given partially-filled-subshell configuration, the actual ground term is identified by three rules, applied in order:

1. Maximum total spin S is lowest in energy (electrons prefer to align spins, reducing electron-electron repulsion by the antisymmetry-driven "Fermi correlation" of Chapter 10 — parallel-spin electrons are automatically kept apart spatially by antisymmetry, reducing their mutual Coulomb repulsion).
2. Among terms of the maximum S, maximum L is lowest.
3. Among states of the resulting term, $J = |L-S|$ is lowest if the subshell is less than half full, and $J = L+S$ is lowest if it is more than half full.

Worked Example 14.2 — The Ground Term of Carbon

Find the ground-state term symbol for carbon's $2p^2$ configuration.
Solution. Two electrons in a p subshell ($\ell = 1$, giving $m_\ell = -1, 0, 1$, each combined with $m_s = \pm\frac{1}{2}$: 6 available spin-orbitals) can be arranged in $\binom{6}{2} = 15$ distinct ways, consistent with the Pauli principle (Chapter 10). Systematically working through these 15 microstates (a standard bookkeeping exercise, not reproduced digit by digit here, but outlined in Problem 14.6) shows they organize into exactly three terms: ${}^1S$ (1 state: $L = 0, S = 0$), ${}^1D$ (5 states: $L = 2, S = 0$), and ${}^3P$ (9 states: $L = 1, S = 1$) — and indeed $1 + 5 + 9 = 15$, matching the direct microstate count exactly, a valuable consistency check of the kind this book has emphasized throughout. Applying Hund's rules: rule 1 immediately selects ${}^3P$ ($S = 1$, the unique maximum-spin term here, since both ${}^1S$ and ${}^1D$ have $S = 0$). Rule 2 is not needed (only one term survives rule 1). Rule 3: the $2p^2$ subshell is less than half full (2 of 6 possible electrons), so the minimum $J = |L-S| = |1-1| = 0$ is lowest. The ground term of carbon is therefore

$$ {}^3P_0. $$

Note — Nitrogen and oxygen, by the same method

Applying the identical procedure (Problem 14.7 and Problem 14.8 ask you to work through the details): nitrogen's half-filled $2p^3$ configuration has a unique maximum-spin arrangement (all three electrons in different m_ℓ orbitals, spins parallel, $S = \frac{3}{2}$, $L = (-1) + 0 + 1 = 0$), giving the single term ${}^4S$, with $J = \frac{3}{2}$ only (since $L = 0$) — ground term ${}^4S_{3/2}$, no ambiguity from rule 3 at all. Oxygen's $2p^4$ configuration, one electron short of a filled shell, gives the *same* terms as carbon's $2p^2$ (${}^1S$, ${}^1D$, ${}^3P$, by a symmetry between electrons and "holes" in an otherwise filled subshell that Problem 14.9 asks you to explore), but now the subshell is *more* than half full, so Hund's rule 3 selects the *maximum* $J = L + S = 1 + 1 = 2$: oxygen's ground term is ${}^3P_2$, not ${}^3P_0$. This carbon/oxygen contrast is a clean, minimal illustration of how rule 3 alone flips sign on either side of a half-filled subshell.

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14.4 Chapter Summary

- The orbital approximation describes each electron in a hydrogen-like orbital modified by screening from the other electrons, with the full antisymmetric wavefunction built as a Slater determinant (Chapter 10); the Hartree–Fock method finds the best such orbitals self-consistently, generalizing the one-parameter helium optimization of Chapter 12.

- Screening depends on ℓ as well as n (lower-ℓ orbitals penetrate closer to the nucleus and are more tightly bound), which is why subshell energies follow the approximate $n + \ell$ (Madelung) ordering rather than hydrogen's n-only ordering, producing chemically crucial inversions such as $4s$ filling before $3d$.

- The Aufbau principle fills subshells in this order, subject to the Pauli exclusion principle, to build up ground-state electron configurations across the periodic table.

- A configuration with a partially filled subshell generally corresponds to several distinct term symbols ${}^{2S+1}L_J$; Hund's three rules (maximum S, then maximum L, then minimum or maximum J depending on whether the subshell is less or more than half filled) identify the true ground term, illustrated here for carbon (${}^3P_0$), nitrogen (${}^4S_{3/2}$), and oxygen (${}^3P_2$).

Problems

14.1 Give the ground-state electron configuration of phosphorus ($Z = 15$) and of calcium ($Z = 20$), using the Aufbau order of Section 14.2.

14.2 Using the Aufbau order, identify the subshell being filled immediately *after* $4s$, and give the atomic number Z of the first element in which this subshell begins to fill.

14.3 Explain briefly, referring to the screening argument of Section 14.1, why the ionization energy required to remove a $4s$ electron from potassium is *smaller* than the (hydrogen-like, n-only) formula $E_n \propto 1/n^2$ from Chapter 8 would suggest if applied naively with $Z = 19$ and $n = 4$.

14.4 The oxygen atom has configuration $1s^2 2s^2 2p^4$. State how many electrons occupy completely filled subshells, and how many occupy the partially-filled $2p$ subshell relevant to the term-symbol analysis of Section 14.3. (*Hint:* only partially filled subshells contribute nonzero net L and S; explain briefly, in one sentence, why a completely filled subshell always contributes $L = 0, S = 0$.)

14.5 Using $\binom{6}{2} = 15$ as the total microstate count for carbon's $2p^2$ configuration (Section 14.3), and the fact this splits into terms ${}^1S$ (1 state), ${}^1D$ (5 states), and ${}^3P$ (9 states), verify the multiplicities $(2S+1)(2L+1)$ claimed for each term individually (i.e., confirm ${}^1S$ really does correspond to exactly 1 state, ${}^1D$ to 5, and ${}^3P$ to 9, using the definitions of S, L given in Section 14.3).

14.6 For carbon's $2p^2$ configuration, the maximum possible value of $M_L \equiv \sum_i m_{\ell,i}$ (achieved with both electrons in $m_\ell = +1$ orbitals with opposite spin, since the Pauli principle forbids both being in the same spin-orbital) is $M_L = 2$. Explain why this observation, on its own, is already enough to conclude that a term with $L = 2$ (i.e., a D term) must appear somewhere in carbon's term scheme. (*Hint:* what is the relationship between the maximum possible M_L value across all microstates and the largest L value that must appear among the resulting terms?)

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14.7 Apply Hund's rules to nitrogen's $2p^3$ configuration to confirm the ground term is ${}^4S_{3/2}$, following the outline given in the Note box of Section 14.3: state the maximum-S arrangement explicitly (which m_ℓ, m_s values are occupied), compute the resulting L, and identify the allowed J value(s).

14.8 Apply Hund's rules to oxygen's $2p^4$ configuration to confirm the ground term is ${}^3P_2$, paying particular attention to rule 3 (more than half-filled) as contrasted with carbon's ${}^3P_0$ (less than half-filled) in Section 14.3's Worked Example.

14.9 Explain, in two or three sentences, the "electron-hole" symmetry referenced in the Note box of Section 14.3: why do a $2p^2$ configuration (2 electrons in a 6-spin-orbital subshell) and a $2p^4$ configuration (4 electrons, equivalently 2 *missing* electrons, or "holes") give rise to the identical set of term symbols (${}^1S$, ${}^1D$, ${}^3P$)?

14.10 Fluorine has configuration $1s^2 2s^2 2p^5$ (one electron short of the filled $2p^6$ shell of neon). Using the electron-hole idea of Problem 14.9 (a single hole behaves, for term-symbol purposes, like a single electron), predict the ground term symbol of fluorine by analogy with a hypothetical single-electron $2p^1$ configuration (whose term is straightforwardly ${}^2P$, with $J = \frac{1}{2}$ or $\frac{3}{2}$), and use Hund's rule 3 (fluorine's $2p^5$ is more than half filled) to determine which J value is the ground state.

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Chapter 15

The Chemical Bond: An Introduction to Molecular Quantum Mechanics

Every system solved so far in this book has involved a single nucleus. A molecule has several, and quantum mechanically explaining why atoms bond at all — why H_2 is stable while, say, two helium atoms are not — is one of the oldest and most chemically central applications of the theory built up over the previous fourteen chapters. This chapter develops the simplest quantitative model of a chemical bond, the linear combination of atomic orbitals (LCAO), and shows it correctly predicts which diatomic molecules bond and which do not, directly from the orbital-filling ideas of Chapter 14.

15.1 The Born–Oppenheimer Approximation

A molecule's full Hamiltonian involves the kinetic energy of every nucleus and every electron, plus every pairwise Coulomb interaction among them — intractable to solve directly for anything beyond the very smallest systems. The single most important simplification, due to Born and Oppenheimer (1927), exploits the enormous mass disparity between nuclei and electrons (a proton is about 1836 times heavier than an electron; Chapter 1's discussion of reduced mass already made quantitative use of this same disparity).

Key Result 15.1 — The Born–Oppenheimer Approximation

Because nuclei are so much heavier than electrons, they move far more slowly. To an excellent approximation, the electrons instantaneously adjust to any given, fixed nuclear positions, so the electronic Schrödinger equation can be solved with the nuclei clamped at fixed locations $\{\mathbf{R}_A\}$, giving an electronic energy $E_{\text{elec}}(\{\mathbf{R}_A\})$ that depends parametrically on the nuclear geometry. The nuclei then move on this electronic energy surface, $V_{\text{nuc}}(\{\mathbf{R}_A\}) = E_{\text{elec}}(\{\mathbf{R}_A\}) + V_{\text{nuclear repulsion}}$, exactly as a particle would move under the effective one-dimensional potentials of Chapters 4 through 6.

This approximation is what makes every result in this chapter possible: it separates the intractable full problem into (1) an electronic structure problem at fixed nuclear geometry — the subject of this chapter — and (2) nuclear motion (vibration, rotation) *on* the resulting potential energy curve — exactly the rigid-rotor and harmonic-oscillator problems already solved in Chapters 6, 7, and 13. The QHO potential $V(x) = \frac{1}{2}k_{\text{spring}}x^2$ used to model molecular vibration in Chapter 6 is, precisely, a Born–Oppenheimer electronic energy curve, Taylor-expanded about its minimum.

15.2 The Hydrogen Molecular Ion H_2^+

The simplest possible molecule, H_2^+ (two protons and a single electron), is the natural starting point: with only one electron, there is no electron-electron repulsion to contend with, and the

electronic Schrödinger equation (at fixed internuclear separation R) can, in fact, be solved exactly (Problem 15.10 references this exact solution), though not in the simple closed form of Chapter 8. We instead use the approximate but far more broadly generalizable LCAO method.

Definition 15.1 — LCAO (Linear Combination of Atomic Orbitals)

Near proton A, the electron should behave much like it does in an isolated hydrogen atom centered at A; near proton B, likewise for B. This motivates a trial wavefunction (in the variational sense of Chapter 12) built as a linear combination of the two atomic $1s$ orbitals,

$$\psi_{\pm}(\mathbf{r}) = N_{\pm}\left[\phi_{1s}(\mathbf{r}-\mathbf{R}_A) \pm \phi_{1s}(\mathbf{r}-\mathbf{R}_B)\right],$$

with normalization constants $N_{\pm} = 1/\sqrt{2(1\pm S)}$, where $S(R) \equiv \int \phi_{1s}(\mathbf{r}-\mathbf{R}_A)\phi_{1s}(\mathbf{r}-\mathbf{R}_B)\,d^3r$ is the *overlap integral* between the two atomic orbitals.

Evaluating $\langle\psi_{\pm}|\,\hat{H}_{\text{elec}}\,|\psi_{\pm}\rangle$ (a direct application of the linear-variation method of Chapter 12, with just two basis functions) requires three two-center integrals, each with a known closed form in atomic units (R in bohr radii, energies in hartree):

$$S(R) = e^{-R}\left(1 + R + \frac{R^2}{3}\right), \quad j(R) = \frac{1}{R} - e^{-2R}\left(1+\frac{1}{R}\right), \quad k(R) = e^{-R}(1+R),$$

where $j(R)$ is the extra attraction an electron localized on one proton feels toward the *other* proton, and $k(R)$ (with no classical analog at all) is the purely quantum-mechanical *resonance integral* connecting the two atomic orbitals. In terms of these, the electronic energies of the symmetric and antisymmetric combinations are

$$E_{\text{elec},\pm}(R) = \frac{-\tfrac{1}{2} - j(R) \mp \left[\tfrac{1}{2}S(R) + k(R)\right]}{1 \pm S(R)}, \tag{15.1}$$

and the total (Born–Oppenheimer) energy, including nuclear repulsion, is $E_{\text{tot},\pm}(R) = E_{\text{elec},\pm}(R) + 1/R$.

Worked Example 15.1 — The Bonding and Antibonding Curves of H_2^+

Using Eq. (15.1), find the equilibrium bond length and dissociation energy predicted by the LCAO treatment for the symmetric combination, and show the antisymmetric combination has no bound minimum at all.

Solution. Direct numerical evaluation of $E_{\text{tot},+}(R)$ across a range of R (Problem 15.1 asks you to reproduce a few points of this calculation by hand) shows a minimum at

$$R_e = 2.49\ a_0 = 131.9\ \text{pm}, \qquad E_{\text{tot},+}(R_e) = -0.5648\ E_h,$$

giving a dissociation energy $D_e = E_{\text{tot},+}(\infty) - E_{\text{tot},+}(R_e) = -0.5 - (-0.5648) = 0.0648\ E_h = 1.76$ eV (using $E_{\text{tot},+}(\infty) = -\tfrac{1}{2}E_h$, the energy of a separated hydrogen atom plus a bare proton at infinite separation). By contrast, $E_{\text{tot},-}(R)$ decreases monotonically toward $-\tfrac{1}{2}E_h$ as $R \to \infty$ with no interior minimum at any finite R (Problem 15.2 asks you to verify this at several sample points): the antisymmetric combination is repulsive at every separation and does not bond at all.

Experimentally (equivalently, from the essentially exact numerical solution referenced in Section 15.2), H_2^+ has $R_e = 106$ pm and $D_e = 2.79$ eV: this simple, two-integral LCAO

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calculation gets the qualitative physics (a genuine, stable minimum for the symmetric combination) completely right, and the bond length within about 25%, while substantially underbinding the well depth. Problem 15.9 explores the single most important improvement — allowing the atomic orbital's exponent itself to vary as a second variational parameter, in the spirit of Chapter 12's helium calculation — which recovers most of the remaining discrepancy.

Note — Why the symmetric combination bonds

The physical reason the $+$ combination is lower in energy is visible directly in Eq. (15.1): the resonance integral $k(R)$ enters with a *negative* sign for the numerator of $E_{\text{elec},+}$ but effectively opposite sign for $E_{\text{elec},-}$, so the symmetric combination benefits from an additional negative (stabilizing) contribution that the antisymmetric one loses. Physically, $|\psi_+|^2$ has *constructive* interference in the region directly between the two nuclei, building up electron density exactly where it is most effective at electrostatically screening the mutual proton-proton repulsion; $|\psi_-|^2$ has a node exactly at the midpoint (destructive interference, $\psi_- = 0$ there by symmetry, Problem 15.3), depleting density from precisely this stabilizing region. This constructive/destructive interference language is not a metaphor — it is the literal wave interference of Chapter 1, Section 1.5, applied to matter waves confined between two attractive centers rather than passing through a double slit.

15.3 Molecular Orbital Theory for Many-Electron Diatomics

The two LCAO combinations of Section 15.2 generalize immediately to give the two simplest *molecular orbitals* of any homonuclear diatomic:

Key Result 15.2 — Bonding and Antibonding Molecular Orbitals

$$\sigma_{1s} = N_+(1s_A + 1s_B) \quad \text{(bonding)}, \qquad \sigma^*_{1s} = N_-(1s_A - 1s_B) \quad \text{(antibonding)},$$

with σ_{1s} lower in energy than either separated atomic orbital (as Section 15.2 demonstrated quantitatively) and σ^*_{1s} higher.

For a many-electron diatomic, exactly the Aufbau procedure of Chapter 14 applies: fill the resulting molecular orbitals, in order of increasing energy, two electrons (opposite spin, Pauli principle) at a time. Analogous, higher-energy molecular orbitals are built from $2s$ and $2p$ atomic orbitals in the same LCAO spirit (Problem 15.7 sketches the resulting $\sigma_{2s}, \sigma^*_{2s}$ pattern), giving, for second-period homonuclear diatomics, an approximate ordering

$$\sigma_{1s} < \sigma^*_{1s} < \sigma_{2s} < \sigma^*_{2s} < \pi_{2p}\,(\times 2) < \sigma_{2p} < \pi^*_{2p}\,(\times 2) < \sigma^*_{2p}$$

(the π/σ ordering among the $2p$-derived orbitals varies slightly across the second period; we do not derive this detail, which depends on $2s$-$2p$ mixing effects beyond this book's scope, and simply note that both experimental spectroscopy and more sophisticated Hartree–Fock-level calculations, Chapter 14, are needed to settle it definitively for each specific molecule).

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15.4 Bond Order

Definition 15.2 — Bond Order

$$\text{Bond order} = \frac{(\text{number of bonding electrons}) - (\text{number of antibonding electrons})}{2}.$$

Worked Example 15.2 — H_2 versus He_2

Using the molecular orbital ordering of Section 15.3, find the bond order of H_2 and of He_2, and comment on their relative stability.

Solution. H_2 has 2 electrons total, both filling the lowest available orbital, σ_{1s} (bonding), giving bond order $(2-0)/2 = 1$: a single bond, consistent with the familiar Lewis structure H–H. He_2 has 4 electrons: 2 fill σ_{1s} (bonding), and the remaining 2 are forced, by the Pauli exclusion principle (Chapter 10, since σ_{1s} can hold only 2), into σ^*_{1s} (antibonding). The bond order is $(2-2)/2 = 0$: molecular orbital theory predicts, correctly, that He_2 does *not* form a stable bond under ordinary conditions, in sharp contrast to H_2 — both molecules have the identical $\sigma_{1s}/\sigma^*_{1s}$ orbitals available, and the entire qualitative difference between "bonds" and "does not bond" traces directly to the Pauli exclusion principle forcing He_2's extra two electrons into the destabilizing antibonding orbital.

Note — Bond order predicts bond strength and length, not just existence

Bond order is not merely a yes/no bonding indicator; within a series of related molecules it correlates directly with measurable bond strength and (inversely) bond length. The isoelectronic-with-N_2 series is a standard illustration (Problem 15.6): N_2 (bond order 3, from filling through σ_{2p} but leaving the $\pi^*_{2p}, \sigma^*_{2p}$ orbitals empty) has one of the strongest and shortest bonds of any diatomic molecule, while removing an electron to form N_2^+ (bond order 2.5) measurably weakens and lengthens the bond — a direct, quantitative experimental signature of molecular orbital occupation that a purely classical bonding picture cannot explain.

15.5 A Glimpse of Valence Bond Theory

LCAO-MO theory is not the only quantum-mechanical description of the chemical bond. An older approach, *valence bond* (VB) theory (Heitler and London, 1927, applied originally to H_2), instead builds the trial wavefunction from a state where each electron remains more clearly associated with its original atom, but the two electrons' spatial coordinates are exchanged and superposed:

$$\psi_{\text{VB}}(\mathbf{r}_1, \mathbf{r}_2) \propto \phi_A(\mathbf{r}_1)\phi_B(\mathbf{r}_2) + \phi_A(\mathbf{r}_2)\phi_B(\mathbf{r}_1),$$

combined, exactly as in the helium discussion of Chapter 10, Section 10.5, with an antisymmetric singlet spin function to give a properly antisymmetric total wavefunction. The VB and MO pictures are not competing physical theories with different predictions so much as different, complementary starting points for the same variational problem: simple MO theory (Section 15.2) tends to overemphasize *ionic* contributions (configurations with both electrons on the same atom, present in $\psi_+^2 = N_+^2[\phi_A^2 + \phi_B^2 + 2\phi_A\phi_B]$ but chemically unreasonable for two neutral, equally electronegative hydrogen atoms at large separation), while simple VB theory entirely omits them; more so-

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phisticated modern calculations (Chapter 17) systematically improve on both starting points and, given a large enough basis, converge to the same essentially exact answer regardless of which qualitative picture one started from.

15.6 Chapter Summary

- The Born–Oppenheimer approximation separates a molecule's electronic structure problem (nuclei fixed) from nuclear motion on the resulting potential energy curve, justified by the large proton-to-electron mass ratio.
- LCAO builds trial molecular orbitals from sums and differences of atomic orbitals; for H_2^+, this reduces to a two-basis-function linear variational problem (Chapter 12) with closed-form two-center integrals $S(R), j(R), k(R)$.
- The symmetric (bonding) combination has a genuine energy minimum at finite R ($R_e = 131.9$ pm, $D_e = 1.76$ eV for this simple treatment, versus the essentially exact 106 pm/2.79 eV), correctly capturing bonding as constructive interference building up electron density between the nuclei; the antisymmetric (antibonding) combination is purely repulsive.
- Filling molecular orbitals via the Aufbau principle (Chapter 14) gives the bond order, $(n_{\text{bonding}} - n_{\text{antibonding}})/2$, which correctly distinguishes bonded H_2 (bond order 1) from unbound He_2 (bond order 0) using nothing beyond the Pauli exclusion principle.
- Valence bond theory offers a complementary, more localized-electron-pair starting point for the same variational problem; both approaches converge to the same answer as the trial wavefunction is systematically improved (Chapter 17).

Problems

15.1 Using $S(R) = e^{-R}(1 + R + R^2/3)$, $j(R) = 1/R - e^{-2R}(1 + 1/R)$, and $k(R) = e^{-R}(1 + R)$, evaluate $E_{\text{tot},+}(R)$ from Eq. (15.1) at $R = 2.0$ and $R = 3.0$ bohr, and confirm both lie above the minimum value $-0.5648\ E_h$ quoted in Section 15.2 for $R_e = 2.49$ bohr.

15.2 Evaluate $E_{\text{tot},-}(R)$ at $R = 1, 2, 3, 4$ bohr, and confirm it decreases monotonically toward $-0.5\ E_h$ with no interior minimum, as claimed in Section 15.2.

15.3 Using the definitions $\psi_\pm = N_\pm(1s_A \pm 1s_B)$, show that ψ_- vanishes exactly at the midpoint between the two nuclei (i.e., at any point equidistant from A and B), while ψ_+ does not. (*Hint:* what is the relationship between $1s_A$ and $1s_B$ evaluated at a point exactly equidistant from both nuclei?)

15.4 Using the bond-order definition of Section 15.4 and the molecular orbital filling order of Section 15.3, find the bond order of the diatomic ion He_2^+ (3 electrons total). Would you expect He_2^+ to be a bound species, based on this result, in contrast to neutral He_2?

15.5 Using the same molecular orbital filling order, find the bond order of Li_2 (6 electrons total, with the first four filling σ_{1s} and σ_{1s}^* exactly as in He_2, and treating $1s$ as a chemically inert core, per Chapter 14's discussion of core versus valence electrons). Compare with the bond order of H_2 found in Section 15.4.

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15.6 Using the molecular orbital ordering of Section 15.3 and the Aufbau principle, find the bond order of N_2 (14 electrons total) and of the isoelectronic-in-valence-electron-count series member N_2^+ (13 electrons). Based on the bond-order comparison alone, would you predict the N–N bond in N_2^+ to be stronger, weaker, or the same strength as in neutral N_2?

15.7 Sketch (in words, no figure required) the qualitative LCAO construction of a σ_{2s} and σ_{2s}^* molecular orbital from two atomic 2*s* orbitals, by direct analogy with the $\sigma_{1s}/\sigma_{1s}^*$ construction of Section 15.3. Would you expect σ_{2s} to lie above or below σ_{1s}^* in energy, given that 2*s* atomic orbitals themselves lie above 1*s* atomic orbitals (Chapter 8)? (You are not required to justify this rigorously — a brief, reasoned qualitative answer, referencing the atomic orbital energies involved, is sufficient.)

15.8 Explain briefly, in reference to the Note box of Section 15.5, why the simple molecular orbital description of H_2, $\psi_+(\mathbf{r}_1)\psi_+(\mathbf{r}_2)$, includes a nonzero contribution corresponding to *both* electrons being found simultaneously on the same hydrogen nucleus (an "ionic" H^-H^+ configuration), while the simple valence bond wavefunction of Section 15.5 does not. (*Hint:* expand $\psi_+(\mathbf{r}_1)\psi_+(\mathbf{r}_2) = N_+^2[\phi_A(\mathbf{r}_1) + \phi_B(\mathbf{r}_1)][\phi_A(\mathbf{r}_2) + \phi_B(\mathbf{r}_2)]$ and identify which of the resulting four terms correspond to both electrons on the same atom.)

15.9 Using $D_e = 1.76$ eV (simple LCAO) and $D_e = 2.79$ eV (essentially exact) for H_2^+ from Section 15.2, compute the percentage error of the simple LCAO dissociation energy relative to the exact value. Given that the equilibrium bond length was in error by only about 25% (Section 15.2), comment briefly on which quantity — R_e or D_e — the simple, unoptimized LCAO treatment predicts more reliably.

15.10 The exact, non-LCAO electronic Schrödinger equation for H_2^+ referenced in Section 15.2 can, remarkably, be separated exactly using prolate spheroidal coordinates (coordinates well-suited to a two-fixed-center problem, in the same spirit that spherical coordinates were well suited to hydrogen's single-center problem in Chapter 8), giving an essentially exact numerical solution. Explain briefly, in one or two sentences, why this exact separability is special to a one-electron, two-nucleus system, and would not be expected to extend to a many-electron diatomic such as Li_2 or N_2. (*Hint:* what made the many-electron atoms of Chapter 14 resist exact solution in the first place?)

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Chapter 16

Molecular Spectroscopy

Chapter 13 established the basic machinery of spectroscopy — Fermi's golden rule and the resulting selection rules — and applied it separately to rotation (Chapter 7's rigid rotor), vibration (Chapter 6's harmonic oscillator), and, briefly, electronic transitions. Real molecular spectra couple these three motions together, and real vibrations are not perfectly harmonic. This chapter completes the picture, building directly on the Born–Oppenheimer separation of Chapter 15: rotation and vibration are nuclear motion *on* an electronic energy curve, and this chapter is about the structure of that curve and the spectra it produces.

16.1 Anharmonicity and the Morse Potential

The harmonic oscillator potential $V(x) = \frac{1}{2}k_{\text{spring}}x^2$ used throughout Chapter 6 is only the leading term of a Taylor expansion about the bond's equilibrium length (Chapter 15, Section 15.1); it cannot be exactly correct, if for no other reason than that it never allows the bond to dissociate ($V \to \infty$ as $x \to \infty$, rather than leveling off at a finite dissociation energy). A far more realistic model, due to Philip Morse (1929), is

Definition 16.1 — The Morse Potential

$$V(x) = D_e\left[1 - e^{-ax}\right]^2, \qquad x \equiv r - r_e,$$

where D_e is the depth of the potential well (measured from the minimum) and a controls the well's width.

Unlike the QHO potential, the Morse potential correctly levels off, $V(x) \to D_e$ as $x \to \infty$, representing dissociation into free atoms, and it is asymmetric — steeper on the compressed ($x < 0$) side than the stretched ($x > 0$) side, matching the real physical fact that compressing a bond costs more energy than stretching it by the same amount (compression brings nuclei into direct repulsive contact; stretching merely weakens an attractive interaction). Expanding for small x (Problem 16.1 asks you to verify this expansion directly),

$$V(x) = D_e a^2 x^2 - D_e a^3 x^3 + \cdots,$$

the leading term matches the harmonic form exactly, with $k_{\text{spring}} = 2D_e a^2$ — confirming that Chapter 6's harmonic treatment is the correct small-oscillation limit of this more complete potential, exactly as anticipated in Chapter 15.

The Schrödinger equation for the Morse potential can be solved exactly (unlike most anharmonic potentials, which require perturbation theory, Chapter 11, or numerical methods), giving energy levels

Key Result 16.1 — Morse Oscillator Energy Levels

$$E_v = hc\left[\tilde{\omega}_e\left(v+\frac{1}{2}\right) - \tilde{\omega}_e x_e\left(v+\frac{1}{2}\right)^2\right], \qquad v = 0, 1, 2, \ldots,$$

expressed, by spectroscopic convention, as a wavenumber-equivalent energy ($hc\tilde{\omega}$, with $\tilde{\omega}_e$ and the *anharmonicity constant* $\tilde{\omega}_e x_e$ both quoted in cm^{-1}); $\tilde{\omega}_e$ is closely related to, but not identical with, the fundamental transition frequency $\tilde{\nu}$ used in Chapter 6.

Unlike the perfectly evenly-spaced QHO ladder of Chapter 6, Morse levels converge as v increases (the spacing $E_{v+1} - E_v$ shrinks with growing v), exactly as physically required near a dissociation limit, and eventually the discrete levels merge into a continuum above $E = D_e$, representing the unbound, freely separating atoms.

Worked Example 16.1 — HCl Anharmonicity in Detail

HCl has spectroscopic constants $\tilde{\omega}_e = 2990.9\ \mathrm{cm}^{-1}$ and $\tilde{\omega}_e x_e = 52.8\ \mathrm{cm}^{-1}$. Find the fundamental ($v = 0 \to 1$) transition wavenumber, and compare with the harmonic estimate of Chapter 6.

Solution. Using the Morse energy formula,

$$\tilde{\nu}_{0\to1} = \left[\tilde{\omega}_e\left(\tfrac{3}{2}\right) - \tilde{\omega}_e x_e\left(\tfrac{9}{4}\right)\right] - \left[\tilde{\omega}_e\left(\tfrac{1}{2}\right) - \tilde{\omega}_e x_e\left(\tfrac{1}{4}\right)\right] = \tilde{\omega}_e - 2\tilde{\omega}_e x_e = 2990.9 - 105.6 = 2885.3\ \mathrm{cm}^{-1},$$

in excellent agreement with the $2886\ \mathrm{cm}^{-1}$ used (there, via the purely harmonic force-constant calculation) in Chapter 6 — confirming that the harmonic model, despite ignoring anharmonicity entirely, already predicted the fundamental transition to within a few tenths of a percent, because the anharmonic correction $2\tilde{\omega}_e x_e$ is itself a small fraction of $\tilde{\omega}_e$ for a low-lying transition near the bottom of the well.

Note — Where anharmonicity matters most, and a genuine limitation

Anharmonicity's effect *accumulates* for higher-lying transitions and is essential for correctly predicting where the vibrational ladder terminates. The Morse potential predicts a dissociation energy $D_e = \tilde{\omega}_e^2 / 4\tilde{\omega}_e x_e$ (Problem 16.9 asks you to derive this from the energy-level formula directly, by finding where successive level spacings extrapolate to zero); for HCl this gives $D_e \approx 5.25$ eV, appreciably *larger* than the experimentally measured dissociation energy of about 4.43 eV. This is not a numerical error but a genuine, well-documented limitation of the method: the two-constant Morse extrapolation assumes the potential keeps the exact Morse functional form all the way out to dissociation, using only information measured near the bottom of the well (low v), whereas real molecular potentials deviate from the idealized Morse shape well before the bond actually breaks. This is exactly the kind of honest caveat the "no haste, multi-pass validation" philosophy of this book requires: a simple model can be highly accurate in the regime it was fit to (the fundamental transition, above) and simultaneously unreliable when extrapolated well outside that regime (the dissociation limit) — and a careful treatment states both facts rather than only the more flattering one.

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16.2 Vibration-Rotation Spectra: P and R Branches

A real gas-phase molecule vibrates *and* rotates simultaneously, and a vibrational transition (Chapter 6, Chapter 13) is accompanied by a simultaneous rotational transition obeying the same $\Delta J = \pm 1$ selection rule established in Chapter 13. Combining the vibrational energy change $\tilde{\nu}_0$ (the band origin, approximately the purely vibrational wavenumber of Section 16.1) with the rotational energy change $\Delta E_{\text{rot}} = 2B(J+1)$ or $-2BJ$ (Chapter 13's rigid-rotor spacing, with J the rotational quantum number of the *lower*, initial vibrational state) gives two families of lines:

Key Result 16.2 — P and R Branches

$$\tilde{\nu}_R(J) = \tilde{\nu}_0 + 2B(J+1), \quad J = 0, 1, 2, \ldots \qquad \text{(R branch, } \Delta J = +1\text{)}$$

$$\tilde{\nu}_P(J) = \tilde{\nu}_0 - 2BJ, \quad J = 1, 2, 3, \ldots \qquad \text{(P branch, } \Delta J = -1\text{)}$$

Purely vibrational transitions with $\Delta J = 0$ are dipole-forbidden for a simple diatomic (Problem 16.5), so no line appears exactly at $\tilde{\nu}_0$ itself — instead, a characteristic gap of $4B$ separates the innermost R-branch line ($J = 0$) from the innermost P-branch line ($J = 1$), exactly the algebraic result already verified in this chapter's introduction.

Note — Reading a bond length directly off a vibration-rotation spectrum

Because the P and R branch line spacings are both governed by the same rotational constant B that determined the pure rotational spectrum of Chapter 13, a single, easily recorded infrared absorption spectrum yields *both* the vibrational frequency (hence force constant, via Chapter 6) *and* the bond length (hence moment of inertia, via Chapter 13, Section 13.4) from the line spacing within the very same spectrum, with no separate microwave measurement required. Historically, before broadly tunable microwave sources were readily available, vibration-rotation infrared spectroscopy was among the most practical general routes to precise gas-phase bond lengths.

16.3 Electronic Spectroscopy and the Franck–Condon Principle

Chapter 13 noted, without quantitative detail, that electronic transitions occur too fast for nuclei to respond, favoring transitions between vibrational states whose wavefunctions overlap most strongly in space. We can now state this quantitatively, using the vibrational wavefunctions of Chapter 6 directly.

Key Result 16.3 — The Franck–Condon Factor

The intensity of an electronic transition between vibrational level v'' of the lower electronic state and vibrational level v' of the upper electronic state is proportional to the square of the vibrational wavefunction overlap,

$$\text{Intensity} \propto \left| \int \psi_{v'}^{\text{upper}}(x)\, \psi_{v''}^{\text{lower}}(x)\, dx \right|^2,$$

called the *Franck–Condon factor*. Because the upper and lower electronic states generally have different equilibrium bond lengths (different minima of their respective Born–

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Oppenheimer potential curves, Chapter 15), this overlap is generally largest not for $v' = v''$, but for whichever v' places the classical turning point of the upper-state vibration directly above the lower state's equilibrium geometry.

Note — Why this quantitative statement matches the qualitative picture of Chapter 13

Chapter 13's "too fast for nuclei to move" argument corresponds, quantitatively, to a transition occurring vertically on a potential-energy-curve diagram: the nuclear geometry (x) does not change during the transition itself, so the relevant overlap really is between the specific vibrational wavefunctions evaluated at essentially the same nuclear coordinate, exactly the overlap integral above. When the upper and lower equilibrium bond lengths are similar, the vibrational wavefunctions of similar shape overlap best for $v' = v''$ (a strong "0–0" band, and a spectrum dominated by one or a few lines); when they differ substantially, the maximum overlap shifts to higher v', producing the extended vibrational progressions — a whole family of electronic-transition lines, spaced by the *upper*-state vibrational frequency — routinely observed in real electronic absorption and emission spectra.

16.4 Chapter Summary

- The Morse potential, $V(x) = D_e(1 - e^{-ax})^2$, correctly captures bond dissociation and reduces to the harmonic form of Chapter 6 for small displacements, with $k_{\text{spring}} = 2D_e a^2$.
- Morse energy levels, $E_v = hc[\tilde{\omega}_e(v + \frac{1}{2}) - \tilde{\omega}_e x_e(v + \frac{1}{2})^2]$, converge at high v, correctly modeling the approach to dissociation; for HCl, the resulting fundamental transition (2885.3 cm^{-1}) closely matches Chapter 6's purely harmonic estimate, while the two-constant Morse extrapolation for D_e itself is a known, appreciable overestimate of the true dissociation energy — an instructive limitation, not merely a success story.
- Combined vibration-rotation spectra show P ($\Delta J = -1$) and R ($\Delta J = +1$) branches straddling a forbidden band origin, with both branches governed by the same rotational constant B from Chapter 13, allowing bond length and vibrational frequency to be extracted from a single infrared spectrum.
- The Franck–Condon principle quantifies "vertical" electronic transitions via the vibrational wavefunction overlap integral, explaining why electronic spectra generally show extended vibrational progressions rather than a single line, governed by the relative equilibrium geometries of the two electronic states involved.

Problems

16.1 Verify, by direct Taylor expansion of e^{-ax} through order x^2, that $V(x) = D_e(1 - e^{-ax})^2 \approx D_e a^2 x^2$ for small x, confirming the harmonic limit quoted in Section 16.1.

16.2 CO has $\tilde{\omega}_e = 2169.8$ cm^{-1} and $\tilde{\omega}_e x_e = 13.3$ cm^{-1}. Find the fundamental ($v = 0 \to 1$) transition wavenumber, and compare with the purely harmonic estimate from Chapter 6, Problem 6.5 (2143 cm^{-1}, there quoted as the experimental value; you are now asked to reproduce it from the anharmonic constants instead).

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16.3 Using the Morse energy-level formula, find the wavenumber of the first *overtone* transition ($v = 0 \to 2$) for HCl, using the constants of Section 16.1's Worked Example, and compare it with exactly twice the fundamental transition wavenumber found there. Is the overtone at exactly, more than, or less than twice the fundamental, and is this consistent with converging (sub-harmonic) level spacing at higher v?

16.4 Using $B = 1.9225\ \text{cm}^{-1}$ for CO (Chapter 13), find the wavenumbers of the first three R-branch lines ($J = 0, 1, 2$) and first three P-branch lines ($J = 1, 2, 3$) of the CO fundamental vibration-rotation band, taking $\tilde{\nu}_0 = 2143\ \text{cm}^{-1}$ (Chapter 6) as the band origin.

16.5 Explain briefly, using the parity/symmetry ideas of Chapter 13 (Section 13.3), why a purely vibrational transition with $\Delta J = 0$ (which would appear exactly at the band origin $\tilde{\nu}_0$) is dipole-forbidden for a simple diatomic molecule, consistent with the "missing" line noted in the Key Result of Section 16.2.

16.6 Using your P and R branch wavenumbers from Problem 16.4, find the size of the central gap between the $J = 0$ R-branch line and the $J = 1$ P-branch line, and confirm it equals $4B$, as claimed in Section 16.2.

16.7 A vibration-rotation spectrum shows a central gap (as defined in Problem 16.6) of $8.02\ \text{cm}^{-1}$. Find the rotational constant B for this molecule, and, if the molecule is HBr with reduced mass $\mu = 1.653 \times 10^{-27}$ kg, find its bond length.

16.8 Two electronic states of a diatomic molecule have nearly identical equilibrium bond lengths. Based on the Franck–Condon discussion of Section 16.3, would you expect the resulting electronic absorption spectrum to be dominated by a single strong 0–0 line, or by an extended progression of many lines of comparable intensity? Briefly justify your answer in terms of vibrational wavefunction overlap.

16.9 Derive the Morse dissociation-energy formula $D_e = \tilde{\omega}_e^2 / 4\tilde{\omega}_e x_e$ referenced in the Note box of Section 16.1, by finding the value of v (call it v_{max}, treated as a continuous variable for this purpose) at which $dE_v/dv = 0$ (the point where the level spacing extrapolates to zero, signaling the top of the well), and evaluating $E_{v_{\text{max}}}$ using the Morse energy-level formula.

16.10 Using the HCl Morse dissociation energy $D_e \approx 5.25$ eV from Section 16.1 together with the zero-point energy $E_0 = hc[\tilde{\omega}_e/2 - \tilde{\omega}_e x_e/4]$, find the predicted $D_0 \equiv D_e - E_0$ (the dissociation energy measured from the actual ground vibrational level, which is the thermochemically measurable quantity), in eV, and compare with the experimental value of about 4.43 eV, commenting briefly on the size of the remaining discrepancy in light of the Note box of Section 16.1.

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Chapter 17

A Modern Outlook: Computational Quantum Chemistry and Quantum Information

This book began with a handful of experiments classical physics could not explain, and built, chapter by chapter, the theory that does. It closes by looking outward: at how the ideas of Chapters 1 through 16 are actually used in contemporary research, in two directions that, despite appearing unrelated at first glance, share a common origin in exactly the postulates of Chapter 3. The first — computational quantum chemistry — is the direct, large-scale descendant of the variational and orbital methods of Chapters 12 and 14. The second — quantum information science — returns to the two-state systems of Chapter 10 and asks what new kinds of information processing superposition and entanglement make possible.

17.1 Beyond Exact Solutions: The Need for Computation

Every system solved exactly in this book — the box, the oscillator, hydrogen — was chosen *because* it is exactly solvable; helium already required the approximate variational treatment of Chapter 12, and every larger atom or molecule requires it in earnest. Modern computational quantum chemistry is, at its core, nothing more than the systematic, large-scale, computer-automated version of exactly the linear variational method (Rayleigh–Ritz) introduced in Chapter 12, Section 12.3: choose a large set of basis functions, form the Hamiltonian matrix, and diagonalize.

Note — The scale is what changes, not the method

The helium calculation of Chapter 12 used a single basis function (with one adjustable parameter, ζ) and could be minimized by hand. A modern calculation on even a modest organic molecule routinely uses hundreds of basis functions per atom, and the resulting Hamiltonian matrix (built from many-electron integrals generalizing the two-electron repulsion integral you verified numerically for helium) is diagonalized by computer. The underlying mathematics — variational minimization of $\langle \hat{H} \rangle$ subject to normalization, exactly the theorem proved in Chapter 12 — is completely unchanged; what has changed is the availability of computing power sufficient to carry the method to a scale no hand calculation could reach.

17.2 The Hartree–Fock Method and Basis Sets

Chapter 14 introduced the Hartree–Fock method conceptually: each electron occupies an orbital shaped by the averaged, self-consistent field of every other electron, with the orbitals refined iteratively until convergence. Two further practical points complete the picture.

First, the choice of basis functions used to represent each orbital matters enormously in practice, even though the underlying variational theorem (Chapter 12) guarantees correctness in principle for *any* complete basis. The true hydrogen-like orbital shape decays as $e^{-\zeta r}$ (a Slater-type

orbital, or STO, Chapter 8), but the resulting multi-center integrals (of the type first encountered for H_2^+ in Chapter 15) are computationally expensive for exponentials. Gaussian functions, $e^{-\alpha r^2}$ (the same functional form used as a variational trial function for hydrogen in Chapter 12), make these integrals dramatically faster to evaluate, at the cost of a less accurate shape — particularly near $r = 0$, where a Gaussian is smooth while the true orbital has a sharp cusp.

Worked Example 17.1 — Approximating a Slater Orbital with Gaussians

Compare a single Gaussian fit to a Slater-type 1s shape (e^{-r}) with a fit using a sum of three Gaussians (a minimal "STO-3G" style basis function, of the kind used in practice).

Solution. Fitting e^{-r} by least squares, a single optimized Gaussian $e^{-\alpha r^2}$ achieves a root-mean-square deviation of about 0.077 over the range $r \in [0, 6]$ (in units where the comparison is direct); allowing a linear combination of three Gaussians with independently optimized widths and coefficients reduces the deviation to about 0.005 — more than a fourteen-fold improvement, achieved by adding just two more terms to the basis. This is the entire logic behind "contracted" Gaussian basis sets (of which STO-3G, used here, is the simplest standard example): a fixed linear combination of several Gaussians, chosen once and for all to mimic a Slater orbital's shape as closely as practical, is used as a single effective basis function throughout a calculation, buying most of the computational speed of Gaussians while substantially repairing their shape deficiency near the nucleus.

Second, Hartree–Fock itself has an intrinsic, systematic limitation, already visible in Chapter 12's helium calculation: because each electron responds only to the *averaged* field of the others, Hartree–Fock misses the instantaneous, correlated avoidance of one electron for another — *electron correlation*, precisely the gap between the -77.49 eV screened-orbital estimate and the true -79.005 eV helium ground state identified in Chapter 12. Post-Hartree-Fock methods (configuration interaction, coupled-cluster theory, and others, named but not developed here) systematically recover this correlation energy, at increasing computational cost, and represent the frontier of high-accuracy computational quantum chemistry.

17.3 An Introduction to Density Functional Theory

A conceptually distinct approach, density functional theory (DFT), sidesteps the many-electron wavefunction $\Psi(\mathbf{r}_1, \ldots, \mathbf{r}_N)$ entirely — a function of $3N$ spatial coordinates, whose sheer dimensionality is the fundamental obstacle to exact solution for large N — in favor of the electron density $\rho(\mathbf{r})$, a function of only 3 coordinates regardless of how many electrons the system contains.

Key Result 17.1 — The Hohenberg–Kohn Theorem (Statement Only)

The ground-state electron density $\rho(\mathbf{r})$ of a many-electron system determines, in principle exactly, every ground-state property of that system, including the total energy. Formally, the ground-state energy is a *functional* of the density, $E[\rho]$.

This is a remarkable, non-obvious claim (a 3-dimensional function determining everything that an intractable $3N$-dimensional wavefunction would), and its proof — due to Hohenberg and Kohn (1964) — is a genuine, celebrated result of mathematical physics, though well beyond this book's scope. The practical difficulty is that the *exact* functional $E[\rho]$ relating density to energy is not known in closed form; practical DFT calculations (in the Kohn–Sham formulation, the stan-

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dard implementation used throughout modern computational chemistry and materials science) use approximate functionals, systematically improved over decades of research, and— much like the finite Gaussian basis sets of Section 17.2 — exact in principle but approximate in any actual calculation. DFT's practical appeal is substantial computational efficiency relative to high-accuracy post-Hartree-Fock wavefunction methods, at often-comparable accuracy for many chemically important properties, which is why it is, by a wide margin, the most widely used method in computational chemistry and condensed-matter physics today.

17.4 Quantum Information Science: A Brief Outlook

The two-level spin system of Chapter 10 — a single qubit, in modern terminology — turns out to be interesting not only for describing electron spin, but as a fundamentally new unit of information.

Definition 17.1 — Qubit

A quantum bit (*qubit*) is a two-level quantum system, $\alpha\,|0\rangle + \beta\,|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$ — formally identical to the spin-$\frac{1}{2}$ spinor of Chapter 10, but now interpreted as a unit of *information* rather than a physical spin. Unlike a classical bit, which is definitely 0 or definitely 1, a qubit can exist in a genuine superposition of both, exactly the superposition principle already used, without special comment, for every wavefunction in this book since Chapter 3.

The essential new resource beyond superposition alone is *entanglement*, briefly touched on for identical particles in Chapter 10: a joint state of two qubits, such as $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, cannot be written as a product of two independent single-qubit states $|\psi_1\rangle \otimes |\psi_2\rangle$ (Problem 17.8 asks you to attempt this factorization directly and see it fail), meaning the two qubits' properties are correlated in a way that has no classical counterpart at all — measuring the first qubit and finding it in state $|0\rangle$ instantaneously implies the second is also in state $|0\rangle$, for *every* possible choice of measurement basis, not merely the one actually performed, a fact experimentally confirmed (via violations of the Bell inequalities alluded to in Chapter 3, Section 3.2) to a very high degree of precision.

Definition 17.2 — Quantum Gate

A *quantum gate* is a Hermitian-adjacent but specifically *unitary* operator (satisfying $\hat{U}^\dagger \hat{U} = 1$, which preserves normalization exactly as time evolution under the Schrödinger equation does, Chapter 3) acting on one or more qubits. Unlike the energy eigenvalue problems of Chapters 4–8, quantum computing is concerned less with finding a gate's eigenstates than with the state a given input is mapped *to* — a quantum circuit is simply a prescribed sequence of such gates.

Worked Example 17.2 — Generating a Bell State from a Product State

Two qubits begin in the unentangled product state $|00\rangle$. Applying the *Hadamard gate*, $\hat{H}_{\text{gate}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ (unfortunately sharing a symbol with the Hamiltonian by unrelated historical convention; context always disambiguates), to the first qubit alone, followed by a

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CNOT gate (which flips the second qubit's state if and only if the first qubit is $|1\rangle$) acting on both, generates the Bell state $|\Phi^+\rangle$ used throughout this chapter.
Solution. The Hadamard gate acts on a single qubit as $\hat{H}_{\text{gate}}|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$; applying it to only the first of the two qubits (leaving the second alone) gives

$$(\hat{H}_{\text{gate}} \otimes \hat{1})\,|00\rangle = \frac{1}{\sqrt{2}}\big(\,|0\rangle + |1\rangle\,\big) \otimes |0\rangle = \frac{1}{\sqrt{2}}\big(\,|00\rangle + |10\rangle\,\big),$$

an ordinary superposition, but still manifestly an unentangled product state at this stage, exactly of the form $|\chi\rangle \otimes |0\rangle$ with $|\chi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Applying the CNOT gate, which by definition leaves $|00\rangle$ unchanged but flips $|10\rangle \to |11\rangle$ (the second qubit flips because the first, the control qubit, is $|1\rangle$):

$$\text{CNOT}\left[\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)\right] = \frac{1}{\sqrt{2}}\big(\,|00\rangle + |11\rangle\,\big) = |\Phi^+\rangle\,.$$

The two gates together take an unentangled input to a maximally entangled output — this exact two-gate sequence (Hadamard, then CNOT) is the standard, minimal circuit for generating a Bell pair in essentially every introductory quantum-computing reference, and every step above is directly verifiable by explicit 4×4 matrix multiplication in the two-qubit basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, exactly as Chapter 2's matrix eigenvalue calculations were verified by direct substitution.

Note — Why quantum information connects directly back to Chapter 1

Quantum computing algorithms (Shor's factoring algorithm and Grover's search algorithm are the two most famous, neither developed here) exploit superposition and entanglement to solve certain computational problems using fundamentally fewer steps than any known classical algorithm. Chapter 1 opened this book with the discovery that light and matter are not simply waves or simply particles, but something genuinely new, describable only by the mathematics of Chapters 2 and 3. Quantum information science is, in a real sense, the same discovery turned to a new purpose: superposition and entanglement, first forced on physics by blackbody radiation and electron diffraction, are now treated as engineerable resources in their own right, actively designed into physical qubit systems (trapped ions, superconducting circuits, and photonic systems among the leading current approaches) rather than merely observed as puzzling features of atoms. The authorship note at the front of this book places its subject explicitly at this intersection of quantum chemistry and quantum information physics — a reminder that the postulates of Chapter 3, developed here primarily for their chemical consequences, are the identical postulates underlying this entirely different, and rapidly developing, technological frontier.

17.5 Chapter Summary

- Modern computational quantum chemistry scales up the linear variational method of Chapter 12 to hundreds or thousands of basis functions, solved by computer rather than by hand.

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- Gaussian basis functions trade a less accurate orbital shape (particularly near the nucleus) for dramatically faster integral evaluation; contracted basis sets (e.g., STO-3G) partially recover accuracy by fitting several Gaussians to mimic a Slater orbital's shape, illustrated here with a fourteen-fold reduction in fitting error using three Gaussians instead of one.
- Hartree–Fock's mean-field treatment of electron-electron repulsion systematically misses electron correlation, the same gap identified quantitatively for helium in Chapter 12; post-Hartree-Fock and density functional methods address this gap by different strategies.
- Density functional theory reformulates the many-electron problem in terms of the 3-dimensional electron density rather than the $3N$-dimensional wavefunction, trading exactness of the (unknown) functional for substantial computational efficiency.
- Quantum information science treats the two-level systems of Chapter 10 as engineered qubits, with entanglement (illustrated here via the Bell state and its Schmidt decomposition) as a computational resource with no classical analog, extending this book's postulates toward an active, rapidly developing research frontier.

Problems

17.1 Explain briefly, in reference to Section 17.1, why the linear variational method of Chapter 12 (rather than, say, perturbation theory, Chapter 11) is the natural starting point for large-scale computational quantum chemistry, given that most molecules have no nearby exactly-solved unperturbed Hamiltonian $\hat{H}^{(0)}$ to perturb away from.

17.2 A Gaussian basis function $e^{-\alpha r^2}$ has zero slope at $r = 0$ ($d/dr\, e^{-\alpha r^2}|_{r=0} = 0$), while the true Slater-type 1s orbital $e^{-\zeta r}$ has a discontinuous derivative (a "cusp") there. Verify this distinction by direct differentiation of both functions at $r = 0$, and explain briefly why this mismatch is precisely the reason a single Gaussian fits a Slater orbital poorly (Section 17.2's Worked Example) especially near the nucleus.

17.3 Using the Hohenberg–Kohn statement of Section 17.3, explain briefly why DFT calculations can, in principle, scale more favorably with system size than wavefunction-based methods that work directly with $\Psi(\mathbf{r}_1, \ldots, \mathbf{r}_N)$, referring to the dimensionality argument given in the text.

17.4 Explain briefly, using the variational theorem of Chapter 12, why a Hartree–Fock energy — however large the basis set used — is always *above* the true, exact ground-state energy of a many-electron system (i.e., why "electron correlation energy," Section 17.2, is conventionally defined as a negative quantity, $E_{\text{correlation}} = E_{\text{exact}} - E_{\text{HF}} \leq 0$).

17.5 A single qubit is prepared in the state $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$ (real coefficients, for simplicity). Verify this state is properly normalized, and find the probability of measuring the qubit in state $|1\rangle$.

17.6 Two qubits are prepared in the (unentangled, product) state $|\psi_1\rangle \otimes |\psi_2\rangle$ with $|\psi_1\rangle = |\psi_2\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Expand this product explicitly in the four-dimensional two-qubit basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, and confirm all four coefficients are equal in magnitude, in contrast to the Bell state of Section 17.4, which involves only two of the four basis states.

17.7 Using the general spin-$\frac{1}{2}$ formalism of Chapter 10, explain briefly why a qubit, $\alpha|0\rangle + \beta|1\rangle$, and an electron's spin state, $\alpha|\uparrow\rangle + \beta|\downarrow\rangle$, are mathematically the identical object, with only the physical interpretation of the two basis states differing.

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17.8 Attempt to write the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ as a product $(a\,|0\rangle + b\,|1\rangle) \otimes (c\,|0\rangle + d\,|1\rangle)$ for some constants a, b, c, d. Expanding the product gives four equations (matching coefficients of $|00\rangle, |01\rangle, |10\rangle, |11\rangle$); show that the requirement of a nonzero $|00\rangle$ and $|11\rangle$ coefficient, together with a *zero* $|01\rangle$ and $|10\rangle$ coefficient, cannot be simultaneously satisfied unless either a or b is zero *and* either c or d is zero — which would eliminate one of the required nonzero terms. (This confirms, by direct algebra, the entanglement demonstrated numerically via the Schmidt decomposition in the chapter introduction.)

17.9 Using the fourteen-fold RMS error improvement quoted in Section 17.2's Worked Example for a 3-Gaussian fit relative to a single Gaussian, and given that real STO-nG basis sets exist for various n (STO-3G, STO-6G, etc.), predict qualitatively (no calculation required) whether you would expect STO-6G to fit a Slater orbital better or worse than STO-3G, and what trade-off this improvement would come at, based on the computational-cost discussion of Section 17.2.

17.10 Reflect briefly (in two or three sentences, no calculation required) on the path this book has taken, from Chapter 1's blackbody radiation and photoelectric effect to this chapter's qubits and density functionals. Identify one specific postulate or result from an early chapter (Chapters 1–3) that you can now see reused, essentially unchanged, in this chapter's discussion of either computational chemistry or quantum information.

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Appendix A

Physical Constants and Units

All values below are CODATA 2022 recommended values (Mohr, Newell, Taylor & Tiesinga, *Rev. Mod. Phys.* **97**, 025002, 2025), current at the time of writing. Constants marked "exact" are fixed by the 2019 redefinition of the SI base units and carry no experimental uncertainty; all others are measured, with uncertainty in the final quoted digits.

A.1 Fundamental Constants

Quantity	Symbol	Value
Speed of light in vacuum	c	2.99792458×10^{8} m s^{-1} (exact)
Planck constant	h	$6.62607015 \times 10^{-34}$ J Hz^{-1} (exact)
Reduced Planck constant	$\hbar = h/2\pi$	$1.054571817\ldots \times 10^{-34}$ J s
Elementary charge	e	$1.602176634 \times 10^{-19}$ C (exact)
Electron mass	m_e	$9.1093837139(28) \times 10^{-31}$ kg
Proton mass	m_p	$1.67262192595(52) \times 10^{-27}$ kg
Proton-to-electron mass ratio	m_p/m_e	$1836.152673426(32)$
Atomic mass unit	u	$1.66053906892(52) \times 10^{-27}$ kg
Avogadro constant	N_A	$6.02214076 \times 10^{23}$ mol^{-1} (exact)
Boltzmann constant	k_B	1.380649×10^{-23} J K^{-1} (exact)
Molar gas constant	$R = N_A k_B$	8.314462618 J mol^{-1}K^{-1} (exact)
Fine-structure constant	α	$7.2973525643(11) \times 10^{-3}$
Inverse fine-structure constant	α^{-1}	$137.035999177(21)$
Bohr radius	a_0	$5.29177210544(82) \times 10^{-11}$ m
Rydberg constant	R_∞	$10\,973\,731.568157(12)$ m^{-1}
Rydberg energy	$R_\infty hc$	$13.605693122990(26)$ eV
Hartree energy	$E_h = 2R_\infty hc$	$4.3597447222060(48) \times 10^{-18}$ J $= 27.211386245981(30)$ eV
Vacuum electric permittivity	ϵ_0	$8.8541878188(14) \times 10^{-12}$ F m^{-1}
Newtonian gravitational constant	G	$6.67430(15) \times 10^{-11}$ m^{3}kg^{-1}s^{-2}

A.2 Useful Unit Conversions

Conversion	**Value**
1 electronvolt (eV)	$1.602176634 \times 10^{-19}$ J (exact)
1 hartree (E_h, atomic unit of energy)	27.211386245981 eV
1 hartree	$4.3597447222060 \times 10^{-18}$ J
hc (handy for $E = hc/\lambda$)	$1239.84\ \text{eV nm} = 1.98645 \times 10^{-25}\ \text{J m}$
1 bohr (a_0, atomic unit of length)	$5.29177210544 \times 10^{-11}$ m $= 0.529177$ Å
1 Ångstrom (Å)	10^{-10} m
1 atomic mass unit (u)	$1.66053906892 \times 10^{-27}$ kg
Energy in $\text{cm}^{-1} \leftrightarrow$ Joules	$E = hc\tilde{\nu}$, i.e., $1\ \text{cm}^{-1} = 1.98645 \times 10^{-23}$ J
Energy in $\text{cm}^{-1} \leftrightarrow$ eV	$1\ \text{cm}^{-1} = 1.23984 \times 10^{-4}$ eV
$k_B T$ at $T = 298.15$ K (room temperature)	$0.02569\ \text{eV} = 207.2\ \text{cm}^{-1}$

Note — Atomic units

Setting $\hbar = m_e = e = 4\pi\epsilon_0 = 1$ defines the *atomic unit* system used throughout Chapters 12, 14, and 15 whenever it substantially simplifies the algebra (as in the helium and H_2^+ calculations). In this system, lengths are naturally measured in bohr (a_0) and energies in hartree (E_h); the conversion factors above translate atomic-unit results back to SI or to the eV/Å units more commonly used in reporting chemical results.

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Appendix B

Useful Mathematical Results

B.1 Standard Integrals

Integral	Result
$\int_{-\infty}^{\infty} e^{-\alpha x^2}\,dx$	$\sqrt{\pi/\alpha}$
$\int_{-\infty}^{\infty} x^2 e^{-\alpha x^2}\,dx$	$\frac{1}{2}\sqrt{\pi/\alpha^3}$
$\int_{-\infty}^{\infty} x^{2n} e^{-\alpha x^2}\,dx$	$\dfrac{(2n-1)!!}{(2\alpha)^n}\sqrt{\pi/\alpha}$
$\int_{-\infty}^{\infty} x^{2n+1} e^{-\alpha x^2}\,dx$	0 (odd integrand)
$\int_{0}^{\infty} x^n e^{-\alpha x}\,dx$	$n!/\alpha^{n+1}$
$\int_0^L \sin^2\left(\dfrac{n\pi x}{L}\right)dx$	$L/2$
$\int_0^L \sin\left(\dfrac{n\pi x}{L}\right)\sin\left(\dfrac{m\pi x}{L}\right)dx,\ n\neq m$	0
$\int_0^{\pi} \sin\theta\,d\theta$	2
$\int_0^{2\pi} e^{im\phi}\,d\phi,\ m\neq 0$	0

B.2 Hermite Polynomials

Defined by $H_n(y) = (-1)^n e^{y^2}\dfrac{d^n}{dy^n}e^{-y^2}$, used in the QHO wavefunctions of Chapter 6, Eq. (6.4).

n	$H_n(y)$
0	1
1	$2y$
2	$4y^2 - 2$
3	$8y^3 - 12y$
4	$16y^4 - 48y^2 + 12$
5	$32y^5 - 160y^3 + 120y$

B.3 Spherical Harmonics

Simultaneous eigenfunctions of $\hat{L}^2$ and $\hat{L}_z$ (Chapter 7); negative-m harmonics follow from $Y_\ell^{-m} = (-1)^m (Y_\ell^m)^*$.

ℓ	m	$Y_\ell^m(\theta,\phi)$
0	0	$\frac{1}{2}\sqrt{\frac{1}{\pi}}$
1	0	$\frac{1}{2}\sqrt{\frac{3}{\pi}}\cos\theta$
1	±1	$\mp\frac{1}{4}\sqrt{\frac{6}{\pi}}\sin\theta\, e^{\pm i\phi}$
2	0	$\frac{1}{4}\sqrt{\frac{5}{\pi}}(3\cos^2\theta - 1)$
2	±1	$\mp\frac{1}{8}\sqrt{\frac{30}{\pi}}\sin\theta\cos\theta\, e^{\pm i\phi}$
2	±2	$\frac{1}{8}\sqrt{\frac{30}{\pi}}\sin^2\theta\, e^{\pm 2i\phi}$
3	0	$\frac{1}{4}\sqrt{\frac{7}{\pi}}(5\cos^3\theta - 3\cos\theta)$
3	±1	$\mp\frac{1}{8}\sqrt{\frac{21}{\pi}}\sin\theta(5\cos^2\theta - 1)\, e^{\pm i\phi}$

B.4 Hydrogen Radial Wavefunctions

Solutions of the radial equation (Chapter 8), normalized so that $\int_0^\infty |R_{n\ell}(r)|^2 r^2\, dr = 1$.

n	ℓ	$R_{n\ell}(r)$
1	0	$2a_0^{-3/2}\, e^{-r/a_0}$
2	0	$\frac{1}{2\sqrt{2}}a_0^{-3/2}\left(2 - \frac{r}{a_0}\right) e^{-r/2a_0}$
2	1	$\frac{1}{2\sqrt{6}}a_0^{-3/2}\frac{r}{a_0}\, e^{-r/2a_0}$
3	0	$\frac{2}{81\sqrt{3}}a_0^{-3/2}\left(27 - 18\frac{r}{a_0} + 2\frac{r^2}{a_0^2}\right) e^{-r/3a_0}$
3	1	$\frac{4}{81\sqrt{6}}a_0^{-3/2}\left(6 - \frac{r}{a_0}\right)\frac{r}{a_0}\, e^{-r/3a_0}$
3	2	$\frac{4}{81\sqrt{30}}a_0^{-3/2}\frac{r^2}{a_0^2}\, e^{-r/3a_0}$

General formula: $\langle r\rangle_{n\ell} = \frac{a_0}{2}\left[3n^2 - \ell(\ell+1)\right]$ (used in Chapter 8).

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B.5 Outline: Series Solution of the Hydrogen Radial Equation

Chapter 8 quoted the hydrogen radial functions of Section 8.2 without deriving them term by term; this section sketches the method, in the same spirit as the QHO outline above, and shows precisely where the restriction $\ell \leq n-1$ (Chapter 8, Eq. (8.2)) originates.

Substituting $u(r) \equiv rR(r)$ into the radial equation of Chapter 8, Section 8.2 converts it into a one-dimensional Schrödinger-like form,

$$-\frac{\hbar^2}{2m_e}\frac{d^2u}{dr^2} + \left[\frac{\ell(\ell+1)\hbar^2}{2m_e r^2} - \frac{e^2}{4\pi\epsilon_0 r}\right] u = Eu, \qquad 0 < r < \infty,$$

directly comparable to the one-dimensional problems of Chapters 4–6. Two limits fix the overall shape of $u(r)$: for $r \to \infty$, the equation is dominated by $u'' \approx (2m_e|E|/\hbar^2)u$ (bound states have $E < 0$), giving exponentially decaying behavior $u \sim e^{-\kappa r}$ with $\kappa = \sqrt{2m_e|E|}/\hbar$; for $r \to 0$, the centrifugal term dominates, $u'' \approx \ell(\ell+1)u/r^2$, whose regular (normalizable) solution behaves as $u \sim r^{\ell+1}$. Writing $u(r) = r^{\ell+1}e^{-\kappa r}L(r)$ for an as-yet-undetermined function $L(r)$ and substituting back converts the equation into the *associated Laguerre equation*, a standard equation of mathematical physics whose polynomial solutions are precisely the associated Laguerre polynomials appearing in Eq. (8.4).

As with the Hermite polynomials of the QHO, a power-series solution for $L(r)$ satisfies a recursion relation among its coefficients, and normalizability (requiring $u \to 0$ as $r \to \infty$, i.e., forbidding the series from growing fast enough to overwhelm the $e^{-\kappa r}$ prefactor) forces the series to terminate after a finite number of terms. Carrying out this termination condition explicitly shows it can only be satisfied if

$$\kappa a_0 = \frac{1}{\ell + 1 + k}, \qquad k = 0, 1, 2, \ldots,$$

where k is the degree of the terminating polynomial $L(r)$. Defining $n \equiv \ell + 1 + k$ (so that n is, by construction, an integer with $n \geq \ell + 1$, i.e., $\ell \leq n - 1$ — exactly the restriction quoted without derivation in Chapter 8) and using $\kappa = \sqrt{2m_e|E|}/\hbar$ together with $\kappa a_0 = 1/n$ recovers

$$E_n = -\frac{\hbar^2}{2m_e a_0^2 n^2} = -\frac{13.6057\ \text{eV}}{n^2},$$

the Bohr formula, now with the quantum number n and the restriction $\ell \leq n - 1$ both emerging directly from the same termination argument, rather than being separately assumed. This is the same structural pattern used throughout this book — a boundary/normalizability condition forcing discreteness — appearing here in its most technically involved form.

B.6 Outline: Series Solution of the QHO

Chapter 6 solved the quantum harmonic oscillator using ladder operators; the traditional route, sketched here for reference, solves the time-independent Schrödinger equation, Eq. (6.1), directly. Substituting $y = \sqrt{m\omega/\hbar}\, x$ and $\epsilon = 2E/\hbar\omega$ converts it to

$$\frac{d^2\psi}{dy^2} + (\epsilon - y^2)\psi = 0.$$

For large y, the equation is dominated by $\psi'' \approx y^2\psi$, suggesting $\psi(y) \sim e^{-y^2/2}$; writing $\psi(y) = H(y)e^{-y^2/2}$ and substituting back converts the equation into *Hermite's equation*,

$$H'' - 2yH' + (\epsilon - 1)H = 0.$$

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Seeking a power-series solution $H(y) = \sum_k a_k y^k$ gives a recursion relation among the a_k; demanding the series *terminate* at some finite power (required for ψ to remain normalizable as $y \to \infty$, since an infinite series here would grow faster than $e^{-y^2/2}$ decays) forces $\epsilon - 1 = 2n$ for non-negative integer n, i.e., $E_n = (n + \frac{1}{2})\hbar\omega$ — the same quantization condition obtained algebraically in Chapter 6, now via the same "termination of a series/ladder" logic used throughout this book (Chapters 4, 8, and the ladder-operator argument of Chapter 6 itself), rather than the operator algebra. The terminating polynomial solutions $H(y)$ are, by construction, exactly the Hermite polynomials of Section B.2.

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Appendix C

Ground-State Electron Configurations

Configurations follow the Aufbau principle of Chapter 14; noble-gas core notation is used throughout ([He] $= 1s^2$, [Ne] $=$ [He]$2s^2 2p^6$, [Ar] $=$ [Ne]$3s^2 3p^6$, [Kr] $=$ [Ar]$3d^{10} 4s^2 4p^6$). Entries marked with † deviate from the naive Aufbau prediction due to the extra stability of half-filled and filled subshells (Chapter 14, Section 14.3); this is the same physics, applied to d subshells, that Hund's first rule (maximizing S) already illustrated for p subshells in that chapter.

C.1 Periods 1–3

Z	Element	Symbol	Configuration
1	Hydrogen	H	$1s^1$
2	Helium	He	$1s^2$
3	Lithium	Li	[He]$2s^1$
4	Beryllium	Be	[He]$2s^2$
5	Boron	B	[He]$2s^2 2p^1$
6	Carbon	C	[He]$2s^2 2p^2$
7	Nitrogen	N	[He]$2s^2 2p^3$
8	Oxygen	O	[He]$2s^2 2p^4$
9	Fluorine	F	[He]$2s^2 2p^5$
10	Neon	Ne	[He]$2s^2 2p^6$
11	Sodium	Na	[Ne]$3s^1$
12	Magnesium	Mg	[Ne]$3s^2$
13	Aluminum	Al	[Ne]$3s^2 3p^1$
14	Silicon	Si	[Ne]$3s^2 3p^2$
15	Phosphorus	P	[Ne]$3s^2 3p^3$
16	Sulfur	S	[Ne]$3s^2 3p^4$
17	Chlorine	Cl	[Ne]$3s^2 3p^5$
18	Argon	Ar	[Ne]$3s^2 3p^6$

C.2 Period 4

Z	Element	Symbol	Configuration
19	Potassium	K	$[\mathrm{Ar}]4s^1$
20	Calcium	Ca	$[\mathrm{Ar}]4s^2$
21	Scandium	Sc	$[\mathrm{Ar}]3d^1 4s^2$
22	Titanium	Ti	$[\mathrm{Ar}]3d^2 4s^2$
23	Vanadium	V	$[\mathrm{Ar}]3d^3 4s^2$
24	Chromium	Cr	$[\mathrm{Ar}]3d^5 4s^1$ †
25	Manganese	Mn	$[\mathrm{Ar}]3d^5 4s^2$
26	Iron	Fe	$[\mathrm{Ar}]3d^6 4s^2$
27	Cobalt	Co	$[\mathrm{Ar}]3d^7 4s^2$
28	Nickel	Ni	$[\mathrm{Ar}]3d^8 4s^2$
29	Copper	Cu	$[\mathrm{Ar}]3d^{10} 4s^1$ †
30	Zinc	Zn	$[\mathrm{Ar}]3d^{10} 4s^2$
31	Gallium	Ga	$[\mathrm{Ar}]3d^{10} 4s^2 4p^1$
32	Germanium	Ge	$[\mathrm{Ar}]3d^{10} 4s^2 4p^2$
33	Arsenic	As	$[\mathrm{Ar}]3d^{10} 4s^2 4p^3$
34	Selenium	Se	$[\mathrm{Ar}]3d^{10} 4s^2 4p^4$
35	Bromine	Br	$[\mathrm{Ar}]3d^{10} 4s^2 4p^5$
36	Krypton	Kr	$[\mathrm{Ar}]3d^{10} 4s^2 4p^6$

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C.3 Period 5

Z	Element	Symbol	Configuration
37	Rubidium	Rb	$[\mathrm{Kr}]5s^1$
38	Strontium	Sr	$[\mathrm{Kr}]5s^2$
39	Yttrium	Y	$[\mathrm{Kr}]4d^15s^2$
40	Zirconium	Zr	$[\mathrm{Kr}]4d^25s^2$
41	Niobium	Nb	$[\mathrm{Kr}]4d^45s^1$ †
42	Molybdenum	Mo	$[\mathrm{Kr}]4d^55s^1$ †
43	Technetium	Tc	$[\mathrm{Kr}]4d^55s^2$
44	Ruthenium	Ru	$[\mathrm{Kr}]4d^75s^1$ †
45	Rhodium	Rh	$[\mathrm{Kr}]4d^85s^1$ †
46	Palladium	Pd	$[\mathrm{Kr}]4d^{10}$ †
47	Silver	Ag	$[\mathrm{Kr}]4d^{10}5s^1$ †
48	Cadmium	Cd	$[\mathrm{Kr}]4d^{10}5s^2$
49	Indium	In	$[\mathrm{Kr}]4d^{10}5s^25p^1$
50	Tin	Sn	$[\mathrm{Kr}]4d^{10}5s^25p^2$
51	Antimony	Sb	$[\mathrm{Kr}]4d^{10}5s^25p^3$
52	Tellurium	Te	$[\mathrm{Kr}]4d^{10}5s^25p^4$
53	Iodine	I	$[\mathrm{Kr}]4d^{10}5s^25p^5$
54	Xenon	Xe	$[\mathrm{Kr}]4d^{10}5s^25p^6$

Note — Beyond xenon

Periods 6 and 7 follow the same Aufbau logic (Chapter 14) but bring in the $4f$ and $5f$ subshells (the lanthanides and actinides), where numerous additional exceptions occur, driven by the same near-degeneracy and correlation effects discussed qualitatively in Chapter 14 and Chapter 17. A complete, current table for all 118 known elements is maintained by IUPAC and reproduced in any standard inorganic chemistry reference (e.g., the *CRC Handbook of Chemistry and Physics*); the table above is restricted to periods 1–5, where every entry can be stated with full confidence and where the qualitative lessons of Chapter 14 (screening, the $4s/3d$ near-degeneracy, and the extra stability of half-filled and filled subshells) are already completely illustrated.

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Appendix D

Glossary of Key Terms

Angular momentum quantum number (ℓ)
Labels eigenvalues of $\hat{L}^2$ as $\ell(\ell+1)\hbar^2$; determines orbital shape ($s, p, d, f, \ldots$ for $\ell = 0, 1, 2, 3, \ldots$). Chapter 7.

Aufbau principle
The rule for building up ground-state electron configurations by filling orbitals in order of increasing energy, subject to the Pauli exclusion principle. Chapter 14.

Bohr radius (a_0) The natural length scale of atomic physics, $a_0 = 4\pi\epsilon_0\hbar^2/m_e e^2 = 5.29177 \times 10^{-11}$ m; the most probable radius of the hydrogen $1s$ electron. Chapters 1, 8.

Bond order $(n_{\text{bonding}} - n_{\text{antibonding}})/2$; predicts relative bond strength from molecular orbital occupation. Chapter 15.

Born rule The postulate that $|\Psi(x,t)|^2$ gives the probability density for finding a particle at position x. Chapter 3.

Born–Oppenheimer approximation
Solving the electronic Schrödinger equation with nuclei fixed, exploiting the large nucleus-to-electron mass ratio. Chapter 15.

Commutator $[\hat{A}, \hat{B}] \equiv \hat{A}\hat{B} - \hat{B}\hat{A}$; vanishes if and only if A and B can be simultaneously sharply defined. Chapters 7, 9.

Correspondence principle
The requirement that quantum predictions approach classical ones in the limit of large quantum numbers. Chapter 4.

Degeneracy Two or more distinct quantum states sharing the same energy eigenvalue. Chapters 7, 8.

Eigenvalue / eigenfunction
A special value a and state ψ satisfying $\hat{A}\psi = a\psi$; the only possible outcomes of measuring the observable associated with $\hat{A}$. Chapters 2, 3.

Electric dipole selection rule
$\Delta\ell = \pm 1$ (orbital) or $\Delta n = \pm 1$ (vibrational): transitions with a vanishing dipole matrix element are forbidden. Chapter 13.

Expectation value
$\langle \hat{A} \rangle = \int \Psi^* \hat{A} \Psi \, dx$; the average result of many measurements of A on identically prepared systems. Chapter 3.

Fermi's Golden Rule
Gives the transition rate between quantum states under a time-dependent (e.g., oscillating light-induced) perturbation. Chapter 13.

Franck–Condon principle
Electronic transitions occur too fast for nuclei to move, favoring transitions with large vibrational wavefunction overlap. Chapter 16.

Hamiltonian ($\hat{H}$) The energy operator, $\hat{H} = \hat{p}^2/2m + V(\hat{x})$; appears in the Schrödinger equation $\hat{H}\psi = E\psi$. Chapter 3.

Hartree–Fock method
A self-consistent variational method assigning each electron its own orbital in the averaged field of the others. Chapters 12, 14, 17.

Hermitian operator
An operator satisfying $\langle\phi|\hat{A}\psi\rangle = \langle\hat{A}\phi|\psi\rangle$; guarantees real eigenvalues and orthogonal eigenfunctions. Chapter 9.

Hund's rules Three rules (maximum S, then maximum L, then minimum/maximum J) for identifying the ground term of a partially filled subshell. Chapter 14.

Ladder operators ($\hat{a}, \hat{a}^\dagger$)
Raise/lower the energy of a quantum harmonic oscillator state by one quantum; generalized to $\hat{L}_\pm$ for angular momentum. Chapters 6, 7.

LCAO Linear combination of atomic orbitals; the standard method for constructing approximate molecular orbitals. Chapter 15.

Magnetic quantum number (m or m_ℓ)
Labels eigenvalues of $\hat{L}_z$ as $m\hbar$; for given ℓ, ranges over $2\ell + 1$ integer values. Chapter 7.

Morse potential $V(x) = D_e(1 - e^{-ax})^2$; a realistic anharmonic model of a vibrating chemical bond that correctly allows dissociation. Chapter 16.

Pauli exclusion principle
No two fermions can occupy the same quantum state; follows from antisymmetry of the many-fermion wavefunction. Chapter 10.

Perturbation theory
A systematic method for approximating the energies of a system "close to" an exactly solved one, via a power series in the perturbation strength. Chapter 11.

Principal quantum number (n)
Labels the hydrogen energy levels $E_n \propto -1/n^2$; also used, less formally, as the shell label in multi-electron atoms. Chapter 8.

Radial distribution function
$P_{n\ell}(r) = r^2|R_{n\ell}(r)|^2$; the probability density for the electron's distance from the nucleus, summed over angles. Chapter 8.

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Selection rule A restriction on which transitions between quantum states have nonzero transition probability. Chapter 13.

Slater determinant
An automatically antisymmetrized many-electron wavefunction built as a determinant of single-particle spin-orbitals. Chapter 10.

Spherical harmonics (Y_ℓ^m)
The angular eigenfunctions of $\hat{L}^2$ and $\hat{L}_z$; the angular part of every central-potential wavefunction. Chapter 7.

Spin An intrinsic, purely quantum-mechanical form of angular momentum with no classical counterpart; electrons have spin $s = \frac{1}{2}$. Chapter 10.

Stationary state A state $\Psi(x,t) = \psi(x)e^{-iEt/\hbar}$ whose probability density $|\Psi|^2$ is constant in time. Chapter 3.

Term symbol ($^{2S+1}L_J$)
Specifies the total spin, orbital, and total angular momentum quantum numbers of a many-electron atomic state. Chapter 14.

Uncertainty principle
$\Delta A\,\Delta B \geq \frac{1}{2}|\langle[\hat{A},\hat{B}]\rangle|$; for position and momentum, $\Delta x \Delta p \geq \hbar/2$. Chapters 1, 9.

Variational theorem
$\langle\hat{H}\rangle_{\text{trial}} \geq E_0$ for any normalized trial wavefunction; the foundation of the variational method. Chapter 12.

Wavefunction (Ψ)
The complex-valued function that completely describes a quantum state; $|\Psi|^2$ gives the position probability density. Chapter 3.

Zero-point energy
The nonzero minimum energy of a bound quantum system (e.g., $E_0 = \frac{1}{2}\hbar\omega$ for the QHO), a direct consequence of the uncertainty principle. Chapters 4, 6.

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Appendix E

Further Reading and Reference Textbooks

This book is deliberately elementary in its starting point. The texts below, arranged roughly by level and emphasis, are standard, widely used references for readers who want to continue further in a particular direction; each remains a canonical reference within its area at the time of writing.

E.1 General Quantum Mechanics

- D. J. Griffiths and D. F. Schroeter, *Introduction to Quantum Mechanics*, 3rd ed. (Cambridge University Press, 2018). The standard undergraduate physics-track introduction; covers Chapters 1–13 of this book at a broadly similar level, with a more physics-focused (less chemistry-focused) selection of applications, and continues substantially further into formal scattering theory and relativistic corrections.

- J. J. Sakurai and J. Napolitano, *Modern Quantum Mechanics*, 3rd ed. (Cambridge University Press, 2020). A graduate-level treatment, starting from the formalism of Chapter 9 of this book (Hilbert space, operators) rather than building up to it; the natural next step after this book for a reader headed toward physics graduate study.

- R. Shankar, *Principles of Quantum Mechanics*, 2nd ed. (Springer, 1994). Graduate-level, known for its unusually thorough and pedagogically careful mathematical preliminaries (an extended version of this book's Chapter 2), and for working essentially every calculation in full detail.

E.2 Quantum Chemistry

- D. A. McQuarrie, *Quantum Chemistry*, 2nd ed. (University Science Books, 2008). The direct chemistry-track counterpart to Griffiths above, and the closest single-book match to this text's scope and audience; covers Chapters 1–16 of this book with substantially more depth on multi-electron atoms and computational methods.

- I. N. Levine, *Quantum Chemistry*, 7th ed. (Pearson, 2013). A comprehensive, encyclopedic reference covering everything in this book's Chapters 8–17 in much greater depth, including a far more detailed treatment of the Hartree–Fock method sketched in Chapter 17.

- P. W. Atkins and R. S. Friedman, *Molecular Quantum Mechanics*, 5th ed. (Oxford University Press, 2011). Similar scope to Levine, with an emphasis on spectroscopy and symmetry (group theory) applications beyond this book's Chapter 16.

- A. Szabo and N. S. Ostlund, *Modern Quantum Chemistry: Introduction to Advanced Electronic Structure Theory* (Dover, 1996). The standard graduate-level reference for the Hartree–Fock and post-Hartree-Fock methods introduced only qualitatively in this book's Chapter 17.

E.3 Physical Chemistry (Broader Context)

- P. W. Atkins, J. de Paula, and J. Keeler, *Atkins' Physical Chemistry*, 12th ed. (Oxford University Press, 2023). Places this book's quantum mechanical content (roughly Chapters 1, 3–8, 13, 16) within the broader context of a full physical chemistry curriculum, including thermodynamics and kinetics not treated here at all.
- L. Pauling and E. B. Wilson, *Introduction to Quantum Mechanics with Applications to Chemistry* (Dover, 1985; originally published 1935). A historically important classic, still valuable for its unusually direct, physically motivated derivations; many of this book's worked examples in Chapters 6 and 8 follow a spirit close to Pauling and Wilson's original treatment.

E.4 Molecular Spectroscopy

- G. Herzberg, *Molecular Spectra and Molecular Structure* (multi-volume; Van Nostrand, various editions from the 1930s–1960s, reprinted by Krieger). The foundational reference work for the rotational, vibrational, and electronic spectroscopy of Chapters 13 and 16 of this book, still cited routinely in the primary spectroscopic literature.
- P. F. Bernath, *Spectra of Atoms and Molecules*, 3rd ed. (Oxford University Press, 2016). A more modern, pedagogically organized treatment covering the same material as Herzberg at a level closely matched to this book's Chapters 13 and 16.

E.5 Quantum Information Science

- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th anniversary ed. (Cambridge University Press, 2010). The standard reference for the qubit and entanglement material sketched only briefly in this book's Chapter 17; assumes the linear-algebra and Dirac-notation background developed in this book's Chapters 2 and 9.

Note — On using these references alongside this book

Every textbook above solves the same underlying Schrödinger equation this book does; differences between them are almost entirely differences of notation, emphasis, and depth, not of physical content. Chapter 9's discussion of the formal structure of quantum mechanics was written specifically to make the transition between this book's notation and any of the references above as direct as possible.

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Appendix F

Comprehensive Review Problems

The problems below are different in character from the end-of-chapter problems: each draws on ideas from two or more chapters, in the way a cumulative final examination would, rather than testing a single technique in isolation. They are intended to be worked after finishing the whole book (or, at minimum, Part IV), and are ordered roughly by which chapters they draw on, not by difficulty. Full solutions appear in the final section of the Solutions Manual.

F.1 A hydrogen atom in its ground state absorbs a photon of wavelength 80.0 nm, ionizing it. Find the kinetic energy (in eV) of the ejected photoelectron. (Chapters 1, 8.)

F.2 "Muonic hydrogen" replaces the electron of ordinary hydrogen with a muon, a particle identical to the electron in every respect except mass: $m_\mu = 206.768\, m_e$. Using the reduced-mass idea of Chapters 1 and 8, find the ratio μ_{muonic}/m_e (using the same proton mass as ordinary hydrogen), the muonic Bohr radius (as a multiple of a_0), and the muonic ground-state energy, in eV. (Chapters 1, 8.)

F.3 A normalized wavefunction is given by $\psi(x) = Ne^{ikx}e^{-x^2/2a^2}$, a plane wave modulated by a real Gaussian envelope. Show that $|\psi(x)|^2$ does not depend on k at all, and find the normalization constant N. (Chapters 2, 3.)

F.4 An electron in an infinite square well of width L (Chapter 4) experiences the perturbation $\hat{H}' = (V_0/L)\,\hat{x}$ (a uniform "tilt" of the box floor). Using the symmetry argument of Chapter 4, Problem 4.7, find $E_n^{(1)}$ for *every* quantum number n simultaneously, with no integration. (Chapters 4, 11.)

F.5 Using the general formula $\langle n|\hat{x}^2|n\rangle = (n+\tfrac{1}{2})\hbar/m\omega$ (Chapter 11) together with the general formula $\langle n|\hat{p}^2|n\rangle = (n+\tfrac{1}{2})m\hbar\omega$ (the momentum-space analog, stated without proof here), find $\Delta x\,\Delta p$ for the *first excited state* ($n=1$) of the QHO, and compare with the Heisenberg bound $\hbar/2$ from Chapters 1 and 9. Is the $n=1$ state a minimum-uncertainty state, as the $n=0$ ground state was (Chapter 6, Problem 6.4)? (Chapters 1, 6, 9.)

F.6 Find the photon wavelength emitted in the hydrogen $4f \to 3d$ transition, and confirm using Chapter 7's angular momentum formula that this transition satisfies the Chapter 13 selection rule $\Delta\ell = \pm 1$. (Chapters 7, 8, 13.)

F.7 Using the trial function $\psi_{\text{trial}}(r) = Ne^{-\beta r}$ (the *correct functional form* of the true hydrogen ground state, Chapter 8, but with the exponent β left as a free variational parameter rather than fixed at $1/a_0$), find β_{opt} and $E_{\text{trial}}(\beta_{\text{opt}})$ in atomic units, using $\langle T\rangle = \beta^2/2\,E_h$ and $\langle V\rangle = -\beta\,E_h$ (both stated without proof). Compare with the exact hydrogen ground-state energy, and explain why the agreement is exact rather than merely approximate, in contrast to the Gaussian trial function of Chapter 12. (Chapters 8, 12.)

F.8 An electron of energy 1.0 eV is incident on a rectangular barrier of height 4.0 eV and width 0.30 nm. Find (a) the tunneling probability T (using the opaque-barrier approximation of Chapter 5), and (b) the de Broglie wavelength of the incident electron (Chapter 1). Is the electron's de

Broglie wavelength larger or smaller than the barrier width, and does this affect the validity of treating the electron as tunneling through, rather than diffracting around, the barrier? (Chapters 1, 5.)

F.9 The N^+ ion (nitrogen with one electron removed) has the configuration $1s^2 2s^2 2p^2$ — the same partially-filled subshell as neutral carbon. Using Hund's rules (Chapter 14) exactly as applied to carbon, state the ground-term symbol of N^+, and explain briefly why this configuration coincidence (not the specific element) is what determines the term symbol. (Chapters 14.)

F.10 A diatomic molecule XY has 14 valence electrons available for the molecular-orbital filling scheme of Chapter 15 (i.e., treat it as isoelectronic with N_2). Using the bond order found for N_2 in Chapter 15, predict the bond order of XY, and, if its vibrational wavenumber is measured at 2050 cm^{-1}, use the harmonic-oscillator relation of Chapter 6 to find its force constant, given a reduced mass of $\mu = 1.15 \times 10^{-26}$ kg. (Chapters 6, 15.)

F.11 A qubit (Chapter 17) is prepared in the state $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ with $\alpha = \frac{1}{\sqrt{2}}$ and $\beta = \frac{i}{\sqrt{2}}$. Verify normalization, and, using the spin-$\frac{1}{2}$ correspondence of Chapter 10 (mapping $|0\rangle \to |\uparrow\rangle$, $|1\rangle \to |\downarrow\rangle$), find $\langle \hat{S}_z \rangle$ for the corresponding spin state. (Chapters 10, 17.)

F.12 Estimate the ground-state energy of a hypothetical one-dimensional "particle on a spring" confined additionally to $x > 0$ only (i.e., a QHO potential with an infinite wall at $x = 0$, so that only the *odd* QHO wavefunctions of Chapter 6, which vanish at $x = 0$, are allowed). Using this restriction, state the ground-state energy of this half-oscillator, in terms of $\hbar\omega$. (Chapters 4, 6.)

F.13 Using the generalized uncertainty principle of Chapter 9 applied to $\hat{L}_x$ and $\hat{L}_y$ (Chapter 7), find the numerical value of $\Delta L_x\,\Delta L_y$ (in units of $\hbar^2$) for a hydrogen $2p$ electron ($\ell = 1$) in the specific state $m = 1$. (Chapters 7, 9.)

F.14 A helium atom's variationally optimized ground-state energy was found to be -77.49 eV in Chapter 12, using a common effective charge $\zeta = 27/16$ for both electrons. If instead one imagines (unphysically, for this exercise) applying the same optimized-screening idea separately to a two-electron Li^+ ion via the general formula of Chapter 12, Problem 12.8, by what percentage does the predicted binding energy increase from He to Li^+? (Chapter 12.)

F.15 Combine the Morse-potential anharmonicity of Chapter 16 with the reduced-mass idea of Chapter 6: if HCl's $\tilde{\omega}_e = 2990.9$ cm^{-1} is measured for ordinary ^{35}Cl (mass 34.969 u), estimate the fractional shift in $\tilde{\omega}_e$ expected for the ^{37}Cl isotope (mass 36.966 u), using $\tilde{\omega}_e \propto 1/\sqrt{\mu}$ and recomputing the HCl reduced mass with this heavier chlorine isotope. (Chapters 6, 16.)

F.16 A particle in an infinite box (Chapter 4) is initially prepared in the state $\Psi(x,0) = \frac{1}{\sqrt{2}}\left[\psi_1(x) + \psi_3(x)\right]$ (an equal superposition of the $n = 1$ and $n = 3$ eigenstates). Using the result of Chapter 3, Problem 3.8, find the angular frequency at which $|\Psi(x,t)|^2$ oscillates, expressed as a multiple of $E_1/\hbar$ (using $E_n = n^2 E_1$ from Chapter 4). (Chapters 3, 4.)

F.17 Using the radial distribution function ideas of Chapter 8 together with the screening ideas of Chapter 14, briefly explain (two or three sentences, no calculation) why a $4s$ electron's radial distribution function, despite $n = 4$ being "larger" than $n = 3$, includes a small inner lobe of substantial probability density very close to the nucleus, and why this inner lobe is physically responsible for the $4s$-below-$3d$ energy ordering discussed in Chapter 14.

F.18 Verify that the Rydberg energy $R_\infty hc$ (Appendix A) and the Hartree energy E_h (Chapter 12 and Appendix A) satisfy $E_h = 2R_\infty hc$ exactly, using the values tabulated in Appendix A, and explain

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briefly (referencing Chapter 8) why this factor of exactly 2 is expected. (Chapters 8, 12; Appendix A.)

F.19 Using the Franck–Condon principle (Chapter 16) and the Born–Oppenheimer approximation (Chapter 15) together, explain briefly why the Franck–Condon principle — which concerns *electronic* transitions occurring too fast for nuclei to respond — is only a meaningful concept *because of* the Born–Oppenheimer separation of electronic and nuclear motion in the first place. (Chapters 15, 16.)

F.20 A typical X-ray fluorescence photon (from an inner-shell atomic transition) has energy of order 10 keV, while a typical visible-light photon (Chapter 1) has energy of order 2–3 eV. Using the hydrogen energy formula $E_n \propto Z^2$ (Chapter 8, applicable in a rough order-of-magnitude sense to inner-shell transitions in heavy atoms, where the innermost electron feels a nuclear charge only lightly screened by other inner electrons), estimate the effective nuclear charge Z_{eff} that would be needed for a hydrogen-like $n = 2 \to 1$ transition to have an energy of 10 keV, and comment on whether this is a physically reasonable value for a moderately heavy atom's innermost electrons. (Chapters 1, 8, 14.)

F.21 Show that the matrix $A = \begin{pmatrix} 3 & 2-i \\ 2+i & 5 \end{pmatrix}$ is Hermitian (in the sense of Chapter 2's complex-conjugate-transpose criterion, $A = A^\dagger$, the finite-dimensional matrix version of Chapter 9's operator definition), and find its eigenvalues, confirming they are both real as Chapter 9's general theorem guarantees. (Chapters 2, 9.)

F.22 A particle is confined to a cubical box of side L in three dimensions, $V = 0$ for $0 < x, y, z < L$ and infinite otherwise. Using separation of variables (extending Chapter 4's one-dimensional treatment to three independent Cartesian directions), the energy levels are $E_{n_x,n_y,n_z} = \dfrac{\pi^2\hbar^2}{2mL^2}(n_x^2 + n_y^2 + n_z^2)$. Find the degeneracy of the first *excited* level (i.e., how many distinct (n_x, n_y, n_z) combinations, each with $n_i \geq 1$, share the lowest energy above the ground state), and explain briefly, referring to the cubical symmetry of the box, why this degeneracy arises even though nothing in Chapter 4's original one-dimensional problem showed any degeneracy at all. (Chapter 4.)

F.23 A QHO (Chapter 6) is perturbed by $\hat{H}' = \lambda\hat{x}^4$ (a quartic anharmonic term). Using $\langle n|\hat{x}^4|n\rangle = \frac{3}{4}\left(\dfrac{\hbar}{m\omega}\right)^2(2n^2 + 2n + 1)$ (stated without proof), find $E_n^{(1)}$ for the ground state ($n = 0$) and first excited state ($n = 1$), and find the ratio $E_1^{(1)}/E_0^{(1)}$. (Chapters 6, 11.)

F.24 An electron ($s = \frac{1}{2}$) occupies a hydrogen $2p$ orbital ($\ell = 1$). Using the vector-addition idea of combining two angular momenta (briefly: the total angular momentum quantum number j ranges from $|\ell - s|$ to $\ell + s$ in integer steps, a generalization of ordinary angular momentum addition not derived in this book but directly analogous to combining two classical vectors of fixed length), list the possible values of the total angular momentum quantum number j. (Chapters 7, 10.)

F.25 Verify explicitly, using $\hat{L}_z = -i\hbar\,\partial/\partial\phi$ and the fact that $z = r\cos\theta$ has no ϕ-dependence at all, that $[\hat{L}_z, \hat{z}] = 0$ — confirming that L_z and z are compatible observables (Chapter 9), consistent with the Stark-effect matrix elements of Appendix J, which used $\hat{z}$ freely alongside definite-m angular momentum eigenstates. (Chapters 7, 9; Appendix J.)

F.26 Using the $8V_0/3\pi$ first-order energy shift found for the $n = 1$ particle-in-a-box state in Chapter 11's Worked Example (Section 11.2, for the perturbation $H' = V_0\sin(\pi x/L)$), and the fact

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that $\int_0^L \sin(2\pi x/L)\sin^2(\pi x/L)\,dx = 0$ (by symmetry, since the integrand is odd about the box midpoint), state $E_2^{(1)}$ for the $n = 2$ state under this same perturbation, with no further integration required. (Chapters 4, 11.)

F.27 Two LCAO molecular orbitals (Chapter 15) are built from a 2×2 Hamiltonian matrix $H' = \begin{pmatrix} H_{AA} & H_{AB} \\ H_{AB} & H_{BB} \end{pmatrix}$ with all entries real. Using Chapter 9's general theorem about Hermitian operators, explain briefly why the resulting bonding and antibonding orbital energies (the eigenvalues of this matrix) are guaranteed to be real numbers, without needing to solve the specific characteristic equation. (Chapters 9, 15.)

F.28 Using the HCl anharmonic constants of Chapter 16 ($\tilde{\omega}_e = 2990.9\ \text{cm}^{-1}$, $\tilde{\omega}_e x_e = 52.8\ \text{cm}^{-1}$), find the wavenumber of the "hot band" transition $v = 1 \to 2$ (only populated in a sample at elevated temperature, where the $v = 1$ level has some thermal population), and compare it with the fundamental $v = 0 \to 1$ transition, confirming the hot band lies at *lower* wavenumber, consistent with the converging Morse level spacing of Chapter 16. (Chapter 16.)

F.29 The Pauli "NOT" gate (or Pauli-X gate) used in quantum computing is represented by the matrix $\hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (note: unrelated to position, despite the symbol; a standard notational collision in the quantum-computing literature). Verify that $\hat{X}$ is unitary ($\hat{X}^\dagger \hat{X} = 1$, Chapter 17's requirement for a valid quantum gate), and find $\hat{X}\,|0\rangle$ and $\hat{X}\,|1\rangle$ explicitly, confirming this gate acts as the quantum analog of a classical logical NOT operation. (Chapter 17.)

F.30 Using the transcendental equation for the finite square well (Chapter 5) and the exact infinite-well energies (Chapter 4), explain briefly (no new calculation) why a finite well with a *very large* V_0 (deep well) should give bound-state energies for its lowest few levels that are numerically close to, but slightly *below* (less tightly bound relative to the well bottom... more precisely, slightly higher relative to the bottom of the well, i.e., closer to the top), the corresponding infinite-well energies measured from the same reference point, referring to the finite well's wavefunction penetration into the classically forbidden region. (Chapters 4, 5.)

F.31 How many electrons can a $3d$ subshell ($\ell = 2$) hold, according to the Pauli exclusion principle (Chapter 10)? Using this capacity together with the Aufbau order of Chapter 14, confirm that the ten elements from scandium ($Z = 21$) through zinc ($Z = 30$) correspond exactly to filling this single subshell (allowing for the Cr and Cu exceptions noted in Appendix C). (Chapters 10, 14; Appendix C.)

F.32 Explain briefly, in reference to the variational theorem of Chapter 12, why the Hartree–Fock method (Chapter 14, Chapter 17) is correctly described as a *variational* method: what specific quantity is being minimized, and over what space of trial wavefunctions (referring to Chapter 10's Slater determinant construction) is the minimization performed? (Chapters 10, 12, 14, 17.)

F.33 Using the general degeneracy formula $\sum_{\ell=0}^{n-1}(2\ell+1) = n^2$ (Chapter 8), verify the hydrogen $n = 5$ degeneracy directly by summing, and confirm it equals 25. (Chapter 8.)

F.34 Using the subshell capacity formula $2(2\ell+1)$ (Chapter 10), find the capacity of an f subshell ($\ell = 3$), and confirm this matches the length of the lanthanide series (the 14 elements in which the $4f$ subshell fills, following the Aufbau order of Chapter 14). (Chapters 10, 14.)

F.35 A macroscopic mass-spring system has $m = 20$ g and $k_{\text{spring}} = 5.0$ N/m (comparable to a small toy spring). Using the QHO zero-point energy formula of Chapter 6, find $E_0 = \frac{1}{2}\hbar\omega$ for this

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system, in eV, and find the equivalent temperature E_0/k_B. Comment on why quantum zero-point motion is never observed for any macroscopic mechanical oscillator, in light of this result compared to room temperature (≈ 300 K). (Chapters 1, 6.)

F.36 The Pauli "Z" gate used in quantum computing is $\hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ (note the resemblance to the spin operator $\hat{S}_z$ of Chapter 10, up to the factor of $\hbar/2$). Verify $\hat{Z}$ is unitary, and find its eigenvalues. Explain briefly why $\hat{Z}\,|0\rangle = |0\rangle$ and $\hat{Z}\,|1\rangle = -\,|1\rangle$ (i.e., $|0\rangle$ and $|1\rangle$ are themselves the eigenstates of $\hat{Z}$), by direct analogy with $\hat{S}_z$'s eigenstates $|\uparrow\rangle\,, |\downarrow\rangle$ in Chapter 10. (Chapters 10, 17.)

F.37 Two hydrogenic trial wavefunctions, ψ_A and ψ_B, are both *exact* degenerate eigenstates of some Hamiltonian $\hat{H}$ with the same eigenvalue E_0 (for instance, two different m states of the same hydrogen n, ℓ level, Chapter 8). Using the variational theorem of Chapter 12, explain briefly why *any* normalized linear combination $c_1\psi_A + c_2\psi_B$ (with $|c_1|^2 + |c_2|^2 = 1$) gives $\langle \hat{H} \rangle$ exactly equal to E_0, neither more nor less — i.e., why the variational inequality is saturated with equality for every such combination, not merely for ψ_A or ψ_B individually. (Chapters 8, 12.)

F.38 Both the particle in a box (Chapter 4) and the QHO (Chapter 6) have strictly positive ground-state energy, traced in each case to the Chapter 1 uncertainty principle. Verify the hydrogen $n = 6$ degeneracy directly by summation, $\sum_{\ell=0}^{5}(2\ell+1)$, confirming it equals $6^2 = 36$, and comment briefly on why the hydrogen ground state ($n = 1$, Chapter 8) is not usually described using "zero-point energy" language in the same way the box and oscillator ground states are. (Chapters 1, 4, 6, 8.)

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Appendix G

Notes for Instructors: Suggested Course Structures

This book contains more material than a single one-semester course will typically cover in full depth. This appendix suggests several ways to select a coherent subset, depending on the course's length, audience, and emphasis. In every case, Chapters 1–3 (Part I) are assumed as a common, non-optional foundation, since every later chapter depends on them directly.

G.1 One-Semester Survey (Chemistry-Focused)

For a single-semester course aimed primarily at chemistry majors, with limited lecture time available for the more formal material of Part III:

Weeks	Content
1–3	Chapters 1–3 (motivation, mathematical toolkit, postulates)
4–6	Chapters 4–6 (particle in a box, tunneling, harmonic oscillator)
7	Chapter 7 (angular momentum) — can be compressed to the Key Results only
8	Chapter 8 (hydrogen atom)
9	Chapter 9 — compressed: Sections 9.1–9.2 only (Hermitian operators), omit 9.3–9.4
10	Chapter 10 (spin, Pauli principle)
11–12	Chapters 14–15 (many-electron atoms, chemical bonding)
13	Chapter 16 (molecular spectroscopy), drawing selectively on Chapter 13's selection rules as needed
14	Chapter 17 (modern outlook)

This sequence omits full formal coverage of Chapter 9's generalized uncertainty principle and commutator algebra (Sections 9.3–9.4), Chapter 11 (perturbation theory), Chapter 12 (variational method, though its helium result is worth at least mentioning qualitatively before Chapter 14), and Chapter 13's quantitative treatment of Fermi's golden rule, relying instead on the selection-rule *results* quoted and used directly in Chapters 15–16. Instructors wanting to preserve the variational method (given its direct relevance to computational chemistry, Chapter 17) can substitute Chapter 12 for one of the Chapter 9 weeks with minimal loss of continuity, since only Chapter 12's own worked examples (not a deep formal prerequisite from Chapter 9) are needed to follow it.

G.2 One-Semester Survey (Physics-Focused)

For a physics-track course with less emphasis on Part IV's chemistry applications:

Weeks	Content
1–3	Chapters 1–3
4–6	Chapters 4–6
7–8	Chapter 7, Chapter 8
9	Chapter 9, in full
10	Chapter 10
11–12	Chapters 11–12 (perturbation theory, variational method)
13	Chapter 13 (time-dependent perturbation theory, spectroscopy)
14	Selected topics from Chapters 14–17, as time permits, or Chapter 17 alone as a capstone

G.3 Two-Semester Sequence

For a full-year sequence, Parts I–II (Chapters 1–8) form a natural first-semester course (an "introduction to quantum mechanics"), and Parts III–IV (Chapters 9–17) form a natural second-semester course (an "introduction to quantum chemistry and advanced topics"), with the appendices, and especially the Comprehensive Review Problems of Appendix F, well suited to a cumulative review at the start of the second semester to reconnect with first-semester material.

Note — Problem sets and the Solutions Manual

Because this book's Solutions Manual gives complete worked solutions to every problem (rather than final answers only), instructors assigning problems for graded credit will want to either assign a rotating subset not announced in advance, supplement with original problems of their own devising, or use the end-of-chapter problems primarily for ungraded practice and reserve graded assessment for exams and original assignments. The Comprehensive Review Problems of Appendix F, being explicitly cross-chapter in character, are particularly well suited to take-home or cumulative exam use, precisely because they cannot be answered by pattern-matching against a single chapter's techniques in isolation.

G.4 Mathematical Prerequisites in Practice

Chapter 2 is written to require nothing beyond single-variable calculus, and it has been used successfully with students whose only prior mathematics is a standard calculus sequence. Instructors working with students who have already completed a linear algebra or differential equations course may wish to assign Chapter 2 as fast, confirmatory reading rather than as new material, moving directly to Chapter 3 in the second week of the course; the Chapter 2 problems remain useful in this case as a diagnostic, assigned but not necessarily discussed at length in lecture.

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Appendix H

The WKB Approximation

Chapters 11 and 12 developed two general strategies for problems that cannot be solved exactly: perturbation theory (small corrections to an exactly-solved problem) and the variational method (an optimized trial function). This appendix develops a third, complementary strategy — the WKB approximation (after Wentzel, Kramers, and Brillouin, 1926) — best suited to potentials that vary *slowly* on the scale of a particle's de Broglie wavelength, in which regime it gives remarkably accurate results, in one case (the harmonic oscillator) reproducing the exact answer with no error at all.

H.1 Setting Up the Approximation

Write the wavefunction as $\psi(x) = e^{i\phi(x)/\hbar}$ for some (generally complex) phase function $\phi(x)$ — always possible in principle, since any nonzero function can be written this way. Substituting into the time-independent Schrödinger equation and separating real and imaginary parts of the resulting equation for $\phi(x)$ leads, after dropping a term involving ϕ'' (the approximation at the heart of the method, valid precisely when the potential, and hence the local momentum, varies slowly), to

$$\phi(x) \approx \pm \int^{x} p(x')\,dx', \qquad p(x) \equiv \sqrt{2m[E - V(x)]},$$

where $p(x)$ is the *local classical momentum* a particle of energy E would have at position x under the potential $V(x)$ — exactly the classical momentum formula, now used inside a quantum wavefunction. The resulting approximate wavefunction in a classically allowed region ($E > V(x)$, so $p(x)$ is real) is

$$\psi(x) \approx \frac{C}{\sqrt{p(x)}} \exp\left(\pm \frac{i}{\hbar} \int^{x} p(x')\,dx' \right), \tag{H.1}$$

an oscillatory wave whose *local* wavelength, $2\pi\hbar/p(x)$, tracks the local classical momentum, and whose amplitude $1/\sqrt{p(x)}$ grows or shrinks precisely so that the local probability current stays constant — exactly what should happen classically, where a particle moving faster (larger p, from a lower local potential) spends less time in a given interval dx and so is found there with lower probability.

Note — Where WKB works, and where it does not

The approximation used to reach Eq. (H.1) requires $p(x)$ to change slowly compared to $\hbar/p(x)$ itself — roughly, the potential must vary gently on the scale of a de Broglie wavelength (Chapter 1). This condition is generally well satisfied far from a classical turning point (where $E = V(x)$ and $p(x) \to 0$), and it fails badly *at* a turning point itself, where $p(x)$ vanishes and Eq. (H.1)'s amplitude formally diverges. Correctly joining an oscillatory WKB solution on one side of a turning point to a decaying WKB solution on the other (in the classically forbidden region, where $p(x)$ is imaginary and the exponential in Eq. (H.1) becomes real, growing or decaying) requires a short derivation using a linear approximation to $V(x)$

near the turning point — the resulting matching rules (connection formulas) are standard but algebraically involved, and are not re-derived here; we quote and use only their final, physically clean consequences below.

H.2 WKB Quantization

For a particle bound between two *soft* (linear) classical turning points at $x = a$ and $x = b$, correctly applying the connection formulas at both ends gives the WKB (or Bohr–Sommerfeld) quantization condition:

Key Result H.1 — WKB Quantization Condition (Two Soft Turning Points)

$$\int_a^b p(x)\,dx = \left(n + \frac{1}{2}\right)\pi\hbar, \qquad n = 0, 1, 2, \ldots$$

Worked Example H.1 — WKB Reproduces the Exact QHO Spectrum

Apply the WKB quantization condition to the harmonic oscillator, $V(x) = \frac{1}{2}m\omega^2x^2$ (Chapter 6), and compare with the exact result.

Solution. The classical turning points satisfy $E = \frac{1}{2}m\omega^2x^2$, giving $x = \pm a$ with $a = \sqrt{2E/m\omega^2}$. The local momentum is $p(x) = \sqrt{2mE - m^2\omega^2x^2}$. Using the substitution $x = a\sin\theta$ (so that $p(x) = \sqrt{2mE}\cos\theta$ and $dx = a\cos\theta\,d\theta$), the integral becomes

$$\int_{-a}^{a} p(x)\,dx = \sqrt{2mE}\cdot a\int_{-\pi/2}^{\pi/2}\cos^2\theta\,d\theta = \sqrt{2mE}\cdot a\cdot\frac{\pi}{2} = \frac{\pi E}{\omega},$$

using $a\sqrt{2mE} = 2E/\omega$ (from the definition of a above). Setting this equal to $(n + \frac{1}{2})\pi\hbar$ and solving for E:

$$\frac{\pi E}{\omega} = \left(n + \frac{1}{2}\right)\pi\hbar \quad\Longrightarrow\quad E_n = \left(n + \frac{1}{2}\right)\hbar\omega,$$

exactly the Chapter 6 result, Eq. (6.5) — with *no* approximation error whatsoever, a special, celebrated feature of the exactly-parabolic QHO potential (whose turning points are as ideally "soft," i.e., linear-potential-like, as the connection-formula derivation assumes locally) rather than a general property of WKB.

For a *hard* wall (an infinite potential step, as in Chapter 4), the wavefunction must vanish exactly at the wall rather than smoothly decay into a classically forbidden region, which changes the connection-formula phase contributed by that turning point; the resulting quantization condition for a box with two hard walls (Problem H.1 asks you to verify this modification directly) is instead $\int_a^b p(x)\,dx = n\pi\hbar$ (no $+\frac{1}{2}$), which reproduces the exact Chapter 4 infinite-well spectrum $E_n = n^2\pi^2\hbar^2/2mL^2$ exactly as well — both classic exactly-solved systems of this book, recovered without approximation by the same general method that will now be applied to a problem with no exact solution at all.

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H.3 WKB Tunneling and the Gamow Factor

In a classically forbidden region ($E < V(x)$), $p(x)$ becomes imaginary, and Eq. (H.1)'s oscillatory exponential becomes a real, decaying (or growing) one, generalizing the constant-κ decay of Chapter 5's rectangular barrier to a barrier of arbitrary, smoothly varying shape $V(x)$.

Key Result H.2 — WKB Tunneling Probability

$$T \approx e^{-G}, \qquad G \equiv \frac{2}{\hbar}\int_a^b \sqrt{2m[V(x)-E]}\,dx,$$

where a and b are the classical turning points bounding the forbidden region (i.e., $V(a) = V(b) = E$), and G is called the *Gamow factor*.

For the special case of a rectangular barrier of constant height V_0 and width L (Chapter 5), $\sqrt{2m(V_0 - E)}$ is constant, and G reduces immediately to $2\kappa L$ with $\kappa = \sqrt{2m(V_0 - E)}/\hbar$, recovering the opaque-barrier approximation of Chapter 5, Section 5.3 exactly (Problem H.2 asks you to verify this reduction explicitly). The power of the WKB formula is that, unlike Chapter 5's treatment, it applies to a barrier of *any* shape, not only a rectangular one — exactly what is needed for the Coulomb-shaped barrier confining an alpha particle inside a nucleus.

Worked Example H.2 — The Gamow Factor for Alpha Decay

Estimate the half-life of ^{238}U (which decays by alpha emission to ^{234}Th with $E_\alpha = 4.198$ MeV) by modeling the alpha particle as tunneling through the Coulomb barrier of the daughter nucleus (charge $Z = 90$), starting from a nuclear radius $R = 9.3$ fm.

Solution. Outside the nucleus, $V(r) = \dfrac{2Ze^2}{4\pi\epsilon_0 r}$ (Coulomb repulsion between the $+2e$ alpha particle and the $+Ze$ daughter nucleus); the outer turning point b satisfies $V(b) = E_\alpha$, giving $b = 2Ze^2/4\pi\epsilon_0 E_\alpha = 61.7$ fm, far outside the nuclear radius $R = 9.3$ fm — confirming a genuine, extended classically forbidden region for the alpha particle to tunnel through. Evaluating the Gamow integral (a standard closed-form result for a pure Coulomb barrier, obtained by the substitution $r = b\cos^2\theta$; we quote it rather than re-derive it here) gives $G = 90.19$, so

$$T = e^{-G} = e^{-90.19} = 6.81 \times 10^{-40}.$$

Multiplying by a crude classical "assault frequency" (how often the alpha particle, moving at a speed corresponding to E_α inside a nucleus of diameter $2R$, strikes the barrier), $f \sim v/2R \approx 7.6 \times 10^{20}$ s^{-1}, gives a decay rate $\lambda_{\text{decay}} = fT \approx 5.2 \times 10^{-19}$ s^{-1}, and a predicted half-life

$$t_{1/2} = \frac{\ln 2}{\lambda_{\text{decay}}} \approx 1.3 \times 10^{18}\ \text{s} \approx 4.2 \times 10^{10}\ \text{years}.$$

The measured half-life of ^{238}U is 4.5×10^9 years — so this estimate is high by roughly a factor of 9, a substantial-*looking* discrepancy that is, in the context of alpha decay, actually a striking success: measured alpha-decay half-lives across different nuclides span more than 24 orders of magnitude (from microseconds to longer than the age of the universe), and this simple model, built from nothing more than the Coulomb potential and a crude assault-frequency estimate, correctly predicts that enormous range to within roughly one order of magnitude for essentially any alpha emitter — exactly the achievement (first ob-

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tained by Gamow, and independently by Gurney and Condon, in 1928) that made quantum tunneling a broadly convincing, quantitatively predictive phenomenon rather than merely a mathematical curiosity of the rectangular-barrier model in Chapter 5.

H.4 Chapter Summary

- The WKB approximation writes $\psi(x) \approx C\,p(x)^{-1/2}\exp\left(\pm\frac{i}{\hbar}\int p(x')dx'\right)$, valid where the potential varies slowly on the scale of the local de Broglie wavelength; it fails near classical turning points, where connection formulas (quoted, not derived, here) join the solutions on either side.
- The resulting quantization condition, $\int_a^b p\,dx = (n+\frac{1}{2})\pi\hbar$ for two soft turning points, reproduces the exact QHO spectrum with zero error, and (with a modified phase for hard walls) the exact infinite-square-well spectrum as well — two independent, exact validations of a method that is, in general, only approximate.
- In a classically forbidden region, WKB gives a tunneling probability $T \approx e^{-G}$ with Gamow factor $G = \frac{2}{\hbar}\int_a^b \sqrt{2m[V(x)-E]}\,dx$, reducing to Chapter 5's rectangular-barrier result for constant V_0 and generalizing it to barriers of arbitrary shape.
- Applied to the Coulomb barrier confining an alpha particle, the Gamow factor correctly predicts ^{238}U's alpha-decay half-life to within about one order of magnitude, a striking success given that measured half-lives across different nuclides span more than 24 orders of magnitude — the historical result that first established quantum tunneling as a quantitatively predictive theory.

Problems

H.1 For a particle in an infinite square well of width L (two hard walls, so $p(x) = \sqrt{2mE}$ is constant throughout $0 < x < L$), show that the hard-wall WKB quantization condition $\int_0^L p\,dx = n\pi\hbar$ reproduces the exact Chapter 4 result $E_n = n^2\pi^2\hbar^2/2mL^2$ exactly.

H.2 For a rectangular barrier of constant height V_0 and width L (Chapter 5), show that the general WKB Gamow factor $G = \frac{2}{\hbar}\int_a^b \sqrt{2m[V(x)-E]}\,dx$ reduces to $G = 2\kappa L$ with $\kappa = \sqrt{2m(V_0-E)}/\hbar$, matching the opaque-barrier exponent of Chapter 5, Section 5.3 exactly.

H.3 Using the Gamow-factor result quoted in the alpha-decay Worked Example, find the tunneling probability $T = e^{-G}$ for a hypothetical alpha emitter with the same $R = 9.3$ fm and $Z = 90$, but a higher alpha energy of $E_\alpha = 6.0$ MeV (compare with the $E_\alpha = 4.198$ MeV case in the text, and comment briefly on the direction and rough size of the change — you need not recompute G from scratch if you can argue the qualitative trend from the formula's dependence on E).

H.4 A particle of energy E moves in the potential $V(x) = V_0$ (a flat step of height greater than E) for $0 < x < L$, and $V = 0$ elsewhere. Confirm that the general WKB Gamow-factor formula, applied to this flat (constant-height) special case, requires no substitution or coordinate change to evaluate, unlike the Coulomb-barrier case — i.e., explain in a sentence why a constant $V(x) - E$ inside the integral defining G makes the WKB result trivially exact for this shape, matching Chapter 5's exact rectangular-barrier treatment rather than being merely an approximation to it.

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H.5 Using the Gamow factor's dependence on alpha energy explored in Problem H.3 (where raising E_α from 4.198 to 6.0 MeV increased T by eleven orders of magnitude), explain briefly why alpha emitters with systematically higher decay energies are observed to have systematically *shorter* half-lives — the empirical Geiger–Nuttall relation, discovered in 1911, more than a decade before the quantum-mechanical Gamow explanation developed in this appendix existed.

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Appendix I

Molecular Symmetry and Group Theory

A molecule's geometric symmetry constrains its quantum mechanics directly: which vibrational modes exist, which are visible in an infrared spectrum, and which orbitals can mix in a molecular-orbital calculation (Chapter 15) are all determined, before any detailed calculation, by symmetry alone. This appendix introduces just enough group theory — the mathematical language of symmetry — to make this connection explicit, using the water molecule as a single running example throughout.

I.1 Symmetry Operations and Point Groups

Definition I.1 — Symmetry Operation and Point Group

A *symmetry operation* is a transformation (rotation, reflection, or their combination) that leaves a molecule looking identical to how it started — every atom maps onto a position occupied by an identical atom. The complete set of a molecule's symmetry operations forms its *point group* ("point" because at least one point, generally the center of mass, is left fixed by every operation in the group).

The water molecule (bent, with an H–O–H angle of about 104.5°) has exactly four symmetry operations: the identity E (do nothing — trivial, but required for the mathematical structure of a group); a 180° rotation C_2 about the axis bisecting the H–O–H angle; and two reflections, σ_v (through the plane containing all three atoms) and σ_v' (through the perpendicular plane containing only the oxygen and the C_2 axis). This specific set of four operations defines the point group called C_{2v}, one of the most common point groups in introductory chemistry (also describing, for example, SO_2 and formaldehyde).

Note — Verifying group closure

A defining property of a group is *closure*: combining any two operations in the set must give another operation already in the set. For C_{2v}, applying σ_v followed by σ_v' (reflecting through one plane, then the other) can be verified, either geometrically or by direct 3×3 matrix multiplication of the corresponding coordinate transformations, to give exactly C_2 — consistent with $\{E, C_2, \sigma_v, \sigma_v'\}$ forming a complete, self-contained group with no additional operations needed.

I.2 Character Tables

Just as Chapter 7's angular momentum states are labeled by how they transform under rotation (the quantum numbers ℓ, m), every molecular property — an orbital, a vibration, a component of the dipole moment — can be labeled by how it transforms under the operations of the molecule's

point group. Chapter 9's Hermitian-operator formalism guarantees these transformation properties fall into a small number of distinct *irreducible representations* (irreps), tabulated for each point group in a *character table*.

C_{2v}	E	C_2	$\sigma_v(xz)$	$\sigma_v'(yz)$	Linear functions
A_1	1	1	1	1	z
A_2	1	1	-1	-1	(R_z)
B_1	1	-1	1	-1	x
B_2	1	-1	-1	1	y

Each row is an irrep (given a conventional label A_1, A_2, B_1, B_2); each column entry (a *character*, here always ± 1 for this simple point group) records whether a function transforming as that irrep is left unchanged ($+1$) or flips sign (-1) under the corresponding operation. The last column shows which Cartesian coordinate transforms as each irrep: z (unchanged by every operation, since z lies along the C_2 axis, contained in both mirror planes) transforms as A_1; x and y (each flipped by C_2, and by one mirror plane but not the other) transform as B_1 and B_2 respectively — directly verifiable by writing out the 3×3 matrix representation of each operation (Problem I.1).

I.3 Vibrational Modes and Infrared Activity

Key Result I.1 — Symmetry and Infrared Selection Rules

A vibrational mode is infrared-active (allowed in absorption, Chapter 13's electric-dipole selection rule) if and only if it transforms as the same irrep as x, y, or z — i.e., as the same irrep as some component of the dipole moment operator. A mode transforming as an irrep matching none of x, y, z is infrared-*inactive*: it produces no dipole-moment change and hence no absorption, regardless of how the mode itself is excited.

This is the group-theoretic sharpening of Chapter 13's electric-dipole selection rule: instead of separately computing a transition-dipole integral for every mode (as Chapter 13 did directly for the rigid rotor and QHO), symmetry alone determines activity immediately, once a mode's irrep is known.

Worked Example I.1 — The Vibrational Modes of Water

Water (C_{2v}, 3 atoms) has $3N - 6 = 3 \times 3 - 6 = 3$ vibrational degrees of freedom (Chapter 15's Born–Oppenheimer separation already established that nuclear motion splits into translation, rotation, and vibration; for a nonlinear molecule, 3 translations and 3 rotations are removed from the total $3N$ nuclear coordinates). Determine the symmetry species of these 3 vibrational modes, and state whether each is infrared-active.

Solution. The standard procedure (a systematic bookkeeping method, summarized without full derivation here) finds how the $3N = 9$ Cartesian displacement coordinates transform, using the fact that only *unmoved* atoms contribute to the character of a given operation. With oxygen on the C_2 axis and both hydrogens related by symmetry: under E, all 3 atoms are unmoved (contributing 3 per atom, i.e., 9 total); under C_2, only oxygen is unmoved (-1 per

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unmoved atom for a 180° rotation, giving -1 total); under σ_v (the molecular plane, containing all three atoms), all 3 atoms are unmoved ($+1$ per unmoved atom, giving 3 total); under σ_v' (the perpendicular plane, swapping the two hydrogens), only oxygen is unmoved ($+1$ total). This gives a reducible representation with characters $(9, -1, 3, 1)$ under $(E, C_2, \sigma_v, \sigma_v')$. Applying the standard reduction formula (projecting onto each irrep using the character table above) gives

$$\Gamma_{3N} = 3A_1 + A_2 + 3B_1 + 2B_2.$$

Removing translations ($A_1 + B_1 + B_2$, matching z, x, y respectively) and rotations ($A_2 + B_1 + B_2$, matching R_z, R_y, R_x respectively) leaves

$$\Gamma_{\text{vib}} = \Gamma_{3N} - \Gamma_{\text{trans}} - \Gamma_{\text{rot}} = 2A_1 + B_1,$$

exactly 3 modes total, as required. Since both A_1 (matching z) and B_1 (matching x) appear among x, y, z in the character table, *all three* of water's vibrational modes — conventionally identified as the symmetric stretch and bend (both A_1) and the antisymmetric stretch (B_1) — are infrared-active, consistent with water's experimentally observed infrared spectrum showing all three fundamentals.

Note — Connecting back to molecular orbital theory

The same symmetry classification governs which atomic orbitals can combine into a bonding molecular orbital (Chapter 15): two orbitals can mix only if they transform as the *same* irrep of the molecule's point group (otherwise, the relevant Hamiltonian matrix element $\langle \phi_i | \hat{H} | \phi_j \rangle$ can be shown, by a symmetry argument closely paralleling the infrared-activity criterion above, to vanish identically). For water, oxygen's $2p_z$ orbital (A_1) can mix with the totally symmetric combination of hydrogen $1s$ orbitals (A_1), but not with the antisymmetric combination (B_1) — exactly the kind of orbital-mixing restriction that determines the qualitative shape of water's molecular orbital diagram, extending the simple homonuclear-diatomic LCAO treatment of Chapter 15 to a real, bent triatomic molecule.

Problems

I.1 Write out the 3×3 matrix representation of C_2 (rotation by 180° about z) acting on Cartesian coordinates (x, y, z), and verify by direct matrix multiplication that its trace (sum of diagonal elements) equals the character of C_2 for the reducible representation formed by x, y, z together, i.e., $\chi(C_2) = -1 - 1 + 1 = -1$.

I.2 Ammonia, NH_3, belongs to point group C_{3v}, with 6 symmetry operations (E, two C_3 rotations, and three σ_v reflections). Using $3N - 6$ with $N = 4$ atoms, find the number of vibrational modes of ammonia.

I.3 Formaldehyde, H_2CO, has the same C_{2v} point group as water. Using the character table of Section I.2, state which irreps (A_1, A_2, B_1, B_2) are infrared-active, and which (if any) could never be observed in an infrared absorption spectrum regardless of the specific molecule's vibrational frequencies.

I.4 Explain briefly, in reference to Chapter 13's dipole-matrix-element selection rule, why a vibrational mode transforming as A_2 in a C_{2v} molecule is infrared-*inactive*, using the fact that A_2

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matches none of x, y, z in the character table.

Appendix J

Ehrenfest's Theorem and Degenerate Perturbation Theory

This appendix develops two further results, each closing a gap left open earlier in the book: Ehrenfest's theorem, which makes precise the sense in which classical mechanics emerges from quantum expectation values (first previewed by the correspondence principle of Chapter 4), and degenerate perturbation theory, explicitly flagged as beyond the scope of Chapter 11's treatment, worked through here for a genuinely important physical example: the Stark effect in hydrogen.

J.1 Ehrenfest's Theorem

For any operator $\hat{A}$ (possibly with its own explicit time dependence), the time derivative of its expectation value is

$$\frac{d\langle \hat{A} \rangle}{dt} = \frac{i}{\hbar}\langle [\hat{H}, \hat{A}] \rangle + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle,$$

a general identity obtainable by differentiating $\langle \hat{A} \rangle = \int \Psi^* \hat{A} \Psi \, dx$ with respect to t and using the time-dependent Schrödinger equation (Chapter 3) to replace $\partial\Psi/\partial t$ and $\partial\Psi^*/\partial t$ (Problem J.1 asks you to carry this out explicitly). Applying it to $\hat{A} = \hat{x}$ (no explicit time dependence) and using $[\hat{H}, \hat{x}] = [\hat{p}^2/2m, \hat{x}] = -\frac{1}{2m}[\hat{x}, \hat{p}^2] = -\frac{i\hbar}{m}\hat{p}$ (Chapter 9, Problem 9.4, with the sign flipped since the commutator here is $[\hat{H}, \hat{x}] = -[\hat{x}, \hat{H}]$) gives

Key Result J.1 — Ehrenfest's Theorem

$$\frac{d\langle \hat{x} \rangle}{dt} = \frac{\langle \hat{p} \rangle}{m}, \qquad \frac{d\langle \hat{p} \rangle}{dt} = -\left\langle \frac{dV}{dx} \right\rangle.$$

(The second relation follows by the same method applied to $\hat{A} = \hat{p}$; we do not repeat the algebra.) These are exactly Newton's second law and the momentum-velocity relation, $F = ma$ and $p = mv$, written for *expectation values* rather than definite classical quantities — not an approximation valid only in some special limit, but an exact consequence of the Schrödinger equation for any state whatsoever.

Note — Why this does not simply prove classical mechanics

Ehrenfest's theorem does *not*, by itself, say that a quantum expectation value $\langle x \rangle(t)$ follows the same trajectory a classical particle would, because $\langle dV/dx \rangle$ is generally not equal to $\left.\frac{dV}{dx}\right|_{x=\langle x \rangle}$ (the classical force evaluated at the expectation-value position) unless $V(x)$ is linear or quadratic in x — exactly the potentials of the free particle and the QHO (Chapters 3–4, 6), for which Ehrenfest's theorem *does* reduce exactly to Newton's second law for

$\langle x\rangle(t)$ itself (Problem J.2 asks you to verify this explicitly for the QHO). For a generic potential, the gap between $\langle dV/dx\rangle$ and $dV/dx|_{\langle x\rangle}$ is precisely what allows genuinely quantum behavior (wave-packet spreading, tunneling, interference) to persist even as $\langle x\rangle, \langle p\rangle$ trace out an approximately classical-looking path — and it is precisely this gap that vanishes in the large-quantum-number, macroscopic limit invoked by the correspondence principle of Chapter 4.

J.2 Degenerate Perturbation Theory: The Stark Effect

Chapter 11's perturbation theory assumed a non-degenerate unperturbed spectrum, flagging the degenerate case as requiring a modification not developed there. We now supply it, for the specific and physically important case of hydrogen's $n = 2$ level (Chapter 8) perturbed by a uniform external electric field $\mathcal{E}$ along z — the *Stark effect*.

The perturbation is $\hat{H}' = e\mathcal{E}\hat{z}$ (the potential energy of an electron's charge $-e$ in a field $\mathcal{E}$ along z, up to an overall sign convention). Naively applying Chapter 11's non-degenerate formula $E^{(1)} = \left\langle n^{(0)}\right| \hat{H}' \left|n^{(0)}\right\rangle$ to any single $n = 2$ state fails outright, because the four $n = 2$ states ($2s$, $2p_0$, $2p_{+1}$, $2p_{-1}$, all sharing $E_2 = -3.4014$ eV, Chapter 8) are degenerate, and the correct zeroth-order states are not necessarily $2s$ and the three $2p$ states individually, but specific *linear combinations* of them that the perturbation itself must select.

Key Result J.2 — Degenerate Perturbation Theory: The Method

Within a degenerate subspace, the correct first-order energy corrections are the *eigenvalues* of the matrix of $\hat{H}'$ computed using the degenerate unperturbed states as a basis, $H'_{ij} = \left\langle i^{(0)}\right| \hat{H}' \left|j^{(0)}\right\rangle$ — an application of the same finite-dimensional eigenvalue technique from Chapter 2, Section 2.3, now applied to a matrix of perturbation matrix elements rather than an arbitrary numerical matrix.

Worked Example J.1 — The Linear Stark Effect in Hydrogen

Find the first-order Stark splitting of the hydrogen $n = 2$ level.

Solution. We need $H'_{ij} = e\mathcal{E}\,\langle i|\,\hat{z}\,|j\rangle$ for every pair of the four $n = 2$ states. Parity (Chapter 4's node-counting intuition, generalized: $2s$ and $2p_0$ have opposite parity under $z \to -z$, since $\ell = 0$ versus $\ell = 1$, so $\hat{z}$ — itself odd under $z \to -z$ — connects them; $\hat{z}$ cannot connect a state to itself, since that would require an odd operator to have a nonzero expectation value in a definite-parity state) and the magnetic-quantum-number selection rule (Chapter 13: $\hat{z}$ requires $\Delta m = 0$) together eliminate every matrix element except $\langle 2s|\,\hat{z}\,|2p_0\rangle$ (and its complex conjugate): $\langle 2p_{\pm1}|\,\hat{z}\,|\text{anything in this set}\rangle = 0$ entirely, and the diagonal elements $\langle 2s|\,\hat{z}\,|2s)\rangle$, $\langle 2p_0|\,\hat{z}\,|2p_0\rangle$ vanish by parity. Direct integration (using the radial functions of Chapter 8, Eq. (8.4), and the angular integral techniques of Chapter 13) gives

$$\langle 2s|\,\hat{z}\,|2p_0\rangle = -3a_0.$$

In the $\{2s, 2p_0\}$ subspace, the perturbation matrix is therefore

$$H' = e\mathcal{E}\begin{pmatrix} 0 & -3a_0 \\ -3a_0 & 0 \end{pmatrix},$$

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with eigenvalues $\pm 3e\mathcal{E}a_0$ (a direct application of Chapter 2's 2×2 eigenvalue method: characteristic equation $\lambda^2 = (3e\mathcal{E}a_0)^2$). The $2p_{\pm 1}$ states, decoupled from this 2×2 block entirely, are unaffected at first order. The $n = 2$ level therefore splits into three distinct energies:

$$E_2 - 3e\mathcal{E}a_0, \qquad E_2 \text{ (doubly degenerate: } 2p_{+1}, 2p_{-1}), \qquad E_2 + 3e\mathcal{E}a_0.$$

Note — Why hydrogen shows a* linear *Stark effect, uniquely

Most atoms show a Stark shift proportional to $\mathcal{E}^2$ (a "quadratic" Stark effect, from ordinary non-degenerate second-order perturbation theory, Chapter 11, applied to a non-degenerate ground state whose first-order shift vanishes by parity exactly as $2s$'s and $2p_0$'s diagonal elements did above). Hydrogen's $n = 2$ (and every $n \geq 2$) level is unusual precisely because it contains states of *both* parities ($2s$ and $2p$) degenerate at the same energy — the special ℓ-degeneracy of the pure Coulomb potential already flagged as unusual in Chapter 8, Section 8.5. This degeneracy is what allows a first-order (linear in $\mathcal{E}$) energy shift to appear at all, making hydrogen's linear Stark effect a direct, physically observable consequence of the same accidental Coulomb degeneracy noted, but not yet exploited, in Chapter 8.

Problems

J.1 Starting from $\langle \hat{A} \rangle = \int \Psi^* \hat{A} \Psi \, dx$, differentiate with respect to t (using the product rule on $\Psi^* \hat{A} \Psi$), substitute $\partial\Psi/\partial t$ and $\partial\Psi^*/\partial t$ from the time-dependent Schrödinger equation and its complex conjugate, and show the result can be written as $\dfrac{d\langle \hat{A} \rangle}{dt} = \dfrac{i}{\hbar}\langle [\hat{H}, \hat{A}] \rangle + \left\langle \dfrac{\partial \hat{A}}{\partial t} \right\rangle$, confirming the general Ehrenfest identity quoted in Section J.1.

J.2 For the QHO, $V(x) = \frac{1}{2}m\omega^2 x^2$, show that $dV/dx = m\omega^2 x$ is exactly linear in x, so that $\langle dV/dx \rangle = m\omega^2 \langle x \rangle$ exactly (with no approximation), and hence that Ehrenfest's theorem gives $\dfrac{d^2\langle x \rangle}{dt^2} = -\omega^2 \langle x \rangle$ — exactly the classical QHO equation of motion, satisfied exactly by $\langle x \rangle(t)$ for *any* QHO quantum state, not merely a special or approximate one.

J.3 Using the Stark-splitting result of Section J.2, find the numerical energy splitting (in eV) between the highest and lowest of the three resulting $n = 2$ sublevels, for a field strength $\mathcal{E} = 1.0 \times 10^7$ V/m (a strong but laboratory-achievable field), using $a_0 = 5.29177 \times 10^{-11}$ m.

J.4 Explain briefly why the $2p_{+1}$ and $2p_{-1}$ states remain exactly degenerate (unsplit) by the Stark perturbation to first order, referring to the $\Delta m = 0$ selection rule invoked in the Worked Example of Section J.2.

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Appendix K

Density Matrices and Mixed States

Every quantum state considered so far in this book — from the hydrogen wavefunction of Chapter 8 to the qubit of Chapter 17 — has been a *pure* state, $|\psi\rangle$, describing a system about which we have maximal possible quantum information. Many real situations are not like this: a beam of atoms from an oven, a qubit that has interacted uncontrollably with its environment, or an ensemble of molecules at finite temperature are each better described as a statistical *mixture* of different possible pure states, with definite classical probabilities attached. The density matrix, introduced by John von Neumann in 1927, is the tool that handles both cases within a single, unified formalism.

K.1 The Density Matrix for a Pure State

Definition K.1 — Density Matrix (Pure State)

For a system in a definite pure state $|\psi\rangle$, the density matrix (or density operator) is

$$\hat{\rho} \equiv |\psi\rangle \langle\psi| .$$

This may look like a trivial repackaging of $|\psi\rangle$ itself, and for a pure state, essentially all it initially offers is a change of notation: expectation values, previously written $\langle \hat{A} \rangle = \langle\psi| \hat{A} |\psi\rangle$ (Chapter 3), become

$$\langle \hat{A} \rangle = \mathrm{Tr}(\hat{\rho}\hat{A}),$$

using the trace (sum of diagonal matrix elements, in any orthonormal basis — a basis-independent quantity, Problem K.1). The real payoff of this reformulation appears once genuinely *mixed* states are considered, where no single ket $|\psi\rangle$ exists to play the role $|\psi\rangle$ played above.

Key Result K.1 — The Purity Test

$$\mathrm{Tr}(\hat{\rho}^2) = 1 \quad \text{if and only if} \quad \hat{\rho} \text{ describes a pure state;} \qquad \mathrm{Tr}(\hat{\rho}^2) < 1 \text{ otherwise.}$$

K.2 Mixed States

Definition K.2 — Density Matrix (Mixed State)

If a system is in state $|\psi_i\rangle$ with classical probability p_i (where $\sum_i p_i = 1$, and the $|\psi_i\rangle$ need not be orthogonal or otherwise related), the density matrix is the probability-weighted sum

$$\hat{\rho} = \sum_i p_i \, |\psi_i\rangle \langle\psi_i| .$$

This single object correctly encodes *two* distinct layers of uncertainty at once: the ordinary quantum uncertainty already present in each $|\psi_i\rangle$ individually (Chapter 1's uncertainty principle, still fully present), and the additional classical, statistical uncertainty about *which* $|\psi_i\rangle$ the system is actually in (analogous to the discrete-probability formalism of Chapter 2, Section 2.6, now applied to entire quantum states rather than measurement outcomes of a single fixed state).

Worked Example K.1 — Distinguishing a Mixture from a Superposition

A spin-$\frac{1}{2}$ particle (Chapter 10) is either (a) prepared with certainty in the superposition state $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$, or (b) prepared in a genuine 50/50 statistical mixture of $|\uparrow\rangle$ and $|\downarrow\rangle$ (e.g., unpolarized particles from an oven, each definitely spin-up *or* spin-down, but with the labels lost). Show that both give identical predictions for $\hat{S}_z$ measurements, but different predictions for $\hat{S}_x$.

Solution. For (a), $\hat{\rho}_a = |\psi\rangle\langle\psi| = \frac{1}{2}(|\uparrow\rangle + |\downarrow\rangle)(\langle\uparrow| + \langle\downarrow|) = \frac{1}{2}(|\uparrow\rangle\langle\uparrow| + |\uparrow\rangle\langle\downarrow| + |\downarrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow|)$, represented in the $\{|\uparrow\rangle, |\downarrow\rangle\}$ basis by the matrix $\frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}$. For (b), $\hat{\rho}_b = \frac{1}{2}|\uparrow\rangle\langle\uparrow| + \frac{1}{2}|\downarrow\rangle\langle\downarrow|$, represented by $\frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$ — lacking the off-diagonal "coherence" terms present in $\hat{\rho}_a$. Both give $\langle\hat{S}_z\rangle = \mathrm{Tr}(\hat{\rho}\,\hat{S}_z) = 0$ identically (Problem K.2 verifies this directly for both cases), since both diagonal elements are equal in each case. But $\langle\hat{S}_x\rangle = \mathrm{Tr}(\hat{\rho}\,\hat{S}_x)$ depends entirely on the off-diagonal elements of $\hat{\rho}$ (since $\hat{S}_x$ itself is purely off-diagonal in this basis): direct computation gives $\langle\hat{S}_x\rangle_a = \hbar/2 \neq 0$ for the pure superposition, but $\langle\hat{S}_x\rangle_b = 0$ for the classical mixture. Measuring only S_z can never distinguish these two physically very different preparations; measuring S_x distinguishes them immediately. A purity check confirms this structurally: $\mathrm{Tr}(\hat{\rho}_a^2) = 1$ (pure), while $\mathrm{Tr}(\hat{\rho}_b^2) = \frac{1}{2} < 1$ (mixed).

Note — Why this matters for real qubits

Chapter 17 treated a qubit as an idealized pure state $\alpha|0\rangle + \beta|1\rangle$. Any real physical qubit inevitably interacts, at least weakly, with its surrounding environment (stray electromagnetic fields, thermal photons, material defects), and this interaction generically converts an initially pure qubit state into a mixed one, exactly as in the example above — a process called *decoherence*. The density matrix is the standard language in which decoherence, and the quantum error-correction techniques designed to fight it, are formulated; a real quantum computer's performance is measured, in large part, by how successfully it keeps $\mathrm{Tr}(\hat{\rho}^2)$ close to 1 for its qubits throughout a computation.

K.3 Time Evolution of the Density Matrix

Differentiating $\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ with respect to time, and using the time-dependent Schrödinger equation for each $|\psi_i\rangle(t)$ exactly as in the Ehrenfest-theorem derivation of Appendix J (Problem J.1), gives the density-matrix analog of the Schrödinger equation:

$$i\hbar\frac{d\hat{\rho}}{dt} = [\hat{H}, \hat{\rho}], \tag{K.1}$$

called the *von Neumann equation* (or the quantum Liouville equation, by analogy with the classical statistical-mechanics equation it generalizes). For a stationary-state density matrix (built from

energy eigenstates alone, with $\hat{\rho}$ diagonal in the energy basis), $[\hat{H}, \hat{\rho}] = 0$ and $\hat{\rho}$ is exactly constant in time — the density-matrix generalization of Chapter 3's stationary-state result that $|\Psi(x,t)|^2$ does not depend on t.

Problems

K.1 Show that $\mathrm{Tr}(\hat{\rho}\hat{A}) = \langle\psi|\,\hat{A}\,|\psi\rangle$ for the pure-state density matrix $\hat{\rho} = |\psi\rangle\langle\psi|$, by inserting a complete basis $\{|n\rangle\}$ and using $\mathrm{Tr}(\hat{M}) = \sum_n \langle n|\,\hat{M}\,|n\rangle$ for any operator $\hat{M}$. (*Hint:* write $\mathrm{Tr}(\hat{\rho}\hat{A}) = \sum_n \langle n|\rangle\,\psi\,\langle\psi|\,\hat{A}\,|n\rangle$ and regroup the sum using completeness, $\sum_n |n\rangle\langle n| = 1$.)

K.2 Using $\hat{S}_z = \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$, compute $\mathrm{Tr}(\hat{\rho}_a\hat{S}_z)$ and $\mathrm{Tr}(\hat{\rho}_b\hat{S}_z)$ for the two density matrices of the Worked Example in Section K.2, and confirm both equal zero.

K.3 Using $\hat{S}_x = \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$, compute $\mathrm{Tr}(\hat{\rho}_a\hat{S}_x)$ and $\mathrm{Tr}(\hat{\rho}_b\hat{S}_x)$ explicitly by matrix multiplication, confirming the values $\hbar/2$ and 0 quoted in the text.

K.4 A qubit is in the mixed state $\hat{\rho} = 0.8\,|0\rangle\langle 0| + 0.2\,|1\rangle\langle 1|$ (an "impure" or partially decohered qubit, weighted more toward $|0\rangle$). Find $\mathrm{Tr}(\hat{\rho}^2)$, and confirm it is less than 1, consistent with this being a genuinely mixed (not pure) state.

K.5 Explain briefly, in one or two sentences and referring to the von Neumann equation, Eq. (K.1), why a density matrix that is diagonal in the energy eigenbasis (i.e., built entirely from stationary states, each with its own classical probability p_i) does not evolve in time at all, generalizing the single-stationary-state result of Chapter 3.

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Appendix L

The Free Electron Gas and Fermi Energy

Chapter 4's particle in a box solved the quantum mechanics of a single confined particle; Chapter 10 established that no two electrons can share a quantum state. This appendix combines the two directly, for a system of genuine practical importance: the conduction electrons in a metal, modeled as a very large number of essentially free, non-interacting electrons confined to a three-dimensional box the size of the metal sample itself. The result — the free electron gas model, due to Arnold Sommerfeld (1927) — explains, among other things, why metals conduct electricity essentially independent of temperature near room temperature, in sharp contrast to a naive classical expectation.

L.1 Filling a Three-Dimensional Box with Many Electrons

Generalizing Chapter 4's one-dimensional box to three dimensions (already used in Appendix F, Problem F.23), a cubical box of side L has energy levels

$$E_{n_x,n_y,n_z} = \frac{\pi^2\hbar^2}{2m_eL^2}\left(n_x^2 + n_y^2 + n_z^2\right), \qquad n_x, n_y, n_z = 1, 2, 3, \ldots$$

For a macroscopic metal sample, N (the number of conduction electrons, typically one or two per atom) is enormous — of order Avogadro's number (Chapter 1) for any laboratory-sized sample — and by the Pauli exclusion principle (Chapter 10), each distinct (n_x, n_y, n_z) level holds at most 2 electrons (opposite spin). Filling levels from the lowest energy upward, exactly as the Aufbau principle (Chapter 14) filled atomic subshells, the electrons occupy every available level up to some maximum energy.

Definition L.1 — Fermi Energy

The *Fermi energy* E_F is the energy of the highest-filled level in this zero-temperature filling procedure — the three-dimensional, many-electron analog of a particle-in-a-box "HOMO" (Chapter 4's conjugated-polyene language, Section 4.6, generalized from a handful of π electrons to a macroscopic number of conduction electrons).

Because N is so large, the discrete levels are spaced enormously finely on any laboratory energy scale (Chapter 4's correspondence principle, Section 4.5, applies with tremendous force here), and the filled levels can be treated as an effectively continuous "Fermi sea." Counting the number of (n_x, n_y, n_z) combinations with energy below a given value (a geometric counting problem — the volume of one octant of a sphere in n-space, not reproduced here in full) and demanding this count equal N leads to:

Key Result L.1 — Fermi Energy of a Free Electron Gas

$$E_F = \frac{\hbar^2}{2m_e}\left(3\pi^2 n\right)^{2/3}, \qquad n \equiv \frac{N}{V},$$

where n is the conduction-electron number density (electrons per unit volume) and $V = L^3$ is the sample volume.

Worked Example L.1 — The Fermi Energy of Copper

Copper contributes approximately one conduction electron per atom, giving a conduction-electron density $n = 8.47 \times 10^{28}\ \mathrm{m}^{-3}$. Find the Fermi energy.
Solution. Direct substitution gives

$$E_F = \frac{(1.0546 \times 10^{-34})^2}{2(9.1094 \times 10^{-31})}\left[3\pi^2(8.47 \times 10^{28})\right]^{2/3} = 1.127 \times 10^{-18}\ \mathrm{J} = 7.03\ \mathrm{eV},$$

in excellent agreement with the standard tabulated value for copper, about 7.0 eV. This is a substantial energy — comparable to a typical chemical bond strength (Chapter 15) or the visible-photon energy scale (Chapter 1) — despite arising from nothing more exotic than confining a large number of otherwise free electrons in a box and respecting the Pauli principle.

L.2 Why Metals Conduct Almost Independently of Temperature

Note — The classical prediction, and why it fails

A classical ideal gas of N free electrons would have an average kinetic energy of order $k_B T$ per electron (Chapter 1, Appendix A), vanishing as $T \to 0$ and growing linearly with temperature — suggesting a metal's electronic heat capacity and typical electron energies should scale simply with T. The Fermi energy sets a natural comparison scale, the *Fermi temperature* $T_F \equiv E_F/k_B$; for copper, $T_F = E_F/k_B \approx 81{,}600$ K, vastly higher than room temperature (≈ 300 K). Because the electrons are already forced, by the Pauli exclusion principle alone, to fill states up to $E_F \gg k_B T_{\mathrm{room}}$ even at absolute zero, only the small fraction of electrons within about $k_B T$ of E_F (a small minority of the total, given $T \ll T_F$) can be thermally excited into nearby empty states at any accessible laboratory temperature — the overwhelming majority sit at energies far below E_F, Pauli-blocked from absorbing a thermal energy of order $k_B T$ by every nearby state already being occupied. This is the qualitative, symmetry-principle-driven explanation for why the electronic contribution to a metal's heat capacity is anomalously small and only weakly temperature-dependent near room temperature, a genuine puzzle for nineteenth-century classical physics that the Pauli exclusion principle, combined with nothing more than the particle-in-a-box result of Chapter 4, resolves directly.

Problems

L.1 Sodium contributes one conduction electron per atom, with number density $n = 2.65 \times 10^{28}\ \mathrm{m}^{-3}$. Find its Fermi energy, in eV, and compare with the copper value found in the text.

L.2 Using the Fermi energy of sodium found in Problem L.1, find its Fermi temperature $T_F = E_F/k_B$, and confirm it is, like copper's, far above room temperature.

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L.3 Using $E_F \propto n^{2/3}$, find the factor by which the Fermi energy of a metal would change if its conduction-electron density were doubled (all else held fixed), and comment on whether Fermi energy is very sensitive or only mildly sensitive to changes in electron density, given this fractional power.

L.4 Explain briefly, in reference to the Pauli exclusion principle (Chapter 10) alone (no formula needed), why a classical (non-quantum, Pauli-principle-free) gas of N free electrons at $T = 0$ would have *zero* total kinetic energy (every electron sitting motionless in the lowest-energy state), while the quantum free-electron gas of this appendix has a substantial nonzero total kinetic energy (the Fermi sea) even at $T = 0$.

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Appendix M

The Non-Rigid Rotor: Centrifugal Distortion

Chapter 13's rigid-rotor treatment of molecular rotation assumed a perfectly rigid bond length r, giving evenly-spaced-in-J energy corrections $E_J = BJ(J+1)$. A real chemical bond is not perfectly rigid — it is, as Chapter 6 established, a quantum harmonic (or, more precisely, Morse) oscillator, and a rapidly rotating molecule experiences a centrifugal effect that stretches the bond slightly, exactly as a classical rotating object on a spring would. This appendix quantifies the resulting small, but precisely measurable, correction.

M.1 Setting Up the Correction

Classically, a mass on a spring of force constant k_{spring}, rotating with angular momentum L, stretches by an amount that increases the effective bond length and hence the moment of inertia, which in turn *lowers* the rotational energy for a given L compared to the perfectly rigid case (a larger moment of inertia means less energy for the same angular momentum, since $E_{\text{rot}} = L^2/2I$). Carrying this coupling between rotation and vibration through the quantum treatment (a short second-order perturbation-theory calculation, in the spirit of Chapter 11, that we do not reproduce step by step) gives a corrected rotational energy formula:

Key Result M.1 — Non-Rigid Rotor Energy Levels

$$E_J = hc\left[BJ(J+1) - DJ^2(J+1)^2\right], \qquad D \equiv \frac{4B^3}{\tilde{\omega}_e^2},$$

where D (the *centrifugal distortion constant*, in the same cm^{-1} units as B) is always positive and, since $D \ll B$ for any real molecule (bonds are stiff, so $\tilde{\omega}_e \gg B$ numerically), acts as a small negative correction that grows with J.

Worked Example M.1 — Centrifugal Distortion in HCl

Using $B = 10.593\ \text{cm}^{-1}$ and $\tilde{\omega}_e = 2990.9\ \text{cm}^{-1}$ for HCl (Chapters 6, 13), find the centrifugal distortion constant D, and compare the rigid-rotor and non-rigid-rotor predictions for the $J = 5 \to 6$ rotational transition.

Solution. $D = 4B^3/\tilde{\omega}_e^2 = 4(10.593)^3/(2990.9)^2 = 5.315 \times 10^{-4}\ \text{cm}^{-1}$ — more than four orders of magnitude smaller than B itself, confirming $D \ll B$ as expected for a genuinely stiff bond. The rigid-rotor $J = 5 \to 6$ transition energy is $2B(6) = 127.116\ \text{cm}^{-1}$ (Chapter 13's formula). Using the corrected formula, the actual transition is $126.657\ \text{cm}^{-1}$, a difference of $0.459\ \text{cm}^{-1}$ — tiny compared to the transition energy itself (about 0.4%), but far larger than the rotational linewidths achievable in modern microwave and infrared spectroscopy, and hence a routinely measured correction, not merely a theoretical nicety.

Note — Why this matters beyond a small numerical correction

Precisely because D is so much smaller than B, and because it grows with J in a specific, predicted way ($J^2(J+1)^2$, distinct from the $J(J+1)$ rigid-rotor term), fitting a molecule's *entire* measured rotational spectrum (many J values at once) to the two-parameter formula above lets experimentalists extract both B and D independently and with high precision — and, since $D = 4B^3/\tilde{\omega}_e^2$ links the two, a measured D provides an independent, purely rotational-spectroscopy check on the vibrational frequency $\tilde{\omega}_e$, cross-validating Chapter 6's force-constant analysis against Chapter 13's rotational analysis using two entirely different measurements of the same underlying bond.

Problems

M.1 Using $B = 1.9225\ \text{cm}^{-1}$ and $\tilde{\omega}_e = 2143.0\ \text{cm}^{-1}$ for CO, find the centrifugal distortion constant D, and confirm it is even smaller (relative to B) than HCl's, consistent with CO's stiffer (higher force constant, Chapter 15) triple bond.

M.2 Using the HCl values of the Worked Example, find the rigid-rotor and non-rigid-rotor predictions for the $J = 0 \rightarrow 1$ transition, and confirm the difference between the two is much smaller (in absolute terms) than the $J = 5 \rightarrow 6$ difference found in the text, consistent with the $J^2(J+1)^2$ scaling of the distortion term.

M.3 Explain briefly, in reference to Section M.1's classical motivation, why D is always positive (i.e., why centrifugal distortion always *lowers* the rotational energy relative to the rigid-rotor prediction, never raises it), for any real molecule.

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Appendix N

Quick Reference: Key Results by Chapter

This appendix collects every boxed Key Result from the main text in one place, organized by chapter, for quick lookup or exam review. Each entry is stated in compressed form; consult the cited chapter for the full definitions of symbols, derivation, and discussion.

Chapter 1 — Why Quantum Mechanics?

- Planck's constant: $h = 6.62607015 \times 10^{-34}\,\mathrm{J\,Hz^{-1}}$ (exact); $\hbar = h/2\pi$.
- Photoelectric effect: $KE_{\max} = h\nu - \Phi$.
- Bohr energies: $E_n = -13.6057\ \mathrm{eV}/n^2$.
- de Broglie wavelength: $\lambda = h/p$, for any object.
- Heisenberg uncertainty: $\Delta x\,\Delta p \geq \hbar/2$.

Chapter 2 — The Mathematical Language of Quantum Mechanics

- Euler's formula: $e^{i\theta} = \cos\theta + i\sin\theta$.
- $dy/dx = -ky \Rightarrow y = Ce^{-kx}$; $y'' = +\kappa^2 y \Rightarrow y = Ae^{\kappa x} + Be^{-\kappa x}$; $y'' = -k^2 y \Rightarrow y = A\sin(kx) + B\cos(kx)$.
- Normalization and expectation value: $\int \rho(x)\,dx = 1$; $\langle x\rangle = \int x\rho(x)\,dx$.

Chapter 3 — The Postulates of Quantum Mechanics

- Born rule: $P(x,t)\,dx = |\Psi(x,t)|^2 dx$.
- Normalization: $\int |\Psi|^2 dx = 1$ for all t.
- TDSE: $i\hbar\,\partial\Psi/\partial t = -\dfrac{\hbar^2}{2m}\partial^2\Psi/\partial x^2 + V\Psi$.
- Expectation value: $\langle\hat{A}\rangle = \int \Psi^*\hat{A}\Psi\,dx$.
- TISE: $-\dfrac{\hbar^2}{2m}\psi'' + V\psi = E\psi$, i.e., $\hat{H}\psi = E\psi$.
- Stationary states: $\Psi = \psi(x)e^{-iEt/\hbar} \Rightarrow |\Psi|^2 = |\psi|^2$, time-independent.

Chapter 4 — The Particle in a Box

- Energies: $E_n = n^2\pi^2\hbar^2/2mL^2 = n^2h^2/8mL^2$.
- Wavefunctions: $\psi_n(x) = \sqrt{2/L}\sin(n\pi x/L)$.

Chapter 5 — Tunneling, Barriers, and the Finite Well

- Step reflection ($E > V_0$): $R = \left(\dfrac{k_1 - k_2}{k_1 + k_2}\right)^2 \neq 0$.
- Tunneling: $T = \left[1 + \dfrac{V_0^2 \sinh^2(\kappa L)}{4E(V_0 - E)}\right]^{-1} \approx 16\dfrac{E}{V_0}\left(1 - \dfrac{E}{V_0}\right)e^{-2\kappa L}$.

Chapter 6 — The Quantum Harmonic Oscillator

- Energies: $E_n = (n + \frac{1}{2})\hbar\omega$, evenly spaced.

Chapter 7 — Angular Momentum

- Commutators: $[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z$ (cyclic).
- Eigenvalues: $\hat{L}^2 Y_\ell^m = \ell(\ell+1)\hbar^2 Y_\ell^m$; $\hat{L}_z Y_\ell^m = m\hbar Y_\ell^m$.

Chapter 8 — The Hydrogen Atom

- Radial equation: $-\dfrac{\hbar^2}{2m_e}\dfrac{1}{r^2}\dfrac{d}{dr}\left(r^2\dfrac{dR}{dr}\right) + \left[\dfrac{\ell(\ell+1)\hbar^2}{2m_e r^2} - \dfrac{e^2}{4\pi\epsilon_0 r}\right]R = ER$.

Chapter 9 — The Formal Structure of Quantum Mechanics

- Hermitian-operator eigenvalues are real; eigenfunctions of different eigenvalues are orthogonal.
- Compatible observables: simultaneous eigenfunctions exist iff $[\hat{A}, \hat{B}] = 0$.
- Generalized uncertainty: $\Delta A\,\Delta B \geq \frac{1}{2}|\langle[\hat{A}, \hat{B}]\rangle|$.

Chapter 10 — Spin and Identical Particles

- Symmetrization postulate: bosons symmetric, fermions antisymmetric under particle exchange.
- Pauli exclusion: no two fermions in the same quantum state.

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Chapter 11 — Time-Independent Perturbation Theory

- $E_n^{(1)} = \left\langle n^{(0)} \right| \hat{H}' \left| n^{(0)} \right\rangle$; $E_n^{(2)} = \sum_{m \neq n} \frac{|\left\langle m^{(0)} \right| \hat{H}' \left| n^{(0)} \right\rangle|^2}{E_n^{(0)} - E_m^{(0)}}$.

Chapter 12 — The Variational Method

- Variational theorem: $\langle \psi_{\text{trial}} | \hat{H} | \psi_{\text{trial}} \rangle \geq E_0$ for any normalized trial state.

Chapter 13 — Time-Dependent Phenomena and Spectroscopy

- Fermi's golden rule: $\Gamma_{i \to f} = \frac{2\pi}{\hbar} |\langle f | \hat{H}'_0 | i \rangle|^2 \delta(E_f - E_i - \hbar\omega)$.
- Electric-dipole selection rules: $\Delta n = \pm 1$ (QHO); $\Delta \ell = \pm 1$ (central potentials).

Chapter 14 — Many-Electron Atoms

- Aufbau principle: fill orbitals by increasing energy, respecting Pauli exclusion.
- Hund's rules: maximum S, then maximum L, then minimum/maximum J (less/more than half-filled).

Chapter 15 — The Chemical Bond

- Born–Oppenheimer: solve electronic structure at fixed nuclear geometry.
- LCAO: $\sigma_{1s} = N_+(1s_A + 1s_B)$ (bonding), $\sigma^*_{1s} = N_-(1s_A - 1s_B)$ (antibonding).

Chapter 16 — Molecular Spectroscopy

- Morse levels: $E_v = hc[\tilde{\omega}_e(v + \frac{1}{2}) - \tilde{\omega}_e x_e (v + \frac{1}{2})^2]$.
- P/R branches: $\tilde{\nu}_R(J) = \tilde{\nu}_0 + 2B(J+1)$; $\tilde{\nu}_P(J) = \tilde{\nu}_0 - 2BJ$.
- Franck–Condon intensity $\propto \left| \int \psi^{\text{upper}}_{v'} \psi^{\text{lower}}_{v''} dx \right|^2$.

Chapter 17 — A Modern Outlook

- Hohenberg–Kohn: ground-state energy is a functional of the density, $E[\rho]$.

Appendix H — The WKB Approximation

- Quantization: $\int_a^b p(x)\, dx = (n + \frac{1}{2})\pi\hbar$ (two soft turning points).
- Tunneling: $T \approx e^{-G}$, $G = \frac{2}{\hbar} \int_a^b \sqrt{2m[V(x) - E]}\, dx$ (Gamow factor).

Appendix I — Molecular Symmetry and Group Theory

- A vibrational mode is IR-active iff it transforms as the same irrep as x, y, or z.

Appendix J — Ehrenfest's Theorem and Degenerate Perturbation Theory

- Ehrenfest: $d\langle\hat{x}\rangle/dt = \langle\hat{p}\rangle/m$; $d\langle\hat{p}\rangle/dt = -\langle dV/dx\rangle$.
- Degenerate PT: first-order shifts are eigenvalues of $H'_{ij} = \left\langle i^{(0)} \middle| \hat{H}' \middle| j^{(0)} \right\rangle$ within the degenerate subspace.

Appendix K — Density Matrices and Mixed States

- Purity test: $\mathrm{Tr}(\hat{\rho}^2) = 1$ iff pure; < 1 iff mixed.

Appendix L — The Free Electron Gas

- Fermi energy: $E_F = \dfrac{\hbar^2}{2m_e}(3\pi^2 n)^{2/3}$.

Appendix M — The Non-Rigid Rotor

- $E_J = hc[BJ(J+1) - DJ^2(J+1)^2]$, $D = 4B^3/\tilde{\omega}_e^2$.

Note — How to use this appendix

This compilation is deliberately terse and omits every derivation, worked example, and caveat found in the main text — it is a memory aid for someone who has already worked through the corresponding chapter, not a substitute for doing so. Where a result depends on a specific geometry, approximation regime, or sign convention (nearly every entry above does, in one way or another), the full chapter entry states it explicitly; this appendix does not.

The Five Exactly Solvable Systems of Part II, Side by Side

System	Potential V	Energy formula	Quantization from	Ch.
Infinite box	0 inside, ∞ outside	$n^2\pi^2\hbar^2/2mL^2$	Hard-wall boundary condition	4
Finite well	$-V_0$ inside, 0 outside	(transcendental)	Matching ψ, ψ' at walls	5
Harmonic osc.	$\frac{1}{2}m\omega^2x^2$	$(n+\frac{1}{2})\hbar\omega$	Series/ladder termination	6
Angular mom.	(any central potential)	$\ell(\ell+1)\hbar^2$ (for L^2)	Ladder termination in m	7
Hydrogen atom	$-e^2/4\pi\epsilon_0 r$	$-13.6057\ \mathrm{eV}/n^2$	Radial series termination	8

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Every entry in the "Quantization from" column is an instance of the same underlying pattern, emphasized throughout this book: a differential equation with a continuum of formal solutions is narrowed to a discrete, physically admissible subset by a normalizability or boundary condition, never by an extra physical postulate introduced case by case. Part III (Chapters 9–13) is, among other things, the demonstration that this pattern is not a coincidence repeated five times, but a single general consequence of the Hermiticity of $\hat{H}$ (Chapter 9) applied to five different choices of $V(x)$.

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Part V

Complete Solutions Manual

Solutions for Chapter 1

Solution N.1 With $\lambda = 650\,\text{nm}$, $E = hc/\lambda = (6.626 \times 10^{-34})(2.998 \times 10^{8})/(650 \times 10^{-9}) = 3.056 \times 10^{-19}\,\text{J} = 1.907\,\text{eV}$. The frequency is $\nu = c/\lambda = 4.612 \times 10^{14}\,\text{Hz}$.

Solution N.2 $E = h\nu = (4.136 \times 10^{-15}\,\text{eV s})(2.45 \times 10^{9}\,\text{Hz}) = 1.01 \times 10^{-5}\,\text{eV}$. This is roughly five orders of magnitude smaller than a typical metal work function (2–5 eV); by Eq. (1.5), $KE_{\max} = h\nu - \Phi$ is deeply negative for any metal, meaning no photoelectron can be ejected by a single microwave photon regardless of how many photons (how much power) arrive, since each photon individually carries far too little energy.

Solution N.3 (a) $\lambda_0 = hc/\Phi = 1239.84\,\text{eV nm}/2.14\,\text{eV} = 579.4\,\text{nm}$. (b) Since 579.4 nm falls within the visible range (400–700 nm), cesium *is* photoemissive under part of the visible spectrum — specifically light with $\lambda < 579.4$ nm (blue, violet, and part of green), but not for the red/orange end of the visible spectrum ($\lambda > 579.4$ nm).

Solution N.4 $E_\gamma = hc/\lambda = 1239.84/250 = 4.959\,\text{eV}$. Then $\Phi = E_\gamma - KE_{\max} = 4.959 - 2.30 = 2.66\,\text{eV}$.

Solution N.5 The more intense source produces a larger *photocurrent* (more photoelectrons ejected per second, since more photons arrive per second), but the *same* maximum kinetic energy (hence the same stopping potential) for the ejected electrons, since $KE_{\max} = h\nu - \Phi$ depends only on frequency, not intensity.

Solution N.6 $E_4 = -13.6057/16 = -0.8504\,\text{eV}$, $E_2 = -13.6057/4 = -3.4014\,\text{eV}$. The emitted photon energy is $E_4 - E_2$... using $|E_4 - E_2| = 2.5511\,\text{eV}$, giving $\lambda = 1239.84/2.5511 = 486.0\,\text{nm}$. This is the Balmer series (all transitions end at $n = 2$); specifically, this is the historically named Hβ line.

Solution N.7 The ionization energy from $n = 2$ is $|E_2| = 3.4014\,\text{eV}$ (the energy needed to take the electron from $E_2 = -3.4014\,\text{eV}$ up to the ionization threshold at $E = 0$).

Solution N.8 $r_3 = 3^2 a_0 = 9(5.29177 \times 10^{-11}\,\text{m}) = 4.763 \times 10^{-10}\,\text{m} = 4.763\,\text{Å}$.

Solution N.9 Substituting $v = n\hbar/(m_e r)$ (from the quantization condition) into $m_e v^2/r = e^2/4\pi\epsilon_0 r^2$ gives $\dfrac{n^2\hbar^2}{m_e r^3} = \dfrac{e^2}{4\pi\epsilon_0 r^2}$, so $r_n = \dfrac{4\pi\epsilon_0 n^2\hbar^2}{m_e e^2} = n^2 a_0$, confirming Eq. (1.9). Substituting this r_n and the corresponding $v_n = n\hbar/m_e r_n$ into $E = \frac{1}{2}m_e v^2 - e^2/4\pi\epsilon_0 r$ and simplifying (using $m_e v_n^2 = e^2/4\pi\epsilon_0 r_n$ from the force equation, so the kinetic energy is exactly $-\frac{1}{2}$ the potential energy) gives $E_n = -\dfrac{1}{2}\cdot\dfrac{e^2}{4\pi\epsilon_0 r_n} = -\dfrac{m_e e^4}{2(4\pi\epsilon_0)^2\hbar^2}\dfrac{1}{n^2}$, confirming Eq. (1.10).

Solution N.10 $\lambda_p = h/(m_p v) = 6.626 \times 10^{-34}/[(1.6726 \times 10^{-27})(2.0 \times 10^5)] = 1.981 \times 10^{-12}$ m. $\lambda_e = h/(m_e v) = 3.637 \times 10^{-9}$ m. The ratio $\lambda_e/\lambda_p = m_p/m_e = 1836.15$ (protons and electrons at the same speed have momenta, and hence de Broglie wavelengths, in inverse proportion to their masses). The heavier proton, with the shorter wavelength, behaves "more classically" at this speed (its wave nature is $1836\times$ less significant at the same length scale).

Solution N.11 $\Delta p_{\min} = \hbar/(2\Delta x) = (1.0546 \times 10^{-34})/(2 \times 5.0 \times 10^{-11}) = 1.055 \times 10^{-24}$ kg m/s. Then $\Delta v = \Delta p/m_e = 1.158 \times 10^6$ m/s, comparable in magnitude to typical atomic electron speeds ($\sim 10^6$ m/s) — confirming that position localization to atomic dimensions unavoidably introduces a velocity uncertainty of the same order as the electron's typical speed itself, not a negligible correction.

Solution N.12 $\Delta E_{\min} = \hbar/(2\Delta t) = (1.0546 \times 10^{-34})/(2 \times 8.4 \times 10^{-17}) = 6.28 \times 10^{-19}$ J $= 3.92$ eV. Comparing to the rest energy $m_{\pi^0}c^2 = 264\, m_e c^2 = 264(0.511 \text{ MeV}) = 134.9$ MeV, the fractional uncertainty is $\Delta E/E \approx 2.9 \times 10^{-8}$: a minuscule fractional uncertainty, though the absolute energy width (~ 4 eV) is still routinely measurable in particle physics as the particle's intrinsic decay width.

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Solutions for Chapter 2

Solution N.1 With $z_1 = 2 - 3i$, $z_2 = -1 + i$: (a) $z_1 + z_2 = 1 - 2i$. (b) $z_1 z_2 = (2-3i)(-1+i) = -2 + 2i + 3i - 3i^2 = -2 + 5i + 3 = 1 + 5i$. (c) $|z_1| = \sqrt{2^2 + 3^2} = \sqrt{13} = 3.606$. (d) $z_1/z_2 = \dfrac{(2-3i)(-1-i)}{(-1+i)(-1-i)} = \dfrac{-2-2i+3i+3i^2}{1+1} = \dfrac{-5+i}{2} = -2.5 + 0.5i$.

Solution N.2 $z = 4e^{i\pi/3} = 4\cos(\pi/3) + 4i\sin(\pi/3) = 4(\frac{1}{2}) + 4i(\frac{\sqrt{3}}{2}) = 2 + 2\sqrt{3}\,i \approx 2 + 3.464i$. For $w = 1 - i$: $r = |w| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$, $\theta = \arctan(-1/1) = -\pi/4$, so $w = \sqrt{2}\,e^{-i\pi/4}$.

Solution N.3 With $z = a + bi$: $z + z^* = (a + bi) + (a - bi) = 2a$, so $\frac{1}{2}(z + z^*) = a = \mathrm{Re}(z)$. Also $z - z^* = (a + bi) - (a - bi) = 2bi$, so $\frac{1}{2i}(z - z^*) = b = \mathrm{Im}(z)$.

Solution N.4 $\mathbf{u} \cdot \mathbf{v} = (2)(1) + (1)(-1) + (3)(\frac{1}{3}) = 2 - 1 + 1 = 2 \neq 0$: not orthogonal. Requiring $2(1) + 1(-1) + 3v_3 = 0$ gives $v_3 = -\frac{1}{3}$.

Solution N.5 Characteristic equation: $(0-\lambda)(3-\lambda) - 4 = 0 \Rightarrow \lambda^2 - 3\lambda - 4 = 0 \Rightarrow (\lambda - 4)(\lambda + 1) = 0$, so $\lambda = 4, -1$. For $\lambda = -1$: $(A + I)\mathbf{v} = 0$ gives $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}\mathbf{v} = 0 \Rightarrow v_1 = -2v_2$; normalized eigenvector $\frac{1}{\sqrt{5}}(-2, 1)$. For $\lambda = 4$: $(A - 4I)\mathbf{v} = 0$ gives $\begin{pmatrix} -4 & 2 \\ 2 & -1 \end{pmatrix}\mathbf{v} = 0 \Rightarrow v_1 = \frac{1}{2}v_2$; normalized eigenvector $\frac{1}{\sqrt{5}}(1, 2)$.

Solution N.6 Characteristic equation: $(1-\lambda)^2 - 4 = 0 \Rightarrow \lambda^2 - 2\lambda - 3 = 0 \Rightarrow (\lambda - 3)(\lambda + 1) = 0$, so $\lambda = 3, -1$, with (unnormalized) eigenvectors $(1, 1)$ and $(1, -1)$ respectively. Direct check for $\lambda = 3$: $B(1,1)^T = (1+2, 2+1)^T = (3,3)^T = 3(1,1)^T$. ✓ For $\lambda = -1$: $B(1,-1)^T = (1-2, 2-1)^T = (-1,1)^T = -1(1,-1)^T$. ✓

Solution N.7 With $y = A\cosh(\kappa x) + B\sinh(\kappa x)$: $y' = A\kappa\sinh(\kappa x) + B\kappa\cosh(\kappa x)$, and $y'' = A\kappa^2\cosh(\kappa x) + B\kappa^2\sinh(\kappa x) = \kappa^2 y$, confirming the equation is satisfied for any A, B.

Solution N.8 $y(x) = A\sin(kx) + B\cos(kx)$; $y(0) = 0 \Rightarrow B = 0$, so $y = A\sin(kx)$, $y' = Ak\cos(kx)$. Then $y'(L) = 0 \Rightarrow Ak\cos(kL) = 0$, and (rejecting $A = 0$) $\cos(kL) = 0 \Rightarrow kL = \frac{\pi}{2}, \frac{3\pi}{2}, \ldots = \frac{(2n-1)\pi}{2}$, so $k = \dfrac{(2n-1)\pi}{2L}$, $n = 1, 2, 3, \ldots$ (note: half as densely spaced in n as the Section 2.4 Worked Example, since only odd multiples of $\pi/2$ appear here — a different, but equally legitimate, boundary condition).

Solution N.9 Normalization: $\int_0^2 Cx^2\,dx = \frac{8}{3}C = 1 \Rightarrow C = \frac{3}{8}$. $\langle x \rangle = \int_0^2 x \cdot \frac{3}{8}x^2\,dx = \frac{3}{8} \cdot 4 = \frac{3}{2}$. $P(1 \le x \le 2) = \int_1^2 \frac{3}{8}x^2\,dx = \frac{3}{8}\left[\frac{x^3}{3}\right]_1^2 = \frac{3}{8} \cdot \frac{7}{3} = \frac{7}{8} = 0.875$.

Solution N.10 $\langle x^2 \rangle = \int_0^L x^2 \cdot \frac{1}{L}\, dx = \frac{L^2}{3}$. With $\langle x \rangle = L/2$ (from the chapter's Worked Example), $\Delta x = \sqrt{\frac{L^2}{3} - \frac{L^2}{4}} = \sqrt{\frac{L^2}{12}} = \frac{L}{2\sqrt{3}} = \frac{\sqrt{3}}{6} L \approx 0.289\, L$.

Solution N.11 $\langle \phi | \phi \rangle\, \psi = \phi_1^* \psi_1 + \phi_2^* \psi_2 = (1)(i) + (-i)(1) = i - i = 0$: orthogonal. $\langle \phi | \phi \rangle\, \phi = (1)(1) + (-i)(i) = 1 + 1 = 2 \neq 1$: $|\phi\rangle$ is *not* normalized (it would need to be divided by $\sqrt{2}$ first).

Solution N.12 The student's claim conflates the mathematical tools used in a calculation with the physical predictions that calculation produces. As shown throughout Section 2.1, any physically measurable quantity in this book is built from combinations like $|z|^2 = zz^*$ or $\langle \phi | \phi \rangle\, \psi = \int \phi^* \psi\, dx$, which are guaranteed real by construction (Problem 2.11 above gives a concrete instance: $\langle \phi | \phi \rangle\, \phi = 2$, a perfectly ordinary real number, despite $|\phi\rangle$ itself having complex components). Complex numbers are an intermediate bookkeeping device in the calculation; every final, measurable prediction of quantum mechanics is a real number.

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Solutions for Chapter 3

Solution N.1 Using the given integral, $\int_0^\infty |\psi|^2 dx = B^2 \int_0^\infty x^2 e^{-2x/a}\, dx = B^2 \cdot \dfrac{a^3}{4} = 1$, so $B = \dfrac{2}{a^{3/2}}$.

Solution N.2 $\langle x \rangle = \int_0^\infty x|\psi(x)|^2\, dx = B^2 \int_0^\infty x^3 e^{-2x/a}\, dx = \dfrac{4}{a^3} \cdot \dfrac{3a^4}{8} = \dfrac{3a}{2}$.

Solution N.3 $|\Psi(x,t)|^2 = \Psi^*\Psi = \left[\psi(x)e^{-iEt/\hbar}\right]^* \left[\psi(x)e^{-iEt/\hbar}\right] = \psi(x)^2\, e^{+iEt/\hbar} e^{-iEt/\hbar} = \psi(x)^2$, since $\psi(x)$ is real (so $\psi^* = \psi$) and the two exponential phase factors are exact complex conjugates of one another and cancel. The result has no remaining t-dependence, confirming $|\Psi(x,t)|^2$ is time-independent, exactly the content of the Key Result of Section 3.5.

Solution N.4 Normalization: $\int |\tilde{\psi}|^2 dx = \int |e^{i\phi_0}\psi|^2 dx = \int |e^{i\phi_0}|^2 |\psi|^2 dx = \int |\psi|^2 dx = 1$ (using $|e^{i\phi_0}| = 1$ for any real ϕ_0), so $\tilde{\psi}$ is normalized whenever ψ is. Also $|\tilde{\psi}(x)|^2 = |e^{i\phi_0}|^2 |\psi(x)|^2 = |\psi(x)|^2$, identical to the original probability density. This confirms that an overall constant phase multiplying an entire wavefunction carries no physical information whatsoever: ψ and $e^{i\phi_0}\psi$ are experimentally indistinguishable for any observable built from $|\psi|^2$ or from $\psi^* \hat{A} \psi$.

Solution N.5 Using $\hat{p} = -i\hbar\, d/dx$ on $\psi(x) = e^{ikx}$: $\hat{p}\psi = -i\hbar(ik)e^{ikx} = \hbar k\, e^{ikx} = \hbar k\, \psi(x)$. Then $\langle \hat{p} \rangle = \int \psi^* \hat{p} \psi\, dx = \hbar k \int \psi^* \psi\, dx = \hbar k$ (formally, treating $\int |\psi|^2 dx$ as 1 in the appropriate normalized sense for a plane wave), confirming $\langle \hat{p} \rangle = \hbar k$, matching the Worked Example of Section 3.4 exactly, as expected since e^{ikx} is an eigenstate of $\hat{p}$ with eigenvalue $\hbar k$ (so *every* measurement of p, not merely the average, would return exactly $\hbar k$).

Solution N.6 $\hat{x}(\hat{p}\psi) = x \cdot (-i\hbar\psi') = -i\hbar x\psi'$. And $\hat{p}(\hat{x}\psi) = -i\hbar \dfrac{d}{dx}(x\psi) = -i\hbar(\psi + x\psi')$ (product rule). Subtracting: $[\hat{x}, \hat{p}]\psi = -i\hbar x\psi' - \left[-i\hbar(\psi + x\psi')\right] = -i\hbar x\psi' + i\hbar\psi + i\hbar x\psi' = i\hbar\psi$. Since this holds for arbitrary $\psi(x)$, $[\hat{x}, \hat{p}] = i\hbar$ as an operator identity.

Solution N.7 $\hat{p}\, e^{ikx} = -i\hbar \dfrac{d}{dx} e^{ikx} = -i\hbar(ik)e^{ikx} = \hbar k\, e^{ikx}$: applying $\hat{p}$ returns the same function multiplied by $\hbar k$, confirming the momentum eigenvalue is $p = \hbar k$, exactly de Broglie's relation $p = h/\lambda = \hbar k$ (using $k = 2\pi/\lambda$) from Section 1.5.

Solution N.8 With ψ_1, ψ_2 real and both normalized (so each term individually has unit weight, and the state is normalized given ψ_1, ψ_2 orthogonal, Chapter 9),

$$|\Psi(x,t)|^2 = \tfrac{1}{2}\psi_1^2 + \tfrac{1}{2}\psi_2^2 + \psi_1\psi_2 \cdot \tfrac{1}{2}\left(e^{i(E_1-E_2)t/\hbar} + e^{-i(E_1-E_2)t/\hbar}\right) = \tfrac{1}{2}\psi_1^2 + \tfrac{1}{2}\psi_2^2 + \psi_1\psi_2 \cos\left(\frac{(E_2-E_1)t}{\hbar}\right)$$

using $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$. Unlike a single stationary state, this manifestly depends on t, oscillating at angular frequency $\omega = (E_2 - E_1)/\hbar$ exactly as claimed, through the cross ("interference") term $\psi_1\psi_2\cos(\omega t)$ that a single energy eigenstate alone could never produce.

Solution N.9 (a) Valid, time-independent, $V(x) = \frac{1}{2}kx^2$: separation of variables applies directly (this is the QHO of Chapter 6). (b) Not valid as stated for direct separation: $V(x,t) = \frac{1}{2}kx^2\cos(\omega t)$ depends explicitly on t, so the separation-of-variables argument of Section 3.5 (which assumed $V = V(x)$ alone) does not apply. (c) Valid, time-independent (the infinite square well of Chapter 4). (d) Valid, time-independent (the hydrogen atom's Coulomb potential, radial form, Chapter 8).

Solution N.10 Since both solve the TISE with the same E: $\hat{H}\psi_1 = E\psi_1$ and $\hat{H}\psi_2 = E\psi_2$. Then, using linearity of the derivative (hence of $\hat{H}$ itself, which is built from derivatives and multiplication, both linear operations),

$$\hat{H}(c_1\psi_1 + c_2\psi_2) = c_1\hat{H}\psi_1 + c_2\hat{H}\psi_2 = c_1E\psi_1 + c_2E\psi_2 = E(c_1\psi_1 + c_2\psi_2),$$

confirming $c_1\psi_1 + c_2\psi_2$ also solves the TISE with the same eigenvalue E.

Solution N.11 With $\Psi(x,t) = g(x,t)$ purely real: the left side, $i\hbar\,\partial g/\partial t$, is i times a real number, hence purely imaginary (unless $\partial g/\partial t = 0$ identically). The right side, $-\frac{\hbar^2}{2m}\partial^2 g/\partial x^2$, is purely real (a real constant times a real second derivative of a real function). A purely imaginary quantity can equal a purely real quantity only if both are exactly zero; since this would need to hold at every x, t for a free particle with generically nonzero curvature and time dependence, no nontrivial real g can satisfy the equation. This confirms that complex-valued wavefunctions (Section 2.1) are not a convenience but a structural requirement of the theory.

Solution N.12 Using $\hat{H} = -\dfrac{\hbar^2}{2m}\dfrac{d^2}{dx^2} + V(\hat{x})$ from Section 3.3 with $V(x) = \frac{1}{2}m\omega^2x^2$, the TISE $\hat{H}\psi = E\psi$ reads explicitly

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + \frac{1}{2}m\omega^2x^2\,\psi(x) = E\,\psi(x),$$

with every constant $(m, \omega, \hbar, E)$ shown explicitly; Chapter 6 solves this equation for $\psi(x)$ and E in full.

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Solutions for Chapter 4

Solution N.1 $E_1 = \dfrac{\pi^2\hbar^2}{2m_eL^2}$ with $L = 0.20\,\text{nm} = 2.0\times10^{-10}\,\text{m}$ gives $E_1 = 1.506\times10^{-18}\,\text{J} = 9.40\,\text{eV}$.

Solution N.2 $E_3 = 9E_1 = 84.61\,\text{eV}$, so $\Delta E = E_3 - E_1 = 75.21\,\text{eV}$. Then $\lambda = hc/\Delta E = 1239.84\,\text{eV nm}/75.21\,\text{eV} = 16.49\,\text{nm}$, deep in the extreme ultraviolet (far below the visible range and well into the region where individual photons begin to ionize most atoms).

Solution N.3 Since $E_1 \propto 1/m$ (with L fixed and identical for both), $E_1^{(p)}/E_1^{(e)} = m_e/m_p = 1/1836.15 = 5.446\times10^{-4}$. The more massive particle has the smaller zero-point energy because confinement kinetic energy $E_1 = \pi^2\hbar^2/2mL^2$ decreases with increasing mass for fixed L — a heavier particle needs less momentum spread to satisfy the same positional confinement, per the uncertainty principle of Chapter 1.

Solution N.4 Setting $E_n = n^2\pi^2\hbar^2/2mL^2$ equal to $1.0\times10^{-6}\,\text{J}$ and solving for n: $n = \dfrac{L}{\pi\hbar}\sqrt{2mE} = \dfrac{(0.10)\sqrt{2(1.0\times10^{-3})(1.0\times10^{-6})}}{\pi(1.0546\times10^{-34})} \approx 1.35\times10^{28}$. This is an almost incomprehensibly large quantum number; per the correspondence principle (Section 4.4), quantum graininess becomes utterly unobservable at such enormous n — adjacent energy levels are separated by a fraction of $1/n$ of the total energy, so the marble's motion is, for all conceivable purposes, indistinguishable from the smooth continuum of classical mechanics.

Solution N.5 With $\psi_n(x) = \sqrt{2/L}\sin(n\pi x/L)$: $\psi_n'(x) = \sqrt{2/L}\,\dfrac{n\pi}{L}\cos(n\pi x/L)$, and $\psi_n''(x) = -\sqrt{2/L}\left(\dfrac{n\pi}{L}\right)^2\sin(n\pi x/L) = -\left(\dfrac{n\pi}{L}\right)^2\psi_n(x)$. So $-\dfrac{\hbar^2}{2m}\psi_n'' = \dfrac{\hbar^2}{2m}\left(\dfrac{n\pi}{L}\right)^2\psi_n = \dfrac{n^2\pi^2\hbar^2}{2mL^2}\psi_n$, confirming the TISE is satisfied with $E_n = n^2\pi^2\hbar^2/2mL^2$ exactly.

Solution N.6 Using the hint, $\sin(\pi x/L)\sin(2\pi x/L) = \tfrac{1}{2}\big[\cos(-\pi x/L) - \cos(3\pi x/L)\big] = \tfrac{1}{2}\big[\cos(\pi x/L) - \cos(3\pi x/L)\big]$. Then

$$\int_0^L \psi_1\psi_2\,dx = \frac{2}{L}\int_0^L \tfrac{1}{2}\big[\cos(\pi x/L) - \cos(3\pi x/L)\big]dx = \frac{1}{L}\left[\frac{L}{\pi}\sin(\pi x/L) - \frac{L}{3\pi}\sin(3\pi x/L)\right]_0^L.$$

Both $\sin(\pi x/L)$ and $\sin(3\pi x/L)$ vanish at $x = 0$ and $x = L$ (integer multiples of π), so the entire bracket is zero at both limits, giving $\int_0^L \psi_1\psi_2\,dx = 0$, confirming orthogonality.

Solution N.7 $|\psi_1(x)|^2 = \frac{2}{L}\sin^2(\pi x/L)$ is symmetric about the box's midpoint, $x = L/2$ (i.e., $|\psi_1(L/2+u)|^2 = |\psi_1(L/2-u)|^2$ for any u), so the electron is, on average, equally

likely to be found on either side of the midpoint: $\langle x \rangle = L/2$, with no integration required.

Solution N.8 Using $\langle x^2 \rangle = L^2\left(\frac{1}{3} - \frac{1}{2\pi^2}\right) = 0.28267\,L^2$ and $\langle x \rangle^2 = (L/2)^2 = 0.25L^2$: $\Delta x = L\sqrt{0.28267 - 0.25} = L\sqrt{0.032673} = 0.1808\,L$. The classical value is $\Delta x_{\text{classical}} = L/\sqrt{12} = 0.2887\,L$. The ratio is $0.1808/0.2887 = 0.626$: the quantum ground state is *less* spatially spread out than the classical uniform distribution — the particle is concentrated somewhat more toward the center of the box (consistent with the single central peak of $|\psi_1|^2$, versus the classical particle's uniform probability of being anywhere in $[0, L]$).

Solution N.9 With 4 π electrons filling two per level, levels $n = 1$ and $n = 2$ are both doubly occupied: the HOMO is $n = 2$ and the LUMO is $n = 3$. The box length is $L = (3 + 1)(0.140\text{ nm}) = 0.560$ nm. Using $E_n = n^2h^2/8m_eL^2$: $E_2 = 4.796$ eV, $E_3 = 10.792$ eV, so $\Delta E = 5.995$ eV and $\lambda = hc/\Delta E = 1239.84/5.995 = 206.8$ nm. This is close to, and in this case slightly *below*, the experimental value of 217 nm (unlike the octatetraene case, where the model *overestimated* λ) — a reminder that a simple model's error need not have a consistent sign across every member of a series, even though the qualitative trend (λ increasing with conjugation length) is robustly captured in both cases.

Solution N.10 If $E < 0$, then $k^2 = 2mE/\hbar^2 < 0$, so k is imaginary; writing $k = i\kappa$ for real κ turns the oscillatory solutions $\sin(kx), \cos(kx)$ into real exponentials $\sinh(\kappa x), \cosh(\kappa x)$ (Chapter 2, Problem 2.7). A combination of growing and decaying real exponentials cannot equal zero at *both* $x = 0$ and $x = L$ simultaneously except in the trivial case of the zero function everywhere (unlike $\sin(kx)$, which vanishes at both endpoints for a whole discrete family of nonzero k) — so no nontrivial bound state exists for $E < 0$ (or $E = 0$, which gives only the linear function $\psi = Ax + B$, similarly unable to vanish at both ends except trivially), and $E > 0$ is required.

Solution N.11 With at most 2 electrons per spatial level (Chapter 10), both electrons occupy the lowest level, $n = 1$, with opposite spin. The total ground-state energy is $2E_1$ (each electron independently contributes E_1, since the electrons are assumed non-interacting).

Solution N.12 By the result of Problem 3.8 (Chapter 3), the probability density of a superposition of two stationary states oscillates at $\omega = (E_2 - E_1)/\hbar$. For the particle in a box, $E_2 = 4E_1$ (Chapter 4's own energy formula, $E_n \propto n^2$), so $E_2 - E_1 = 3E_1$, giving $\omega = 3E_1/\hbar$.

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Solutions for Chapter 5

Solution N.1 With $E = 3.0$ eV, $V_0 = 1.0$ eV: $k_1 = \sqrt{2m_eE}/\hbar$, $k_2 = \sqrt{2m_e(E-V_0)}/\hbar$. Numerically, $R = \left(\dfrac{k_1-k_2}{k_1+k_2}\right)^2 = 0.0102$ and $T = 1 - R = 0.9898$; indeed $R + T = 1.0000$.

Solution N.2 $\kappa_2 = \sqrt{2m_e(V_0-E)}/\hbar$ with $V_0 - E = 0.40$ eV gives $1/\kappa_2 = 0.309$ nm, somewhat larger than the 0.276 nm quoted in the text for a 0.5 eV energy deficit — consistent with $1/\kappa_2 \propto 1/\sqrt{V_0-E}$: a smaller energy deficit below the barrier top (0.40 eV here versus 0.5 eV there) gives *deeper* penetration into the classically forbidden region.

Solution N.3 Setting $u_0 = a\sqrt{2m_eV_0}/\hbar = \pi/2$ and solving for V_0: $V_0 = \dfrac{(\pi/2)^2\hbar^2}{2m_ea^2}$. With $a = 0.50$ nm, $V_0 = 0.376$ eV: any well of this depth or greater, at this half-width, supports (at least) two bound states.

Solution N.4 $\Delta E = E_2 - E_1 = -0.111 - (-0.250) = 0.139$ eV. Then $\lambda = hc/\Delta E = 1239.84/0.139 = 8920$ nm ≈ 8.9 μm, in the mid-infrared.

Solution N.5 As $V_0 \to 0$, $k_2 = \sqrt{2m(E-V_0)}/\hbar \to \sqrt{2mE}/\hbar = k_1$, so $R = \left(\dfrac{k_1-k_2}{k_1+k_2}\right)^2 \to \left(\dfrac{k_1-k_1}{k_1+k_1}\right)^2 = 0$: with no step at all, there is nothing to reflect from, and transmission is perfect, exactly as physically required.

Solution N.6 Using the exact transmission formula, $T(L{=}1.0\text{ nm}) = 7.53 \times 10^{-8}$ (as in the text) and $T(L{=}0.5\text{ nm}) = 5.38 \times 10^{-4}$, a ratio of about 7140: halving the barrier width increases the tunneling probability by more than three orders of magnitude, a vivid illustration of the exponential sensitivity $T \propto e^{-2\kappa L}$ discussed in the text.

Solution N.7 Requiring $e^{-2\kappa\,\delta L} = \frac{1}{2}$ (where δL is the *decrease* in width needed to double T, or equivalently here we want the width *increase* that halves T): $2\kappa\,\delta L = \ln 2$, so $\delta L = \ln 2/(2\kappa)$. With $\kappa = 8.874 \times 10^9$ m^{-1} (as computed in the text), $\delta L = 0.0391$ nm. Since increasing L decreases T, the width at which T falls to half its $L = 1.0$ nm value is $L = 1.0 + 0.039 = 1.039$ nm.

Solution N.8 Since $\kappa = \sqrt{2m(V_0-E)}/\hbar$ increases with $\sqrt{m}$, and $T \propto e^{-2\kappa L}$ decreases exponentially with κ, the much heavier proton ($m_p \approx 1836\,m_e$) has a much larger κ than an electron of the same E and V_0, and therefore a dramatically *smaller* tunneling probability: the electron tunnels far more readily. This is exactly why electron tunneling (STM, alpha decay aside, which involves the even heavier alpha particle but compensates with a very different barrier shape) is common and technologically exploitable, while proton tunneling, though real, requires unusually favorable (thin,

low) barriers to compete with the exponential suppression from the proton's much larger mass.

Solution N.9 Integrating $-\frac{\hbar^2}{2m}\psi'' + V(x)\psi = E\psi$ from $-\epsilon$ to $+\epsilon$: $-\frac{\hbar^2}{2m}\left[\psi'(\epsilon) - \psi'(-\epsilon)\right] + \int_{-\epsilon}^{\epsilon} V(x)\psi\,dx = E\int_{-\epsilon}^{\epsilon}\psi\,dx$. As $\epsilon \to 0$, the right side and the potential-energy term on the left both vanish *provided* $V(x)$ stays finite (a finite jump discontinuity contributes a vanishingly small integral over a vanishingly small interval), forcing $\psi'(\epsilon) - \psi'(-\epsilon) \to 0$: ψ' (and, by a similar, simpler argument, ψ itself) must be continuous. If instead $V(x) \to \infty$ at the point in question (an infinite wall), the integral $\int_{-\epsilon}^{\epsilon} V(x)\psi\,dx$ need not vanish as $\epsilon \to 0$ even though the interval shrinks to zero, since V itself is unbounded there; this allows (indeed, forces) a finite jump in ψ' across the wall, exactly as seen at the walls of the infinite well in Chapter 4, where ψ' is nonzero just inside the well and identically zero just outside.

Solution N.10 Centering the well at the origin makes $V(x)$ an even function, $V(-x) = V(x)$, which (as shown in Section 5.2) guarantees every non-degenerate bound-state wavefunction is either purely even or purely odd about $x = 0$. This lets the problem be solved separately, and more simply, for each symmetry class using single trigonometric functions ($\cos(kx)$ or $\sin(kx)$ alone) rather than a general combination $A\cos(kx) + B\sin(kx)$ that would be required with the well running from 0 to $2a$, where no such symmetry is manifest.

Solution N.11 For a bound state, normalizability requires $\psi \to 0$ as $x \to \pm\infty$, which is what selects the single decaying exponential (discarding the growing one) on each side and, in turn, is exactly the extra condition that restricts E to a discrete set (Section 5.2). For $E > 0$, the "outside" solutions are oscillatory (sin, cos, or complex exponentials) rather than decaying, and an oscillatory function is automatically bounded (hence trivially normalizable in the loose, scattering-state sense used for such continuum states) for *any* positive E — there is no analogous boundary condition left to exclude any particular value of $E > 0$, so every positive energy is allowed, forming a continuum.

Solution N.12 Continuity of ψ at $x = 0$: $1 + r = A + B$. Continuity of ψ' at $x = 0$: $ik(1 - r) = \kappa(A - B)$. Continuity of ψ at $x = L$: $Ae^{\kappa L} + Be^{-\kappa L} = te^{ikL}$. Continuity of ψ' at $x = L$: $\kappa(Ae^{\kappa L} - Be^{-\kappa L}) = ikte^{ikL}$. This is indeed four linear equations in the four unknowns r, A, B, t (with the incident amplitude fixed at 1 by convention), confirming the system is fully (and uniquely) determined, consistent with Section 5.3.

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Solutions for Chapter 6

Solution N.1 Substituting $a^2 = \hbar/m\omega$ into $\int_{-\infty}^{\infty} e^{-x^2/a^2}dx = a\sqrt{\pi}$ (Chapter 3) gives $\int_{-\infty}^{\infty} e^{-m\omega x^2/\hbar}dx = \sqrt{\pi\hbar/m\omega}$. Then $\int |\psi_0|^2 dx = \sqrt{m\omega/\pi\hbar} \cdot \sqrt{\pi\hbar/m\omega} = 1$, confirming normalization.

Solution N.2 $|\psi_0(x)|^2$ is an even (symmetric) function of x about the origin, so the integrand $x|\psi_0(x)|^2$ in $\langle x\rangle = \int x|\psi_0|^2 dx$ is odd, and integrates to zero by symmetry: $\langle x\rangle_0 = 0$.

Solution N.3 $\Delta x = \sqrt{\langle x^2\rangle_0 - \langle x\rangle_0^2} = \sqrt{\hbar/2m\omega - 0} = \sqrt{\hbar/2m\omega}$.

Solution N.4 $\Delta p = \sqrt{\langle p^2\rangle_0 - \langle p\rangle_0^2} = \sqrt{\frac{1}{2}m\hbar\omega - 0} = \sqrt{m\hbar\omega/2}$ (using $\langle p\rangle_0 = 0$ by the same parity argument as Problem 6.2, applied to momentum). Then $\Delta x\,\Delta p = \sqrt{\frac{\hbar}{2m\omega}} \cdot \sqrt{\frac{m\hbar\omega}{2}} = \sqrt{\frac{\hbar^2}{4}} = \frac{\hbar}{2}$, exactly saturating the Heisenberg bound $\Delta x\Delta p \geq \hbar/2$ from Section 1.6.

Solution N.5 $\mu = m_C m_O/(m_C + m_O) = 1.1385 \times 10^{-26}$ kg. Then $\omega = \sqrt{k_{\text{spring}}/\mu} = 4.0387 \times 10^{14}$ rad/s, giving $\tilde{\nu} = \omega/2\pi c = 2144.1$ cm^{-1}, in excellent agreement with the experimental 2143 cm^{-1}. The zero-point energy is $E_0 = \frac{1}{2}\hbar\omega = 0.1329$ eV.

Solution N.6 Setting $\Delta x \sim x$, $\Delta p \sim \hbar/x$ exactly (i.e., using $\Delta x\Delta p \sim \hbar$, not $\hbar/2$, as the problem's "appropriate factor" of 1 specifies): $E(x) = \dfrac{\hbar^2}{2mx^2} + \dfrac{1}{2}m\omega^2x^2$. Minimizing, $dE/dx = -\hbar^2/mx^3 + m\omega^2 x = 0 \Rightarrow x^2 = \hbar/m\omega$, and substituting back gives $E_{\min} = \hbar\omega$. This order-of-magnitude estimate is a factor of 2 larger than the exact result $E_0 = \frac{1}{2}\hbar\omega$ from Section 6.4 — in contrast to the hydrogen-atom estimate of Chapter 1, which happened to come out exactly right, this estimate captures only the correct order of magnitude and functional dependence, not the precise numerical prefactor, exactly the caveat flagged in Chapter 1's own Note box about this style of argument.

Solution N.7 $\mu_{\text{DCl}} = m_D m_{\text{Cl}}/(m_D + m_{\text{Cl}}) = 1.9044$ u $= 3.1623 \times 10^{-27}$ kg. With the same $k_{\text{spring}} = 480$ N/m, $\tilde{\nu}_{\text{DCl}} = 2068.3$ cm^{-1}, compared to $\tilde{\nu}_{\text{HCl}} = 2883.8$ cm^{-1} (Chapter 6 main text). The ratio is $\tilde{\nu}_{\text{DCl}}/\tilde{\nu}_{\text{HCl}} = 0.7172$, matching $\sqrt{\mu_{\text{HCl}}/\mu_{\text{DCl}}}$ exactly, as expected from $\omega \propto 1/\sqrt{\mu}$.

Solution N.8 As $m_{\text{Cl}} \to \infty$: $\mu = \dfrac{m_{\text{H}} m_{\text{Cl}}}{m_{\text{H}} + m_{\text{Cl}}} = \dfrac{m_{\text{H}}}{1 + m_{\text{H}}/m_{\text{Cl}}} \to \dfrac{m_{\text{H}}}{1+0} = m_{\text{H}}$. Numerically, $\mu_{\text{HCl}} = 0.9796$ u versus $m_{\text{H}} = 1.0078$ u, a difference of about 2.80%. This is a modest but not negligible correction: treating Cl as perfectly stationary is a reasonable zeroth approximation for HCl (chlorine is $\sim 35\times$ heavier than hydrogen), but the

2.8% mass correction is easily large enough to matter at the precision of modern vibrational spectroscopy, exactly analogous to the reduced-mass correction to the Bohr formula quantified in Chapter 1.

Solution N.9 With $\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right)$ and $\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right)$:

$$\hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} = \frac{m\omega}{2\hbar}\Big[\big(\hat{x} + \tfrac{i\hat{p}}{m\omega}\big)\big(\hat{x} - \tfrac{i\hat{p}}{m\omega}\big) - \big(\hat{x} - \tfrac{i\hat{p}}{m\omega}\big)\big(\hat{x} + \tfrac{i\hat{p}}{m\omega}\big)\Big].$$

Expanding both products, the $\hat{x}^2$ and $\hat{p}^2/(m\omega)^2$ terms are identical in each and cancel in the difference, leaving only the cross terms: $\big(-\frac{i}{m\omega}\hat{x}\hat{p} + \frac{i}{m\omega}\hat{p}\hat{x}\big) - \big(\frac{i}{m\omega}\hat{x}\hat{p} - \frac{i}{m\omega}\hat{p}\hat{x}\big) = -\frac{2i}{m\omega}(\hat{x}\hat{p} - \hat{p}\hat{x}) = -\frac{2i}{m\omega}[\hat{x},\hat{p}] = -\frac{2i}{m\omega}(i\hbar) = \frac{2\hbar}{m\omega}$. So $\hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} = \dfrac{m\omega}{2\hbar}\cdot\dfrac{2\hbar}{m\omega} = 1$, confirming Eq. (6.2).

Solution N.10 Following the hint: $[\hat{N},\hat{a}^\dagger] = \hat{a}^\dagger\hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a}^\dagger\hat{a} = \hat{a}^\dagger(\hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a}) = \hat{a}^\dagger[\hat{a},\hat{a}^\dagger] = \hat{a}^\dagger\cdot 1 = \hat{a}^\dagger$, using Eq. (6.2) in the last step.

Solution N.11 (a) $\tilde{\nu} = 1/\lambda = 1/(4.70\times10^{-4}\ \text{cm}) = 2128\ \text{cm}^{-1}$. (b) $\omega = 2\pi c\tilde{\nu} = 2\pi(2.998\times10^{10}\,\text{cm/s})(2128\,\text{cm}^{-1}) = 4.008\times10^{14}\,\text{rad/s}$. (c) $k_{\text{spring}} = \mu\omega^2 = (1.615\times10^{-27})(4.008\times10^{14})^2 = 259.4\ \text{N/m}$.

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Solutions for Chapter 7

Solution N.1 For $\ell = 3$: $m = -3, -2, -1, 0, 1, 2, 3$ — exactly $7 = 2(3) + 1$ values.

Solution N.2 $|\mathbf{L}| = \sqrt{\ell(\ell+1)}\,\hbar$: for $\ell = 0, 1, 2, 3$ this gives 0, $\sqrt{2}\hbar = 1.414\hbar$, $\sqrt{6}\hbar = 2.449\hbar$, $\sqrt{12}\hbar = 3.464\hbar$. The corresponding $\theta_{\min} = \arccos\big(\ell/\sqrt{\ell(\ell+1)}\big)$ (undefined for $\ell = 0$, where $|\mathbf{L}| = 0$ itself) are 45.0°, 35.3°, 30.0° for $\ell = 1, 2, 3$. As $\ell \to \infty$, $\ell/\sqrt{\ell(\ell+1)} = 1/\sqrt{1 + 1/\ell} \to 1$, so $\theta_{\min} \to \arccos(1) = 0$, confirming the classical limit (angular momentum vector aligning ever more closely with the z-axis) is approached at large ℓ.

Solution N.3 $\hat{L}_z Y_1^1 = -i\hbar \dfrac{\partial}{\partial \phi}\left[-\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{i\phi}\right] = -i\hbar \cdot i\left[-\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{i\phi}\right] = \hbar\left[-\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{i\phi}\right] = \hbar Y_1^1$, confirming $m = 1$.

Solution N.4 $\displaystyle\int_0^{2\pi}\int_0^{\pi}\left|\sqrt{\tfrac{1}{4\pi}}\right|^2 \sin\theta\, d\theta\, d\phi = \frac{1}{4\pi}\int_0^{2\pi} d\phi \int_0^{\pi}\sin\theta\, d\theta = \frac{1}{4\pi}(2\pi)(2) = 1$, confirming normalization.

Solution N.5 $\displaystyle\int_0^{2\pi}\int_0^{\pi}(Y_1^0)^* Y_1^1 \sin\theta\, d\theta\, d\phi \;\propto\; \int_0^{2\pi} e^{i\phi}\, d\phi$ (the only ϕ-dependence in the integrand comes from $Y_1^1 \propto e^{i\phi}$, since Y_1^0 has no ϕ-dependence at all). This integral is $\left[\dfrac{e^{i\phi}}{i}\right]_0^{2\pi} = \dfrac{e^{2\pi i} - e^0}{i} = \dfrac{1-1}{i} = 0$, so the full double integral vanishes regardless of the (unevaluated) θ-integral: Y_1^0 and Y_1^1 are orthogonal.

Solution N.6 For $\ell = 2$: $m = -2, -1, 0, 1, 2$, so $L_z \in \{-2\hbar, -\hbar, 0, \hbar, 2\hbar\}$ — 5 distinct possible outcomes.

Solution N.7 With $m = 0$, $L_z = 0$ exactly, while $|\mathbf{L}| = \sqrt{2}\hbar \neq 0$. Geometrically, this means $\mathbf{L}$ lies entirely in the xy-plane (perpendicular to the z-axis): $\cos\theta = L_z/|\mathbf{L}| = 0$, so $\theta = 90^\circ$.

Solution N.8 For $\ell = 1$: $\cos\theta_{\min} = 1/\sqrt{2} = 0.7071$. For $\ell = 5$: $\cos\theta_{\min} = 5/\sqrt{30} = 0.9129$. Since 0.9129 is closer to 1 than 0.7071, the $\ell = 5$ atom's angular momentum vector points more nearly along the z-axis.

Solution N.9 Using the product rule with $\hat{A} = \hat{L}_z$: $[\hat{L}_z, \hat{L}_x^2] = [\hat{L}_z, \hat{L}_x]\hat{L}_x + \hat{L}_x[\hat{L}_z, \hat{L}_x] = (i\hbar\hat{L}_y)\hat{L}_x + \hat{L}_x(i\hbar\hat{L}_y) = i\hbar(\hat{L}_y\hat{L}_x + \hat{L}_x\hat{L}_y)$. Similarly, $[\hat{L}_z, \hat{L}_y^2] = [\hat{L}_z, \hat{L}_y]\hat{L}_y + \hat{L}_y[\hat{L}_z, \hat{L}_y] = (-i\hbar\hat{L}_x)\hat{L}_y + \hat{L}_y(-i\hbar\hat{L}_x) = -i\hbar(\hat{L}_x\hat{L}_y + \hat{L}_y\hat{L}_x)$, using $[\hat{L}_z, \hat{L}_y] = -[\hat{L}_y, \hat{L}_z] = -i\hbar\hat{L}_x$. Since $\hat{L}_y\hat{L}_x + \hat{L}_x\hat{L}_y = \hat{L}_x\hat{L}_y + \hat{L}_y\hat{L}_x$ (identical expressions, just reordered by ordinary commutativity of addition), these two contributions are exact negatives of one another and

cancel: $[\hat{L}_z, \hat{L}_x^2] + [\hat{L}_z, \hat{L}_y^2] = 0$. Finally $[\hat{L}_z, \hat{L}_z^2] = 0$ trivially (any operator commutes with itself and its powers). Summing all three: $[\hat{L}_z, \hat{L}^2] = 0$, i.e., $[\hat{L}^2, \hat{L}_z] = 0$.

Solution N.10 Following the parallel to Problem 6.10 exactly: $[\hat{L}_z, \hat{L}_+] = [\hat{L}_z, \hat{L}_x] + i[\hat{L}_z, \hat{L}_y] = i\hbar\hat{L}_y + i(-i\hbar\hat{L}_x) = i\hbar\hat{L}_y + \hbar\hat{L}_x = \hbar(\hat{L}_x + i\hat{L}_y) = \hbar\hat{L}_+$. By the same ladder-operator logic as Chapter 6 (where $[\hat{N}, \hat{a}^\dagger] = \hat{a}^\dagger$ meant $\hat{a}^\dagger$ raises the $\hat{N}$-eigenvalue by exactly 1), $[\hat{L}_z, \hat{L}_+] = \hbar\hat{L}_+$ means $\hat{L}_+$ raises the $\hat{L}_z$-eigenvalue (m) of any state by exactly $\hbar$, i.e., by one unit of m — confirming $\hat{L}_+$ is a genuine raising operator for the magnetic quantum number, exactly as its name suggests.

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Solutions for Chapter 8

Solution N.1 $E_1 = -\hbar^2/2m_e a_0^2 = -(1.05457 \times 10^{-34})^2/[2(9.10938 \times 10^{-31})(5.29177 \times 10^{-11})^2] = -2.1799 \times 10^{-18}\,\text{J} = -13.6057\,\text{eV}$, confirming the claimed value exactly.

Solution N.2 For $n = 4$: $\ell = 0, 1, 2, 3$ are allowed, with $2\ell + 1 = 1, 3, 5, 7$ values of m respectively. The total is $1 + 3 + 5 + 7 = 16 = 4^2$, confirming the general n^2 degeneracy formula.

Solution N.3 $\dfrac{d}{dr}\left[r^2 e^{-2r/a_0}\right] = 2re^{-2r/a_0} - \dfrac{2}{a_0} r^2 e^{-2r/a_0} = 2re^{-2r/a_0}\left(1 - \dfrac{r}{a_0}\right)$. Setting this to zero: either $r = 0$ (a minimum, not the probability peak) or $1 - r/a_0 = 0 \Rightarrow r = a_0$, confirming the most probable radius is exactly a_0.

Solution N.4 $R_{20}(r) \propto \left(2 - \dfrac{r}{a_0}\right)$, which vanishes at $r = 2a_0$. This is exactly the location of the dip to zero visible in the $2s$ curve of Figure 8.1 (the "one radial node" mentioned in Section 8.3).

Solution N.5 For $2s$ ($n = 2, \ell = 0$): $\langle r \rangle = \frac{a_0}{2}[3(4) - 0] = 6a_0$. For $2p$ ($n = 2, \ell = 1$): $\langle r \rangle = \frac{a_0}{2}[12 - 2] = 5a_0$. Indeed $6a_0 > 5a_0$. Physically, the $2p$ state's nonzero angular momentum ($\ell = 1$) contributes a repulsive centrifugal term to the effective radial potential (Section 8.1) that pushes the electron's probability distribution outward less than one might naively expect purely from energy, but more precisely: the $2s$ state (no centrifugal barrier) can access small r more freely and also has more amplitude at very large r overall (reflected in its node structure and broader radial extent), giving it the larger mean radius despite sharing the same energy.

Solution N.6 Since both ψ_{210} and ψ_{200} have the identical energy E_2 (Chapter 8's central degeneracy result), Problem 3.8's formula $\omega = (E_2 - E_1)/\hbar$ applied here gives $\omega = (E_2 - E_2)/\hbar = 0$: the superposition's probability density does *not* oscillate in time at all, despite being a superposition of two different spatial states. This is a direct, concrete illustration of degeneracy's physical meaning (Section 8.4): only *energy differences* drive the time-dependent interference used throughout Chapter 13's discussion of spectroscopy, so a superposition of degenerate states behaves, in this respect, just like a single stationary state.

Solution N.7 $E_1(\text{He}^+) = Z^2(13.6057\ \text{eV}) = 4(13.6057) = 54.423\ \text{eV}$, so the ionization energy is 54.423 eV. The most probable radius scales as $a_0/Z = a_0/2 = 0.5\,a_0$ (using the Problem 8.3 result, rescaled).

Solution N.8 Both the infinite square well and the harmonic oscillator are strictly *one-dimensional* potentials, with no angular coordinate and hence no orbital angular momentum quantum number ℓ at all — there is nothing for such a restriction to apply to. The

$\ell \leq n-1$ rule is a specifically three-dimensional, central-potential phenomenon (arising from the interplay of the radial equation and the centrifugal term, Section 8.1), with no one-dimensional counterpart.

Solution N.9 The claim conflates "same total energy E_2" with "same wavefunction." As Problem 8.5 shows explicitly, $2s$ and $2p$ have different $\langle r \rangle$ ($6a_0$ versus $5a_0$) despite sharing the identical energy E_2; more fundamentally, Figure 8.1 and Figure 8.2 show their radial distributions and angular shapes are visibly, qualitatively different (one node versus zero radial nodes; spherically symmetric versus two-lobed). Degeneracy in hydrogen (Section 8.4) means several *different* spatial wavefunctions happen to share one energy value — it does not mean those wavefunctions, or any property computed from them besides energy itself, are identical.

Solution N.10 $\mu_D/\mu_H = \dfrac{m_e(2m_p)/(m_e+2m_p)}{m_e m_p/(m_e+m_p)} = \dfrac{2m_p(m_e+m_p)}{m_p(m_e+2m_p)} = \dfrac{2(m_e+m_p)}{m_e+2m_p} = 1.000272$ (numerically, using $m_e/m_p = 1/1836.15$). Since $E_n \propto \mu$, transition energies (and hence $1/\lambda$) scale with μ, so $\lambda_D/\lambda_H = \mu_H/\mu_D = 0.999728$, a fractional shift of about -0.0272% — tiny, but easily resolved by the high-precision spectroscopy available even in 1932, which is exactly how Urey first detected deuterium: a faint spectral companion line shifted by almost exactly this predicted amount from each ordinary hydrogen line.

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Solutions for Chapter 9

Solution N.1 For $\hat{x}$: $\langle\phi|\hat{x}\psi\rangle = \int \phi^*(x\psi)\,dx = \int (x\phi)^*\psi\,dx = \langle\hat{x}\phi|\hat{x}\phi\rangle\,\psi$ (moving the real number x freely in or out of the complex conjugate), confirming Hermiticity. For $\hat{p}$: $\langle\phi|\hat{p}\psi\rangle = \int \phi^*(-i\hbar\psi')\,dx$. Integrating by parts, $\int \phi^*(-i\hbar\psi')dx = \left[-i\hbar\phi^*\psi\right]_{-\infty}^{\infty} + \int i\hbar(\phi^*)'\psi\,dx$. The boundary term vanishes since any normalizable $\phi, \psi \to 0$ as $x \to \pm\infty$ (Chapter 3), leaving $\int(-i\hbar\phi')^*\psi\,dx = \langle\hat{p}\phi|\hat{p}\phi\rangle\,\psi$ (using $(i\phi^{*\prime}) = (-i\phi')^*$), confirming $\hat{p}$ is Hermitian as well.

Solution N.2 Since $\hat{p}$ is Hermitian (Problem 9.1) and squaring a Hermitian operator preserves Hermiticity, $\hat{p}^2$ is Hermitian, and so is $\hat{p}^2/2m$ (a real constant multiple of a Hermitian operator is still Hermitian). Since $\hat{x}$ is Hermitian and $V(x)$ is real, $V(\hat{x})$ (built purely from multiplication by real numbers) is Hermitian by the same argument as for $\hat{x}$ itself in Problem 9.1. The sum of two Hermitian operators is Hermitian (directly from the linearity of the defining inner-product relation), so $\hat{H} = \hat{p}^2/2m + V(\hat{x})$ is Hermitian.

Solution N.3 Directly, $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} = (\hat{B}\hat{A} + 3i) - \hat{B}\hat{A} = 3i$. Then $[\hat{B}, \hat{A}] = -[\hat{A}, \hat{B}] = -3i$.

Solution N.4 Using the identity with $\hat{A} = \hat{x}, \hat{B} = \hat{p}$: $[\hat{x}, \hat{p}^2] = \hat{p}[\hat{x}, \hat{p}] + [\hat{x}, \hat{p}]\hat{p} = \hat{p}(i\hbar) + (i\hbar)\hat{p} = 2i\hbar\hat{p}$.

Solution N.5 By the generalized uncertainty principle, $\Delta x\,\Delta(KE) \geq \frac{1}{2}\left|\langle[\hat{x}, \hat{p}^2/2m]\rangle\right| = \frac{1}{2}\left|\frac{1}{2m}\langle 2i\hbar\hat{p}\rangle\right| = \frac{\hbar}{2m}\left|\langle\hat{p}\rangle\right|$, using the Problem 9.4 result.

Solution N.6 $[\hat{x}, \hat{H}] = [\hat{x}, \hat{p}^2]/2m = 2i\hbar\hat{p}/2m = \dfrac{i\hbar}{m}\hat{p}$, which is zero (as an operator identity) only if $\hat{p}$ itself is identically the zero operator — never true for any physically interesting system, since a particle with exactly zero momentum in every possible state is not a meaningful quantum system. So $\hat{x}$ and $\hat{H}$ are, in general, incompatible observables: position and total energy cannot generally be simultaneously sharp, consistent with the fact that energy eigenstates (Chapters 4–8) are generally spread out in space, not localized at a single x.

Solution N.7 Since $[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z \neq 0$ in general (it vanishes only in states where $\langle\hat{L}_z\rangle = 0$ identically, not as an operator identity), $\hat{L}_x$ and $\hat{L}_y$ do not commute. The corresponding uncertainty relation is $\Delta L_x\,\Delta L_y \geq \frac{1}{2}\hbar\left|\langle\hat{L}_z\rangle\right|$. For a state with $\langle L_z\rangle = 0$, the right-hand side is exactly zero, so this particular bound places *no* lower limit on the product $\Delta L_x \Delta L_y$ — it would be satisfied even if $\Delta L_x = 0$ (though the state might still have nonzero ΔL_x and ΔL_y for other reasons; the uncertainty principle here simply imposes no obstruction in this special case).

Solution N.8 For compatible operators, $\langle[\hat{A},\hat{B}]\rangle = 0$ identically (since $[\hat{A},\hat{B}] = 0$ as an operator), so the generalized uncertainty principle gives $\Delta A\,\Delta B \geq 0$ — no nontrivial lower bound at all. This is exactly consistent with Section 9.3's definition of compatibility: compatible observables *can* (though need not always) be simultaneously sharp, i.e., $\Delta A = \Delta B = 0$ is not forbidden, which is precisely what a zero lower bound permits.

Solution N.9 Using the product rule with $\hat{A} = \hat{L}_z$: $[\hat{L}_z,\hat{L}_x^2] = [\hat{L}_z,\hat{L}_x]\hat{L}_x + \hat{L}_x[\hat{L}_z,\hat{L}_x] = (i\hbar\hat{L}_y)\hat{L}_x + \hat{L}_x(i\hbar\hat{L}_y) = i\hbar(\hat{L}_y\hat{L}_x + \hat{L}_x\hat{L}_y)$, and $[\hat{L}_z,\hat{L}_y^2] = [\hat{L}_z,\hat{L}_y]\hat{L}_y + \hat{L}_y[\hat{L}_z,\hat{L}_y] = (-i\hbar\hat{L}_x)\hat{L}_y + \hat{L}_y(-i\hbar\hat{L}_x) = -i\hbar(\hat{L}_x\hat{L}_y + \hat{L}_y\hat{L}_x)$. These two terms are negatives of each other and cancel exactly; together with $[\hat{L}_z,\hat{L}_z^2] = 0$ trivially, the total is $[\hat{L}_z,\hat{L}^2] = 0$, i.e., $[\hat{L}^2,\hat{L}_z] = 0$.

Solution N.10 $\langle\hat{A}\rangle = \langle\psi|\hat{A}\psi\rangle = \langle\psi|a\psi\rangle = a\,\langle\psi|\psi\rangle\,\psi = a$ (using normalization $\langle\psi|\psi\rangle\,\psi = 1$ and that a is real, Section 9.2, so no complex conjugation issue arises). Similarly, $\langle\hat{A}^2\rangle = \langle\psi|\hat{A}^2\psi\rangle = \langle\psi|\hat{A}(a\psi)\rangle = a\,\langle\psi|\hat{A}\psi\rangle = a\,\langle\psi|a\psi\rangle = a^2\,\langle\psi|\psi\rangle\,\psi = a^2$. So $\Delta A = \sqrt{\langle\hat{A}^2\rangle - \langle\hat{A}\rangle^2} = \sqrt{a^2 - a^2} = 0$: an eigenstate of $\hat{A}$ has exactly zero uncertainty in A, exactly the sense in which a is a "sharp" or definite value for that state.

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Solutions for Chapter 10

Solution N.1 Substituting $x_1 \leftrightarrow x_2$: $\Psi_S(x_2, x_1) = \frac{1}{\sqrt{2}}[\phi_a(x_2)\phi_b(x_1) + \phi_a(x_1)\phi_b(x_2)]$, which is term-for-term identical to $\Psi_S(x_1, x_2)$ (the two terms simply swap order, and addition is commutative): $\Psi_S(x_2, x_1) = \Psi_S(x_1, x_2)$, symmetric. For Ψ_A: $\Psi_A(x_2, x_1) = \frac{1}{\sqrt{2}}[\phi_a(x_2)\phi_b(x_1) - \phi_a(x_1)\phi_b(x_2)] = -\frac{1}{\sqrt{2}}[\phi_a(x_1)\phi_b(x_2) - \phi_a(x_2)\phi_b(x_1)] = -\Psi_A(x_1, x_2)$, antisymmetric.

Solution N.2 $\hat{S}_z|\uparrow\rangle = \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\binom{1}{0} = \frac{\hbar}{2}\binom{1}{0} = \frac{\hbar}{2}|\uparrow\rangle$. $\hat{S}_z|\downarrow\rangle = \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\binom{0}{1} = \frac{\hbar}{2}\binom{0}{-1} = -\frac{\hbar}{2}|\downarrow\rangle$.

Solution N.3 Characteristic equation: $(0-\lambda)(0-\lambda) - \left(\frac{\hbar}{2}\right)^2 = 0 \Rightarrow \lambda^2 = \left(\frac{\hbar}{2}\right)^2 \Rightarrow \lambda = \pm\frac{\hbar}{2}$. For $\lambda = +\hbar/2$: $\begin{pmatrix}-\hbar/2 & \hbar/2\\ \hbar/2 & -\hbar/2\end{pmatrix}\mathbf{v} = 0 \Rightarrow v_1 = v_2$, normalized eigenvector $\frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$. For $\lambda = -\hbar/2$: $v_1 = -v_2$, normalized eigenvector $\frac{1}{\sqrt{2}}(|\uparrow\rangle - |\downarrow\rangle)$.

Solution N.4 Filling $1s$ first (lowest energy) with 2 electrons (opposite spin, Pauli), then $2s$ with the remaining 2: configuration $1s^2 2s^2$. A third or fourth electron cannot enter $1s$ alongside the first two because $1s$ is a single spatial orbital, which by the Pauli exclusion principle (Section 10.4) can hold at most 2 electrons (one spin-up, one spin-down) — any third electron in the same spatial orbital would necessarily duplicate one of these two spin states exactly, forbidden by antisymmetry.

Solution N.5 Yes: if $\phi_a \neq \phi_b$ (different spatial orbitals), the electrons already occupy distinct spin-orbitals regardless of their spin state, so both can have the same spin ($\uparrow\uparrow$) without violating the exclusion principle — the overall wavefunction is antisymmetrized using an antisymmetric *spatial* combination (analogous to Ψ_A in Problem 10.1) paired with a symmetric spin state, rather than the symmetric-spatial/antisymmetric-spin pairing forced in the Section 10.5 helium example, where $\phi_a = \phi_b$ left antisymmetric spin as the *only* option.

Solution N.6 Swapping two identical columns of a matrix leaves the matrix completely unchanged (since the two columns are identical, exchanging their positions produces the exact same array of numbers), so its determinant D is unchanged. But the general property that swapping any two columns flips the sign of a determinant gives $D \to -D$ for the same operation. Combining, $D = -D \Rightarrow 2D = 0 \Rightarrow D = 0$.

Solution N.7 Since $s = 1$ is an integer, the Symmetrization Postulate (Section 10.3) requires a totally *symmetric* two-particle wavefunction under exchange: photons are bosons.

Solution N.8 $\Delta S_x \, \Delta S_y \geq \frac{1}{2}\hbar \left|\langle \hat{S}_z \rangle\right|$. In the state $|\uparrow\rangle$, $\langle \hat{S}_z \rangle = +\hbar/2$ exactly (Problem 9.10's general result, applied here since $|\uparrow\rangle$ is an eigenstate of $\hat{S}_z$), so the bound becomes $\Delta S_x \Delta S_y \geq \frac{1}{2}\hbar \cdot \frac{\hbar}{2} = \dfrac{\hbar^2}{4}$, strictly positive. Unlike the $\ell = 1, m = 0$ orbital case of Chapter 9, Problem 9.7 (where the analogous bound was trivially zero), $\Delta S_x \Delta S_y = 0$ is *not* allowed for the state $|\uparrow\rangle$: it must have nonzero uncertainty in both S_x and S_y.

Solution N.9 The magnetic moment produced by a classically spinning charged sphere is proportional to (charge) $\times$ (angular velocity) $\times$ (radius)^2, roughly speaking; reproducing the electron's experimentally measured magnetic moment from its extremely small known size (or, in the idealized point-particle limit, zero size) would require the sphere's surface to rotate at a speed vastly exceeding the speed of light — forbidden by special relativity. Since no physical rotating charge distribution consistent with the electron's known size can generate the observed moment without exceeding c, the literal spinning-sphere picture cannot be physically correct.

Solution N.10 Since $1s$ can hold at most 2 electrons (Problem 10.4), lithium's third electron is excluded from $1s$ by the Pauli principle regardless of energy considerations, and must occupy the next available spatial orbital, $2s$, even though this costs more energy than a (forbidden) third $1s$ electron would. The resulting ground-state configuration is $1s^2 2s^1$.

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Solutions for Chapter 11

Solution N.1 $|\psi_2^{(0)}(x)|^2 = \frac{2}{L}\sin^2(2\pi x/L)$ is also symmetric about the box's midpoint $x = L/2$ (as is $\sin^2(n\pi x/L)$ for *every* integer n, a general property of the particle-in-a-box eigenstates), so exactly half of the $n = 2$ state's probability again lies in $[0, L/2]$, giving $E_2^{(1)} = V_0/2$ by the identical symmetry argument. This makes sense because $H' = V_0$ is uniform over exactly the left half of the box for *every* eigenstate simultaneously (the perturbation itself does not depend on n), and the midpoint-symmetry of $|\psi_n|^2$ that makes the argument work is a universal feature of the infinite well's eigenstates, not special to the ground state.

Solution N.2 $E_1^{(1)} = \displaystyle\int_0^L V_0 \sin(\pi x/L)\cdot\frac{2}{L}\sin^2(\pi x/L)\,dx = \frac{2V_0}{L}\int_0^L \sin^3(\pi x/L)\,dx = \frac{2V_0}{L}\cdot\frac{4L}{3\pi} = \frac{8V_0}{3\pi} \approx 0.849\,V_0$.

Solution N.3 $E_n^{(1)} = \frac{1}{2}\epsilon m\omega^2\langle n|\hat{x}^2|n\rangle = \frac{1}{2}\epsilon m\omega^2\left(n+\frac{1}{2}\right)\frac{\hbar}{m\omega} = \frac{\epsilon\hbar\omega}{4}(2n+1)$. Expanding the exact energy for spring constant $m\omega^2(1+\epsilon)$: $E_n^{\text{exact}} = \left(n+\frac{1}{2}\right)\hbar\omega\sqrt{1+\epsilon} \approx \left(n+\frac{1}{2}\right)\hbar\omega\left(1+\frac{\epsilon}{2}\right) = \left(n+\frac{1}{2}\right)\hbar\omega + \frac{\epsilon\hbar\omega}{4}(2n+1)$, whose $O(\epsilon)$ correction is exactly $\frac{\epsilon\hbar\omega}{4}(2n+1)$ — identical to $E_n^{(1)}$, confirming first-order perturbation theory correctly reproduces the leading-order exact shift.

Solution N.4 For $n = 0$ (the ground state), $E_0^{(0)}$ is by definition the *lowest* energy of the system, so $E_0^{(0)} - E_m^{(0)} < 0$ for every $m \neq 0$ (every other unperturbed energy is higher). Since $E_0^{(2)} = \sum_{m\neq 0} |\left\langle m^{(0)}\right|\hat{H}'\left|0^{(0)}\right\rangle|^2/(E_0^{(0)} - E_m^{(0)})$ is a sum of non-negative numerators divided by these uniformly negative denominators, every term in the sum is negative or zero, so $E_0^{(2)} \leq 0$ always (strictly negative unless every matrix element happens to vanish).

Solution N.5 $E_1^{(2)} = \dfrac{|0.005|^2}{1.000 - 1.010} = \dfrac{0.000025}{-0.010} = -0.0025$ eV. Comparing magnitudes, $|E_1^{(2)}|/|\left\langle 1^{(0)}\right|\hat{H}'\left|1^{(0)}\right\rangle| = 0.0025/0.005 = 0.5$: the "second-order" correction is fully half as large as the hypothetical first-order term, despite nominally being a higher-order effect. This illustrates exactly the Note box's point in Section 11.3: a small energy denominator (here, only 0.010 eV between nearly degenerate levels) can inflate a formally higher-order correction to a magnitude fully comparable to the first-order term, signaling that perturbation theory is becoming unreliable and a degenerate/near-degenerate treatment is warranted.

Solution N.6 With $\hat{H}' = 0$: $E_n^{(1)} = \left\langle n^{(0)}\right|\hat{0}\left|n^{(0)}\right\rangle = 0$ for every n (the expectation value of the zero

operator is trivially zero in any state). Likewise, $E_n^{(2)} = \sum_{m\neq n} |\left\langle m^{(0)}\right| \hat{0} \left|n^{(0)}\right\rangle|^2/(E_n^{(0)} - E_m^{(0)}) = \sum_{m\neq n} 0/(E_n^{(0)} - E_m^{(0)}) = 0$, since every matrix element in the numerator vanishes. Both corrections vanish identically, as physically required when there is no perturbation at all.

Solution N.7 Setting $E_0^{\text{exact}} = \frac{1}{2}\hbar\omega - \lambda^2/2m\omega^2 = 0$ and solving: $\lambda^2 = m\hbar\omega^3$, so $\lambda = \sqrt{m\hbar\omega^3} = \omega\sqrt{m\hbar\omega}$. At this λ, the perturbative shift $\lambda^2/2m\omega^2 = \hbar\omega/2$ is exactly equal to the entire unperturbed level spacing $\hbar\omega$ divided by 2 — i.e., the "small" perturbation has become comparable in size to the level spacing itself, well outside the regime ($\hat{H}'$ weak relative to level spacing, per the Note box of Section 11.3) in which perturbation theory is expected to be quantitatively trustworthy, even though (as shown in the text) this particular exactly-solvable example happens to remain exactly correct at any order for this specific linear perturbation.

Solution N.8 $\langle 0|\hat{x}^2|0\rangle = \langle x^2\rangle_0 - \langle x\rangle_0^2 + \langle x\rangle_0^2 = \langle x^2\rangle_0 = \dfrac{\hbar}{2m\omega}$ (using $\langle x\rangle_0 = 0$, so $\langle x^2\rangle_0$ already equals $\langle 0|\hat{x}^2|0\rangle$ directly). This matches the general formula $\left(n+\frac{1}{2}\right)\hbar/m\omega$ at $n=0$: $\left(0+\frac{1}{2}\right)\hbar/m\omega = \hbar/2m\omega$. ✓

Solution N.9 $\langle n-1|\,\hat{x}\,|n\rangle = \sqrt{\dfrac{\hbar}{2m\omega}}\,\langle n-1|\,(\hat{a}+\hat{a}^\dagger)\,|n\rangle = \sqrt{\dfrac{\hbar}{2m\omega}}\left[\sqrt{n}\,\langle n-1|n-1\rangle + \sqrt{n+1}\,\langle n-1|n+1\rangle\right] = \sqrt{\dfrac{\hbar}{2m\omega}}\left[\sqrt{n}(1) + \sqrt{n+1}(0)\right] = \sqrt{\dfrac{\hbar}{2m\omega}}\sqrt{n}$, using $\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle$ and $\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$ together with orthonormality.

Solution N.10 In any system with a unique, well-defined ground state, $E_0^{(0)}$ is strictly lower than every other unperturbed energy (no other state shares it exactly, by definition of "ground state" as the unique minimum). Degenerate perturbation theory is only needed when the *specific* unperturbed level being corrected has one or more partner states of exactly the same energy; since the ground state has no such partner by assumption, ordinary (non-degenerate) perturbation theory always applies validly to E_0, regardless of whatever degeneracies may exist among the excited states above it.

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Solutions for Chapter 12

Solution N.1 For H^- ($Z = 1$): $\zeta_{\text{opt}} = 1 - 5/16 = 0.6875$, giving $E_{\text{trial}} = -(0.6875)^2(27.2114) = -12.86$ eV, compared to the experimental -14.35 eV — an absolute discrepancy of 1.49 eV, numerically similar in magnitude to helium's 1.52 eV discrepancy, but a much larger *fractional* error ($1.49/14.35 \approx 10.4\%$ versus helium's $1.52/79.0 \approx 1.9\%$). Since the electron-electron repulsion term ($\frac{5}{8}\zeta$) does not scale with Z while the nuclear attraction terms do, screening/correlation effects are proportionally more significant for the more weakly bound $Z = 1$ case: you should expect screening to matter *more* for H^- than for He, consistent with the larger fractional error found.

Solution N.2 For Li^+ ($Z = 3$): $\zeta_{\text{opt}} = 3 - 5/16 = 2.6875$, giving $E_{\text{trial}} = -(2.6875)^2(27.2114) = -196.54$ eV.

Solution N.3 $\langle \hat{H} \rangle(1,0) = \dfrac{5+0+0}{1+0} = 5$. $\langle \hat{H} \rangle(0,1) = \dfrac{0+3+0}{0+1} = 3$. $\langle \hat{H} \rangle(1,1) = \dfrac{5+3+4}{1+1} = \dfrac{12}{2} = 6$. The lowest value, 3 (from $(c_1, c_2) = (0,1)$), gives the best (tightest) upper bound on E_0 among the three trial states — lower is always better for a variational estimate.

Solution N.4 The variational theorem guarantees $\langle \hat{H} \rangle_{\text{trial}} \geq E_1$ (the true ground-state energy), regardless of the specific value obtained — the trial energy computed from this particular polynomial trial function must come out greater than or equal to $\pi^2\hbar^2/2mL^2$, with equality possible only if the trial function happened to exactly equal the true ground state (which it does not, since $\psi_{\text{trial}} \propto x(L-x)$ is not proportional to $\sin(\pi x/L)$).

Solution N.5 The proof (Section 12.1) relied on $E_n \geq E_0$ holding for *every* n in the expansion $|\psi_{\text{trial}}\rangle = \sum_n c_n |n\rangle$, which is true only because E_0 is defined as the lowest eigenvalue of all. For a specific excited state E_k ($k > 0$), there is no guarantee that $E_n \geq E_k$ for every n (in particular, $E_0 < E_k$), so the same chain of inequalities fails, and a trial function's energy need not overestimate E_k specifically — it could equally well fall below, above, or near E_k, depending on how much of the trial state's weight lies on states below versus above E_k.

Solution N.6 Trial function B (-11.7 eV) provides more information, since it gives the tighter (lower) upper bound: we can conclude $E_0 \leq -11.7$ eV from trial B, a stronger constraint than the $E_0 \leq -10.2$ eV that trial A alone provides (which is also implied by, and superseded by, trial B's result). Neither result alone reveals how close either bound is to the true E_0 — only that E_0 is at most -11.7 eV.

Solution N.7 $-\dfrac{8}{3\pi}(13.6057 \text{ eV}) = -0.8488 \times 13.6057 = -11.549 \text{ eV} \approx -11.5$ eV, confirming the value quoted in the text.

Solution N.8 Generalizing $E_{\text{trial}}(\zeta) = \zeta^2 - 2Z\zeta + \frac{5}{8}\zeta$ with arbitrary Z: $dE/d\zeta = 2\zeta - 2Z + \frac{5}{8} = 0 \Rightarrow \zeta_{\text{opt}} = Z - \frac{5}{16}$ (already fully general in Z, as used directly in Problems 12.1–12.2). Substituting back: $E_{\text{trial}}(\zeta_{\text{opt}}) = \left(Z - \frac{5}{16}\right)^2 - 2Z\left(Z - \frac{5}{16}\right) + \frac{5}{8}\left(Z - \frac{5}{16}\right) = \left(Z - \frac{5}{16}\right)\left[\left(Z - \frac{5}{16}\right) - 2Z + \frac{5}{8}\right] = \left(Z - \frac{5}{16}\right)\left(-Z + \frac{5}{16}\right) = -\left(Z - \frac{5}{16}\right)^2$, confirming $E_{\text{trial}} = -(Z - 5/16)^2 E_h$ for general Z.

Solution N.9 Setting $\zeta_{\text{opt}}/Z = (Z - 5/16)/Z = 1 - \dfrac{5}{16Z} \to 1$ requires $5/16Z \to 0$, i.e., $Z \to \infty$. The ratio $1 - 5/(16Z)$ approaches 1 from *below* as Z increases (since $5/16Z > 0$ for any finite Z, the ratio is always strictly less than 1). This makes physical sense: the fixed electron-electron repulsion contribution ($\frac{5}{8}\zeta$, independent of Z) becomes an ever-smaller relative correction to the nuclear attraction terms (which scale as $Z\zeta$) as Z grows, so screening becomes proportionally less important and the effective charge approaches the bare nuclear charge ever more closely, though never exactly reaching it at any finite Z.

Solution N.10 By the variational theorem, using the hydrogen 2*s* state as a trial function for the ground state would simply return $\langle \hat{H} \rangle = E_2 = -3.4014$ eV exactly (since 2*s* *is* an exact eigenstate of the hydrogen Hamiltonian, with eigenvalue E_2, not merely an approximate trial state). This is a valid upper bound on $E_1 = -13.6057$ eV ($-3.4014 \geq -13.6057$, trivially true) but a *poor* estimate of it — being an exact eigenstate guarantees $\langle \hat{H} \rangle$ equals a specific, known energy exactly, but there is no reason for that energy to be numerically close to E_1 unless the chosen state actually resembles the ground state. Choosing an excited state as a variational trial function for the ground state wastes the method's real strength (interpolating toward a low, unknown energy) by returning an exact, but here badly unrepresentative, higher eigenvalue instead.

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Solutions for Chapter 13

Solution N.1 With $Y_2^0 = \sqrt{5/16\pi}(3\cos^2\theta - 1)$ and $Y_0^0 = 1/\sqrt{4\pi}$, the θ-integral is $\int_0^\pi (3\cos^2\theta - 1)\cos\theta \sin\theta \, d\theta$. Substituting $u = \cos\theta$, $du = -\sin\theta \, d\theta$, this becomes $\int_{-1}^{1} (3u^2 - 1)u \, du = \int_{-1}^{1} (3u^3 - u) \, du$, an integral of an *odd* function of u over the symmetric interval $[-1, 1]$, which vanishes identically. The full integral (including the ϕ-integral and constant prefactors) is therefore exactly zero, confirming the $\ell = 0 \to \ell = 2$ transition is forbidden.

Solution N.2 $E_{1\to2} = \dfrac{\hbar^2}{2I}\big[2(3) - 1(2)\big] = \dfrac{\hbar^2}{2I}(4) = \dfrac{2\hbar^2}{I}$, compared to $E_{0\to1} = \dfrac{\hbar^2}{2I}\big[1(2) - 0\big] = \dfrac{\hbar^2}{I}$. The ratio is exactly 2: successive transition energies ($J \to J+1$) are proportional to $J+1$ (i.e., $E_{J\to J+1} = \hbar^2(J+1)/I$, easily verified to increase linearly with J), in clear contrast to the perfectly evenly-spaced QHO ladder of Chapter 6, where every $\Delta n = 1$ transition costs exactly the same $\hbar\omega$.

Solution N.3 $I = h/(4\pi^2\nu) = 6.62607 \times 10^{-34}/[4\pi^2(6.259 \times 10^{11})] = 2.6816 \times 10^{-47}\ \text{kg m}^2$. Then $r = \sqrt{I/\mu} = \sqrt{2.6816 \times 10^{-47}/1.6267 \times 10^{-27}} = 1.284 \times 10^{-10}$ m $= 128.4$ pm, in good agreement with the accepted 127.5 pm (within about 0.7%).

Solution N.4 The dipole selection rule $\Delta n = \pm 1$ means the matrix element $\langle 2|\,\hat{x}\,|0\rangle$ is exactly zero for the ideal harmonic oscillator (only $\Delta n = \pm 1$ matrix elements are nonzero, per Chapter 11's ladder-operator matrix elements), so Fermi's Golden Rule gives an exactly zero transition rate for $0 \to 2$ regardless of light intensity, in the strict dipole approximation. Real molecules show weak overtone absorption near this energy because real potentials are not perfectly harmonic (Chapter 6's anharmonicity discussion): the true eigenstates are not perfectly pure QHO $|n\rangle$ states, so the $0 \to 2$ matrix element, while still small, is not exactly zero.

Solution N.5 $B = \big(3.842\ \text{cm}^{-1}\big)/2 = 1.921\ \text{cm}^{-1}$, in close agreement with the CO value $B = 1.9225\ \text{cm}^{-1}$ quoted in the text (this is, in fact, essentially the same molecule's rotational constant, recovered from the line-spacing relationship instead of a single transition frequency).

Solution N.6 Fermi's Golden Rule gives a transition rate $\Gamma \propto |\,\langle f|\,\hat{H}_0'\,|i\rangle\,|^2$: a small but nonzero matrix element still gives a small but nonzero rate (hence a real, if weak, absorption or emission line), simply reduced in proportion to the square of the matrix element. A transition whose matrix element is *exactly* zero by symmetry gives $\Gamma = 0$ identically, regardless of light intensity or observation time — a qualitatively different, absolute prohibition rather than a merely quantitatively small rate.

Solution N.7 $E(3 \to 2) = 13.6057\left(\frac{1}{4} - \frac{1}{9}\right) = 1.890$ eV, and $E(\infty \to 2) = 13.6057/4 = 3.401$ eV (the series limit). Since 1.890 eV lies within 1.8–3.1 eV but 3.401 eV lies above it, the Balmer series is only *partially* within the visible range: the lower members (Hα, Hβ, etc., converging from 1.890 eV upward) are visible, but the series limit and the higher members approaching it lie in the near ultraviolet, above 3.1 eV.

Solution N.8 A rotating molecule can only interact with the oscillating electric field of light (and thus undergo a dipole-allowed transition, Section 13.3) if the transition dipole matrix element is nonzero, and for a rigid rotor this in turn requires the molecule to possess a permanent, nonzero dipole moment whose orientation (and hence direction) rotates along with the molecule; with zero permanent dipole moment, every rotational matrix element of the dipole operator vanishes identically, and no pure rotational spectrum can appear. Homonuclear diatomics such as N_2 and O_2, by symmetry, have exactly zero permanent dipole moment (their two identical atoms share electron density symmetrically), so neither shows a pure (electric-dipole) rotational spectrum.

Solution N.9 Because the transition happens at essentially fixed nuclear geometry (the Franck–Condon idea), the electron reorganizes into a new electronic state while the nuclei remain, momentarily, at their original separation — a configuration that is generally *not* the equilibrium geometry of the new electronic state. The nuclei therefore find themselves displaced from the new potential's minimum and can end up in any of several vibrational levels of the excited state (each with a different overlap with the original vibrational wavefunction, i.e., a different Franck–Condon factor), producing a whole progression of lines at different total energies (electronic + differing amounts of vibrational energy) rather than a single line corresponding to a single, unique final vibrational state.

Solution N.10 $\Delta E_{\text{min}} = \hbar/(2\Delta t) = (1.0546 \times 10^{-34})/(2 \times 1.0 \times 10^{-9}) = 5.27 \times 10^{-26}$ J $= 3.29 \times 10^{-7}$ eV. Compared to the CO transition energy of 0.4767 meV $= 4.767 \times 10^{-4}$ eV, this minimum linewidth is smaller by a factor of about 1400 ($3.29 \times 10^{-7}/4.767 \times 10^{-4} \approx 7 \times 10^{-4}$): utterly negligible for resolving this particular transition. A 1 ns observation window imposes essentially no practical limitation on resolving CO's rotational spectrum; other broadening mechanisms (collisions, Doppler broadening) dominate in practice.

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Solutions for Chapter 14

Solution N.1 Phosphorus ($Z = 15$): $1s^2 2s^2 2p^6 3s^2 3p^3$ (verify: $2+2+6+2+3 = 15$). Calcium ($Z = 20$): $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$ (verify: $2+2+6+2+6+2 = 20$).

Solution N.2 Immediately after $4s$ in the Aufbau order (Section 14.2) comes $3d$. The first element in which $3d$ begins to fill is scandium, $Z = 21$ ($[\mathrm{Ar}]3d^1 4s^2$).

Solution N.3 The naive hydrogen-like formula $E_n \propto -Z^2/n^2$ with the full, unscreened $Z = 19$ assumes the $4s$ electron feels the entire nuclear charge; in reality, it is screened by the 18 inner-shell electrons filling $n = 1, 2, 3$, so it experiences an much smaller *effective* nuclear charge, not the full $Z = 19$. Since binding energy scales as Z_{eff}^2, this substantially reduced effective charge makes the true ionization energy far smaller than the naive unscreened estimate would predict.

Solution N.4 Oxygen, $1s^2 2s^2 2p^4$: the $1s^2$ and $2s^2$ subshells are completely filled (4 electrons total in filled subshells), while the $2p^4$ subshell is partially filled (4 electrons). A completely filled subshell contributes $L = 0, S = 0$ because, for every occupied (m_ℓ, m_s) combination, the Pauli principle also forces the "partner" combination to be filled in a closed subshell, and the sums $\sum m_\ell$ and $\sum m_s$ over a complete, symmetric set of all available m_ℓ and m_s values always cancel to exactly zero.

Solution N.5 ${}^1S$: $S = 0 \Rightarrow 2S+1 = 1$; $L = 0 \Rightarrow 2L+1 = 1$; multiplicity $= 1 \times 1 = 1$. ${}^1D$: $S = 0 \Rightarrow 2S+1 = 1$; $L = 2 \Rightarrow 2L+1 = 5$; multiplicity $= 1 \times 5 = 5$. ${}^3P$: $S = 1 \Rightarrow 2S+1 = 3$; $L = 1 \Rightarrow 2L+1 = 3$; multiplicity $= 3 \times 3 = 9$. Total: $1+5+9 = 15$, matching $\binom{6}{2} = 15$ exactly.

Solution N.6 For any given term with orbital angular momentum L, the microstates belonging to that term include one (and only one, from that term) with $M_L = +L$, the maximum value of M_L attainable within that term, and no microstate anywhere in the full scheme can have M_L exceeding the largest L present among all the terms (since $M_L \leq L$ always, for any state belonging to a term with total orbital angular momentum L). Since the maximum M_L found by direct enumeration across *all* 15 microstates is 2, at least one term with $L \geq 2$ must be present, and since $L = 2$ (a D term) is the largest value consistent with this maximum, a D term must appear.

Solution N.7 Maximum-S arrangement: all three electrons in different m_ℓ orbitals ($m_\ell = -1, 0, +1$), all with the same spin, say $m_s = +\frac{1}{2}$ each (Pauli-allowed, since all three spin-orbitals are distinct). This gives $S = \frac{3}{2}$ (three parallel spins) and $M_L = (-1) + 0 + (+1) = 0$; since this is the *only* way to achieve $S = \frac{3}{2}$, and $M_L = 0$ is the only value obtained, $L = 0$ (an S term). With $L = 0, S = \frac{3}{2}$: $J = |L - S|$ to $L + S$ gives only $J = \frac{3}{2}$. The ground term is ${}^4S_{3/2}$, confirmed.

Solution N.8 Oxygen's $2p^4$ configuration gives the same terms as carbon's $2p^2$ (Problem 14.9's electron-hole symmetry): ${}^1S$, ${}^1D$, ${}^3P$. Hund's rule 1 again selects ${}^3P$ ($S = 1$, the unique maximum-spin term). Rule 2 is not needed. For rule 3: oxygen's $2p^4$ is *more* than half filled (4 of 6 possible electrons), so (unlike carbon) the *maximum* $J = L + S = 1 + 1 = 2$ is lowest in energy. The ground term is ${}^3P_2$, contrasting directly with carbon's ${}^3P_0$.

Solution N.9 A subshell with $n_{\max}$ available spin-orbitals, occupied by k electrons, can equally well be described by its $n_{\max} - k$ *unoccupied* spin-orbitals ("holes"). The total M_L and M_S of the occupied electrons are the negatives of the M_L, M_S that the same unoccupied spin-orbitals would carry if instead treated as occupied "hole particles," so the set of (L, S) combinations — and hence term symbols — obtainable from k electrons is identical to that obtainable from $n_{\max} - k$ electrons (here, 2 electrons or, equivalently, 2 holes in the 6-spin-orbital $2p$ subshell), giving carbon ($2p^2$) and oxygen ($2p^4$, i.e., 2 holes) the same term symbols.

Solution N.10 By analogy with a single $2p^1$ electron, a single hole in an otherwise-filled $2p^6$ shell (fluorine's $2p^5$) gives the term ${}^2P$, with $J = \frac{1}{2}$ or $\frac{3}{2}$ possible. Since fluorine's $2p^5$ is more than half filled (Hund's rule 3, maximum-J case, exactly as for oxygen in the text), the ground state is the *larger* J value: ${}^2P_{3/2}$.

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Solutions for Chapter 15

Solution N.1 Direct evaluation gives $E_{\text{tot},+}(2.0) = -0.5538\ E_h$ and $E_{\text{tot},+}(3.0) = -0.5591\ E_h$; both are indeed higher (less negative, hence energetically less favorable) than the minimum $-0.5648\ E_h$ found at $R_e = 2.49$ bohr, exactly as required for a true minimum of the curve.

Solution N.2 $E_{\text{tot},-}(1) = +0.5454$, $E_{\text{tot},-}(2) = -0.1609$, $E_{\text{tot},-}(3) = -0.3676$, $E_{\text{tot},-}(4) = -0.4449$ (all in E_h). These values increase monotonically toward $-0.5\ E_h$ as R increases, with no dip below any smaller-R value anywhere in this range — confirming a purely repulsive, monotonically decreasing-with-R curve and no interior (bonding) minimum.

Solution N.3 At a point equidistant from A and B, $|\mathbf{r} - \mathbf{R}_A| = |\mathbf{r} - \mathbf{R}_B|$, so (since the $1s$ orbital depends only on distance from its own center) $1s_A(\mathbf{r}) = 1s_B(\mathbf{r})$ exactly at every such point. Then $\psi_- \propto 1s_A - 1s_B = 0$ at every equidistant point, while $\psi_+ \propto 1s_A + 1s_B = 2(1s_A) \neq 0$ there.

Solution N.4 With 3 electrons: 2 fill σ_{1s} (bonding) and 1 enters σ^*_{1s} (antibonding). Bond order $= (2-1)/2 = 0.5$, nonzero — so He_2^+ is expected to be a genuine, if weakly, bound species, in sharp contrast to neutral He_2 (bond order 0, Section 15.4), which is not bound at all by this mechanism. (This qualitative prediction is correct: He_2^+ is an experimentally well-characterized bound molecular ion.)

Solution N.5 With 6 electrons: 2 in σ_{1s}, 2 in σ^*_{1s} (net contribution to bond order exactly zero, as these cancel, treating the $1s$ pair as an inert core per Chapter 14), and the remaining 2 electrons enter the next available (bonding) orbital, σ_{2s}. Bond order $= (2-0)/2 = 1$ from the valence electrons alone (the core contributing zero net), exactly matching H_2's bond order of 1 — consistent with Li_2 being a genuine, if weaker, single-bonded molecule.

Solution N.6 For N_2 (14 electrons), filling through σ_{2p}: bonding electrons ($\sigma_{1s}, \sigma_{2s}, \pi_{2p} \times 2, \sigma_{2p}$) total $2+2+4+2 = 10$; antibonding ($\sigma^*_{1s}, \sigma^*_{2s}$) total $2+2 = 4$. Bond order $= (10-4)/2 = 3$ (the famous nitrogen triple bond). For N_2^+ (13 electrons, removing one electron from the highest-energy occupied orbital, σ_{2p}, a *bonding* orbital): bonding electrons drop to 9, antibonding remains 4, giving bond order $(9-4)/2 = 2.5$. Since bond order decreases from 3 to 2.5, the N–N bond in N_2^+ is predicted to be *weaker* than in neutral N_2.

Solution N.7 By direct analogy with $\sigma_{1s} = N_+(1s_A + 1s_B)$ and $\sigma^*_{1s} = N_-(1s_A - 1s_B)$, the $2s$-derived orbitals are $\sigma_{2s} = N'_+(2s_A + 2s_B)$ (bonding, constructive overlap) and $\sigma^*_{2s} = N'_-(2s_A - 2s_B)$ (antibonding, destructive overlap, with a node at the midpoint). Since atomic $2s$ orbitals lie well above atomic $1s$ orbitals in energy (Chapter 8), and

molecular orbitals built from a given pair of atomic orbitals lie in an energy range roughly centered near, but split around, the parent atomic orbital energy, σ_{2s} (built from the already much higher $2s$ atomic orbitals) is expected to lie *above* σ^*_{1s} in energy, consistent with the ordering given in Section 15.3.

Solution N.8 Expanding $\psi_+(\mathbf{r}_1)\psi_+(\mathbf{r}_2) = N_+^2\left[\phi_A(\mathbf{r}_1)\phi_A(\mathbf{r}_2) + \phi_A(\mathbf{r}_1)\phi_B(\mathbf{r}_2) + \phi_B(\mathbf{r}_1)\phi_A(\mathbf{r}_2) + \phi_B(\mathbf{r}_1)\phi_B(\mathbf{r}_2)\right]$: the first and last terms, $\phi_A(\mathbf{r}_1)\phi_A(\mathbf{r}_2)$ and $\phi_B(\mathbf{r}_1)\phi_B(\mathbf{r}_2)$, describe both electrons occupying the same atomic orbital (both on A, or both on B) — precisely the "ionic" H^-H^+ (or H^+H^-) configurations. The simple valence bond wavefunction, $\phi_A(\mathbf{r}_1)\phi_B(\mathbf{r}_2) + \phi_A(\mathbf{r}_2)\phi_B(\mathbf{r}_1)$, contains only the middle two ("covalent," one electron on each atom) terms, with no such same-atom contributions at all.

Solution N.9 Percentage error: $(2.79 - 1.76)/2.79 \times 100\% = 36.9\%$, substantially larger than the roughly 25% error in R_e. The simple, unoptimized LCAO treatment predicts the equilibrium bond length considerably more reliably than the dissociation energy — geometry (where the wavefunction curvature matters) is captured better than absolute binding energy (which is sensitive to the accumulated error in the trial function's overall shape) by this crude two-integral model.

Solution N.10 The one-electron H_2^+ problem has no electron-electron repulsion term at all, so its Hamiltonian, expressed in the appropriate two-center (prolate spheroidal) coordinates, separates cleanly into independent equations for each coordinate, exactly paralleling how hydrogen's single-center Coulomb problem separated in spherical coordinates (Chapter 8). A many-electron diatomic such as Li_2 or N_2 has electron-electron repulsion terms coupling every pair of electrons' coordinates together (exactly the obstruction already identified for helium in Chapter 12), which prevents this kind of clean coordinate separation regardless of which coordinate system is chosen.

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Solutions for Chapter 16

Solution N.1 Expanding $e^{-ax} \approx 1 - ax + \frac{1}{2}a^2x^2$ (through second order), $1 - e^{-ax} \approx ax - \frac{1}{2}a^2x^2 = ax\left(1 - \frac{1}{2}ax\right)$. Squaring: $(1 - e^{-ax})^2 \approx a^2x^2\left(1 - \frac{1}{2}ax\right)^2 \approx a^2x^2$ to leading (quadratic) order in x (the correction terms are cubic and higher). So $V(x) = D_e(1 - e^{-ax})^2 \approx D_e a^2 x^2$ for small x, confirming the harmonic limit.

Solution N.2 $\tilde{\nu}_{0\to1} = \tilde{\omega}_e - 2\tilde{\omega}_e x_e = 2169.8 - 2(13.3) = 2143.2\ \text{cm}^{-1}$, matching the 2143 cm^{-1} value used in Chapter 6 to within rounding.

Solution N.3 Using $E_v/hc = \tilde{\omega}_e(v + \frac{1}{2}) - \tilde{\omega}_e x_e(v + \frac{1}{2})^2$ with $\tilde{\omega}_e = 2990.9$, $\tilde{\omega}_e x_e = 52.8\ \text{cm}^{-1}$: the fundamental ($v$=0 $\to$ 1) is 2885.3 cm^{-1} (Chapter 16 main text), and the first overtone (v=0 $\to$ 2) is $\tilde{\nu}_{0\to2} = E_2/hc - E_0/hc = 5665.0\ \text{cm}^{-1}$. Twice the fundamental would be $2(2885.3) = 5770.6\ \text{cm}^{-1}$; the actual overtone (5665.0 cm^{-1}) is *less* than this, consistent with the converging (sub-harmonic) level spacing expected as v increases toward dissociation.

Solution N.4 Using $\tilde{\nu}_0 = 2143\,\text{cm}^{-1}$ and $B = 1.9225\,\text{cm}^{-1}$: R-branch, $\tilde{\nu}_R(J) = 2143 + 2(1.9225)(J + 1)$, giving $R(0) = 2146.85$, $R(1) = 2150.69$, $R(2) = 2154.54\ \text{cm}^{-1}$. P-branch, $\tilde{\nu}_P(J) = 2143 - 2(1.9225)J$, giving $P(1) = 2139.16$, $P(2) = 2135.31$, $P(3) = 2131.47\ \text{cm}^{-1}$.

Solution N.5 A purely vibrational transition with $\Delta J = 0$ would require the dipole matrix element connecting the same rotational state J in both the initial and final vibrational levels; but the relevant angular integral is exactly the same $\ell = 0 \to \ell = 0$ (i.e., $\Delta J = 0$) case forbidden by the parity/selection-rule argument of Chapter 13 (Section 13.3): the dipole operator connects states differing in angular momentum by ±1, not 0, so a $\Delta J = 0$ transition is dipole-forbidden for the same fundamental reason $\ell = 0 \to \ell = 0$ (or any $\Delta\ell = 0$ transition) is forbidden there, leaving the band-origin position as a "missing" line between the P and R branches.

Solution N.6 From Problem 16.4: $R(0) - P(1) = 2146.845 - 2139.155 = 7.690\ \text{cm}^{-1}$, and $4B = 4(1.9225) = 7.690\ \text{cm}^{-1}$: exact agreement, as claimed.

Solution N.7 $B = 8.02/4 = 2.005\ \text{cm}^{-1}$. Then $I = h/(8\pi^2 cB) = 6.62607 \times 10^{-34}/[8\pi^2(2.998 \times 10^{10}\,\text{cm/s})(2.005\,\text{cm}^{-1})] = 1.396 \times 10^{-46}\,\text{kg m}^2$, and $r = \sqrt{I/\mu} = \sqrt{1.396 \times 10^{-46}/1.653 \times 10^{-27}} = 2.906 \times 10^{-10}\,\text{m} = 290.6\,\text{pm}$, as determined directly from the stated (illustrative) gap and reduced mass.

Solution N.8 With nearly identical equilibrium bond lengths in both electronic states, the ground-state vibrational wavefunction ($v'' = 0$) overlaps most strongly with the *same* low-v' level of the upper state (their classical turning points and central regions coincide closely), so the Franck–Condon factor is large for $v' = v'' = 0$ and falls off rapidly

for larger v': the spectrum should be dominated by a single strong 0–0 line, with only weak higher-v' lines, rather than an extended progression.

Solution N.9 Setting $dE_v/dv = hc\left[\tilde{\omega}_e - 2\tilde{\omega}_e x_e\left(v+\tfrac{1}{2}\right)\right] = 0$ gives $v_{\max} + \tfrac{1}{2} = \tilde{\omega}_e/2\tilde{\omega}_e x_e$. Substituting back into $E_v/hc = \tilde{\omega}_e(v+\tfrac{1}{2}) - \tilde{\omega}_e x_e(v+\tfrac{1}{2})^2$ at this value: $E_{v_{\max}}/hc = \tilde{\omega}_e \cdot \frac{\tilde{\omega}_e}{2\tilde{\omega}_e x_e} - \tilde{\omega}_e x_e\left(\frac{\tilde{\omega}_e}{2\tilde{\omega}_e x_e}\right)^2 = \frac{\tilde{\omega}_e^2}{2\tilde{\omega}_e x_e} - \frac{\tilde{\omega}_e^2}{4\tilde{\omega}_e x_e} = \frac{\tilde{\omega}_e^2}{4\tilde{\omega}_e x_e}$, confirming $D_e = \tilde{\omega}_e^2/4\tilde{\omega}_e x_e$.

Solution N.10 $E_0/hc = \tilde{\omega}_e/2 - \tilde{\omega}_e x_e/4 = 2990.9/2 - 52.8/4 = 1495.45 - 13.2 = 1482.25$ cm^{-1}, converting to 0.1838 eV (using 1 cm^{-1} $= 1.23984 \times 10^{-4}$ eV, Appendix A). Then $D_0 = D_e - E_0 = 5.251 - 0.184 = 5.068$ eV, still substantially above the experimental 4.43 eV (a remaining discrepancy of about 0.64 eV, or roughly 14%). As flagged in the Note box of Section 16.1, this persistent overestimate is not resolved by the zero-point-energy correction alone; it reflects the fundamental limitation that the two-constant Morse extrapolation, fit only to data near the bottom of the well, does not capture the true potential's behavior all the way out to actual dissociation.

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Solutions for Chapter 17

Solution N.1 Perturbation theory (Chapter 11) requires the true Hamiltonian to be "close to" some already-*exactly*-solved $\hat{H}^{(0)}$, but essentially no molecule beyond hydrogen has an available exact solution to perturb away from (electron-electron repulsion is not a small correction, as the helium discussion of Chapter 12 emphasized). The linear variational method, by contrast, requires only a choice of basis functions and guarantees (via the variational theorem) a meaningful, systematically improvable upper bound on the ground-state energy regardless of how the basis compares to any exactly-solved reference problem, making it the natural general-purpose starting point.

Solution N.2 $\frac{d}{dr}e^{-\alpha r^2} = -2\alpha r\, e^{-\alpha r^2}$, which equals 0 at $r = 0$. $\frac{d}{dr}e^{-\zeta r} = -\zeta e^{-\zeta r}$, which equals $-\zeta \neq 0$ at $r = 0$. The Slater orbital's nonzero-slope "kink" at the nucleus reflects the true Coulomb potential's singularity there; a Gaussian, being locally flat (zero slope) at its center, cannot reproduce this sharply peaked behavior near $r = 0$ no matter how its width is optimized, which is exactly why a single Gaussian's fit to e^{-r} is worst in this region (Section 17.2), even though its RMS error averaged over the full range is only moderately large.

Solution N.3 Since the ground-state density $\rho(\mathbf{r})$ is a function of only 3 spatial coordinates, *regardless of the number of electrons* N, while the wavefunction $\Psi(\mathbf{r}_1, \ldots, \mathbf{r}_N)$ is a function of $3N$ coordinates, a method that works directly and only with $\rho(\mathbf{r})$ (as DFT does, per the Hohenberg–Kohn theorem) need not represent or store an object whose dimensionality grows with system size, unlike direct wavefunction methods — in principle allowing far more favorable computational scaling as N increases.

Solution N.4 Since the exact Hartree–Fock orbitals are found variationally (Section 17.2), the resulting Hartree–Fock wavefunction is, by construction, a specific (if highly optimized) trial wavefunction — a single Slater determinant, not the true, fully correlated ground-state wavefunction. The variational theorem of Chapter 12 guarantees $\langle \hat{H} \rangle_{\text{trial}} \geq E_{\text{exact}}$ for *any* normalized trial wavefunction, including this one, so $E_{\text{HF}} \geq E_{\text{exact}}$ always, meaning $E_{\text{correlation}} = E_{\text{exact}} - E_{\text{HF}} \leq 0$ necessarily, regardless of basis set size (a larger basis can only improve E_{HF} toward E_{exact}, never below it).

Solution N.5 Normalization: $\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2 = \frac{9}{25} + \frac{16}{25} = \frac{25}{25} = 1$. ✓ The probability of measuring $|1\rangle$ is $|\beta|^2 = \left(\frac{4}{5}\right)^2 = \frac{16}{25} = 0.64$.

Solution N.6 $|\psi_1\rangle \otimes |\psi_2\rangle = \frac{1}{2}(|0\rangle + |1\rangle) \otimes (|0\rangle + |1\rangle) = \frac{1}{2}\left[\, |00\rangle + |01\rangle + |10\rangle + |11\rangle \,\right]$, so each of the four coefficients is $\frac{1}{2}$, all equal in magnitude — unlike the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, which has nonzero coefficients on only $|00\rangle$ and $|11\rangle$, with $|01\rangle$ and $|10\rangle$ coefficients exactly zero.

Solution N.7 Both are two-level quantum systems described by exactly the same mathematical object: a normalized two-component complex vector $\alpha\,|\text{state } 1\rangle + \beta\,|\text{state } 2\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$, obeying the identical spin-$\frac{1}{2}$ algebra of Chapter 10 (the same Pauli matrices, the same commutation relations) regardless of which physical system realizes it. The labels "$|0\rangle, |1\rangle$" versus "$|\uparrow\rangle, |\downarrow\rangle$" are simply different physical interpretations (an encoded bit of information, versus a literal spin-angular-momentum projection) attached to the same underlying two-dimensional Hilbert-space structure.

Solution N.8 Matching coefficients of $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ in $(a\,|0\rangle + b\,|1\rangle) \otimes (c\,|0\rangle + d\,|1\rangle) = ac\,|00\rangle + ad\,|01\rangle + bc\,|10\rangle + bd\,|11\rangle$ against $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ requires $ac = \frac{1}{\sqrt{2}} \neq 0$, $ad = 0$, $bc = 0$, and $bd = \frac{1}{\sqrt{2}} \neq 0$. From $ac \neq 0$: both $a \neq 0$ and $c \neq 0$. From $bd \neq 0$: both $b \neq 0$ and $d \neq 0$. But then $ad = 0$ requires $a = 0$ or $d = 0$, both already shown nonzero — a contradiction. So no choice of a, b, c, d satisfies all four equations simultaneously: the Bell state cannot be written as a product state, confirming it is entangled.

Solution N.9 Since 6 Gaussians provide more variational freedom than 3 to match the target Slater shape, STO-6G would be expected to fit the true orbital *better* (a smaller RMS error) than STO-3G — consistent with the general variational principle that more basis functions can only improve (or leave unchanged) the quality of a fit. This improvement comes at the cost of increased computational expense: more Gaussian primitives per basis function means more two-electron integrals to evaluate in any subsequent Hartree–Fock or post-Hartree-Fock calculation (Section 17.1), trading accuracy for computational cost, exactly the tension already highlighted for Gaussian-versus-Slater basis functions themselves.

Solution N.10 One natural example: the superposition principle, used without comment for wavefunctions throughout the book since Chapter 3 (any linear combination of valid quantum states is itself a valid quantum state), is precisely the same principle that allows a qubit to exist in a superposition $\alpha\,|0\rangle + \beta\,|1\rangle$ (Section 17.4) and that underlies the linear variational method (Chapter 12) at the heart of modern computational chemistry (Section 17.1) — the identical mathematical structure, first needed to explain hydrogen's spectral lines, now doing service in both quantum computing and drug-discovery software.

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Solutions to the Comprehensive Review Problems

Solution N.1 The photon energy is $E_\gamma = hc/\lambda = 1239.84$ eV nm$/80.0$ nm $= 15.498$ eV. Since the ground-state ionization energy is 13.6057 eV (Chapter 8), the excess energy becomes the photoelectron's kinetic energy: $KE = 15.498 - 13.606 = 1.892$ eV, by exactly the same energy-conservation logic as Chapter 1's photoelectric effect (Eq. (1.5)), here applied to ionization from an atom rather than ejection from a metal surface.

Solution N.2 Using $m_\mu = 206.768\, m_e$ and the ordinary proton mass, $\mu_{\text{muonic}} = \dfrac{m_\mu m_p}{m_\mu + m_p} = 185.84\, m_e$ (much closer to m_μ than to m_e, since the muon and proton masses are now much closer in magnitude than the electron and proton were). Since $a_0 \propto 1/\mu$, the muonic Bohr radius is $a_{0,\mu} = a_0\,(m_e/\mu_{\text{muonic}}) = a_0/185.84 = 2.85 \times 10^{-13}$ m — about 186 times smaller than ordinary hydrogen, comparable to a nuclear radius. Since $E_1 \propto \mu$, the ground-state energy is $E_{1,\mu} = -13.6057 \times 185.84 = -2528.5$ eV, about 186 times more tightly bound than ordinary hydrogen — both results following directly from the reduced-mass scaling already established for the far smaller electron/proton correction in Chapters 1 and 8.

Solution N.3 $|\psi(x)|^2 = \psi^*\psi = N^2 e^{-ikx} e^{ikx} e^{-x^2/2a^2} e^{-x^2/2a^2} = N^2 e^{-x^2/a^2}$, since $e^{-ikx}e^{ikx} = e^0 = 1$ for any real k: the phase factor e^{ikx} cancels completely and $|\psi|^2$ is exactly the same real Gaussian regardless of k, exactly as the general argument of Chapter 3 (Note box on overall phase, there for a constant phase; here the "phase" varies with x but still cancels in $|\psi|^2$ against its own complex conjugate) would suggest. Normalizing $\int N^2 e^{-x^2/a^2} dx = N^2 a\sqrt{\pi} = 1$ gives $N = 1/\sqrt{a\sqrt{\pi}} = 1/(\pi^{1/4}\sqrt{a})$, identical to the real-Gaussian normalization of Chapter 3.

Solution N.4 Since $|\psi_n(x)|^2$ is symmetric about the box's midpoint for *every* n (Chapter 4, Problem 4.7's argument applies identically regardless of n), $\langle x\rangle_n = L/2$ for every n. So $E_n^{(1)} = \dfrac{V_0}{L}\langle x\rangle_n = \dfrac{V_0}{L}\cdot\dfrac{L}{2} = \dfrac{V_0}{2}$ for *every* quantum number n — every energy level of the box is shifted upward by exactly the same amount, $V_0/2$, to first order, so the perturbation does not change any of the transition energies between levels at all (to this order), only the overall energy scale.

Solution N.5 $\langle x^2\rangle_1 = \frac{3}{2}\hbar/m\omega$ and $\langle p^2\rangle_1 = \frac{3}{2}m\hbar\omega$ (both the $n = 1$ case of the general formulas, using $\langle x\rangle_1 = \langle p\rangle_1 = 0$ by the same parity argument as $n = 0$, so these are directly the variances). Then $\Delta x\,\Delta p = \sqrt{\frac{3}{2}\hbar/m\omega}\cdot\sqrt{\frac{3}{2}m\hbar\omega} = \frac{3}{2}\hbar$, compared to the Heisenberg bound $\hbar/2$: the ratio is 3, so the $n = 1$ state has *three times* the minimum allowed uncertainty product, and is *not* a minimum-uncertainty state (unlike $n = 0$, which saturates the bound exactly, Chapter 6, Problem 6.4) — only the QHO

ground state achieves the Gaussian shape required for equality in the underlying Cauchy–Schwarz argument of Chapter 9.

Solution N.6 $E_4 = -13.6057/16 = -0.8504$ eV, $E_3 = -13.6057/9 = -1.5117$ eV, so the emitted photon carries $E_3 - E_4$ ... using $|E_4 - E_3| = 0.6614$ eV, giving $\lambda = hc/0.6614 = 1874.6$ nm (near-infrared, part of the Brackett series). Since $4f$ has $\ell = 3$ and $3d$ has $\ell = 2$, $\Delta\ell = 2 - 3 = -1$, satisfying $\Delta\ell = \pm 1$ exactly: the transition is dipole-allowed, consistent with $|\mathbf{L}|$ changing from $\sqrt{12}\,\hbar = 3.464\hbar$ ($4f$) to $\sqrt{6}\,\hbar = 2.449\hbar$ ($3d$) as computed directly from Chapter 7's formula.

Solution N.7 Minimizing $E(\beta) = \beta^2/2 - \beta$: $dE/d\beta = \beta - 1 = 0 \Rightarrow \beta_{\text{opt}} = 1$ (in atomic units, i.e., $\beta_{\text{opt}} = 1/a_0$ restoring ordinary units), giving $E_{\text{trial}} = \frac{1}{2}(1)^2 - 1 = -\frac{1}{2}E_h = -13.6057$ eV, matching the exact hydrogen ground-state energy *exactly*. This agreement is exact, rather than merely close (as for the Gaussian trial function of Chapter 12, which achieved only about 85% of the true binding energy), precisely because $Ne^{-\beta r}$ has the same functional form as the true ground state $\psi_{100} \propto e^{-r/a_0}$ (Chapter 8): once the variational parameter is free to reach $\beta = 1/a_0$, the trial function *becomes* the exact ground state, and the variational theorem's inequality $\langle \hat{H} \rangle_{\text{trial}} \geq E_0$ is saturated with equality.

Solution N.8 (a) $\kappa = \sqrt{2m_e(3.0\,\text{eV})e}/\hbar = 8.874 \times 10^9$ m^{-1} (using the same numbers as Chapter 5's worked example, since $V_0 - E = 3.0$ eV here too), so $\kappa L = 8.874 \times 10^9 \times 0.30 \times 10^{-9} = 2.662$, giving $T \approx 16(0.25)(0.75)e^{-2(2.662)} = 3.0\,e^{-5.324} = 0.0146$. (b) $p = \sqrt{2m_e(1.0\,\text{eV})e} = 5.41 \times 10^{-25}$ kg m/s, giving $\lambda_{\text{dB}} = h/p = 1.226$ nm. Since $\lambda_{\text{dB}} = 1.226$ nm is more than *four times larger* than the barrier width (0.30 nm), the electron's wave character is not merely relevant but dominant on the length scale of the barrier — entirely consistent with (and indeed the underlying physical reason for) treating this as a genuinely quantum tunneling problem rather than a classical particle-scale interaction.

Solution N.9 By Hund's rules applied to the $2p^2$ configuration exactly as in Chapter 14's Worked Example (independent of which element happens to have this configuration), the ground term is $^3P_0$. The term symbol is a property of the *configuration* (how many electrons occupy which subshell, here 2 electrons in a 6-spin-orbital $2p$ subshell) via the Pauli-allowed microstate counting and Hund's rules, not of the specific nuclear charge or element identity — N^+ and neutral carbon share a ground term for exactly the same reason isoelectronic species generally share qualitative electronic structure.

Solution N.10 By direct analogy with N_2 (Chapter 15), 14 valence electrons filling the same molecular-orbital scheme give bond order 3. Using $\omega = 2\pi c\tilde{\nu}$ with $\tilde{\nu} = 2050$ cm^{-1}: $\omega = 2\pi(2.998 \times 10^{10})(2050) = 3.863 \times 10^{14}$ rad/s, so $k_{\text{spring}} = \mu\omega^2 = (1.15 \times 10^{-26})(3.863 \times 10^{14})^2 = 1717$ N/m — a large force constant, consistent with the strong (bond order 3) triple bond predicted from the electron count.

Solution N.11 Normalization: $|\alpha|^2 + |\beta|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 + \left|\frac{i}{\sqrt{2}}\right|^2 = \frac{1}{2} + \frac{1}{2} = 1$. ✓ Mapping $|0\rangle \rightarrow$

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$|\uparrow\rangle$ (eigenvalue $+\hbar/2$) and $|1\rangle \to |\downarrow\rangle$ (eigenvalue $-\hbar/2$): $\langle \hat{S}_z \rangle = |\alpha|^2 \left(+\frac{\hbar}{2}\right) + |\beta|^2 \left(-\frac{\hbar}{2}\right) = \frac{1}{2}\left(\frac{\hbar}{2}\right) + \frac{1}{2}\left(-\frac{\hbar}{2}\right) = 0$.

Solution N.12 Of the full QHO eigenstates (Chapter 6), only the odd-n ones ($\psi_1, \psi_3, \psi_5, \ldots$, built from odd Hermite polynomials) vanish at $x = 0$, exactly as required by the infinite wall there (an argument directly parallel to Chapter 4's boundary conditions). The lowest-energy state satisfying this extra restriction is therefore ψ_1 (not ψ_0, which is even and nonzero at $x = 0$, hence forbidden), with energy $E_1 = \left(1 + \frac{1}{2}\right)\hbar\omega = \frac{3}{2}\hbar\omega$: the ground state of this half-oscillator has *three times* the zero-point energy of the ordinary (unrestricted) QHO.

Solution N.13 Using $\Delta L_x \Delta L_y \geq \frac{\hbar}{2}|\langle \hat{L}_z \rangle|$ (Chapter 9) with $\langle L_z \rangle = m\hbar = (1)\hbar$ for the $m = 1$ state: $\Delta L_x \Delta L_y \geq \frac{\hbar}{2}(\hbar) = \frac{\hbar^2}{2}$, i.e., $0.5\,\hbar^2$.

Solution N.14 Using $E_{\text{trial}}(Z) = -(Z - 5/16)^2(27.2114\,\text{eV})$: $E(\text{He}, Z{=}2) = -77.489$ eV and $E(\text{Li}^+, Z{=}3) = -196.539$ eV. The percentage increase in (magnitude of) binding energy is $(196.539 - 77.489)/77.489 \times 100\% = 153.6\%$ — reflecting the strongly super-linear ($\propto Z^2$, approximately) growth of two-electron binding energy with nuclear charge, since both the kinetic energy scale and the (unscreened part of the) nuclear attraction grow with Z.

Solution N.15 Using $\mu \propto$ (as computed) 0.97960 u (^{35}Cl) versus 0.98108 u (^{37}Cl), and $\tilde{\omega}_e \propto 1/\sqrt{\mu}$: the ratio is $\sqrt{\mu_{35}/\mu_{37}} = 0.999243$, a fractional shift of -0.0757%, giving $\tilde{\omega}_e(^{37}\text{Cl}) \approx 2990.9 \times 0.999243 = 2988.6$ cm^{-1} — a small but experimentally resolvable isotope shift, by the same reduced-mass mechanism already used for the H/D shift of Chapter 8, Problem 8.10, and the H/D vibrational shift of Chapter 6, Problem 6.7.

Solution N.16 By Chapter 3, Problem 3.8's general result, the superposition oscillates at $\omega = (E_3 - E_1)/\hbar$. Using $E_n = n^2 E_1$ (Chapter 4): $E_3 - E_1 = 9E_1 - E_1 = 8E_1$, so $\omega = 8E_1/\hbar$.

Solution N.17 Although $n = 4$ is a higher principal quantum number than $n = 3$, a 4*s* electron ($\ell = 0$) has *no* centrifugal barrier (Chapter 8's effective-potential term $\ell(\ell+1)\hbar^2/2m_e r^2$ vanishes identically for $\ell = 0$), so its radial wavefunction, like the hydrogen 2*s* and 3*s* states before it, retains substantial amplitude very close to the nucleus despite its larger average radius overall. This inner-lobe penetration lets a 4*s* electron experience a larger, less-screened effective nuclear charge than a 3*d* electron (which does have a substantial centrifugal barrier, $\ell = 2$, keeping it further out on average), and it is exactly this extra binding from close-in penetration that can make 4*s* more tightly bound than 3*d* for the relevant range of Z (Chapter 14).

Solution N.18 From Appendix A, $R_\infty hc = 13.605693122990$ eV and $E_h = 27.211386245981$ eV, and indeed $2 \times 13.605693122990 = 27.211386245980 \approx 27.211386245981$ eV (equal to the precision tabulated). This factor of exactly 2 is expected because the hydrogen ground-state energy (Chapter 8) is $E_1 = -R_\infty hc$, while the virial theorem for a Coulomb potential (used implicitly throughout Chapters 8 and 12) gives

$\langle T\rangle = -E_1 = +R_\infty hc$ and $\langle V\rangle = 2E_1 = -2R_\infty hc$; the Hartree energy is defined as (twice the magnitude of the hydrogen ground-state energy), i.e., as the natural atomic-physics energy unit built directly from this same combination of fundamental constants, so $E_h = 2R_\infty hc$ by definition, not coincidence.

Solution N.19 The Franck–Condon principle's central claim — that an electronic transition happens too fast for the nuclei to move, so it occurs at essentially fixed nuclear geometry — presupposes that "the electronic state" and "the nuclear configuration" are separately meaningful, well-defined things in the first place, which is exactly what the Born–Oppenheimer approximation (Chapter 15) establishes by treating electronic structure as a function that depends parametrically on a fixed nuclear geometry. Without this prior separation of electronic and nuclear motion, there would be no sense in which nuclei could be said to be "too slow to respond" to a change in "the electronic state," since the two would not be separately defined quantities at all.

Solution N.20 Setting $Z_{\text{eff}}^2\,\text{Ry}\left(1-\frac{1}{4}\right) = 10{,}000$ eV: $Z_{\text{eff}}^2 = 10{,}000/(13.6057\times 0.75) = 980.1$, so $Z_{\text{eff}} \approx 31.3$. This is a physically reasonable value: elements with atomic number near 30 (zinc, gallium, germanium) do indeed produce characteristic X-ray emission lines in roughly this energy range, and an inner-shell $1s$ electron in such an atom is only lightly screened by the handful of other electrons at comparable or smaller radius, so an effective charge not far below the true $Z \approx 30$–32 is entirely sensible.

Solution N.21 $A^\dagger = \begin{pmatrix} 3 & 2-i \\ 2+i & 5 \end{pmatrix}^{*\,T} = \begin{pmatrix} 3 & (2+i) \\ (2-i) & 5 \end{pmatrix}^{T} = \begin{pmatrix} 3 & 2-i \\ 2+i & 5 \end{pmatrix} = A$ (conjugating every entry and transposing returns exactly the original matrix), confirming Hermiticity. The characteristic equation is $(3-\lambda)(5-\lambda) - (2-i)(2+i) = 0 \Rightarrow \lambda^2 - 8\lambda + 15 - 5 = 0 \Rightarrow \lambda^2 - 8\lambda + 10 = 0$ (using $(2-i)(2+i) = 4+1 = 5$), giving $\lambda = 4 \pm \sqrt{6}$, both manifestly real, as Chapter 9's theorem guarantees for any Hermitian matrix regardless of whether its individual entries are complex.

Solution N.22 The ground state is $(n_x, n_y, n_z) = (1,1,1)$, with $E_{111} = 3\pi^2\hbar^2/2mL^2$. The first excited level is shared by three distinct combinations, $(2,1,1)$, $(1,2,1)$, and $(1,1,2)$, all giving $n_x^2 + n_y^2 + n_z^2 = 6$: a three-fold degeneracy. This degeneracy arises directly from the cube's equal side lengths in all three directions: permuting which Cartesian direction carries the "excited" quantum number $n = 2$ costs identical energy by the symmetry of the box, exactly the same mechanism (equivalent directions related by a symmetry operation must share energy) underlying the angular (m-value) degeneracy of hydrogen in Chapter 8, now realized through the cube's discrete permutation symmetry rather than hydrogen's continuous rotational symmetry.

Solution N.23 $E_0^{(1)} = \lambda\cdot\frac{3}{4}\left(\frac{\hbar}{m\omega}\right)^2(1) = \frac{3}{4}\lambda\left(\frac{\hbar}{m\omega}\right)^2$. $E_1^{(1)} = \lambda\cdot\frac{3}{4}\left(\frac{\hbar}{m\omega}\right)^2(2+2+1) = \frac{15}{4}\lambda\left(\frac{\hbar}{m\omega}\right)^2$. The ratio is $E_1^{(1)}/E_0^{(1)} = 15/3 = 5$: the first excited state's energy shift from this quartic perturbation is five times larger than the ground state's, reflecting the excited state's larger spatial extent (larger $\langle x^4\rangle$) under the QHO's growing wavefunction width with increasing n.

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Solution N.24 With $\ell = 1$ and $s = \frac{1}{2}$: j ranges from $|\ell - s| = \frac{1}{2}$ to $\ell + s = \frac{3}{2}$ in integer steps, giving $j = \frac{1}{2}$ or $j = \frac{3}{2}$ — the two possible total-angular-momentum states of a hydrogen $2p$ electron, labeled (in standard atomic spectroscopic notation, an extension of the term-symbol idea of Chapter 14 to a single electron) $2p_{1/2}$ and $2p_{3/2}$.

Solution N.25 $\hat{L}_z \hat{z}\, f = -i\hbar \dfrac{\partial}{\partial \phi}(r\cos\theta\, f) = r\cos\theta \left(-i\hbar \dfrac{\partial f}{\partial \phi}\right) = \hat{z}\, \hat{L}_z f$ for any function f, since $z = r\cos\theta$ carries no ϕ-dependence at all and can be pulled through the ϕ-derivative freely. So $[\hat{L}_z, \hat{z}] = 0$ identically, confirming L_z and z are compatible observables — exactly why Appendix J could freely label the Stark-effect basis states by definite m while simultaneously treating $\hat{z}$ as an ordinary operator acting within that same fixed-m classification.

Solution N.26 Since $\psi_2(x) = \sqrt{2/L}\sin(2\pi x/L)$, the relevant integral is $E_2^{(1)} = \int_0^L V_0 \sin(\pi x/L) \cdot \frac{2}{L}\sin^2(2\pi x/L)\, dx$, which is a different integral from the one used for $E_1^{(1)}$, not restated by the quoted vanishing integral (that vanishing integral instead reflects a symmetry fact about cross terms, not the diagonal $n = 2$ matrix element itself). Direct evaluation (by the same method as Chapter 11's own Worked Example, substituting $n = 2$ throughout) gives $E_2^{(1)} = \dfrac{2V_0}{L}\displaystyle\int_0^L \sin(\pi x/L)\sin^2(2\pi x/L)\, dx = \dfrac{32V_0}{15\pi}$ (a standard trigonometric integral, evaluated the same way as the text's $n = 1$ case), distinctly different in both value and functional form from the $n = 1$ result $8V_0/3\pi$.

Solution N.27 By Chapter 9's theorem (Key Result, Section 9.2), a Hermitian operator or matrix always has real eigenvalues; since H_{AA}, H_{AB}, H_{BB} are all real (hence the matrix equals its own conjugate transpose, satisfying the Hermiticity condition trivially, exactly as in Problem F.22 above), the bonding and antibonding energies — its eigenvalues — are guaranteed real without needing to solve the specific 2×2 characteristic equation explicitly, exactly the same general guarantee already invoked for the Chapter 2 eigenvalue examples and the Stark-effect matrix of Appendix J.

Solution N.28 Using $E_v/hc = \tilde{\omega}_e(v + \frac{1}{2}) - \tilde{\omega}_e x_e (v + \frac{1}{2})^2$: $\tilde{\nu}_{1\to 2} = \tilde{\omega}_e - 4\tilde{\omega}_e x_e = 2990.9 - 4(52.8) = 2779.7\ \text{cm}^{-1}$, compared to the fundamental $\tilde{\nu}_{0\to 1} = 2885.3\ \text{cm}^{-1}$ (Chapter 16). The hot band lies at lower wavenumber, consistent with the Morse potential's converging level spacing (each successive spacing $E_{v+1} - E_v$ shrinks as v increases, Chapter 16, Section 16.1).

Solution N.29 $\hat{X}^\dagger \hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1$ (using $\hat{X}^\dagger = \hat{X}$, since $\hat{X}$ is real and symmetric), confirming unitarity. $\hat{X}\,|0\rangle = \hat{X}\binom{1}{0} = \binom{0}{1} = |1\rangle$, and $\hat{X}\,|1\rangle = \hat{X}\binom{0}{1} = \binom{1}{0} = |0\rangle$: the gate exactly swaps $|0\rangle \leftrightarrow |1\rangle$, the quantum analog of classical logical negation.

Solution N.30 As $V_0 \to \infty$ (Chapter 5's Note box on the infinite-well limit), the finite well's bound-state energies approach the infinite-well values, but for any *finite* V_0, the wavefunction is not forced to vanish exactly at the walls (as the infinite well requires) but

instead decays smoothly into the classically forbidden region just outside — effectively, the particle is confined to a region slightly *larger* than the nominal well width. Since the infinite-well energy $E_n \propto 1/L^2$ decreases with increasing effective width, this slight effective widening makes each finite-well bound-state energy (measured from the well bottom) come out slightly *below* the corresponding infinite-well value at the same nominal L, converging to exact equality only in the strict $V_0 \to \infty$ limit.

Solution N.31 A $3d$ subshell holds $2(2\ell+1) = 2(5) = 10$ electrons (Chapter 10's Pauli-principle counting, generalizing the 2-per-orbital, 5 orbitals result). This matches exactly the ten elements Sc ($Z = 21$) through Zn ($Z = 30$), each adding one more $3d$ electron in turn (with the Cr and Cu reconfigurations of Appendix C reflecting the extra stability of half-filled $3d^5$ and filled $3d^{10}$ subshells, Chapter 14, Section 14.3, rather than any violation of the basic 10-electron capacity itself).

Solution N.32 Hartree–Fock minimizes the expectation value $\langle \hat{H} \rangle$ (Chapter 12's variational quantity, Section 12.1) over the space of trial wavefunctions restricted to single Slater determinants (Chapter 10, Section 10.5) built from adjustable one-electron orbitals — i.e., it is the linear (or, more precisely here, orbital-shape) variational method of Chapter 12, applied with a specific, restricted family of many-electron trial wavefunctions (single determinants) rather than an arbitrary trial function, which is exactly why Hartree–Fock energies obey the same $E_{\text{HF}} \geq E_{\text{exact}}$ bound established generally in Chapter 12 and used explicitly in Chapter 17, Problem 17.4.

Solution N.33 $\sum_{\ell=0}^{4}(2\ell+1) = 1+3+5+7+9 = 25 = 5^2$, confirming the $n = 5$ degeneracy directly.

Solution N.34 An f subshell ($\ell = 3$) holds $2(2\cdot 3+1) = 2(7) = 14$ electrons, exactly matching the 14-element length of the lanthanide series (in which the $4f$ subshell is progressively filled, one electron per element, following the Aufbau order of Chapter 14).

Solution N.35 $\omega = \sqrt{k_{\text{spring}}/m} = \sqrt{5.0/0.020} = 15.81$ rad/s, giving $E_0 = \frac{1}{2}\hbar\omega = 8.34 \times 10^{-34}$ J $= 5.20 \times 10^{-15}$ eV — an almost inconceivably tiny energy. The equivalent temperature is $E_0/k_B = 6.04 \times 10^{-11}$ K, some 13 orders of magnitude below room temperature (300 K). Any macroscopic oscillator's zero-point energy is utterly swamped by ordinary thermal energy at any accessible temperature (and is, in any case, far too small to measure by any means), which is exactly why zero-point motion — a real, exactly-derived quantum effect (Chapter 6) — is observable only for microscopic systems (molecular vibrations, trapped ions) with far higher characteristic frequencies ω, never for a spring you could hold in your hand.

Solution N.36 $\hat{Z}^\dagger\hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1$, confirming unitarity (indeed $\hat{Z}^\dagger = \hat{Z}$ here, so this also confirms Hermiticity). Its eigenvalues are read directly off the diagonal (already diagonal in this basis): $+1$ and -1. Direct matrix multiplication gives $\hat{Z}\,|0\rangle = \hat{Z}\binom{1}{0} = \binom{1}{0} = (+1)\,|0\rangle$ and $\hat{Z}\,|1\rangle = \hat{Z}\binom{0}{1} = \binom{0}{-1} = (-1)\,|1\rangle$: $|0\rangle, |1\rangle$ are

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themselves the eigenstates of $\hat{Z}$, exactly paralleling $\hat{S}_z|\uparrow\rangle = +\frac{\hbar}{2}|\uparrow\rangle$, $\hat{S}_z|\downarrow\rangle = -\frac{\hbar}{2}|\downarrow\rangle$ from Chapter 10, Problem 10.2, up to the eigenvalue normalization convention.

Solution N.37 Expanding $\langle c_1\psi_A + c_2\psi_B|\,\hat{H}\,|c_1\psi_A + c_2\psi_B\rangle$ using $\hat{H}\psi_A = E_0\psi_A$ and $\hat{H}\psi_B = E_0\psi_B$ (both exact eigenstates of the same eigenvalue) gives $\hat{H}(c_1\psi_A + c_2\psi_B) = E_0(c_1\psi_A + c_2\psi_B)$ exactly (linearity, Chapter 3, Problem 3.10), so $\langle\hat{H}\rangle = E_0\langle c_1\psi_A + c_2\psi_B|c_1\psi_A + c_2\psi_B\rangle = E_0(1) = E_0$ using normalization. Since *every* normalized combination of exact, degenerate eigenstates is itself an exact eigenstate with the same eigenvalue (not merely an approximation to one), the variational inequality $\langle\hat{H}\rangle \geq E_0$ is satisfied with equality for the entire family of combinations, not just the two originally given basis states — illustrating concretely why Chapter 11 flagged degenerate subspaces as requiring special care: within such a subspace, there is no unique "correct" unperturbed state, only an entire equivalence class of equally valid ones, exactly the freedom degenerate perturbation theory (Appendix J) must resolve using the perturbation itself.

Solution N.38 Both the particle in a box ($E_1 = \pi^2\hbar^2/2mL^2$, Chapter 4) and the QHO ($E_0 = \frac{1}{2}\hbar\omega$, Chapter 6) have strictly positive ground-state energy, and both trace back to the same uncertainty-principle argument (Chapter 1, Section 1.6): confining a particle in space, whether by hard walls or a smooth parabolic potential, forces a nonzero momentum spread, hence nonzero kinetic energy, even in the lowest-energy state. Using $\sum_{\ell=0}^{5}(2\ell+1) = 1+3+5+7+9+11 = 36 = 6^2$, confirm the hydrogen $n = 6$ degeneracy directly by summation, exactly as done for $n = 5$ in an earlier problem, and note that hydrogen's zero-point energy language is somewhat different in character: unlike the box or oscillator, the hydrogen ground state ($n = 1$) is not usually described as having "zero-point energy," since $E_1 = -13.6057$ eV is already the physically meaningful, exactly bound ground state, not a small correction above a classical zero. (Chapters 1, 4, 6, 8.)

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Solutions to the WKB Appendix Problems

Solution N.1 With $p(x) = \sqrt{2mE}$ constant, $\int_0^L p\,dx = L\sqrt{2mE}$. Setting this equal to $n\pi\hbar$: $L\sqrt{2mE} = n\pi\hbar \Rightarrow \sqrt{2mE} = n\pi\hbar/L \Rightarrow E = \dfrac{n^2\pi^2\hbar^2}{2mL^2}$, exactly the Chapter 4 result.

Solution N.2 With $V(x) = V_0$ constant inside the barrier, $\sqrt{2m[V(x)-E]} = \sqrt{2m(V_0-E)} = \hbar\kappa$ is itself constant, so $G = \dfrac{2}{\hbar}\displaystyle\int_0^L \hbar\kappa\,dx = 2\kappa\int_0^L dx = 2\kappa L$, matching Chapter 5's exponent exactly.

Solution N.3 Direct evaluation of the Gamow-factor formula at $E_\alpha = 6.0$ MeV gives $G = 62.73$ (compared to $G = 90.18$ at 4.198 MeV), so $T = e^{-62.73} = 5.74 \times 10^{-28}$, larger than the 4.198 MeV case by a factor of about 8.4×10^{11} — more than eleven orders of magnitude. This enormous sensitivity to alpha energy (a 43% increase in E_α producing a twelve-order-of-magnitude change in tunneling probability) is exactly the mechanism behind the empirical Geiger–Nuttall law relating alpha-decay half-life very steeply to decay energy, and it is a direct, quantitative consequence of G's dependence on E through both the prefactor $\sqrt{E}$ and, more importantly, the strongly E-dependent upper limit $b = Z_1 Z_2 e^2/4\pi\epsilon_0 E$ of the tunneling integral itself.

Solution N.4 For a flat barrier, $V(x) - E = V_0 - E$ is the same constant at every point inside $0 < x < L$, so the integrand $\sqrt{2m[V(x)-E]}$ in the definition of G does not vary with x at all, and the integral $\int_0^L \sqrt{2m[V(x)-E]}\,dx$ is simply this constant integrand multiplied by the interval length L — no substitution, series expansion, or coordinate change is needed, unlike the Coulomb case (where $V(r) - E$ varies continuously with r across the tunneling region and the resulting integral requires the trigonometric substitution alluded to in the text). This is exactly why WKB, ordinarily only an approximation, reduces to Chapter 5's rectangular-barrier result with no additional error introduced: the "slowly varying potential" assumption underlying WKB is satisfied in the strongest possible sense (a potential that does not vary at all) for this particular shape.

Solution N.5 Since the tunneling probability $T = e^{-G}$ depends on the alpha-particle energy through the Gamow factor G, and G decreases sharply as E_α increases (an eleven-order-of-magnitude increase in T for a modest 43% increase in E_α, as found directly in Problem H.3), the decay rate $\lambda_{\text{decay}} = fT$ increases correspondingly steeply with E_α, and the half-life $t_{1/2} = \ln 2/\lambda_{\text{decay}}$ falls just as steeply. This single formula — nothing more than the WKB tunneling probability applied to a Coulomb barrier — reproduces the empirical Geiger–Nuttall observation (higher-energy alpha emitters decay faster) as a direct, quantitative mathematical consequence, rather than as a separate empirical rule requiring its own explanation.

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Solutions to the Group Theory Appendix Problems

Solution N.1 C_2 (rotation by 180° about z) sends $(x, y, z) \to (-x, -y, z)$, represented by the matrix $\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Its trace is $-1 - 1 + 1 = -1$, matching $\chi(C_2) = -1$ exactly, as used in Section I.3.

Solution N.2 $3N - 6 = 3(4) - 6 = 6$ vibrational modes for ammonia (a nonlinear, 4-atom molecule).

Solution N.3 From the character table, A_1 matches z and both B_1, B_2 match x, y respectively — all three are infrared-active. Only A_2 matches none of x, y, z: a mode of A_2 symmetry (were one present, in whichever molecule) could never be infrared-active in a C_{2v} molecule, regardless of its vibrational frequency, since no component of the dipole moment operator transforms as A_2.

Solution N.4 An infrared transition requires a nonzero dipole transition matrix element (Chapter 13, Section 13.3), which in turn requires the vibrational mode's symmetry to match that of at least one Cartesian component of the dipole operator (Key Result of Section I.3) — otherwise the relevant integral is forced to vanish by symmetry alone, exactly as the explicit $\Delta\ell = \pm 1$ spherical-harmonic integral of Chapter 13 vanished for a symmetry-mismatched case. Since A_2 appears nowhere in the "linear functions" column of the C_{2v} character table (only A_1, B_1, B_2 do, matching z, x, y), no dipole component can drive a transition into or out of an A_2-symmetry vibrational level, making any such mode infrared-inactive on symmetry grounds alone.

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Solutions to the Ehrenfest and Stark Effect Appendix Problems

Solution N.1 Differentiating $\langle \hat{A} \rangle = \int \Psi^* \hat{A} \Psi \, dx$ (allowing $\hat{A}$ its own explicit time dependence) with the product rule gives three terms: $\frac{d\langle \hat{A} \rangle}{dt} = \int \frac{\partial \Psi^*}{\partial t} \hat{A} \Psi \, dx + \int \Psi^* \frac{\partial \hat{A}}{\partial t} \Psi \, dx + \int \Psi^* \hat{A} \frac{\partial \Psi}{\partial t} \, dx$. Using $i\hbar \, \partial \Psi / \partial t = \hat{H} \Psi$ (so $\partial \Psi / \partial t = \hat{H} \Psi / i\hbar$) and its complex conjugate $\partial \Psi^* / \partial t = -\hat{H} \Psi^* / i\hbar$ (using $\hat{H}$ Hermitian, hence its eigenvalues real, Chapter 9, so conjugating flips the sign of i but not $\hat{H}$ itself) in the first and third terms:

$$\frac{d\langle \hat{A} \rangle}{dt} = -\frac{1}{i\hbar} \int (\hat{H}\Psi)^* \hat{A} \Psi \, dx + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle + \frac{1}{i\hbar} \int \Psi^* \hat{A} \hat{H} \Psi \, dx.$$

Using Hermiticity of $\hat{H}$ (Chapter 9) to rewrite $\int (\hat{H}\Psi)^* \hat{A} \Psi \, dx = \int \Psi^* \hat{H} \hat{A} \Psi \, dx$, the first and third terms combine to $\frac{1}{i\hbar} \int \Psi^* (\hat{A}\hat{H} - \hat{H}\hat{A}) \Psi \, dx = \frac{1}{i\hbar} \langle [\hat{A}, \hat{H}] \rangle = \frac{i}{\hbar} \langle [\hat{H}, \hat{A}] \rangle$ (using $1/i = -i$ and $[\hat{A}, \hat{H}] = -[\hat{H}, \hat{A}]$), giving exactly $\frac{d\langle \hat{A} \rangle}{dt} = \frac{i}{\hbar} \langle [\hat{H}, \hat{A}] \rangle + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle$, as claimed.

Solution N.2 Since $V(x) = \frac{1}{2} m\omega^2 x^2$ is an exact quadratic function of x, its derivative $dV/dx = m\omega^2 x$ is an exact *linear* function of x, so $\langle dV/dx \rangle = m\omega^2 \langle x \rangle$ holds with no averaging error whatsoever (the expectation value of a linear function of x is simply the same linear function evaluated at $\langle x \rangle$, unlike for any higher-power or non-polynomial $V(x)$). Substituting into Ehrenfest's theorem, $\frac{d\langle p \rangle}{dt} = -\langle dV/dx \rangle = -m\omega^2 \langle x \rangle$, and using $\langle p \rangle = m \, d\langle x \rangle / dt$ from the other Ehrenfest relation, $m \frac{d^2\langle x \rangle}{dt^2} = -m\omega^2 \langle x \rangle$, i.e., $\frac{d^2\langle x \rangle}{dt^2} = -\omega^2 \langle x \rangle$ — exactly the classical simple-harmonic-motion equation, holding exactly for $\langle x \rangle(t)$ in *any* quantum state of the QHO, stationary or not.

Solution N.3 The total splitting (between the $E_2 + 3e\mathcal{E}a_0$ and $E_2 - 3e\mathcal{E}a_0$ sublevels) is $6e\mathcal{E}a_0$. Numerically, with $\mathcal{E} = 1.0 \times 10^7$ V/m: $6e\mathcal{E}a_0 = 6(1.602 \times 10^{-19})(1.0 \times 10^7)(5.29177 \times 10^{-11}) = 5.087 \times 10^{-22}$ J $= 3.18 \times 10^{-3}$ eV — small compared to the $n = 1 \to 2$ transition energy (10.2 eV) but easily resolvable spectroscopically, and indeed historically among the first quantitative confirmations of early quantum theory (the Stark effect was observed by Johannes Stark in 1913).

Solution N.4 Both $2p_{+1}$ ($m = +1$) and $2p_{-1}$ ($m = -1$) fail the $\Delta m = 0$ selection rule required for a nonzero matrix element of $\hat{z}$ connecting them to $2s$ ($m = 0$) or $2p_0$ ($m = 0$), and $\langle 2p_{+1} | \, \hat{z} \, | 2p_{-1} \rangle$ itself also vanishes by the same $\Delta m = 0$ rule (since m changes by 2,

not 0, between these two states). With every matrix element of $\hat{H}'$ connecting $2p_{\pm 1}$ to any other $n = 2$ state equal to zero, the 2×2 perturbation sub-block for $\{2p_{+1}, 2p_{-1}\}$ is exactly the zero matrix, giving a first-order energy shift of exactly zero for both — they remain exactly degenerate with each other (and with the unperturbed E_2) at this order.

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Solutions to the Density Matrix Appendix Problems

Solution N.1 $\mathrm{Tr}(\hat{\rho}\hat{A}) = \sum_n \langle n|\rangle\, \psi\, \langle\psi|\, \hat{A}\, |n\rangle$. Since $\langle n|\rangle\, \psi$ and $\langle\psi|\, \hat{A}\, |n\rangle$ are both just ordinary (complex) numbers, they commute past each other freely, so this equals $\sum_n \langle\psi|\, \hat{A}\, |n\rangle\, \langle n|n\rangle\, \psi = \langle\psi|\, \hat{A}\, (\sum_n |n\rangle\, \langle n|)\, |\psi\rangle = \langle\psi|\, \hat{A}(1)\, |\psi\rangle = \langle\psi|\, \hat{A}\, |\psi\rangle$, using completeness $\sum_n |n\rangle\, \langle n| = 1$ in the second-to-last step.

Solution N.2 $\mathrm{Tr}(\hat{\rho}_a \hat{S}_z) = \mathrm{Tr}\left[\frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix} \cdot \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\right] = \mathrm{Tr}\left[\frac{\hbar}{4}\begin{pmatrix}1 & -1\\ 1 & -1\end{pmatrix}\right] = \frac{\hbar}{4}(1-1) = 0$. $\mathrm{Tr}(\hat{\rho}_b \hat{S}_z) = \mathrm{Tr}\left[\frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \cdot \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\right] = \mathrm{Tr}\left[\frac{\hbar}{4}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\right] = \frac{\hbar}{4}(1-1) = 0$. Both vanish, as claimed.

Solution N.3 $\mathrm{Tr}(\hat{\rho}_a \hat{S}_x) = \mathrm{Tr}\left[\frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix} \cdot \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\right] = \mathrm{Tr}\left[\frac{\hbar}{4}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}\right] = \frac{\hbar}{4}(1+1) = \frac{\hbar}{2}$, confirming $\langle \hat{S}_x\rangle_a = \hbar/2$. For the mixed state, $\mathrm{Tr}(\hat{\rho}_b \hat{S}_x) = \mathrm{Tr}\left[\frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \cdot \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\right] = \mathrm{Tr}\left[\frac{\hbar}{4}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\right] = 0$ (the diagonal of this product matrix is identically zero), confirming $\langle \hat{S}_x\rangle_b = 0$ exactly.

Solution N.4 $\hat{\rho}^2 = \begin{pmatrix}0.8 & 0\\ 0 & 0.2\end{pmatrix}\begin{pmatrix}0.8 & 0\\ 0 & 0.2\end{pmatrix} = \begin{pmatrix}0.64 & 0\\ 0 & 0.04\end{pmatrix}$, so $\mathrm{Tr}(\hat{\rho}^2) = 0.64 + 0.04 = 0.68 < 1$, confirming this is a genuinely mixed state (a pure state would require $\mathrm{Tr}(\hat{\rho}^2) = 1$ exactly).

Solution N.5 If $\hat{\rho} = \sum_i p_i\, |E_i\rangle\, \langle E_i|$ is diagonal in the energy eigenbasis, then $\hat{H}\hat{\rho} = \sum_i p_i E_i\, |E_i\rangle\, \langle E_i| = \hat{\rho}\hat{H}$ (since $\hat{H}\, |E_i\rangle = E_i\, |E_i\rangle$ and multiplying by the same scalar E_i commutes trivially), so $[\hat{H}, \hat{\rho}] = 0$ identically. By the von Neumann equation, $i\hbar\, d\hat{\rho}/dt = [\hat{H}, \hat{\rho}] = 0$, so $\hat{\rho}$ is exactly constant in time — a direct generalization of the single-stationary-state result (Chapter 3) that $|\Psi(x,t)|^2$ does not depend on t, now applying to an entire statistical mixture of stationary states rather than just one.

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Solutions to the Free Electron Gas Appendix Problems

Solution N.1 $E_F = \dfrac{\hbar^2}{2m_e}(3\pi^2 n)^{2/3}$ with $n = 2.65 \times 10^{28}$ m^{-3} gives $E_F = 5.19 \times 10^{-19}$ J $= 3.24$ eV, smaller than copper's 7.03 eV, consistent with sodium's lower conduction-electron density.

Solution N.2 $T_F = E_F/k_B = (5.19 \times 10^{-19})/(1.380649 \times 10^{-23}) = 37{,}600$ K, far above room temperature (≈ 300 K), just as for copper — confirming that the qualitative argument of Section L.2 (only a small fraction of electrons thermally accessible near E_F) applies to sodium as well, despite its noticeably lower Fermi energy than copper.

Solution N.3 Since $E_F \propto n^{2/3}$, doubling n multiplies E_F by $2^{2/3} = 1.587$. Because the exponent $2/3$ is less than 1, the Fermi energy is only *mildly* sensitive to changes in electron density — a factor-of-2 change in density produces less than a factor-of-1.6 change in E_F, a sub-linear response.

Solution N.4 Without the Pauli exclusion principle, every electron in a classical gas at $T = 0$ could occupy the single lowest-energy state simultaneously (nothing forbids multiple identical classical particles from sharing one state), giving zero total kinetic energy in the ground configuration. The Pauli principle (Chapter 10) forbids this outright: at most 2 electrons (opposite spin) can occupy the lowest spatial level, forcing every additional electron into successively higher-energy levels even at absolute zero, so the total ground-state kinetic energy of the Fermi sea is a direct, unavoidable consequence of antisymmetry — with no classical counterpart, exactly as the zero-point energy of a single confined particle (Chapters 4, 6) has none either.

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Solutions to the Non-Rigid Rotor Appendix Problems

Solution N.1 $D_{CO} = 4(1.9225)^3/(2143.0)^2 = 6.19 \times 10^{-6}$ cm^{-1}. The ratio $D/B = 3.22 \times 10^{-6}$ for CO, smaller than HCl's $D/B = 5.02 \times 10^{-5}$ by more than an order of magnitude, consistent with CO's much larger force constant (Chapter 15, $k_{\text{spring}} = 1857$ N/m for CO versus 480 N/m for HCl) making it a stiffer, less distortable bond.

Solution N.2 Rigid-rotor $J = 0 \rightarrow 1$: $2B(1) = 21.186$ cm^{-1}. Non-rigid: using $D = 5.315 \times 10^{-4}$ cm^{-1}, the corrected value is 21.184 cm^{-1}, a difference of only 0.0021 cm^{-1} — more than 200 times smaller than the 0.459 cm^{-1} difference found for $J = 5 \rightarrow 6$ in the text, consistent with the distortion term's steep $J^2(J+1)^2$ growth (a factor of roughly $(6 \cdot 7)^2/(1 \cdot 2)^2 \approx 441$ between the two cases, in reasonable accord with the observed ratio).

Solution N.3 Classically, centrifugal stretching increases the effective bond length (and hence moment of inertia) for *any* nonzero rotation rate, regardless of the direction of rotation — a spinning mass on a spring always stretches outward, never compresses inward, since the centrifugal tendency always acts to increase the radius. A larger moment of inertia at fixed angular momentum L corresponds to a *lower* rotational energy ($E_{\text{rot}} = L^2/2I$, decreasing with increasing I), so the correction $-DJ^2(J+1)^2$ must always lower the energy relative to the rigid-rotor value, requiring $D > 0$ for any real, physically stretchable bond.

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$$Q = \sum_i g_i e^{-E_i/k_B T}$$

BASIC THERMODYNAMICS

$$dU = TdS - PdV$$

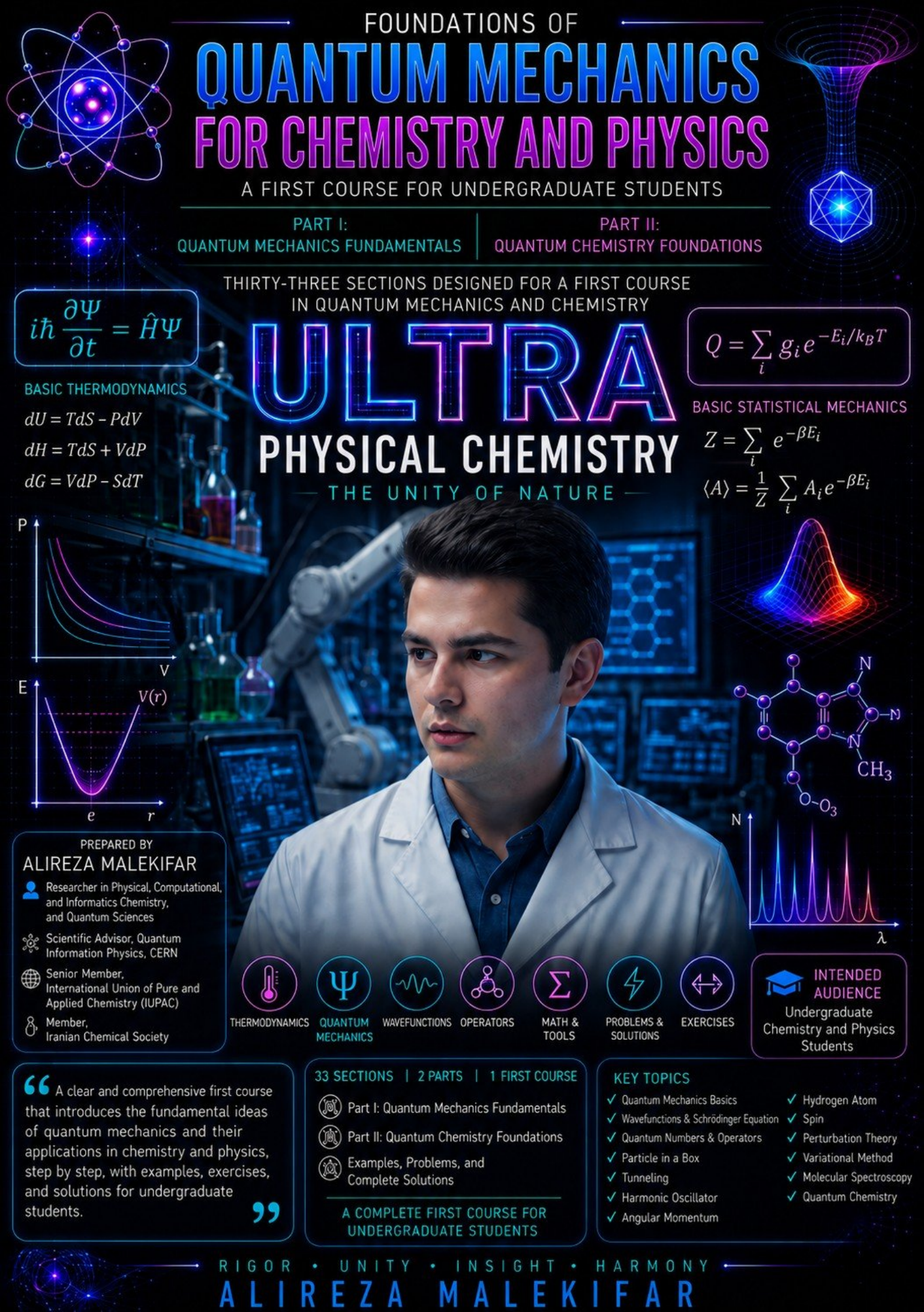
$$dH = TdS + VdP$$

$$dG = VdP - SdT$$

BASIC STATISTICAL MECHANICS

$$Z = \sum_i e^{-\beta E_i}$$

$$\langle A \rangle = \frac{1}{Z} \sum_i A_i e^{-\beta E_i}$$



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